THE COMPLETE

Nursing School BUNDLE®

2024 EDITION

NURSE IN THE MAKING®

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NURSE IN THE MAKING® ICONS TO LOOK FOR:

MEMORY TRICK
Memory tricks help you remember information in a different way. These will help jog your memory by providing another way to recall the information.

STARRED
The NCLEX & nursing exams like to ask about these. So be sure to remember the starred sections!

MONITOR
This is a reminder to monitor symptoms, especially after certain surgeries or in an acute stage. It is especially important as the nurse to look for these symptoms.

EDUCATE
You as the nurse will need to educate your patient in areas such as medication regimens, lifestyle changes/ modifications, and much more!

REPORT
You as the nurse will need to know when to report certain things to the health care provider (HCP).

DIET MODIFICATIONS
A patient may be asked to follow a certain diet depending on their condition. This icon is a reminder that you should know what diet modifications the patient should follow.

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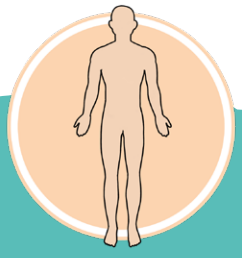

Head-to-Toe Assessment

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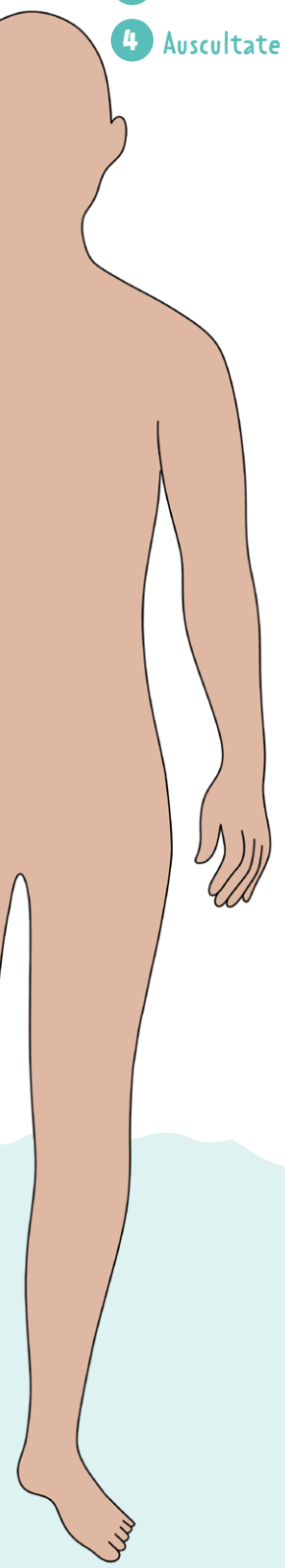
Head-to-Toe Assessment

1. Inspect
2. Palpate
3. Percuss
4. Auscultate



INTRODUCTION

- Knock
- Introduce yourself
- Wash hands
- Provide privacy
- Verify patient's ID using 2 patient identifiers (name & date of birth)
- Explain what you are doing (using non-medical language)

ORIENTATION

- What is your name?
- Do you know where you are?
- Do you know what month it is?
- Who is the current president?
- Do you know what brought you here?
- **A&O X4** = oriented to person, place, time, and situation

"NORMAL" VITAL SIGNS

- **HEART RATE (HR):** 60–100 bpm
- **BLOOD PRESSURE (BP):** 120/80 mmHg
- **OXYGEN SATURATION (SPO_2):** 95–100%
- **TEMPERATURE (T):** 97.8°F–99°F (36.5°C–37.2°C)
- **RESPIRATORY RATE (RR):** 12–20 breaths/min

HEAD & FACE

HEAD

- Inspect head/scalp/hair
- Palpate head/scalp/hair

EYES

- Inspect external eye structures
- Inspect color of conjunctiva and sclera
- **PERRLA**
  - **P**upils
  - **E**qual
  - **R**ound,
  - **R**eactive to **L**ight,
  - & **A**ccommodation

FACE

- Inspect
- Check for symmetry
  - To assess **CRANIAL NERVE VII**, check...

VII: FACIAL

- Raise eyebrows
- Smile
- Frown
- Show teeth
- Puff out cheeks
- Tightly close eyes

EARS

- Symmetry, drainage/discharge, pain, hearing deficits

NOSE

- Symmetry, patency, drainage/discharge, presence of deviated septum

MOUTH

- Color of mucous membranes, moist/dry, lesions, abnormal dental findings, abnormal breath odors

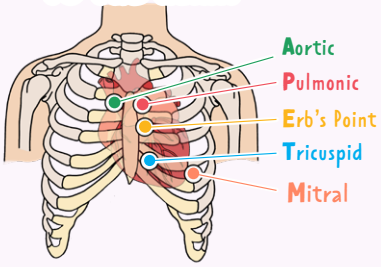
NECK

- Inspect and palpate
- Palpate carotid pulse
- Check skin turgor (under clavicle)

HEART

- Auscultate heart sounds (**A, P, E, T, M**) with diaphragm and bell
  - *Note any murmurs, whooshing, bruits, or muffled heart sounds*

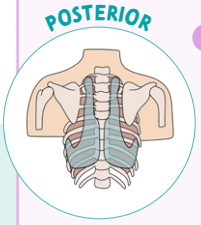
5 AREAS FOR LISTENING TO THE HEART



MNEMONIC

All
People
Enjoy
Time
Magazine

POSTERIOR



POSTERIOR CHEST

- Inspect
- Auscultate lung sounds in posterior and lateral chest
  - *Note any crackles or diminished breath sounds*
- Auscultate in a sequence to compare lungs (right upper, left upper, right middle, left middle, etc.)

ANTERIOR

ANTERIOR CHEST

- Inspect:
  - Use of accessory muscles
  - AP to transverse diameter
  - Sternum configuration
- Palpate: symmetric expansion
- Auscultate lung sounds: anterior and lateral
  - *Note any crackles or diminished breath sounds*

Assess the depth of the respirations:

EFFORT	RHYTHM
Note if it's **labored** or **unlabored**	Note if it's **regular** or **irregular**

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Head-to-Toe Assessment

PERIPHERALS

UPPER EXTREMITIES

- Inspect and palpate
  - *Note any texture, lesions, temperature, moisture, tenderness, & swelling*
- Palpate radial pulses bilaterally

PULSE SCALE:	
0	Pulse is absent
1+	Diminished
2+	Normal
3+	Full
4+	Bounding, strong

SHOULDER

- Inspect, palpate, assess

ELBOWS

- Inspect, palpate, assess

HANDS & FINGERS

- Inspect hands/fingers/nails
- Palpate hands and finger joints
- Check muscle strength of hands bilaterally
  - *Does each hand grip evenly?*

SKIN

- Assess skin turgor by pinching or lifting the skin

INTERPRETATION: If the skin stays elevated and does not return to its original place quickly, it indicates the patient is **dehydrated**

SPINE

- Have the client stand up (if able)
- Inspect the skin on the back
- Inspect spinal curvature (cervical/thoracic/lumbar)
- Palpate spine
  - *Note any lesions, lumps, or abnormalities*

LOWER EXTREMITIES (hips, knees, ankles)

LOWER EXTREMITIES

- Inspect
  - Overall skin coloration
  - Lesions
  - Hair distribution
  - Varicosities
  - Edema
- Palpate: check for edema (pitting or non-pitting)
- Check capillary refill bilaterally

CAPILLARY REFILL TIME (CRT)
Time taken for capillary bed to regain its color after pressure has been applied
NORMAL: < 2–3 SECONDS

HIPS

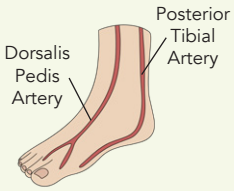
- Inspect and palpate

KNEES

- Inspect and palpate

ANKLES

- Inspect ankles
- Palpate
  - Posterior tibial (PT) & dorsalis pedis (DP) pulse bilaterally



PULSE SCALE:	
0	Pulse is absent
1+	Diminished
2+	Normal
3+	Full
4+	Bounding, strong

ABDOMEN

★ Assess in different order:

1. Inspect
2. Auscultate
3. Percuss
4. Palpate

★ If we were to percuss & palpate before listening (auscultating), we would alter the bowel sounds. This would lead to inaccurate results.

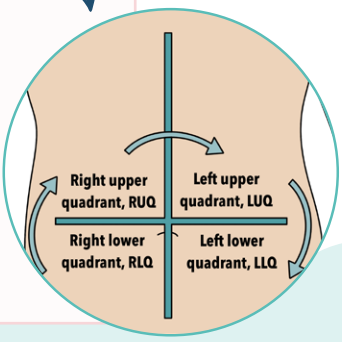
- Inspect
  - Skin color
  - Contour
  - Scars
  - Aortic pulsations
- Auscultate bowel sounds: all 4 quadrants (start in RLQ and go clockwise)
- Light palpation: all 4 quadrants

ABSENT:
Must listen for at least 5 minutes to chart absent bowel sounds

HYPOACTIVE:
One bowel sound every 3–5 minutes

NORMOACTIVE:
Gurgles 5–30 times per minute

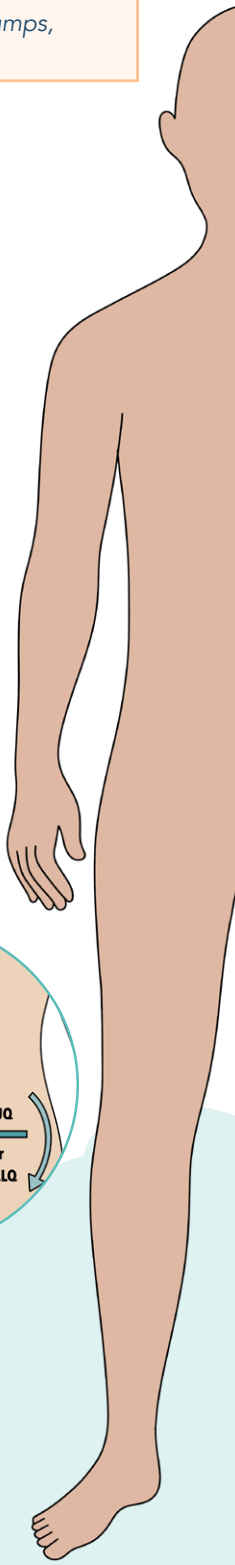
HYPERACTIVE:
Can sometimes be heard without a stethoscope. Constant bowel sounds (> 30 sounds per minute)



OVERALL

During the exam, be sure to:

- Position and drape patient appropriately during exam (gives patient privacy)
- Give patient feedback/instructions
- Exhibit professional manner during exam (treat patient with respect and dignity)
- Be organized and follow a logical sequence (order of exam should "made sense")



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Notes

It's a beautiful thing when a **CAREER** and a **PASSION** come together.

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Dosage Calculation

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Dosage Calc Rules

Medication errors kill; prevention is crucial!

1 Show ALL your work.

2 Leading zeros must be placed before any decimal point.

The decimal point may be missed without the zero

EXAMPLE **.2** mg should be **0.2** mg

WHY? **.2** could appear to be **2**

(0.2 mg of morphine is VERY different than 2 mg of morphine!)

3 No trailing zeros.

EXAMPLE **0.7** mL *NOT* **0.70** mL

1 mg *NOT* **1.0** mg

WHY? **1.0** could appear to be **10**!

4 Do not round until you have the final answer!

HOW TO ROUND YOUR FINAL ANSWER:

If the number in the thousands place is **5 OR GREATER** → The # in the hundredth place is rounded up

EXAMPLES **1.995** mg is rounded to **2** mg

1.985 mg is rounded to **1.99** mg

If the number in the thousands place is **4 OR LESS** → The # is dropped

EXAMPLES **0.992** mg is rounded to **0.99** mg

DECIMAL REFERENCE GUIDE

34.732

tens ones tenths hundredths thousandths

5 Most nursing schools do not give partial credit.

This means every step must be done correctly!

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Abbreviations

TIMES OF MEDICATIONS

ac	before meals
pc	after meals
daily	every day
bid	two times a day
tid	three times a day
qid	four times a day
qh	every hour
ad lib	as desired
stat	immediately
q2h	every 2 hours
q4h	every 4 hours
q6h	every 6 hours
prn	as needed
hs	at bedtime

EXAMPLE

QUESTION: A patient is receiving 1 mg tid.
How many mg will they receive in one day?

Remember: tid = 3X a day

ANSWER: If they are receiving 1 mg for 3X a day, that's 1 mg x 3 = 3 mg per day

ROUTES OF ADMINISTRATION

PO	by mouth
IM	intramuscularly
PR	per rectum
subQ	subcutaneously
SL	sublingual
ID	intradermal
GT	gastrostomy tube
IV	intravenous
IVP	intravenous push
IVPB	intravenous piggyback
NGT	nasogastric tube

DRUG PREPARATION

tab, tabs	tablet
cap, caps	capsule
gtt	drop
EC	enteric coated
CR	controlled release
susp	suspension
elix	elixir
sup, supp	suppository
SR	sustained release
ER/XR	extended release

METRIC

g (gm, Gm)	gram
mg	milligram
mcg	microgram
kg	kilogram
L	liter
mL	milliliter
mEq	milliequivalent

APOTHECARY & HOUSEHOLD

gtt	drop
min, m, mx	minim
tsp	teaspoon
pt	pint
gal	gallon
dr	dram
oz	ounce
T, tbs, tbsp	tablespoon
qt	quart

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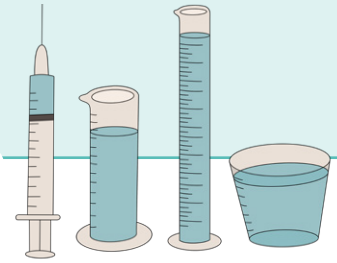
Conversions

Based on Volume



1 mg = 1,000 mcg
1 g = 1,000 mg
1 oz = 30 mL
8 oz = 1 cup
1 tsp = 5 mL
1 dram = 5 mL
1 tbsp = 15 mL
1 tbsp = 3 tsp
1 L = 1,000 mL

THE METRIC SYSTEM

LARGE unit to **SMALL** unit → move decimal to the **RIGHT**

SMALL unit to **LARGE** unit → move decimal to the **LEFT**

MOVING TO A LARGER UNIT?
Move the decimal place to the Left
(Ex: mcg → mg)

MEMORY TRICK

Larger unit think Left

EXAMPLE

1500 mcg = _____mg

A **mg** is **larger** than a **mcg**
Therefore you move decimal 3 places to the Left

1500. mcg = 1.500 mg (1.5 mg)
3 2 1

Based on Weight

1 kg = 2.2 lbs
1 lb = 16 oz

lb → kg
DIVIDE by **2.2**

EXAMPLE 120 **lbs** = _____ **kg**

120 **lbs** / **2.2** = 54.545 **kg**

kg → lb
MULTIPLY by **2.2**

EXAMPLE 45.6 **kg** = _____ **lbs**

45.6 **kg** x **2.2** = 100.32 **lbs**

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Formula Method

For **VOLUME**-Related Dosage Orders:

$$\frac{D}{H} \times V = A$$

D = Desired

EXAMPLE: "The physician orders **120 mg**..."

Some medications like heparin and insulin are prescribed in **units/hour**

H = Dosage of medication available

EXAMPLE: "The medication is supplied as **100 mg**/5 mL"

V = Volume the medication is available in

EXAMPLE: "The medication is supplied as 100 mg/**5 mL**"

A = Amount of Medication required for administration

YOUR ANSWER

All answers should be computed per instructions. This may be **per dose, per shift, per day, etc.** Read the instructions carefully! (Ex. "how many tablets will you give in 24 hours?")"

EXAMPLE 1

Ordered: Drug C 150 mg
Available: Drug C 300 mg/tab
How many tablets should be given?

$$\frac{D}{H} \times V = A$$

What's our desired? Drug C 150mg PO
What do we have? Drug C 300mg/tab
What's our quantity/volume? tablets

150 ~~mg~~ ÷ 300 ~~mg~~ x 1 tab = 0.5 tabs

150 ÷ 300 = 0.5 x 1 = 0.5 tabs

FINAL ANSWER: 0.5 tabs

EXAMPLE 2

Ordered: Drug C 10,000 units SubQ
Available: Drug C 5,000 units/mL
How many mL should be given?

$$\frac{D}{H} \times V = A$$

What's our desired? Drug C 10,000 SubQ
What do we have? Drug C 5,000 units
What's our quantity/volume? 1 mL

10,000 ~~units~~ ÷ 5,000 ~~units~~ x 1 mL = 2 mL

10,000 ÷ 5,000 = 2 x 1 = 2 mL

FINAL ANSWER: 2 mL

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IV Flow Rates

mL / hour

$$\frac{\text{mL of solution}}{\text{total hours}} = \text{mL/hr}$$

What if the question is given in MINUTES?

Since there are 60 minutes in one hour, use this formula:

$$\frac{\text{mL of solution}}{\text{min}} \times 60_{\text{(minutes)}} = \text{mL/hr}$$

If the question is asking for flow rate and you're given units of mL, you need to write the answers in **mL/hr!**

mL/hr is always rounded to the nearest **WHOLE** number!

EXAMPLE #1

ORDERED: 1000 mL D5W to infuse over 3 hours. What will the flow rate be?

$$\frac{1000 \text{ mL}}{3 \text{ hr}} = 333.333 \text{ mL/hr}$$

ANSWER: 333 mL/hr

(rounded to the nearest whole number)

EXAMPLE #2

ORDERED: Infuse 3 grams of Penicillin in 50 mL normal saline over 30 minutes.

$$\frac{50 \text{ mL}}{30 \cancel{\text{min}}} \times 60 \cancel{\text{min}} = 100 \text{ mL/hr}$$

ANSWER: 100 mL/hr

gtt / min

$$\frac{\text{mL of solution}}{\text{total minutes}} \times \text{drop factor} = \text{gtt/min}$$

What if the question is given in HOURS?

Since there are 60 minutes in one hour, use this formula:

Convert hours to minutes!

EXAMPLES:

1 hour = 60 minutes

2.5 hours = 150 minutes

If a drop factor is included, the question is asking for flow rate in gtt/min.

You need to write the answers in **gtt/minute**!

REMEMBER OUR ABBREVIATIONS:

gtt means "drop"!

EXAMPLE #1

ORDERED: 1000 mL of Lactated Ringer's to infuse at 50 mL/hr. Drop factor for tubing is a 5 gtt/mL. *(Convert: 1 hour = 60 min)*

$$\frac{50 \cancel{\text{mL}}}{60 \text{ min}} \times 5 \text{ gtt/}\cancel{\text{mL}} = 4 \text{ gtt/min}$$

50 ÷ 60 = 0.833 x 5 = 4.166

Round to the nearest whole number → 4

FINAL ANSWER: 4 gtt/min

REMEMBER RULE #4 Don't round till the end!

EXAMPLE #2

ORDERED: 100 mL of Metronidazole to infuse over 45 minutes. The tubing you are using has a drop factor of 10 gtt/mL.

$$\frac{100 \cancel{\text{mL}}}{45 \text{ min}} \times 10 \text{ gtt/}\cancel{\text{mL}} = 22 \text{ gtt/min}$$

100 ÷ 45 = 2.222 x 10 = 22.222

Round to the nearest whole number → 22

FINAL ANSWER: 22 gtt/min

REMEMBER RULE #4 Don't round till the end!

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Practice Questions

Do all 10 questions without looking at the correct answers on the following pages. Don't forget to show all your work. After you are done, walk through each question…even the questions you got correct!

1 **ORDERED:** Rosuvastatin 3000 mcg PO ac
AVAILABLE: Rosuvastatin 2 mg tablet (scored)
How many tabs will you administer in 24 hours?

2 **ORDERED:** Tylenol supp 2 g PR q6h
AVAILABLE: Tylenol supp 700 mg
How many supp will you administer? Round to nearest tenth.

3 **ORDERED:** Potassium chloride 0.525 mEq/lb PO dissolved in 6 oz of juice at 0930
AVAILABLE: Potassium chloride 12 mEq/mL
How many mL of potassium chloride will you add to the juice for a 66.75 kg patient? Round to nearest tenth.

4 **1000 mL D5W to infuse over 4 hours. How many mL will infuse per hour?**

5 **150 mL Cipro 250 mcg to infuse over 45 minutes. How many mL per hour will the IV pump be set to?**

6 **250 mL normal saline over 5 hours. Tubing drop factor of 10 gtt/mL.**

7 **Humulin R 200 units in 100 mL of normal saline to infuse at 4 units/hr.**

8 **Dopamine 600 mg in 200 mL of normal saline to infuse at 10mcg/kg/min. Pt weight = 190 lbs.**

9 **2.5 L normal saline to infuse over 48 hours. How many mL per hour will the patient receive?**

10 **ORDERED:** Morphine 100 mg IM q12h prn pain
AVAILABLE: Morphine 150 mg/2.6 mL
How many mL will you administer? Round to nearest hundredth.

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Comprehensive Review

1

ORDERED: Rosuvastatin 3000 mcg PO ac
AVAILABLE: Rosuvastatin 2 mg tablet (scored)

How many tabs will you administer in 24 hours?

STEP 1: CONVERT DATA

mcg → mg

3000 mcg = 3 mg

REMEMBER **SMALL TO BIG:** move the decimal point 3 to the left
unit is getting Larger think Left

STEP 2: READY TO USE DATA

ORDERED: 3 mg
AVAILABLE: 2 mg
VOLUME: 1 tab
ADMINISTERED AC: before each meal
QUESTION IS ASKING: dosage in 24 hours

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$\frac{D}{H} \times V = A$

SHOW YOUR WORK

$\frac{3 \text{ mg}}{2 \text{ mg}} = 1.5$

1.5 x 1 tab = 1.5

1.5 x 3 = 4.5 tabs per day

ROUND: *No rounding necessary*

DON'T FORGET TO CHECK TIMES OF MEDICATION!

The medication is ordered to be given AC, which means before each meal. Since there are 3 meals in a day (24 hours), the answer must be multiplied by 3.

FINAL ANSWER: 4.5 tabs

2

ORDERED: Tylenol supp 2 g PR q6h
AVAILABLE: Tylenol supp 700 mg

How many supp will you administer? Round to nearest tenth.

STEP 1: CONVERT DATA

g → mg

2g = 2000 mg

REMEMBER **BIG TO SMALL:** move the decimal point 3 to the right

STEP 2: READY TO USE DATA

ORDERED: 2000 mg
AVAILABLE: 700 mg
VOLUME: 1 supp

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$\frac{D}{H} \times V = A$

SHOW YOUR WORK

$\frac{2000 \text{ mg}}{700 \text{ mg}} = 2.857$

REMEMBER RULE #4
Don't round till the end!

2.857 x 1 supp = 2.857 supp

ROUND: *Nearest tenth*

2.857 supp → 2.9 supp

FINAL ANSWER: 2.9 supp

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3

ORDERED: Potassium chloride 0.525 mEq/lb PO dissolved in 6 oz of juice at 0930
AVAILABLE: potassium chloride 12 mEq/mL
How many mL of potassium chloride will you add to the juice for a 66.75 kg patient? Round to nearest tenth.

STEP 1: CONVERT DATA

kg → lb

66.75 ~~kg~~ x 2.2 (lb/~~kg~~) = 146.85 lb

! In this case, ordered amount depends on patient weight

mEq/lb → mEq

(0.525 mEq/~~lb~~ x 146.85 ~~lb~~ = 77.096 mEq)

STEP 2: READY TO USE DATA

ORDERED: 77.096 mEq
AVAILABLE: 12 mEq
VOLUME: 1 mL

STEP 3: IRRELEVANT DATA

Dissolved in 12 oz of juice at 0930

! Question asked for "per dose" because no time frame was given

STEP 4: FORMULA USED

$$\frac{D}{H} \times V = A$$

SHOW YOUR WORK

$$\frac{77.096 \text{ mEq}}{12 \text{ mEq}} = 6.424$$

! **REMEMBER RULE #4** Don't round till the end!

6.424 X 1 mL = 6.424 mL

ROUND: *Nearest tenth*

6.424 mL → 6.4 mL

FINAL ANSWER: 6.4 mL

4

1000 mL D5W to infuse over 4 hours.
How many mL will infuse per hour?

STEP 1: CONVERT DATA

N/A

STEP 2: READY TO USE DATA

1000 mL
4 hr

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{\text{mL of solution}}{\text{total hours}} = \text{mL/hr}$$

SHOW YOUR WORK

$$\frac{1000 \text{ mL}}{4 \text{ hr}} = 250 \text{ mL/hr}$$

! **mL/hr** is always rounded to the nearest **WHOLE** number!

ROUND: *No rounding necessary*

FINAL ANSWER: 250 mL/hr

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Comprehensive Review

5

150 mL Cipro 250 mcg to infuse over 45 minutes. How many mL per hour will the IV pump be set to?

If the question is asking for flow rate ("to infuse") and you're given mL of solution, you need to write the answer in **mL/hours!**

STEP 1: CONVERT DATA

N/A

STEP 2: READY TO USE DATA

ML OF SOLUTION: 150 mL
TOTAL HOURS: 45 min

STEP 3: IRRELEVANT DATA

Cipro 250 mcg

IMPORTANT: don't let this information lead you to use the wrong formula. In this example, we're asked for a flow rate which requires mL of solution and total time.

STEP 4: FORMULA USED

$$\frac{\text{mL of solution}}{\text{total minutes}} \times 60 = \text{mL/hr}$$

SHOW YOUR WORK

$$\frac{150 \text{ mL}}{45 \text{ min}} = 3.333 \times 60 = 200 \text{ mL/hr}$$

REMEMBER RULE #4 Don't round till the end!

mL/hr is always rounded to the nearest **WHOLE** number!

ROUND: *No rounding necessary*

FINAL ANSWER: 200mL/hr

6

250 mL normal saline over 5 hours. Tubing drop factor of 10 gtt/mL.

STEP 1: CONVERT DATA

hr → min

1 hour = 60 minutes

$$5 \text{ hr} \times \frac{60 \text{ min}}{1 \text{ hr}} = 300 \text{ min}$$

STEP 2: READY TO USE DATA

ML OF SOLUTION: 250 mL
TOTAL MINUTES: 300 min
DROP FACTOR: 10 gtt/mL

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{\text{mL of IV solution}}{\text{time in minutes}} \times \text{drop factor} = \text{gtt/min}$$

SHOW YOUR WORK

$$\frac{250 \text{ mL}}{300 \text{ min}} = 0.8333 \text{ mL/min}$$

REMEMBER RULE #4 Don't round till the end!

$$0.8333 \text{ mL/min} \times 10 \text{ gtt/mL} = 8.3333 \text{ gtt/min}$$

ROUND: *gtt/mL is always rounded to the nearest whole number!*

8.3333 gtt/min → 8 gtt/min

The question may not specify to round the final answer to a whole number; you are expected to know this with gtt/min units.

FINAL ANSWER: 8 gtt/min

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Comprehensive Review

7

Humulin R 200 units in 100 mL of normal saline to infuse at 4 units/hr.

STEP 1: CONVERT DATA

N/A

STEP 2: READY TO USE DATA

DESIRED: 4 units/hr
AVAILABLE: 200 units
VOLUME: 100 mL

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{D}{H} \times V = A$$

SHOW YOUR WORK

$$\frac{4 \text{ units/hr}}{200 \text{ units}} = 0.02 \text{ /hr}$$

0.02 /hr x 100 mL = 2 mL/hr

mL/hr is always rounded to the nearest **WHOLE** number!

ROUND: *No rounding necessary*

FINAL ANSWER: 2 mL/hr

8

Dopamine 600 mg in 200 mL of normal saline to infuse at 10 mcg/kg/min.
Pt weight = 190 lbs.

If the question is asking for flow rate ("to infuse") and you're given mL of solution, you need to write the answer in **mL/hours!**

STEP 1: CONVERT DATA

mcg → mg

10 mcg = 0.010 mg

REMEMBER

SMALL TO BIG:
move the decimal point 3 to the left
unit is getting Larger think Left

lb → kg

190 lb / 2.2 = 86.363 kg

$$\frac{mg/kg}{min} \rightarrow \frac{mg}{min}$$

In this case, ordered amount depends on patient weight

0.010 mg/~~kg~~/min x 86.363 ~~kg~~ = 0.863 mg/min

STEP 2: READY TO USE DATA

DESIRED: 0.863 mg/min
AVAILABLE: 600 mg
VOLUME: 200 mL

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{D}{H} \times V = A$$

SHOW YOUR WORK

$$\frac{0.863 \text{ mg/min}}{600 \text{ mg}} = 0.00143 \text{ /min}$$

0.00143 /min x 200 mL = 0.2878 mL/min

0.2878 mL/~~min~~ x 60 ~~min~~ = 17.2727 mL/hr

WAIT!
This is in mL/min ... we need units of mL/hr!

ROUND: *mL/hr is always rounded to nearest whole number!*

17.2727 mL/hr → 17 mL/hr

FINAL ANSWER: 17 mL/hr

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9

2.5 L normal saline to infuse over 48 hours.
How many mL per hour will the patient receive?

If the question is asking for flow rate ("to infuse") and you're given mL of solution, you need to write the answer in **mL/hours!**

STEP 1: CONVERT DATA

L → mL

REMEMBER **BIG TO SMALL:** move the decimal point 3 to the right

2.5 L = 2500 mL

STEP 2: READY TO USE DATA

ML OF SOLUTION: 2500 mL
TOTAL HOURS: 48 hr

STEP 3: IRRELEVANT DATA

N/A

STEP 4: FORMULA USED

$$\frac{\text{mL of solution}}{\text{total hours}} = \text{mL/hr}$$

SHOW YOUR WORK

$$\frac{2500 \text{ mL}}{48 \text{ hours}} = 52.0833 \text{ mL/hr}$$

ROUND: *mL/hr is always rounded to nearest whole number!*

52.0833 mL/hr → 52 mL/hr

FINAL ANSWER: 52 mL/hr

10

ORDERED: Morphine 100 mg IM q12h prn pain
AVAILABLE: Morphine 150 mg/2.6 mL
How many mL will you administer?
Round to nearest hundredth.

STEP 1: CONVERT DATA

N/A

STEP 2: READY TO USE DATA

ORDERED: 100 mg
AVAILABLE: 150 mg
VOLUME: 2.6 mL

STEP 3: IRRELEVANT DATA

IM q12h prn pain

Question asked for "per dose" because no time frame was given

STEP 4: FORMULA USED

$$\frac{D}{H} \times V = A$$

SHOW YOUR WORK

$$\frac{100 \text{ mg}}{150 \text{ mg}} = 0.6666$$

0.6666 x 2.6 mL = 1.7333 mL

ROUND: *nearest hundredth*

1.7333 mL → 1.73 mL

FINAL ANSWER: 1.73 mL

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Dosage Calc Template

Use this to walk through any dosage calculation question!

QUESTION:

STEP 1: CONVERT DATA

STEP 2: READY TO USE DATA

STEP 3: IRRELEVANT DATA

STEP 4: FORMULA USED

SHOW YOUR WORK

FINAL ANSWER:

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Notes

You are
CLOSER THAN
you were
YESTERDAY.

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Lab Value Cheat Sheet

with Memory Tricks

NURSE IN THE MAKING

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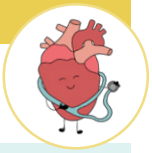
Lab Value Cheat Sheet

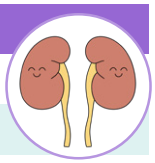
VITAL SIGNS



BLOOD PRESSURE (BP)	SYSTOLIC	120 mmHg
	DIASTOLIC	80 mmHg
HEART RATE (HR)	60 - 100 bpm	
RESPIRATORY RATE (RR)	12 - 20 breaths/min	
TEMPERATURE (T)	97.8 - 99°F (36.5 - 37.2°C)	
OXYGEN SATURATION (SPO_2)	95 - 100%	
SPO_2 IN COPD PT.	as low as 88%	

COPD patients are expected to have low O_2 levels

KIDNEY FUNCTION



CALCIUM	9 - 11 mg/dL
MAGNESIUM	1.5 - 2.5 mg/dL
PHOSPHORUS	2.5 - 4.5 mg/dL
SPECIFIC GRAVITY	1.010 - 1.030
GFR	90 - 120 mL/min/1.73 m^2
BUN	7 - 20 mg/dL
CREATININE	0.6 - 1.2 mg/dL

PANCREAS

AMYLASE	60 - 120 U/dL
LIPASE	0 - 150 U/L

COMPLETE BLOOD COUNT (CBC)

WHITE BLOOD CELLS (WBCs)	4,500 - 11,000 mm^3
RED BLOOD CELLS (RBCs)	4.5 - 5.5 X 10^6 /µL
PLATELETS (PLTs)	150,000 - 450,000 /µL
HEMOGLOBIN (HGB)	FEMALE: 12 - 16 g/dL
	MALE: 13 - 18 g/dL
HEMATOCRIT (HCT)	FEMALE: 36% - 48%
	MALE: 39% - 54%

BASIC METABOLIC PANEL (BMP)

SODIUM	135 - 145 mEq/L
POTASSIUM	3.5 - 5.0 mEq/L
CHLORIDE	95 - 105 mEq/L
CALCIUM	9 - 11 mg/dL
BUN	7 - 20 mg/dL
CREATININE	0.6 - 1.2 mg/dL
ALBUMIN	3.4 - 5.4 g/dL
TOTAL PROTEIN	6.2 - 8.2 g/dL

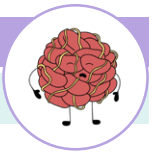
HBA1C

NON-DIABETIC	4 - 5.6%
PRE-DIABETIC	5.7 - 6.4%
DIABETIC	> 6.5%
Goal for diabetic:	< 7%

LIVER FUNCTION TEST (LFT)

ALT	7 - 56 U/L
AST	5 - 40 U/L
ALP	40 - 120 U/L
BILIRUBIN	0.1 - 1.2 mg/dL

COAGs



PT	10 - 13 sec
PTT	25 - 35 sec
aPTT	NOT ON HEPARIN: 30 - 40 secs
	ON HEPARIN: 47 - 70 secs
INR	NOT ON WARFARIN: < 1
	ON WARFARIN: 2 - 3

BMI

UNDERWEIGHT	< 18.5
HEALTHY WEIGHT	18.5 - 24.9
OVERWEIGHT	25.0 - 29.9
OBESITY	> 30.0

ABGs

PH	7.35 - 7.45
$PaCO_2$	35 - 45 mmHg
PaO_2	80 - 100 mmHg
HCO_3	22 - 26 mEq/L

LIPID PANEL

TOTAL CHOLESTEROL	< 200 mg/dL
TRIGLYCERIDE	< 150 mg/dL
LDL	< 100 mg/dL
HDL	> 60 mg/dL

LDL bad cholesterol think we want LOW levels

HDL Happy cholesterol think we want HIGH levels

OTHER

MAP (mean arterial pressure)	70 - 100 mmHg
ICP (intracranial pressure)	5 - 15 mmHg
GLASGOW COMA SCALE	BEST = 15

MILD: 13 - 15 MODERATE: 9 - 12 SEVERE: 3 - 8

Lab values, instruments, and institutions differ based on the facility. Local policy should supersede. Author & publisher intend this reference to be free of errors but no guarantee can be made & assume no responsibility for any outcomes resulting from its use.

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Lab Value Memory Tricks

Electrolytes

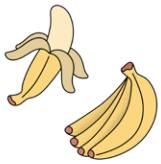
SODIUM: 135 - 145

*Commit to memory!

POTASSIUM: 3.5 - 5

BANANAS:
There are about **3-5** in every bunch & you want them half ripe (½)

So, think **3.5** - **5.0**



PHOSPHORUS: 2.5 - 4.5

PHOR: **4**

US: **2** (me + you = 2)

*don't forget the .5

CALCIUM: 9 - 11

CALL 911

MAGNESIUM: 1.5 - 2.5

MAGnifying glass you see **1.5 - 2.5** bigger than normal

CHLORIDE: 95 - 105

Think of a chlorinated pool that you want to go in when it's SUPER HOT: **95 - 105** °F

Complete Blood Count (CBC)

HEMOGLOBIN (HgB)
FEMALE: **12** - **16** g/dL
MALE: **13** - **18** g/dL

HEMATOCRIT (HCT)
FEMALE: **36**% - **48**%
MALE: **39**% - **54**%

To remember HCT, multiply HgB by 3

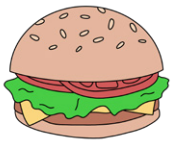
12 X 3 = **36**
16 X 3 = **48**
(FEMALE)

13 X 3 = **39**
18 X 3 = **54**
(MALE)

Basic Metabolic Panel (BMP)

BUN: 7 - 20 mg/dL

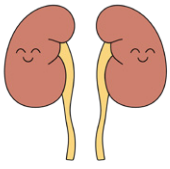
Think hamburger BUNs... Hamburgers can cost anywhere from $**7** - $**20** dollars



CREATININE: 0.6 – 1.2 mg/dL

This is the same value as **LITHIUM's** therapeutic range (**0.6 - 1.2** mmol/L)

Lithium is excreted almost solely by the kidneys... And creatinine is a value that tests how well your kidneys filter



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Notes

It doesn't get
EASIER,
you just get
STRONGER!

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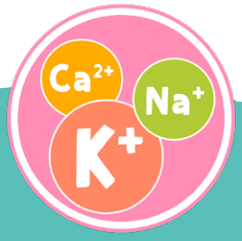

Electrolyte Imbalances

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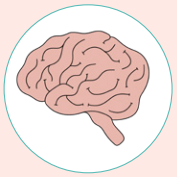
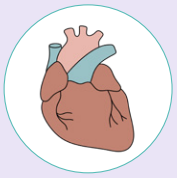
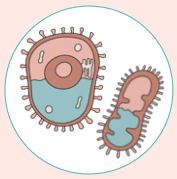
Function of Electrolytes in the Body

ELECTROLYTE	MAIN FUNCTION	NORMAL RANGE	MEMORY TRICK	
SODIUM (Na^+)	Helps maintain blood volume & blood pressure	**135 - 145** mEq/L	**S**odium **S**wells (sodium causes the body to retain water)	**SODIUM** think **BRAIN:** Sodium imbalances can lead to neuro changes; the brain does not like when sodium is out of range! 
POTASSIUM (K^+)	Helps muscles to contract (including the heart muscle)	**3.5 - 5.0** mEq/L	**P**otassium **P**umps (potassium helps the heart muscle to pump blood to the body)	**POTASSIUM** think **HEART:** Potassium imbalances can cause cardiac dysrhythmias that can be life-threatening 
CALCIUM (Ca^+)	Helps with heart function, blood clotting & bone formation	**9 - 11** mg/dL	**C**alcium think **C**reating bone	**CALCIUM** think **BONES:** Calcium imbalances can lead to an increased risk for pathological fractures
MAGNESIUM (Mg)	Helps muscles and nerves stay healthy, helps regulate energy levels	**1.5 - 2.5** mg/dL	**M**agnesium **M**anages **M**uscle	**MAGNESIUM** think **CALCM & SEDATED:** Magnesium acts like a sedative
PHOSPHORUS (P)	Helps create/maintain teeth & bones, helps to repair cells & body tissue	**2.5 - 4.5** mg/dL	**P**hosphorus think **P**erfect teeth	**PHOSPHORUS** think **TEETH:** Phosphorus helps the body to use vitamins to maintain tooth and bone health
CHLORIDE (Cl^-)	Helps maintain acid-base balance, helps to control fluid levels in the cells	**95 - 105** mEq/L	**C**hloride think **C**arrying fluids	**CHLORIDE** think **CELLULAR:** Chloride helps to balance the acids and bases in the body to prevent disturbance; maintains healthy fluid volume in the cells 

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Sodium (Na^+) Imbalance

SIGNS & SYMPTOMS

> 145 mEq/L = HYPERNATREMIA

MNEMONIC **"FRIED SALT"**

- **F** Flushed skin
- **R** Restless, anxious, confused, irritable
- **I** Increased BP & fluid retention
- **E** Edema (pitting)
- **D** Decreased urine output
- **S** Skin is dry
- **A** Agitation
- **L** Low-grade fever
- **T** Thirst (dry mucous membranes)

Sodium imbalances can lead to neuro changes

MEMORY TRICK

Where SODIUM goes, WATER flows!

If there is a lot of sodium in the vessels, there will also be a lot of water in the vessels; this is why these symptoms are seen

< 135 mEq/L = HYPONATREMIA

MNEMONIC **"SALT LOSS"**

- **S** Stupor/coma
- **A** Anorexia (nausea/vomiting)
- **L** Lethargy (weakness/fatigue)
- **T** Tachycardia (thready pulse)
- **L** Limp muscles (muscle weakness)
- **O** Orthostatic hypotension
- **S** Seizures/headache
- **S** Stomach cramping (hyperactive bowels)

Sodium imbalances can lead to neuro changes

TWO TYPES OF HYPONATREMIA:

HYPOVOLEMIC HYPONATREMIA: From ↓ levels of fluid & sodium

HYPERVOLEMIC HYPONATREMIA: From ↑ levels of water in the body which DILUTES sodium

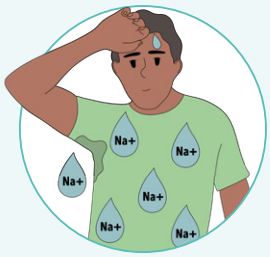
RISK FACTORS

- ↑ Sodium intake
  - Oral ingestion
  - Adm. of IV fluids w/ sodium (hypertonic IV fluids)
- **LOSS OF FLUIDS** from:
  - Fever
  - Burns
  - Diabetes insipidus (DI)

↓ water = ↑ salt (Hemoconcentration)

- **LOSS OF SODIUM** from:
  - **D**iaphoresis
  - **D**iarrhea & vomiting
  - **D**rains (NGT suction)
  - **D**iuretics
  - Sia**D**h (dilution)

5 Ds



- ↑ in water
  - Heart failure

↑ water = ↓ salt (All the water dilutes the sodium - hemodilution)

TREATMENT

- Restrict sodium intake
- Adm. IV fluids if due to fluid loss: (isotonic or hypotonic solutions)

- **Adm. IV fluids** (hypertonic solution)

Hypertonic solutions contain HIGH amounts of salt

TREAT UNDERLYING CAUSE!

- Place patient on seizure precaution
- Place the patient on fluid restriction if due to SIADH (they are in fluid volume overload)
- Place patient on airway protection (NPO)
  - Never give food or water to a patient who is lethargic, confused, or in a comatose state

Risk for aspiration

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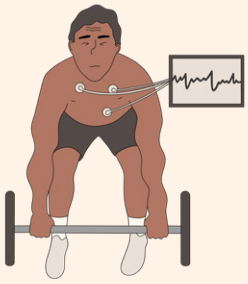
Potassium (K^+) Imbalance

> 5 mEq/L = HYPERKALEMIA

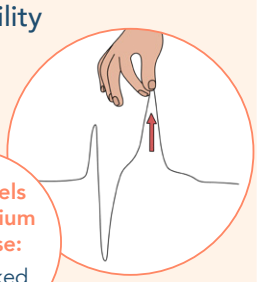
SIGNS & SYMPTOMS

Muscles contract for TOO long
=
Tight & contracted



MNEMONIC **"MURDER"**

- **M** **M**uscle cramps & weakness
- **U** **U**rine abnormalities
- **R** **R**espiratory distress
- **D** **D**ecreased cardiac contractility (↓ HR, ↓ BP)
- **E** **E**KG changes
- **R** **R**eflexes (↓ DTR)



High levels of potassium can cause:
Tall, peaked T-waves

MEMORY TRICK **HYPER**kalemia think **H**igh (peaked)

RISK FACTORS

- Intake of too much potassium (IV fluids with K^+)
- Adrenal gland issues (insufficiency)
- High levels of acid in blood (acidosis)
- Non-steroidal anti-inflammatory drugs (NSAIDs) (ibuprofen, naproxen)
- Potassium-sparing diuretics (spironolactone)

TREATMENT

STOP potassium intake (IV or PO)

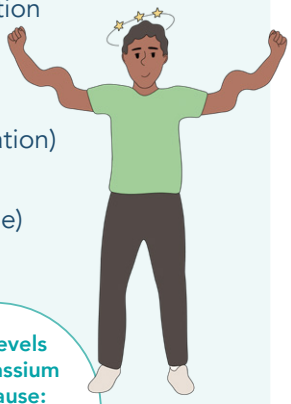
- **Adm. medications**
  - IV sodium bicarbonate
  - IV calcium gluconate

EKG monitoring for both: potassium imbalances can cause cardiac dysrhythmias that can be life-threatening!

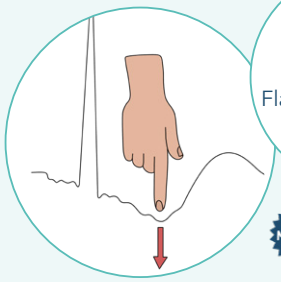
< 3.5 mEq/L = HYPOKALEMIA

SIGNS & SYMPTOMS

Generalized weakness in the muscles
(Example: smooth muscle in bronchi, GI system)



- Weak muscles & LESS contraction
- ↓ Reflexes
- Shallow breathing
- Slowing of GI system (constipation)
- ↓ Blood pressure (especially with position change)
- Nausea, vomiting, bloating
- EKG changes



Low levels of potassium can cause:
Flattened T-wave or inversion of the T-wave

MEMORY TRICK **HYPO**kalemia think **LOW** (flattened or inversion)

RISK FACTORS

- Low potassium intake (not eating, NPO diet)
- Vomiting & diarrhea
- Gastric suction
- Alkalosis
- Potassium-wasting diuretics (loop or thiazide)

TREATMENT

REPLACE the potassium

ORAL POTASSIUM SUPPLEMENT

IV POTASSIUM

- ALWAYS dilute in fluid
- Potassium can burn the vein, therefore infuse **SLOWLY!**

Never administer potassium via IV push

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Calcium (Ca^+) Imbalance

> 11 mg/dL = HYPERCALCEMIA

SIGNS & SYMPTOMS

MNEMONIC "BACKME"

- **B** Bone pain
- **A** Arrhythmias
- **C** Cardiac arrest (bounding pulses)
- **K** Kidney stones → Also called renal calculi
- **M** Muscle weakness
- **E** Excessive urination

RISK FACTORS

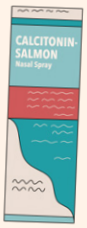
- ↑ Calcium absorption
- ↓ Calcium excretion
- Kidney disease → The kidneys are unable to excrete excess calcium out of the body
- Use of thiazide diuretics
- HYPERparathyroidism & HYPERthyroidism
- Bone breakdown from metastatic cancer
- Highly concentrated blood (hemoconcentration) → Can be from dehydration!

TREATMENT

STOP calcium intake (IV or PO)

- **Adm. medications to ↓ calcium levels**
  - Phosphorus
  - Calcitonin

MEMORY TRICK: CalciTONin helps TONe down calcium levels in the blood



A patient with a calcium imbalance is at risk for a **PATHOLOGICAL FRACTURE**

Move the patient carefully and slowly

- ↑ calcium intake in diet
  - Calcium supplements
  - Vitamin D
  - Calcium gluconate
- Initiate seizure precautions
  - High risk for seizures

< 9 mg/dL = HYPOCALCEMIA

MNEMONIC "CATS GO NUMB"

- **C** Convulsions/seizures
- **A** Arrhythmias
- **T** Tetany
- **S** Spasms & stridor
- **GO NUMB** Numbness in fingers, face, limbs

POSITIVE TROUSSEAU'S:
Carpal spasm caused by inflating a blood pressure cuff

CHVOSTEK'S SIGNS:
Contraction of facial muscles w/ a light tap over the facial nerve

MEMORY TRICK: Think "*C*" for *C*heesy smile

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Magnesium (Mg) Imbalance

> 2.5 mg/dL = HYPERMAGNESEMIA

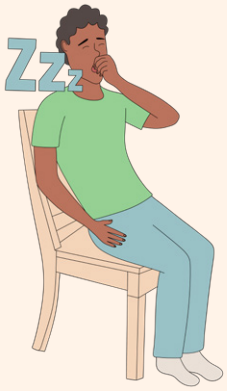
MEMORY TRICK: Magnesium acts like a **SEDATIVE!**

< 1.5 mg/dL = HYPOMAGNESEMIA

SIGNS & SYMPTOMS

LOW ↓ everything – SEDATED

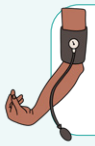
- ↓ DTRs (deep tendon reflex)
- ↓ Energy (drowsiness/coma)
- ↓ HR (bradycardia)
- ↓ BP (hypotension)
- ↓ RR (bradypnea)
- ↓ Respirations (shallow)
- ↓ Bowel sounds



HIGH ↑ everything – NOT SEDATED

- ↑ DTRs (deep tendon reflex) "Hyperreflexia"
- ↑ HR (tachycardia)
- ↑ BP (hypertension)
- Shallow respirations
- Twitches, paresthesias
- Tetany & seizures
- Irritability & confusion

Also seen in hypocalcemia. Ca & Mg rise and fall together!



POSITIVE TROUSSEAU'S: Carpal spasm caused by inflating a blood pressure cuff

CHVOSTEK'S SIGNS: Contraction of facial muscles w/ light tap over the facial nerve

MEMORY TRICK: Think "C" for Cheesy smile

RISK FACTORS

- Increased magnesium intake
  - Magnesium-containing antacids (TUMS) & laxatives
  - Excessive adm. of magnesium IV
- Renal insufficiency
  - ↓ renal excretion of Mg = ↑ Mg in the blood
- Diabetic ketoacidosis (DKA)

- Insufficient magnesium intake
  - Malnutrition/vomiting/diarrhea
  - Malabsorption syndrome
  - Celiac & Crohn's disease
- Increased magnesium excretion
  - Diuretics or chronic alcoholism
- Intracellular movement of magnesium
  - Hyperglycemia & insulin adm.
  - Sepsis

TREATMENT

- Adm. loop diuretics
- IV administration of calcium chloride or calcium gluconate
- Restrict dietary intake of magnesium-containing foods
- Avoid the use of laxatives & antacids containing magnesium
- Use of hemodialysis in severe cases

- Adm. magnesium sulfate IV or PO
- Place patient on seizure precautions
- Instruct patient to ↑ intake of magnesium-containing foods

Nuts & seeds
Legumes
Whole grains
Milk

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Phosphorus (P) Imbalance

> 4.5 mg/dL = HYPERPHOSPHATEMIA

SIGNS & SYMPTOMS

Directly leads to LOW levels of calcium (hypocalcemia)

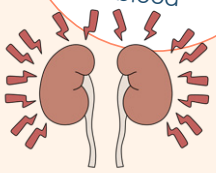
- Muscle spasms & tetany
- Cardiac arrhythmias
- Seizures
- Dry & brittle skin/nails

RISK FACTORS

Kidney dysfunction

- Consistent use of enemas and/or laxatives
- Rhabdomyolysis
- Vitamin D toxicity
- Hypoparathyroidism
- Acromegaly

An injured kidney is unable to filter the excess phosphate, causing levels to rise in the blood



TREATMENT

DIET MODIFICATIONS:

- ↓ Dietary phosphorus
- Use **dialysis** for patients with end-stage renal disease (ESRD)
- Control hypertension to maintain kidney function

Foods that are high in phosphorus
Red meat, beans, dairy products, nuts, & lentils

< 2.5 mg/dL = HYPOPHOSPHATEMIA

SIGNS & SYMPTOMS

- Fatigue & weakness
- Delayed growth & development in children
- Poor bone density & frequent fractures
- Loss of appetite
- Cardiac arrhythmias

RISK FACTORS

- Chronic vomiting or diarrhea
  - *Example:* eating disorders that involve vomiting
- Overconsumption of diuretics
- Patients with significant burn injuries
- Malnutrition & starvation
- ETOH (alcohol) dependency
- Refeeding syndrome

Anything that causes loss of FLUIDS
↑ vomiting, urination, defecation, or fluid loss from burns all cause phosphate to leave the body
=
Lower serum level of phosphate

TREATMENT

DIET MODIFICATIONS:

- ↑ Dietary phosphorus
- Adm. oral or IV phosphate
- Reintroduce nutrients slowly in patients with history of starvation to prevent refeeding syndrome
- Reduce diuretic dosing/use
- Provide proper care & recovery of burns

↑ P = ↓ Ca^+ → INVERSE RELATIONSHIP

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Chloride (Cl) Imbalance

> 105 mEq/L = HYPERCHLOREMIA

SIGNS & SYMPTOMS

- **HYPER**tension
  - Fluid retention
- Generalized swelling
- Peripheral edema
- Cardiac arrhythmias

RISK FACTORS

- Hypernatremia (↑ sodium)
- Overuse of IV sodium chloride
- Metabolic acidosis
- Renal damage
  - The kidneys are not able to filter & excrete excess chloride = ↑ chloride in the body
- Dehydration
  - Decrease water = increased concentration of chloride in the body
- Uncontrolled glucose levels
- Diabetes insipidus (DI)

TREATMENT

DIET MODIFICATIONS:

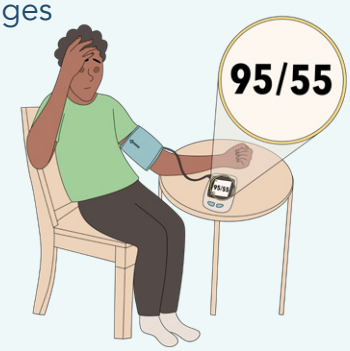
- Sodium restriction
- ↑ Fluids to flush salt
  - Oral or IV fluids

- Start blood glucose management or insulin
- Use dialysis in patients with renal disease

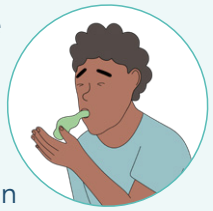
< 95 mEq/L = HYPOCHLOREMIA

- **HYPO**tension & tachycardia
- Mental status changes
- Muscle weakness
- Fatigue

Similar signs & symptoms of **HYPO**natremia (low sodium levels)



- Dehydration related to large fluid volume loss through:
  - Vomiting & diarrhea
  - Syndrome of inappropriate antidiuretic hormone secretion (SIADH)
  - Overuse of diuretics



- Addison's Disease
- Metabolic alkalosis
- Excessive suctioning of gastric contents
- Potassium imbalance

DIET MODIFICATIONS:

- ↑ Dietary salt intake with meals

- Adm. IV potassium
- Adm. IV sodium chloride
- Rehydration
- Limit or reduce diuretic use

↑ Cl = ↑ Na^+
↓ Cl = ↓ Na^+
→ **SAME RELATIONSHIP**

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Fundamentals

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Tips for Fundamentals Class

Don't just memorize, understand!

Fundamentals are truly the foundation on which you build the rest of your nursing knowledge. That's why understanding the basics is so important.

Some things just have to be memorized, but in general, understanding and connecting the dots will help you far more than simply memorizing information.

Learn about nursing itself

Ethical principles, theories in nursing, scope of practice, and delegation are just a few of the topics you'll cover in fundamentals class.

Knowing this information will help you on the NCLEX and once you start working as a nurse!

Make the core information stick

Remember, fundamentals include lab values, vital signs, basic interventions, and more. You'll never stop using this information as a nurse.

Learn the information in fundamentals, and it will continue to expand with each course to come.

Build your skills

Fundamentals class is heavily based on core nursing skills such as checking vital signs and administering oxygen.

Utilize your simulation labs as a safe place for learning the proper steps for each procedure, and continue practicing to improve your skills.

You can find step-by-step nursing skills in The Complete Fundamentals Flashcards!

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Nursing Documentation

Abbreviations

There are many abbreviations, so be sure to follow your facility's policy on approved abbreviations

AAA Abdominal Aortic Aneurysm
Abd Abdomen
Ac Before Meals
ACLS Advanced Cardiac Life Support
AD Admitting Diagnosis or Advance Directives
A&D Admission and Discharge
Ad lib As Desired
ADL Activities of Daily Living
Adm. Administration
AKA Above-the-Knee Amputation or Alcoholic Ketoacidosis
ALL Acute Lymphocytic Leukemia
Amb Ambulation
AP or **A.P.** Appendectomy or Anterior/Posterior
AV Atrioventricular
Bid Twice a Day
BKA Below-the-Knee Amputation
BLS Basic Life Support
BM Bowel Movement
BP Blood Pressure
BPH Benign Prostatic Hyperplasia
BUN Blood Urea Nitrogen
BX Biopsy
CABG Coronary Artery Bypass Graft
CAD Coronary Artery Disease
CBC Complete Blood Count
CCU Cardiac Care Unit/Coronary Care Unit
CF Cystic Fibrosis
CHF Congestive Heart Failure
CKD Chronic Kidney Disease
C/O Complaining Of
COPD Chronic Obstructive Pulmonary Disease
CPR Cardiopulmonary Resuscitation
C&S Culture and Sensitivity
CVA Cerebrovascular Accident (stroke)
CVC Central Venous Catheter
D/C Discontinue or Discharge
D&C Dilatation and Curettage
DI Diabetes Insipidus
DIC Disseminated Intravascular Coagulation
DKA Diabetic Ketoacidosis
DM Diabetes Mellitus
DVT Deep Vein Thrombosis
DX Diagnosis
ECG or **EKG** Electrocardiogram
ED Emergency Department
EENT Eye, Ears, Nose, and Throat
ETT Endotracheal Tube
FBS Fasting Blood Sugar
Fx Fracture
GI Gastrointestinal
Gtt or **G.T.T.** Glucose Tolerance Test
HOB Head of Bed
HR Heart Rate
HS Bedtime
Hx History
ICU Intensive Care Unit
I&O Intake and Output
LES Lower Esophageal Sphincter
LMP Last Menstrual Period
LOC Level of Consciousness
LP Lumbar Puncture
LPN Licensed practical nurse
LVN Licensed Vocational Nurse
MAP Mean Arterial Pressure
MRI Magnetic Resonance Imaging
MVA Motor Vehicle Accident
NGT Nasogastric Tube
NKA No Known Allergies
NPO Nothing by Mouth
O_2 Oxygen
OA Osteoarthritis
OB Obstetrics
OOB Out of Bed
OR Operating Room
Ortho Orthopedics
OT Occupational Therapist or Occupational Therapy
Pc After Meals
PFT Pulmonary Function Test
PLT Platelets
PRBC Packed Red Blood Cells
Pre-op Before Surgery
Prn or **p.r.n.** As Needed
PT Physical Therapist or Physical Therapy
PTCA Percutaneous Transluminal Coronary Angioplasty
PVC Premature Ventricular Contraction
RA Rheumatoid Arthritis
RBC Red Blood Cell(s)
RN Registered nurse
Rom/R.O.M. Range of Motion
RT Respiratory Therapist or Respiratory Therapy
SBAR Situation, Background, Assessment, Recommendation
SIADH Syndrome of Inappropriate Antidiuretic Hormone Secretion
SLE Systemic Lupus Erythematosus
SOB Shortness of Breath
SSE or **S.S.E.** Soap Suds Enema
Stat At Once, Immediately
STD Sexually Transmitted Disease
TB Tuberculosis
TIA Transient Ischemic Attack
Tid Three Times a Day
TPN Total Parenteral Nutrition
T&S Type and Screen
TURP Transurethral Resection of the Prostate
UA Urinalysis
UAP Unlicensed assistive personnel
US Ultrasound
UTI Urinary Tract Infection
VS Vital Signs
WBC White Blood Count
WNL Within Normal Limits

The Nursing Process

MNEMONIC "A Delicious PIE"

1 ASSESS
- Gather information
- Verify that the information collected is clear & accurate

2 DIAGNOSE
- Interpret the information collected
- Identify & prioritize the problem through a nursing diagnosis (be sure it's NANDA-approved)

3 PLAN
- Set goals to solve the problem
- Prioritize the outcomes of care

MNEMONIC Set SMART Goals:
- **S**pecific
- **M**easurable
- **A**chievable
- **R**elevant
- **T**ime frame

4 IMPLEMENT
- Reach those goals through performing the nursing actions
- Implement the goals set above in the planning stage

5 EVALUATE
- Determine the outcome of the goals
- Evaluate the patient's compliance
- Document the patient's response to pain
- Modify & assess the need for changes

CAN BE PROVEN

OBJECTIVE DATA

Unbiased facts: things you can see, hear, feel, auscultate, and measure

*O*bjective think *O*bserve

EXAMPLES:
- ★ Vital signs (except for pain)
  - Blood pressure, respiratory rate, heart rate, oxygen
- Bleeding
- Vomiting

MEMORY TRICK

CANNOT BE PROVEN

SUBJECTIVE DATA

Opinions or biases

*S*ubjective think what the patient *S*ays

EXAMPLES:
- ★ Pain level
  - Past history & emotions

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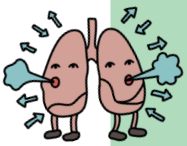
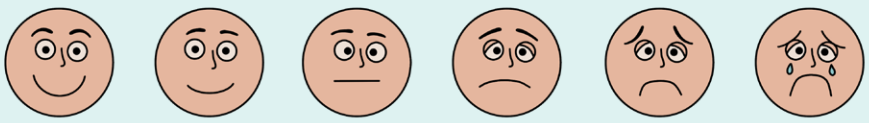
Vital Signs

BLOOD PRESSURE (BP)	**SYSTOLIC** 120 mmHg **DIASTOLIC** 80 mmHg	Hypotension = low blood pressure Hypertension = high blood pressure
HEART RATE (HR)	60 – 100 bpm	Bradycardia = < 60 bpm Tachycardia = > 100 bpm
 **RESPIRATORY RATE (RR)**	12 – 20 breaths/min	Bradypnea = < 12 breaths/min Tachypnea = > 20 breaths/min
TEMPERATURE (T)	97.8°F – 99°F (36.5°C - 37.2°C)	Hypothermia = < 95°F (< 35°C) Hyperthermia = > 104°F (> 40°C)
OXYGEN SATURATION (SpO_2)	95% – 100%	Hypoxemia = low oxygen levels
PAIN	Pain is subjective data given to you by the patient	Can be measured in various ways: Numerical scale, Wong-Baker FACES®, or verbal rating scale

Wong-Baker FACES® Pain Rating Scale



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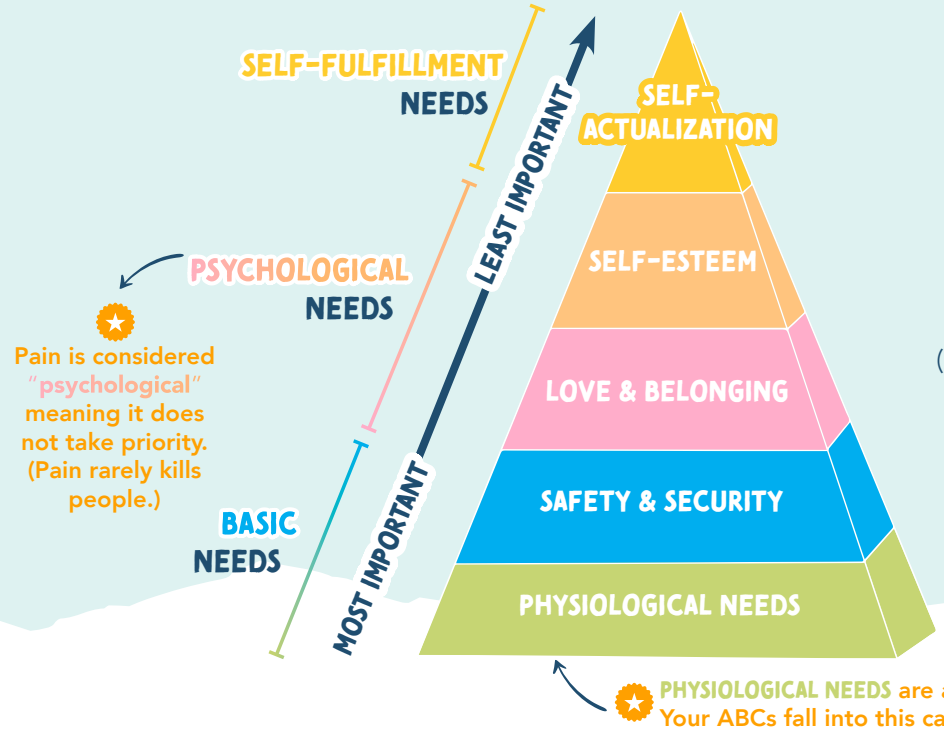
Maslow's Hierarchy of Basic Needs

PRIORITY QUESTIONS

You know you are being asked a **PRIORITY QUESTION** when the question asks:

- What is the **most important**?
- What is the **initial response**?
- Which action should the nurse take **first**?

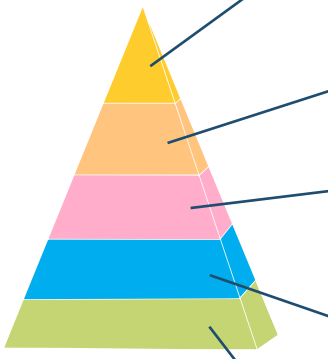
When you see these questions, you should immediately think of **MASLOW'S HIERARCHY OF NEEDS** and **ABCs**!



Pain is considered "psychological" meaning it does not take priority. (Pain rarely kills people.)

This shows the **5** levels of human needs, with **PHYSIOLOGICAL NEEDS** being the most important (oxygen, fluids, nutrition, shelter). **ABCs** fall into Maslow's **PHYSIOLOGICAL NEEDS**!

PHYSIOLOGICAL NEEDS are always a priority (the most important). Your ABCs fall into this category!



SELF-ACTUALIZATION

- Hope
- Spiritual well-being
- Enhanced growth

SELF-ESTEEM

- Control
- Competence
- Positive regard
- Acceptance/worthiness

LOVE & BELONGING

- Maintenance of support systems
- Protection from isolation

SAFETY & SECURITY

- Protection from injury
- Promote a sense of security
- Trust in nurse-patient relationship

PHYSIOLOGICAL NEEDS

- Airway
- Respiratory effort
- Heart rate
- Nutrition
- Elimination

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ABCs

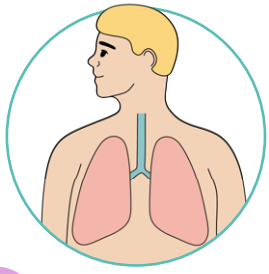
A Airway

#1 Patent Airway

Patent means "open," so the airway is clear

ASK YOURSELF:

Can the patient successfully breathe oxygen in and breathe CO_2 out?



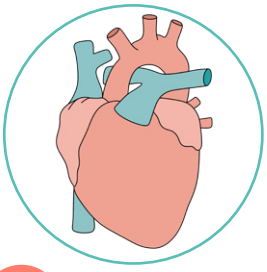
B Breathing

#2 Breathing

Gas exchange is taking place inside the lungs

ASK YOURSELF:

Can gas exchange successfully happen in the patient's lungs?



C Circulation

#3 Circulation

Blood circulating through the body and organs being perfused

ASK YOURSELF:

Is there a reason that the blood isn't pumping/circulating in the patient's body?

(Example: the heart is working to pump the blood to the vital organs)

	Airway	Breathing	Circulation
POSSIBLE PROBLEMS	Obstruction of the airway (tongue, swelling, foreign body fluids)	▪ Pulmonary edema ▪ Asthma ▪ Pneumothorax	▪ Cardiac arrest ▪ Internal/external bleeding (hemorrhage) ▪ Shock (hypovolemic or cardiogenic) ▪ Cardiac arrhythmias
SIGNS & SYMPTOMS	▪ Stridor ▪ Gasping ▪ Cyanosis	▪ Hyperventilation or hypoventilation ▪ Gasping ▪ ↓ O_2 saturation	▪ Weak/thready pulse ▪ ↓ Pulse or no pulse ▪ Pallor ▪ Obvious bleeding
TREATMENT	▪ Reposition the patient to open airway ▪ Perform jaw thrust maneuver or abdominal thrusts to clear airway ▪ Intubation	▪ Administer oxygen ▪ Ventilate ▪ Insertion of chest tube	▪ Administer IV fluids/blood products ▪ Control bleeding ▪ Perform high-quality CPR to restore normal heart rhythm

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Nursing Ethics & Law

Ethical Principles

PRINCIPLE	MEMORY TRICK	QUALITIES

AUTONOMY	Autonomy think All by myself (I have the right to make my own decisions)	▪ Respecting a patient's values & beliefs
		▪ Remembering that patients hold the right to:
		  ▪ Make their own decisions
		  ▪ Make all choices in their care

BENEFICENCE	"BENEfit" = GOOD	▪ Always doing right by the patient & acting with compassion ▪ Always choosing the good for each patient
FIDELITY	Fidelity think Faithful	▪ Remaining loyal and faithful in actions & care ▪ Keeping the promise of safe, well-intentioned care
ACCOUNTABILITY	ACCOUNTABILITY think I am ACCOUNTABLE for my actions	▪ Taking responsibility for all actions ▪ Being accountable for errors or mistakes
JUSTICE	"just" = fair I JUST want FAIR CARE for all	▪ Delivering equal care to all
NONMALEFICENCE	"non" = none "mal" = bad	▪ Doing no harm ▪ Not inflicting any harm on any patient (intentional or unintentional)
VERACITY	VERacity think VERy honest	▪ Being entirely honest with the patient at all times

HIPAA

HIPAA

Health Insurance Portability & Accountability Act

KEY PROVISIONS:

PRIVACY RULE

- Medical records & patient information are protected
- Patient has right to copies of their medical records

ENFORCEMENT RULE

- Those who breach confidentiality will be punished

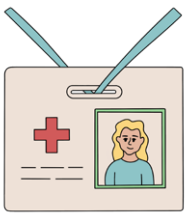
HIPAA compliance must be enforced in all medical settings

SECURITY RULE

- Safety barriers or safeguards are used for all medical records
- Electronic medical records are protected by software & monitoring

UNIQUE IDENTIFIERS:

- Specific to an individual & one of a kind (not to be shared!)
- Full name, date of birth, social security number, medical record number, and reason for care



BEST PRACTICES:

DO NOT:

- ✗ Speak about patient's personal information with others
- ✗ Speak loudly or expose information to others when with patient
- ✗ Share information with patient's family or friends if not listed as authorized
- ✗ Use phone recordings or video that includes any patient information or images
- ✗ Include revealing patient information in case studies (or clinical discussion)
- ✗ Leave computers with open tabs unattended
- ✗ Leave open charts on community surfaces
- ✗ Talk about patient information/identifiers openly at nurse's station

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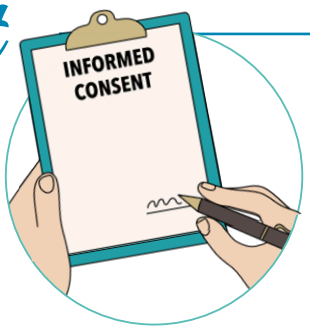
Nursing Ethics & Law

Informed Consent

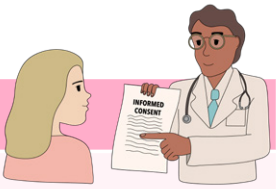
Informed consent is a written legal document between the patient and their healthcare provider that leads to an agreement for a certain treatment, surgery, or care. Every patient has the legal right to refuse a surgical procedure or treatment.



WHEN IS IT NEEDED?

- Before non-emergent surgeries
- Before administration of blood products
- Before procedures that:
  - Require anesthesia, sedation, or radiation
  - Are invasive
  - Can cause risk to the patient

Informed consent may not be necessary for lifesaving operations



PROVIDER'S ROLE

- Explain **ALL** components of the procedure
  - Benefits
  - Risks
  - Complications
  - Recovery
- Answer all questions (before & after the informed consent is given)

It is not the nurse's job to provide an explanation of the treatment being given!

NURSE'S ROLE

- Be present during the patient's signature & act as a witness
- The nurse:
  - ✓ Can clarify what the surgeon said
  - ✗ Can not add new information
- Document & upload the informed consent

The consent is not valid if the patient is taking medications, alcohol, or drugs, or has any disease that impairs their judgment

If the patient has questions before or after signing the consent, the provider must be notified and answer those questions

Advance Directives

ADVANCE DIRECTIVES

Gives **DIRECTION** in **ADVANCE** for personal wishes and medical care for when the patient is not capable of making those decisions on their own

COMPONENTS:

The patient states IN ADVANCE what they want, which may include:

- Resuscitation status (DNR, full code)
- Intubation
- Life-saving measures
- Comfort
- Treatment & hospitalizations
- Delegation of choices (who will be appointed as DPOA)

TYPES OF ADVANCE DIRECTIVES:

LIVING WILL
Exact directions for care if the patient is unable to make choices

DURABLE POWER OF ATTORNEY (DPOA)
Trusted family member/individual makes choices FOR the patient if the patient is unable to make choices

NURSING CONSIDERATIONS:

- All patients should discuss advance directives with their families!
- Discuss this topic during admission for *every* patient regardless of age
- *If a patient has one in place, it must be on file & signed to be active*

PURPOSE:

- Preparation & proactive care
- Steps in place to eliminate confusion or argument if the patient becomes incapacitated
- Clear guidelines for family & medical staff

MNEMONIC **"PLAN"**

Prepare & be **P**roactive
Living will & DPOA
Advise on code status & wishes
Not active if not on file!

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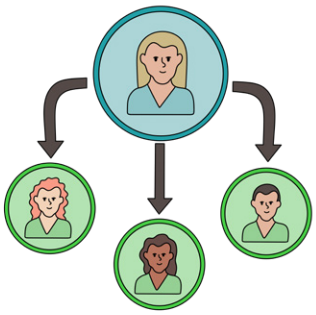
Delegation & Scope of Practice

Delegation:

Handing off the responsibility of a task to another individual



NOTE:

When a registered nurse delegates a task, the task is transferred, but accountability and responsibility are NOT transferred. The original RN is still responsible!

5 RIGHTS OF DELEGATION

1. **RIGHT TASK**
   The task is within the scope of practice for that particular individual
2. **RIGHT CIRCUMSTANCE**
   The patient is stable, and the situation is appropriate
3. **RIGHT PERSON**
   The person accepting the task is appropriate
4. **RIGHT DIRECTIONS & COMMUNICATION**
   The nurse gives clear directions, communicates the time frame, performs the task, and reports back; the UAP notes any concerning findings and reports them immediately
5. **RIGHT SUPERVISION & EVALUATION**
   The nurse follows up, verifies documentation, oversees, and is ready to assist

Scope of Practice:

A collection of tasks specific to a healthcare personnel position that give parameters of what they are allowed to do.

What is in their scope?

RN

Unstable clients

- Starting blood products
- Perform sterile procedures
- Starting IVs & administering IV medications

MNEMONIC "TAPE"

- **T** Initial **T**eaching
- **A** **A**ssessment (admission, post-op & comprehensive)
- **P** **P**lanning
- **E** Discharge/initial **E**ducation/teaching

LPN/LVN

Stable clients

- Monitor RN's findings & gather data
- Reinforce teaching
- Perform routine procedures (urinary catheterization, ostomy care, wound care)
- Monitor IV fluids & blood products
- Maintaining IVs & IV medications (based on the state)
- Place tubes
- Administer enteral feedings
- Perform sterile procedures

SPECIFIC EVALUATIONS

Lung sounds, bowel sounds & neurovascular checks

UAP

- Obtain vital signs
- Document intake & output (I&O)
- Get blood from the blood bank
- Assist with activities of daily living (ADLs)

ADLs

Feeding (not with aspiration risk)

- Positioning
- Ambulation
- Cleaning
- Linen change
- Hygiene care

RN = Registered nurse
LPN = Licensed practical nurse
LVN = Licensed vocational nurse
UAP = Unlicensed assistive personnel
(Example: certified nursing assistant (CNA))

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Infection Control

Chain of Infection

1 INFECTIOUS AGENT
fungi, bacteria,
viruses, parasites, etc

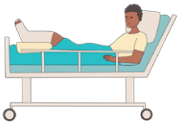
2 RESERVOIR
(where agents are found)
- Animals
- Humans
- Food & water sources
- Environment
  (lakes, soil, woods)

3 PORTAL OF EXIT
(where germs escape from)
- Openings in skin
- Saliva & mucus
- Stool, urine & blood
- Moisture & droplets

4 MODE OF TRANSMISSION
(how agents travel)
- Superficial contact
- Sexual contact
- Aerosolized or droplet particulates
- Underprepared food

5 PORTAL OF ENTRY
(how agents enter the body)
- Openings in skin
- Body cavity
- Mucous membrane
  (eyes, mouth, nose)
- Eating & drinking

6 SUSCEPTIBLE HOST
(whom agent travels to next)
- Anyone!
- Higher risk if
  immunodeficient, young,
  old, underlying conditions
- Animals



Stages of Infection

INCUBATION:
Time it takes for the infection to *GROW* and *PRODUCE SYMPTOMS*

PRODROMAL:
Infection is growing, but body is showing only EARLY signs of illness (contagious)

ILLNESS:
Obvious symptoms (person is clearly sick)

DECLINE:
Amount of infection *DECLINES*
(due to time or medication)
- Symptoms are improving
- Person is at HIGH risk for another infection
  (their immunity is "down")

CONVALESCENCE:
Improvement
(person feels better)
- If chronic issues are present & related, these may last longer or be continuous

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Infection Control

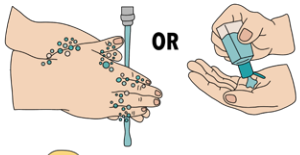
PPE: Personal Protective Equipment

Donning: Putting on PPE

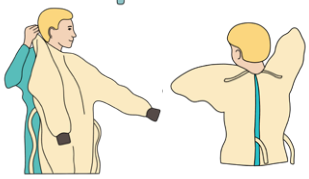
MEMORY TRICK "I am dONning by putting this ON"

- Put on PPE before entering the patient's room
- Do not touch your face while wearing PPE
- Minimize contact with items in the patient's room

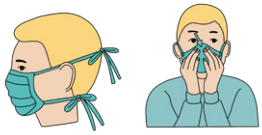
1. Perform hand hygiene



2. Put on gown

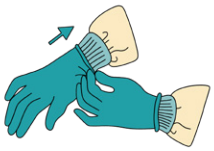


3. Put on mask/respirator



4. Put on goggles/ face shield

5. Put on gloves



Doffing: Removing PPE

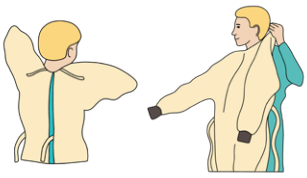
MEMORY TRICK "I am dOFFing by taking this OFF"

- Remove PPE at the patient's doorway or outside the room
- If hands become soiled while removing PPE, stop & perform hand hygiene
- After hand hygiene, continue with PPE removal

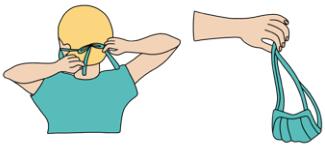
1. Remove gloves

2. Remove protective eyewear

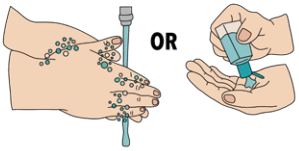
3. Remove gown



4. Remove & discard respirator



5. Perform hand hygiene



COMMON HOSPITAL-ASSOCIATED INFECTIONS (HAIs)

- Also called nosocomial infection or hospital acquired infection.
- An infection that is contracted by the patient during their hospital stay, and that they did not have prior to admission

CAUTI.....Catheter-associated urinary tract infection
SSI.....Surgical site infection
CLABSI.....Central line-associated bloodstream infection
VAP.....Ventilator-associated pneumonia
C. Diff.....Clostridium difficile
MRSA.....Methicillin-resistant Staphylococcus aureus

Meticulous hand hygiene practices & use of chlorhexidine washes help to prevent HAIs

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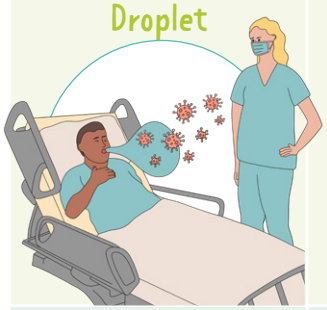
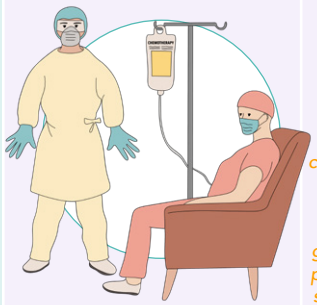
Transmission-Based Precautions

PRECAUTION	DESCRIPTION	EQUIPMENT	USED FOR	NURSING CONSIDERATIONS
Standard or "Universal"	General infection control to protect yourself and others from spread of germs	▪ Non-sterile gloves *If anticipating contact with body fluids, wear extra protection:* ▪ Goggles ▪ Surgical mask ▪ Face shield ▪ Gown	ALL PATIENTS! 	▪ Perform proper hand hygiene ▪ Use hand sanitizers on way in and out of patient rooms ▪ Never use same gloves/ PPE for different patients
Contact	Precautions taken to protect from infection spread mostly by touch/contact	▪ Non-sterile gloves ▪ Surgical gown	▪ Methicillin-resistant Staphylococcus aureus (MRSA) ▪ C. Difficile ▪ Drug-resistant organisms	▪ Private room ▪ Keep patient in room as much as possible ▪ Visitors should avoid direct contact and wear gloves/gown
Droplet 	Precautions taken to prevent infection spread by sneezing, coughing, or talking (anything that can spread via droplet)	▪ Surgical mask	▪ Adenovirus (pediatrics) ▪ Pertussis (whooping cough) ▪ Rubella ▪ Diphtheria ▪ Mumps ▪ Bacterial meningitis	▪ Private room ▪ Door can stay open ▪ Patient must wear surgical mask outside of room (when going for imaging, procedures, etc.) ▪ Visitors should be 3 feet away (droplets can travel!)
Airborne	Precautions taken to protect against infection spread through particles that can stay in the air and travel	▪ Gown ▪ Non-sterile gloves ★ **N95 respirator or respiratory hood**	MEMORY TRICK "MTV" ▪ **M**easles (rubeola) ▪ **T**uberculosis ▪ **V**aricella (chicken pox) & herpes zoster (shingles)	▪ Private room with monitored negative air pressure (must keep the door closed) ▪ N95 respirator or respirator hood (supplies air through a hose) ▪ Patient must wear surgical mask outside of room (when going for imaging, procedures, etc.)
Neutropenic 	Type of isolation used to protect patients with weak immune systems from germs ★ *You may hear this called "reverse isolation" which is designed to protect the patient from other people's germs. Normally, we are protecting others from a sick/contagious patient.*	Healthcare workers, all visitors, and the patient will wear: ▪ Gown ▪ Non-sterile gloves ▪ Surgical mask	▪ ↓ WBC count (neutropenia) ▪ Immunocompromised patients ▪ Oncology (cancer) patients ▪ Patients receiving chemotherapy	**AVOID:** All these things can carry harmful microbes ✗ Fresh flowers ✗ Fresh or raw fruits/vegetables ✗ Undercooked meats ✗ Standing water

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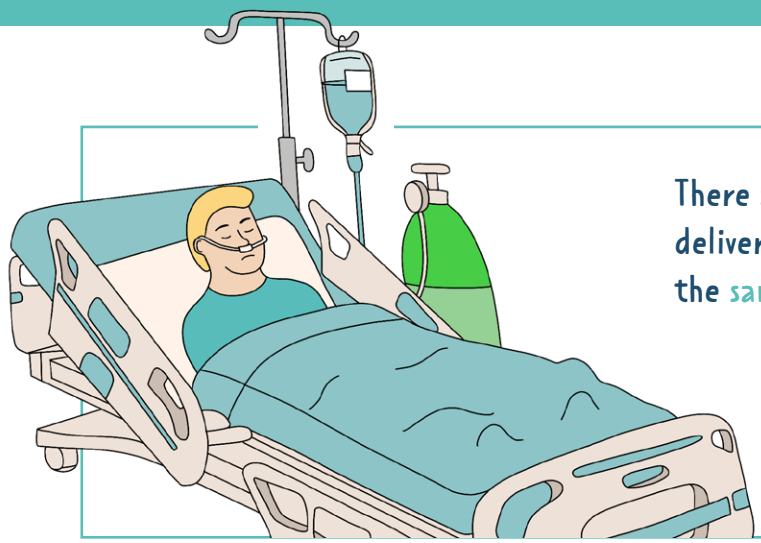
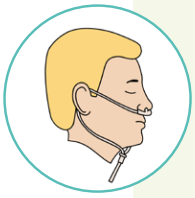
Oxygen Delivery Systems



There are many types of oxygen delivery systems, but they all have the same goal:

They are used to **ADMINISTER, REGULATE,** and **SUPPLEMENT OXYGEN.**

MASK TYPE	FLOW RATE	FiO_2	DESCRIPTION	
 Nasal cannula	**1 – 6** L/min	**24 – 44%**	**Low-flow device** Used for non-acute situations	▪ Two prongs (one for each nostril)—these point down! ▪ Loops behind the ears, not the head! ▪ Can be ordered as humidified to decrease nasal irritation/dryness
Simple mask	**6 – 12** L/min	**35 – 50%**	**Low-flow device** Used for non-acute situations	▪ Air holes on both sides allow for some external air exchange ▪ Fits to the face with a strap around the head
Non-rebreather mask	**10 – 15** L/min	**60 – 90%**	**Low-flow device** Used for acutely ill patients	▪ One-way valves prevent outside air from entering ▪ Reservoir bag: holds oxygen as a "reservoir" source for the patient as they breathe
High-flow oxygen therapy	Up to **60** L/min	**21 – 100%**	**High-flow device** Often a high-flow nasal cannula	▪ Similar setup to nasal cannula, but with high-flow oxygen ▪ Can be ordered as humidified to decrease nasal irritation/dryness
Venturi mask	**2 – 15** L/min	**24 – 50%**	**High-flow device** Best for patients with chronic lung disease	▪ Most precise O_2 delivery without intubation (controlled percentages of O_2) MEMORY TRICK **Venturi mask think Very accurate O_2**
Face tent	at least **10** L/min	**24 – 100%**	**High-flow device** Effective for those who don't tolerate masks well	▪ Good for patients with facial trauma or burns, or patients who will not tolerate a fitted mask ▪ Provides humidification

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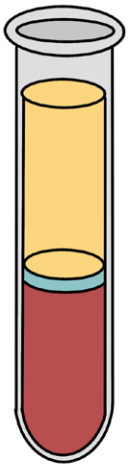
Blood Types

Before a blood transfusion happens, a patient's blood should be sent to the lab to be typed & cross-matched. If a patient receives blood that is not a compatible type, it can lead to a transfusion reaction and potentially death.



Plasma
55% of total blood

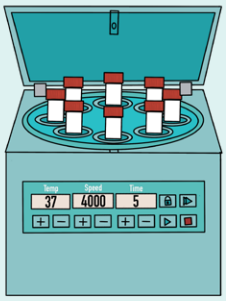
Water
Ions
Proteins
Nutrients
Waste
Gases

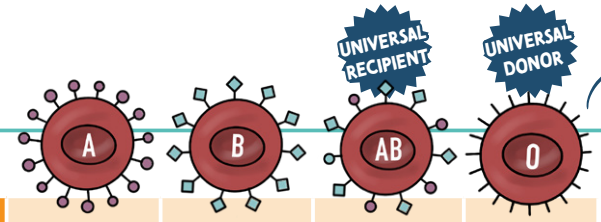
White blood cells & platelets
< 1% of total blood

Erythrocytes
45% of total blood

CENTRIFUGE

This is a device that uses force to separate components of fluids. It separates fluids of different densities and is used by labs to separate blood.





MEMORY TRICK: O think universal dOnOr

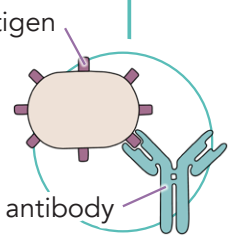
ANTIGEN	A	B	A & B	NONE
ANTIBODIES	B	A	NONE	A & B
RECIPIENT	A, O	B, O	ALL	O
DONOR	A, AB	B, AB	AB	ALL

Can receive any blood type

Compatible with any blood type

ANTIGENS
- Elicits a strong immune response
- Identifies the cell

PLASMA ANTIBODIES
- Protects body from "invaders" (think ANTI)
- Opposite of the type of antigen found on the RBC



RH FACTOR

Rhesus (Rh) factor is an inherited protein found on the surface of red blood cells.

If your blood has the protein, you're Rh **positive**.

If your blood lacks the protein, you're Rh **negative**.

HAS
Rh on surface
CAN RECEIVE
+ −

DOES NOT HAVE
Rh on surface
CAN RECEIVE
−

DONOR BLOOD TYPES

RECIPIENT BLOOD TYPES	O−	O+	A−	A+	B−	B+	AB−	AB+
O−	♥							
O+	♥	♥						
A−	♥		♥					
A+	♥	♥	♥	♥				
B−	♥				♥			
B+	♥	♥			♥	♥		
AB−	♥		♥		♥		♥	
AB+	♥	♥	♥	♥	♥	♥	♥	♥

Always check with your hospital's protocol about blood product administration

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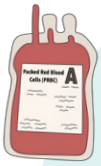
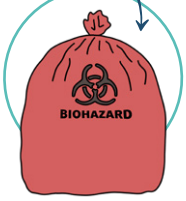
Blood Transfusions

Administration OF BLOOD TRANSFUSION

1. Insert an IV line using a 16G,* 18G, or 20G IV needle
   **commonly used for trauma patients*
2. Run it with normal saline 0.9% (keep-vein-open rate)
   Blood is transfused with a special Y-tubing with an inline filter
3. Begin the transfusion slowly
   - **A** The first 15 minutes are the **MOST CRITICAL**, the RN must stay at bedside
   - **B** After 15 minutes, the flow can be increased (unless a transfusion reaction has occurred)
   - **C** Vital signs are monitored every 30 minutes–1 hour
4. Dispose of the transfusion bag in a red biohazard bag
5. Document the patient's tolerance to the administration of the blood product



*If you use a needle that's too small (i.e., 24G) when administering blood products, it can cause the blood to **LYSE**.*

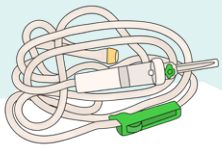


Facts ABOUT BLOOD TRANSFUSIONS

- Administered by the RN
- Normal saline is the only compatible solution to use with blood or blood components

MEMORY TRICK — *NO*rmal Saline think, there's *NO* other way!

- All blood products require a filter
  This is to remove clots, lumps of platelets & WBCS during the infusion

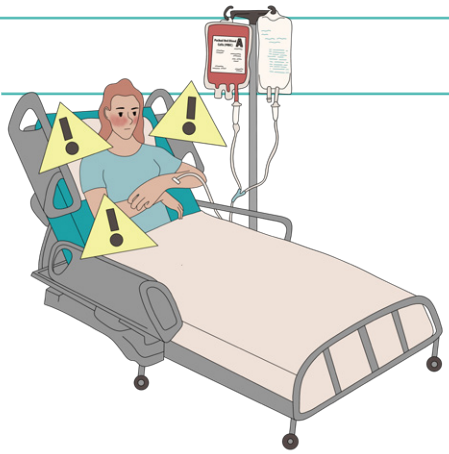


Numbers to Know:

A type & screen and a crossmatch are good for **72 HOURS**	Blood must be hung (started) within **30 MINUTES** from the time the blood is picked up from the blood bank	All blood must be transfused within **4 HOURS** of the time the blood was hung (started)

Transfusion Reaction

A transfusion reaction is an adverse reaction that happens as a result of receiving a blood transfusion.



SIGNS & SYMPTOMS

- ↑ Heart rate (tachycardia)
- ↓ Blood pressure (hypotension)
- ↑ Temperature (fever)
- Itching/urticaria/skin rash
- Wheezing/dyspnea/tachypnea
- Anxiety
- Flushing
- Back pain

OTHER COMPLICATIONS OF BLOOD TRANSFUSIONS

- Septicemia
- Blood that is contaminated with microorganisms
- Circulatory overload
- Blood being infused too rapidly for the client to tolerate

NURSING CONSIDERATIONS

1. **STOP** the transfusion
2. Change the IV tubing down to the IV site
3. Keep the IV open with normal saline
4. Notify the HCP & blood bank
5. Do not leave the patient alone
   (monitor the patient's vital signs & continue to assess)

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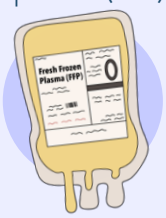
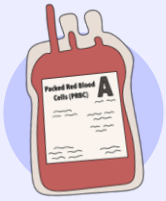
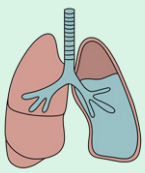
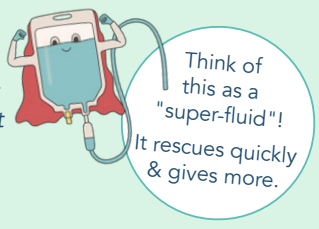
Colloids vs. Crystalloids

	Colloids	Crystalloids
	LARGE MOLECULES Colloids have ***LARGE*** molecules, making them more efficient at increasing fluid volume in the blood	**SMALL MOLECULES** Crystalloids have ***small*** molecules They are less expensive than colloids and provide immediate fluid resuscitation
EXAMPLES	Fresh frozen plasma (FFP)  Albumin  Packed red blood cells (PRBC)	Hypertonic solution Isotonic solution Hypotonic solution
USES	▪ Shock ▪ Burns ▪ Pancreatitis ▪ Excessive bleeding	Depends on the type of crystalloids (***Example:*** use isotonic solution to replace fluid loss; use hypotonic solution for those with hypernatremia or DKA)
HALF-LIFE	Hours or days	30–60 minutes
VEGAN & VEGETARIAN CONSIDERATIONS	Not an ideal choice for vegan or vegetarian patients	Safer choice for vegan & vegetarian patients
ALLERGIC REACTION/ ANAPHYLAXIS	Higher risk	Little to no risk
COMPLICATIONS	Large amounts of infused fluid can cause congestive heart failure (fluid buildup)	Large amounts of infused fluid can cause peripheral edema (swelling of the extremities) & pulmonary edema (fluid in the lungs) 
FLUID REPLACEMENT RATIO	1:1 ratio *Fluid volume out = Fluid volume replaced*	3:1 ratio *Crystalloids are better for rapid volume replacement because they provide MORE with less* 

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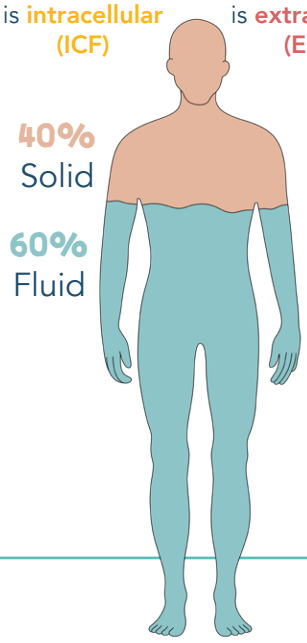
IV Therapy: Basics

Body Fluids Components

Our bodies are made up of about **60% WATER (FLUID)**

OF THIS 60%:

- **2/3** is **intracellular (ICF)**
- **1/3** is **extracellular (ECF)**



Body Fluid Compartments

FLUID in our body is found in **2 PLACES**:

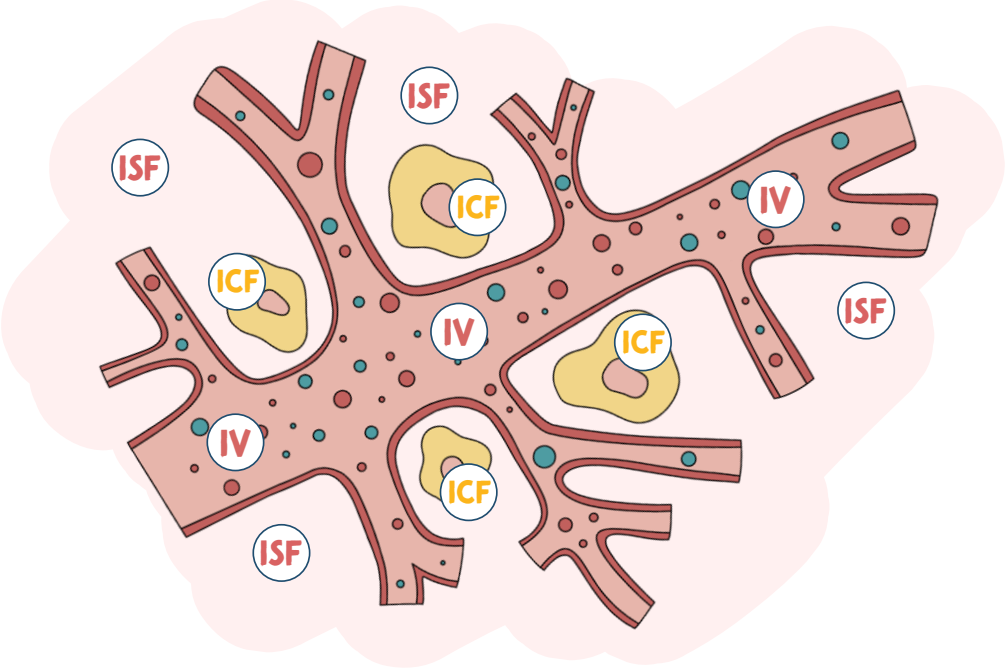
1. **INTRACELLULAR (ICF)**: fluid **INSIDE** the cell

There are millions of cells in our body

2. **EXTRACELLULAR (ECF)**: fluid **OUTSIDE** the cell

INTERSTITIAL FLUID (ISF)
Fluid that surrounds the cells in the **tissues**

INTRAVASCULAR (IV) FLUID
Plasma/fluid in the **blood vessels**



Cells & Homeostasis

The cells in the body love to have everything equal (homeostasis). So when fluids/solutes shift, **DIFFUSION/OSMOSIS** occurs to return to homeostasis again.

TIP
Sodium is a solute!

DIFFUSION

the movement of a **SOLUTE** from an area of

HIGHER

concentration

to an area of

LOWER

concentration

(until there is equal concentration)

OSMOSIS

the movement of **WATER** through a semipermeable membrane from a

LOWER

solute concentration

to a

HIGHER

solute concentration

(until there is equal concentration)

said another way...

from a **HIGHER** water concentration to a **LOWER** water concentration

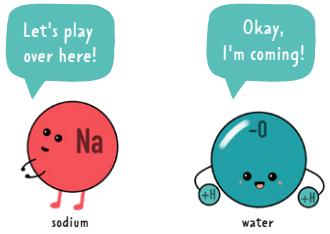
(until there is equal concentration)

Sodium & Water

MEMORY TRICK

Where **SODIUM** goes, **WATER** flows!

Sodium is the cool kid,
so water wants to be his friend.



Example: If sodium shifts into the cell (intracellular space), water will follow and leave the extracellular space (the vessel)

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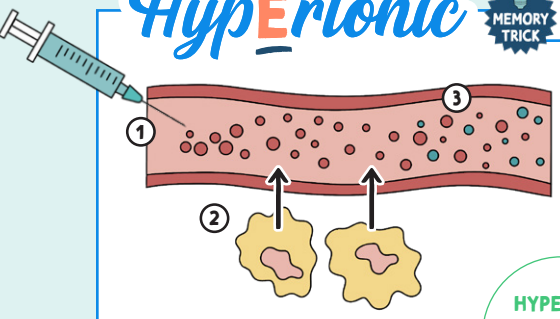
IV Therapy: Types of IV Solutions

HypErtonic

MEMORY TRICK: HypErtonic think Enters the vessel from the cells



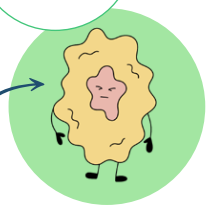
1 IV administration
2 effects of the solution
3 homeostasis after

MORE salt in the solution & **LESS** water in the solution

The vessel becomes **MORE** concentrated than the cell. Water then **LEAVES** the cell.

Therefore, the cells will **SHRINK.**

HYPERtonic = ↑ HIGH SALT concentration



EXAMPLES:

- 5% saline
- 3% saline
- 5% dextrose in 0.9% saline (D5NS)
- 5% dextrose in 0.45% saline (D5 ½ NS)
- 5% dextrose in LR (D5LR)
- 10% dextrose in water (D10W)

MEMORY TRICK: HYPERtonic think HIGH numbers

*The only exception to this memory trick is **5% DEXTROSE IN WATER (D5W)**

5% DEXTROSE IN WATER (D5W)
starts as ISOTONIC and then changes to HYPOTONIC when the dextrose is metabolized

USED FOR:

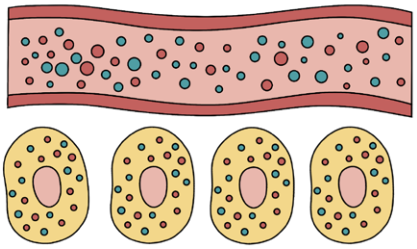
- Cerebral edema
- Hyponatremia (↓ levels of sodium)
- Metabolic alkalosis
- Maintenance fluid
- Hypovolemia

MONITOR FOR:

Fluid volume overload

Isotonic

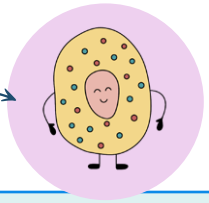
MEMORY TRICK: Isotonic think stays where I put it



EXPANDS intravascular fluid volume & replaces fluid loss

SAME osmolality as body fluids (equal water & particle ratio)

The cells will **STAY THE SAME.**



EXAMPLES:

- 0.9% sodium chloride = **normal saline (NS)**
- 5% dextrose in water (D5W)*
- Lactated Ringer's (LR)

This is the only solution compatible with blood or blood products

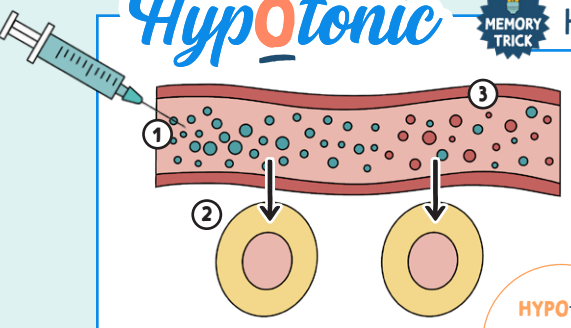
USED FOR:

- Blood loss (hemorrhage, burns, surgery)
- Dehydration (vomiting, diarrhea)
- Fluid maintenance
- Diabetic ketoacidosis (DKA)
  In DKA, there is so much glucose in the cells, they need water!

** See "5% dextrose in water (D5W)" box above*

HypOtonic

MEMORY TRICK: HypOtonic think goes Out of the vessel & into the cell



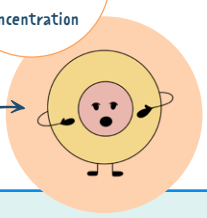
1 IV administration
2 effects of the solution
3 homeostasis after

LESS salt in the solution & **MORE** water in the solution

The vessel becomes **LESS** concentrated than the cell. Water then **ENTERS** the cell.

Therefore, the cells will **SWELL.**

HYPOtonic = ↓ LOW SALT concentration



EXAMPLES:

- 0.45% saline (1/2 NS)
- 0.33% saline (1/3 NS)
- 0.225 saline (1/4 NS)
- 5% dextrose in water (D5W)*

USED FOR:

- Helping kidneys excrete fluids
- Hypernatremia (↑ levels of sodium)

DO NOT GIVE WITH:

- ✘ ↑ ICP
- ✘ Burns
- ✘ Trauma

** See "5% dextrose in water (D5W)" box above*

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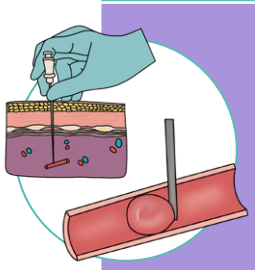
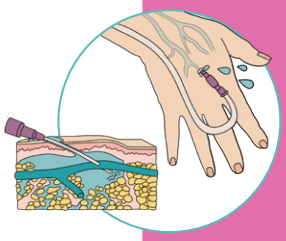
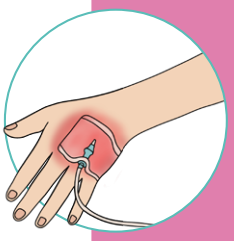
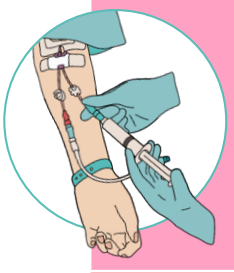
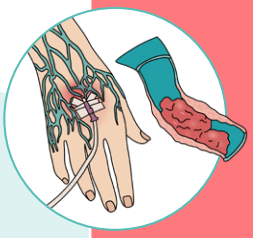
IV Therapy: Complications

	PATHOLOGY	SYMPTOMS	TREATMENT
 **AIR EMBOLISM**	Entry of air into the vein through the IV tubing	▪ Tachycardia ▪ Chest pain ▪ Hypotension ▪ ↓ LOC ▪ Cyanosis ▪ Dyspnea or cough	▪ Clamp the tubing ▪ Turn the patient onto their left side & place in **Trendelenburg position** ▪ Notify the HCP
 **INFILTRATION**	Leaking of IV fluid into surrounding tissue	AT THE SITE: ▪ Pain ▪ Swelling ▪ Coolness ▪ Numbness ▪ No blood return	▪ Remove the IV ▪ Elevate the extremity ▪ Apply a warm or cool compress ▪ Avoid rubbing the area
 **INFECTION**	Entry of microorganism into the body via IV	▪ Tachycardia ▪ Redness ▪ Swelling ▪ Chills & fever ▪ Malaise ▪ Nausea & vomiting	▪ Remove the IV ▪ Obtain cultures ▪ Possibly administer antibiotics
 **CIRCULATORY OVERLOAD**	Administration of fluids too rapidly (fluid volume overload)	▪ ↑ Blood pressure ▪ Distended neck veins ▪ Dyspnea ▪ Wet cough & crackles	▪ ↓ Flow rate (keep-vein-open rate) ▪ Elevate the head of the bed ▪ Keep the patient warm ▪ Notify the HCP
 **PHLEBITIS**	Inflammation of the vein Can lead to a **clot** (thrombophlebitis) 	AT THE SITE: ▪ Heat ▪ Redness ▪ Tenderness ▪ ↓ Flow of IV	▪ Remove the IV ▪ Notify the HCP ▪ Restart the IV on the opposite side
HEMATOMA	Collection of blood in the tissues	AT THE SITE: ▪ Blood ▪ Hard & painful lump ▪ Ecchymosis	▪ Elevate the extremity ▪ Apply pressure & ice

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Hypovolemia vs. Hypervolemia

HYPOVOLEMIA

low ↓ volume ↓ in the blood

ALSO CALLED: Dehydration • Fluid volume deficit (FVD) • Hypovolemic shock

CAUSES

- Loss of fluid from ANYWHERE
  - Thoracentesis
  - Paracentesis
  - Hemorrhage
  - NG tube
  - Trauma
  - GI losses
    - Vomiting
    - Diarrhea
- Third spacing
  - Burns
  - Ascites
- Polyuria (peeing a lot)
  - Diabetes
  - Diuretics
  - Diabetes insipidus

THIRD SPACING shifts the fluids from the **INTRAVASCULAR SPACE** (the vein) into the **INTERSTITIAL SPACE** (third space).

This causes a drop in the circulating blood volume.

SIGNS & SYMPTOMS

- Flat neck veins
- ↑ HR (weak & thready)
- ↑ Respirations
- ↑ Urine specific gravity
- ↓ BP

Less volume = Less pressure

- ↓ CVP
- ↓ Weight
- ↓ Skin turgor
- ↓ Urine output
- Dry mucous membranes
- Thirst

LABS

MEMORY TRICK: Concentrated (dehydrated) makes the #s Curve up

- ↑ Urine specific gravity
- ↑ Hematocrit (%)
- ↑ Serum sodium
- ↑ BUN

NURSING CONSIDERATIONS/ TREATMENT

- Fluid replacement
  - Administer via PO or IV
- Safety precautions
  - Risk for falls due to orthostatic hypotension
- Daily weight & I&Os

Monitor for fluid volume overload

HYPERVOLEMIA

high ↓ volume ↓ in the blood

ALSO CALLED: Overhydrated • Fluid volume excess (FVE)

CAUSES

- Heart failure
- Kidney dysfunction
  - Can't filter the blood = backup of fluids
- Cirrhosis
- ↑ Sodium intake

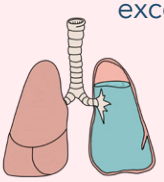
MEMORY TRICK: Where SODIUM goes, WATER flows! Sodium is the cool kid, so water wants to be his friend.

SIGNS & SYMPTOMS

- Jugular vein distention (JVD)
- ↑ HR (bounding)
- ↑ BP
- ↑ Weight
- ↑ CVP

More volume = More pressure

- Wet lung sounds
  - Crackles/dyspnea
  - Due to backflow of fluid from the heart



- ↑ Edema
- ↑ Polyuria
- Kidneys are trying to get rid of the excess fluid

LABS

MEMORY TRICK: DILUTED (overhydrated) makes the #s go DOWN

- ↓ Urine specific gravity
- ↓ Hematocrit (%)
- ↓ Serum sodium
- ↓ BUN

NURSING CONSIDERATIONS/ TREATMENT

- Low-sodium diet
- Daily weight & I&Os
- Diuretics
- High Fowler's or Semi-Fowler's position
  - Easier to breathe

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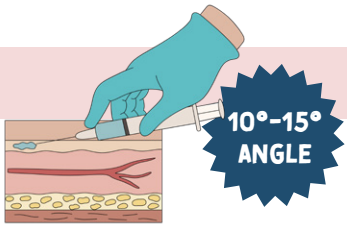
Parenteral Administration

Any route of administration that does not involve drug absorption through the GI tract.
Parenteral means outside of the digestive tract.

SLOWEST ABSORPTION

QUICKEST ABSORPTION

INTRADERMAL (ID)



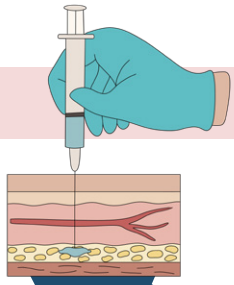
Should form a BLEB

USES:
- TB testing (Mantoux test/PPD)
- Allergy sensitivities

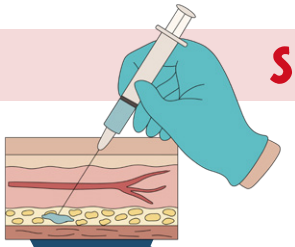
NEEDLE SIZE:
25–27 gauge

USUAL SITE:
Inner forearm

SUBCUTANEOUS (SUBQ)



Normal to overweight patients



Underweight patients

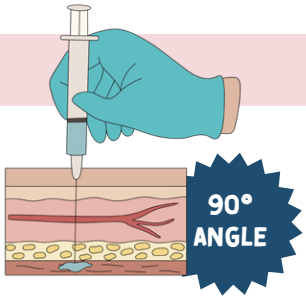
Giving a malnourished/underweight patient a medication at a 90° angle could lead to accidental intramuscular injury!

USES:
Nonirritating, water-soluble medications (insulin & heparin)

NEEDLE SIZE:
23–25 gauge

USUAL SITE:
- Abdomen
- Posterior upper arm
- Thigh

INTRAMUSCULAR (IM)



Use the Z-TRACK METHOD

Do not inject more than 3 mL (2 mL for the deltoid)

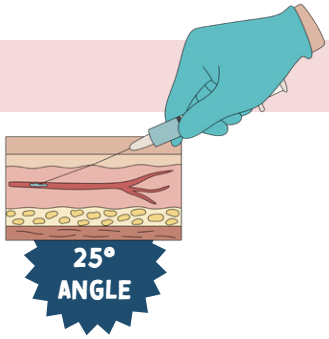
Divide larger volumes into two syringes & use two different sites

USES:
Irritating medications, solutions in oils & aqueous suspensions

NEEDLE SIZE:
22–25 gauge

USUAL SITE:
- Deltoid
- Vastus lateralis
- Ventrogluteal

INTRAVENOUS (IV)



25° angle used when starting an IV

USES:
Administration of medications, fluids & blood products

NEEDLE SIZE:
16 gauge:
patients who have trauma
18 gauge:
surgery & blood administration
22–24 gauge:
children, older adults

USUAL SITE:
- Hand
- Wrist
- Cubital fossa (antecubital)
- Foot
- Scalp

MEMORY TRICK
The Lower the number, the Larger the IV bore

GAUGES & IV USES

16G	18G	20G*	22G	24G
Trauma, surgery, rapid fluid administration (bolus)	Administering blood, rapid infusions (bolus), CT scans with IV dye	Medications, routine therapies, IV fluids	IV fluids, medications	Pediatric patients, elderly patients, very fragile/small veins

LARGEST ← → SMALLEST

Some hospitals allow blood to be administered with 20G

Always check with your hospital's protocol about IV and blood product administration

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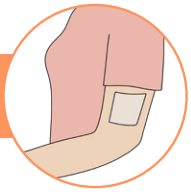
Enteral & Other Routes of Administration

ORAL

- **CONTRAINDICATIONS:** vomiting, aspiration precautions/absence of a gag reflex, decreased LOC, difficulty swallowing
- Have patient sit at 90° angle to help with swallowing
- NEVER crush enteric-coated or time-release medications
- Break or cut **scored** tablets only!

TRANSDERMAL



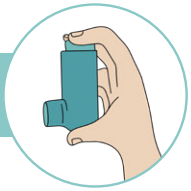
DOs:

- ✓ Rotate the application site
- ✓ Place the patch in a clean, dry, hair-free area
- ✓ Swim or shower with the patch in place

DON'Ts:

- ✗ Don't apply a new patch while an old patch is still applied
- ✗ Don't cut or alter the patch

INHALATION

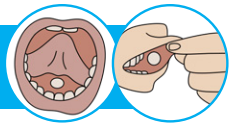


- 20–30 seconds between puffs
- 2–5 minutes between different medications

To prevent thrush:

- Rinse mouth after use of any inhaled medication (esp. steroids)
- Use a spacer (this helps the medication go into your lungs, leaving less in your mouth/throat)

SUBLINGUAL & BUCCAL

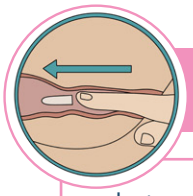


SUBLINGUAL: Under the tongue

BUCCAL: Between the cheek & the gum

Do not swallow!

Keep the medication under the tongue (sublingual) or in between the cheek and gum (buccal) until it has completely dissolved



SUPPOSITORIES

RECTAL

- Lateral or Sims' position
- Use lubrication
- Insert beyond the internal sphincter
- Leave it in for 5 minutes

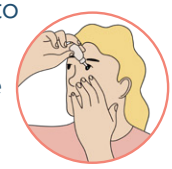
VAGINAL

- Lie supine with knees bent & feet flat on the bed, close to hips
- Insert the suppository along the posterior wall of the vagina (3–4 inches deep)
- Stay supine for at least 5 minutes

INSTALLATION (drops, ointments, sprays)

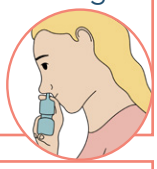
EYES

- If there is a dried section use a moistened sterile gauze & wipe from inner to outer canthus to prevent bacteria from entering the eye
- Have the patient tilt their head back slightly
- Pull the lower eyelid down gently to expose the conjunctival sac
- Hold the dropper 1–2 cm above the conjunctival sac & drop medication directly into the sac
- Close the eyelid & apply gentle pressure on the nasolacrimal duct for 30–60 seconds



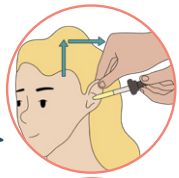
NOSE

- Place the tip of the bottle into one nostril & gently close the other nostril. Breathe through the nose while squeezing the bottle.
- Do not blow nose for 5 minutes after drop instillation



EARS

- Have the patient tilt their head
- Warm the solution before administration to prevent vertigo & dizziness



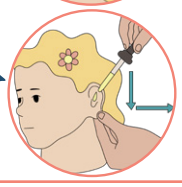
MEMORY TRICK

AD**U**LT → **UP**

ADULTS: pull pinna **UP**ward & outward

CHIL**D** → **DOWN**

CHILDREN < 3 YEARS OF AGE: pull ear lobe **DOWN** & back



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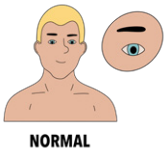
Integumentary (Skin) Overview

Inspection of the Skin

Color changes are more difficult to notice in patients with darker skin

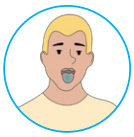
	DESCRIPTION	INDICATION	LOCATIONS
PALLOR	Loss of color	Lack of blood flow, anemia, shock	Face, conjunctiva, nail beds, palms, lips, mucous membranes
ERYTHEMA	Redness *can be blanchable or nonblanchable*	Inflammation, localized vasodilation, sun exposure, rash, hyperthermia	Skin (areas of trauma or pressure)
JAUNDICE	Yellow to orange	Liver dysfunction	Skin, sclera, mucous membranes
CYANOSIS	Bluish or blue-tinged	Hypoxia (due to not enough oxygen or impaired venous return)	Lips, mucous membranes, nail beds, skin

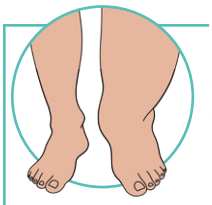
The best way to assess for **JAUNDICE** is to press gently on the forehead or nose. If the skin looks yellow where you applied pressure, it indicates jaundice.



PERIPHERAL CYANOSIS
(cyanosis of the peripherals (fingertips, palms, toes)
Rarely a life-threatening medical emergency

CENTRAL CYANOSIS
(cyanosis around the mouth, tongue, or mucous membranes
⚠ Medical emergency!





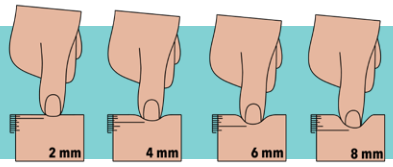
EDEMA is the accumulation of excess fluid in the body's tissues that causes swelling of the skin

WEEPING EDEMA Areas that have pitting edema can leak fluid out directly from the skin

EDEMA can be:
- NON-PITTING
- PITTING

PITTING is when you press the edematous area for a few seconds and it dimples or pits

GRADING PITTING EDEMA



+1 = TRACE | +2 = MILD | +3 = MODERATE | +4 = SEVERE

Types of WOUND DRAINAGE

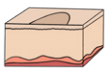
SEROUS	Clear, watery plasma
SEROSANGUINEOUS	Pale, pink, watery mixture of clear and red fluid
SANGUINEOUS	Bright red blood
PURULENT	Thick, yellowish-green. Foul odor.

Indicates active bleeding

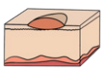
May indicate infection

PRIMARY LESION

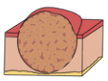
Develops as a result of a disease process



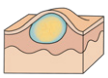
MACULE
Flat discoloration of the skin < 1 cm
Example: freckles



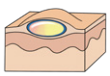
PAPULE
Solid, slightly elevated lesion < 1 cm
Example: moles



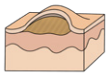
NODULE
Solid, elevated lesion > 1 cm
Example: lipomas



PUSTULE
Enclosed, pus-filled cavity
Example: acne



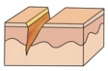
WHEAL
Superficial, raised lesion
Example: allergic reactions



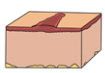
VESICLE
Elevated cavity containing clear fluid
Example: chickenpox, shingles

SECONDARY LESION

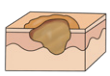
Results from a primary lesion or due to a patient's actions (scratching, picking)



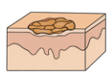
FISSURE
Linear crack/tear with abrupt edge
Example: anal fissures, athlete's foot



SCAR
Area where normal tissue is lost & replaced with connective tissue
Example: healed areas after surgery/injury



EROSION
Scooped-out, shallow depression
Example: severe pressure injuries



SCALE
Compact, flaky skin (silvery or white)
Example: exfoliative dermatitis which is caused by a medication reaction

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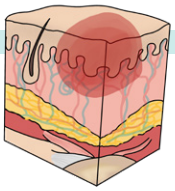
Pressure Injuries (Ulcers)

May also be called Decubitus Ulcers or "Bed Sores"

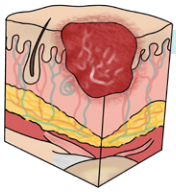
What is a Pressure Injury?

The breakdown of skin (compromised skin integrity) due to unrelieved pressure



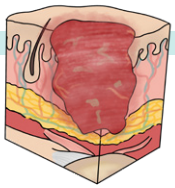
TYPE 1

- Skin is intact (unbroken)
- Nonblanchable redness
- Swollen tissue
- Darker skin → may appear blue/purple



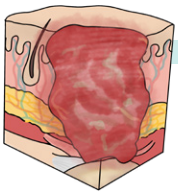
TYPE 2

- Skin is NOT intact
- Partial thickness loss
- No fatty tissue is visible
- Superficial ulcer



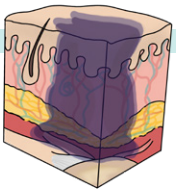
TYPE 3

- Skin is NOT intact
- Full thickness SKIN loss
  - Damage to or necrosis of subQ tissue
  - No bone, muscle, or tendon exposed
- Ulcer extends down to the underlying fascia, but not through it
- Deep crater with or without tunneling



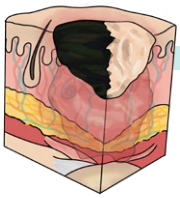
TYPE 4

- Skin is NOT intact
- Full thickness TISSUE loss
  - Destruction of tissue
  - Bone, muscle, or tendon exposed
- Deep pockets of infection & tunneling



DEEP TISSUE INJURY (DTI)

- Skin is intact (unbroken)
- Tissue beneath the surface is damaged
- Appears purple or dark red



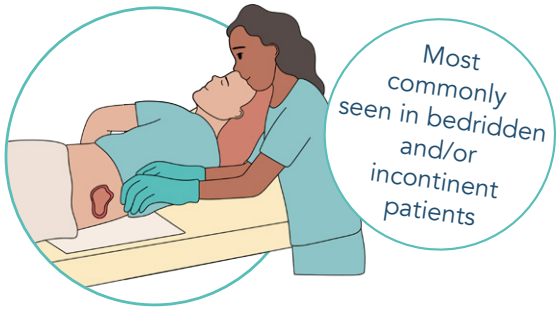
UNSTAGEABLE

Stage cannot be determined due to eschar or slough covering the visibility of the wound

RISK FACTORS

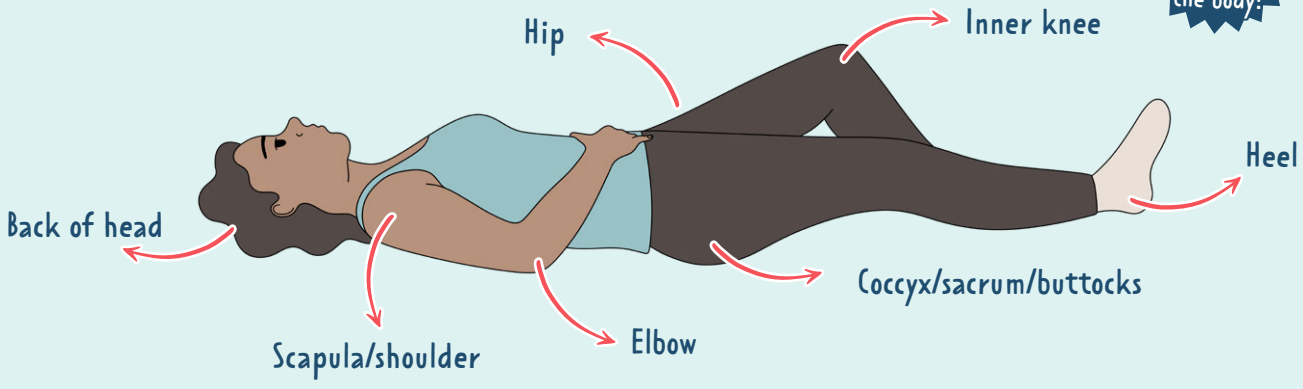
MNEMONIC "AVOIDS PRESS"

- **A** Aging skin
- **V** Vascular disorders
- **O** Obesity
- **I** Immobility & incontinence
- **D** Diabetes
- **S** Skin friction
- **P** Poor nutrition
- **R** Reduced RBCs (anemia)
- **E** Edema
- **S** Sensory deficits
- **S** Sedation



Common Areas for Pressure Injuries

Think about the bony areas of the body!



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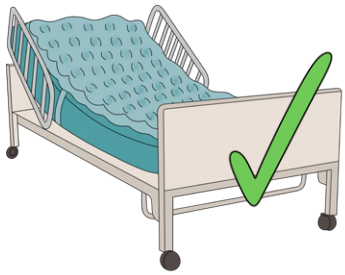
Pressure Injuries (Ulcers)

May also be called Decubitus Ulcers or "Bed Sores"

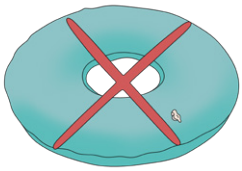
NURSING CONSIDERATIONS

RELIEVE PRESSURE

Apply pressure-relieving devices *(specialty beds, mattress overlays, air cushions, foam-padded seat cushions, float heels etc.)*



Do not use donut-type devices or synthetic sheepskin



SKIN HYGIENE

- Clean skin with mild soap & moisturize for hydration
- Do not scrub or rub bony prominences
- For incontinent patients:
  - Clean regularly & use a barrier cream
- Keep skin dry

Urine & stool on the skin can lead to skin breakdown!

PROPER NUTRITION

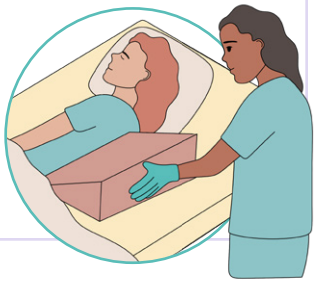
- ↑ Protein intake
- Adequate hydration
- Possible enteral nutrition

Protein promotes wound healing

REPOSITIONING

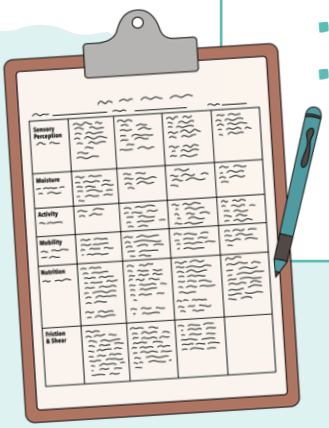
Turn/reposition patient every 2 hours while in the bed

- LIFT, don't PULL
  - Pulling could cause shearing & friction from force



MONITOR

- Size & color of the wound
- Braden Scale scoring (tool for anticipating the risk of pressure ulcers)



BRADEN SCALE

Assess your patient's skin for pressure injuries **EVERY** shift using the Braden Scale!

Looks at **6** categories:

- **SENSORY PERCEPTION**
- **MOISTURE**
- **ACTIVITY**
- **MOBILITY**
- **NUTRITION**
- **FRICTION & SHEAR**

Interpretation

LOW RISK: 22 - 23
LESS RISK: 19 - 21
HIGH RISK: < 18

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Notes

Every **ACCOMPLISHMENT** starts with the decision to **TRY**.

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Mental Health

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Tips for Mental Health Class

Mental health is all about communication

In non-psychiatric conditions, the treatment can be more straightforward. Often, we're able to administer a medication or perform a clinical intervention.

But in mental health, part of the treatment and interventions centers around **communication**.

It is crucial to understand and use therapeutic communication.

Know your interventions

Mental health class will include many diagnoses and conditions that don't all require the same interventions.

Learn about interventions for patients that may include multiple elements such as nutrition, exercise, different types of therapy, or the use of restraints.

One size does not fit all, and understanding the conditions means understanding how to intervene as well.

Pharmacology ties in

Remember that medications are not the only treatment; it's important to learn all associated therapies too. Of course, many conditions do require medications.

It's also important to learn the different medication classes and for which diagnoses they are used for. Be sure to know how they work (their mechanism of action) as well as how the medications interact with one another.

Be familiar with the antidepressant & antipsychotic medication classes, along with which conditions they are used to treat.

Safety, safety, safety

You will hear that safety is always the number one priority. Well, it is!

As you learn about these conditions and answer any mental health questions, keep in mind that patients may be at risk of hurting themselves or others.

Learn what things you can do as the nurse to keep you and your patients safe!

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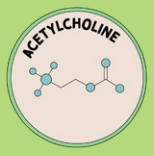
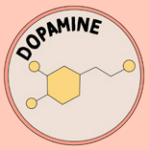
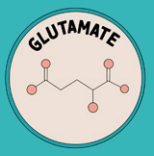
Neurotransmitters Overview

	DESCRIPTION OF FUNCTION	KEY POINT TO KNOW	HOW IT AFFECTS MENTAL HEALTH	
			↓ LOW LEVELS	↑ HIGH LEVELS
 **ACETYLCHOLINE**	▪ Learning & retaining information ▪ Attention span ▪ Muscle movements	★ **MEMORY** Low levels are associated with Alzheimer's and loss of memory	★ **Alzheimer's** ▪ Memory loss ▪ Dementia	▪ Depression ▪ Anxiety ▪ Muscle paralysis
 **DOPAMINE** (catecholamine)	▪ Mood & pleasure sensation ▪ Libido ▪ Physical movement & motions ▪ Sleep habits	★ **ADDICTION** Dopamine release = reward & return (urge to repeat a behavior)	▪ Parkinson's ▪ Depression ▪ Fibromyalgia ▪ Lack of motivation	▪ Schizophrenia ▪ Hallucinations ▪ Bipolar disorder ▪ Manic episodes
NOREPINEPHRINE (catecholamine)	▪ Attention span ▪ Mood ▪ Motivation ▪ Energy ▪ Concentration	★ **BALANCE IS KEY** A swing in either direction can cause anxiety & depressive sensations	▪ Depression ▪ ADHD/ADD ▪ Postpartum depression	▪ Increased anxiety ▪ Panic attacks ▪ Overstimulation
EPINEPHRINE (ADRENALINE) (catecholamine)	▪ Concentration & focus ▪ Survival mechanisms	★ **FIGHT OR FLIGHT** This response kicks in to save your body from dangerous situations	▪ Lethargy ▪ Lack of motivation ▪ Lack of concentration	▪ Anxiety ▪ Panic attacks
GABA (gamma-aminobutyric acid)	▪ Sleep regulation ▪ Feelings of calm ▪ Stimulation & brain activity	★ **CALM** Balanced & elevated levels lead to sensations of calm	▪ Schizophrenia ▪ Anxiety ▪ Panic disorders ▪ PTSD	▪ Improved concentration ▪ Anxiety reduction ▪ Sleep disorders or hypersomnia
 **GLUTAMATE**	▪ Memory ▪ Nerve health & transmission	★ **EXCITEMENT** This excitatory neurotransmitter helps transmit messages in the brain	▪ ADHD ▪ Fatigue ▪ Poor energy	▪ Restlessness ▪ Anxiety & panic ▪ Insomnia ▪ Pain disorders
SEROTONIN	▪ Mood regulation ▪ Sleep ▪ Libido	★ **MOOD** MEMORY TRICK Low Serotonin think Low & Sad High Serotonin think Happy & Smiley	▪ Depression ▪ Fatigue ▪ Anxiety	▪ Improved mood Dangerously high levels = serotonin syndrome

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Therapeutic Communication

✓ Dos

- ✓ Allow the patient to control the discussion
- ✓ Give recognition/validation (not advice)
- ✓ Engage in active listening
- ✓ Use open-ended questions
- ✓ Provide therapeutic touch if appropriate for the situation
- ✓ Use a calm, clear voice
- ✓ Focus on the patient and their feelings, not your own
- ✓ Paraphrase/restate/summarize

So, you are saying you haven't been sleeping well?

Examples:

Is there something you would like to talk about?

So, you are saying you haven't been sleeping well?

Tell me more about that.

Tell me more about_____.

✘ Don'ts

- ✘ Ask why
- ✘ Ask an excessive amount of questions
- ✘ Give advice
- ✘ Give false reassurance
- ✘ Change the conversation topic
- ✘ Give approval or disapproval
- ✘ Use close-ended questions/statements
- ✘ Talk about yourself or your own experiences

Don't worry, everything will be fine.

Don't worry.

I know how you feel — when my mom was diagnosed with cancer...

Examples:

Don't worry!

Don't be silly!

I think you should _____.

That's great!

Types of Communication

	EXPLANATION	EXAMPLE
NONVERBAL	Ability to communicate & provide comfort ***without*** the use of words	▪ Body language ▪ Facial expressions ▪ Eye contact ▪ Posture ▪ Vocal cues (yawning, tone of voice, pitch of voice)
THERAPEUTIC	Helpful & conducive communication methods for healing, trust & relationship establishment	▪ Validation of feelings ▪ Use of a soft, calming voice ▪ Paraphrasing and restating ▪ Open-ended questions
NON-THERAPEUTIC	Communication that creates a barrier to healing or establishing a trusting relationship	▪ Asking why a person feels something ▪ Talking about yourself ▪ Using close-ended statements ▪ Changing topics

ACTIVE LISTENING	Act of listening ***intentionally*** & engaging with the speaker	▪ Nodding
		▪ Maintaining eye contact
		▪ Using phrases like:
		  ▪ "I understand."
		  ▪ "That must make you feel..."

WRITTEN COMMUNICATION (Helpful for nonverbal patients)	Expressing emotion or communication with another person by writing words	▪ Journaling ▪ Sending letters/notes to others ▪ Writing therapeutic notes to your "former self" ▪ Creating lists/schedules ▪ Writing notes on a whiteboard

MOTIVATIONAL INTERVIEWING	Leading a discussion with helpful cues or questions for a patient to reflect upon & answer	▪ Asking/leading a conversation with:
		  ▪ "How does that make you feel?"
		  ▪ "What are the positives/negatives of this?"
		  ▪ "What would you like to change?"
		  ▪ "What seems to help?"
		  ▪ "Can you help me understand this better?"

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Bipolar Disorder

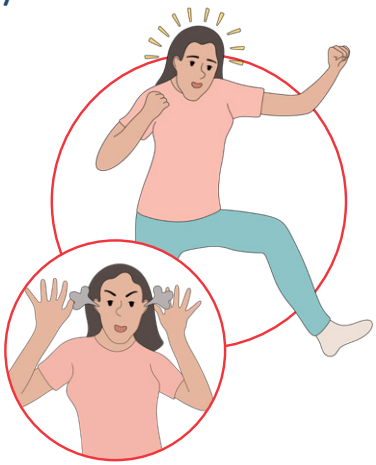
MOOD SWINGS:

Depression to mania with periods of normalcy

swings from

Depressive PHASE
Periods of **LOW** mood

Manic PHASE
Periods of **HIGH** mood



SIGNS & SYMPTOMS:

DEPRESSIVE	MANIC
▪ Depression & sadness ▪ ↓ Energy levels ▪ Sleep disturbances ▪ Too much or too little sleep	▪ Hyper mood ▪ Irritable ▪ Poor judgment ▪ Flight of ideas *(scattered conversation with rapid speech)* ▪ Grandiosity ▪ Impulsivity (Example: spending large amounts of money) ▪ Hallucinations/delusions

This HYPER mood can cause exhaustion, malnutrition & dehydration!

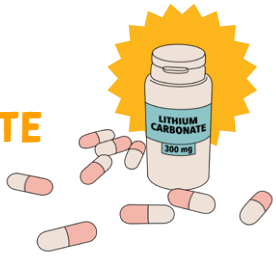
RISK FACTORS:

- Family history of serious mental illness or bipolar disorder
- History of trauma or severe stress (Example: losing a loved one)
- Alcohol or drug abuse

MEDICATION:

Gold standard medication:

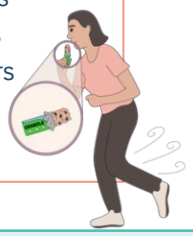
LITHIUM CARBONATE



NURSING CONSIDERATIONS:

- **Offer energy & protein-dense foods that are easily consumed on the go** (especially during a manic phase)
  - Provide a safe environment
    - Remove harmful objects from the room
  - ↓ Stimuli
    - Turn off or turn down the TV & music

- Hamburgers
- Sandwiches
- Granola bars
- Shakes
- Fruit juices



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Anorexia Nervosa (AN)

PATHOLOGY:

Powerful fear of gaining weight → Restriction of caloric intake & obsessive monitoring → Distorted body image

SIGNS & SYMPTOMS:

- Underweight (BMI < 18.5)
- Amenorrhea (females)
- Presence of lanugo

Absence of menstruation due to calorie restriction & excessive exercise, causing hormone levels to decrease

Soft hair covering the body in an attempt to warm itself due to lack of body fat

- Poor skin health/turgor
- Constipation
- Electrolyte imbalances
- Cardiac arrhythmias
- Change in vital signs
  - ↓ Heart rate
  - ↓ Blood pressure
  - ↓ Temperature

Due to dehydration

RISK FACTORS:

- Family history of eating disorders
- Anxiety & panic disorders
- Depression
- Alcohol dependence
- Post Traumatic Stress Disorder (PTSD)
- Obsessive tendencies

TREATMENT:

THERAPIES

- Behavioral therapy
- Supervised eating
- Inpatient admission for those in crisis

MEDICATIONS

- Anxiolytics
- Antidepressants

NURSING CONSIDERATIONS:

SLOW reintroduction to nutrition

REFEEDING SYNDROME:

Caloric intake introduced TOO QUICKLY, leading to:

- Erratic glucose levels
- Major electrolyte imbalance
- Thiamine deficiency
- Death

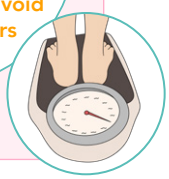
CHECKING WEIGHT

Check the patient's weight post-void, using the same:

- Time each day
- Density of clothing/ limited clothes
- Scale

Hearing or seeing weight & related values can be VERY triggering

★ Position patient BACKWARD to avoid seeing numbers on scale



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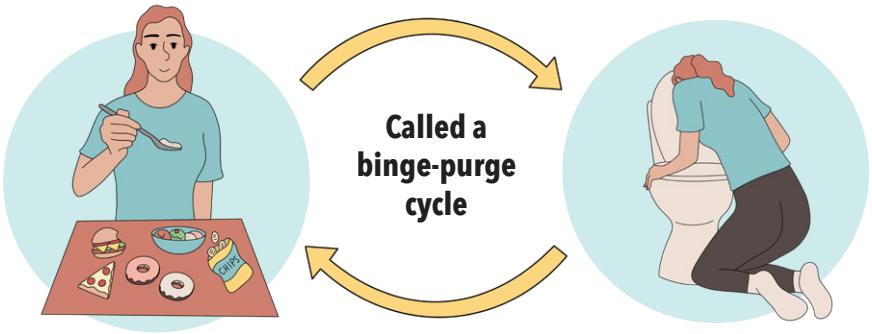
Bulimia Nervosa

PATHOLOGY:



Binge eating

Called a binge-purge cycle

Purging
to compensate for behaviors

Purging can include:
- Causing self to vomit
- Overexercising
- Using rapid-acting laxatives
- Tightly restricting of food

SIGNS & SYMPTOMS:

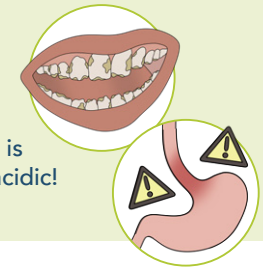
BEHAVIORAL SIGNS

- Taking frequent trips to bathroom (purging)
- Eating alone and avoiding meal time
- Running water when visiting bathroom
- Using laxatives and diuretics

PHYSICAL SYMPTOMS

- Typically normal normal weight to overweight
- Electrolyte imbalances
- Cardiac arrhythmias
- **From vomiting:**
  - **Tooth erosion**
  - **Injury to esophagus**
  - **Bad breath**

Vomit is very acidic!



RISK FACTORS:

↑ dopamine levels & ↓ serotonin levels may be related to episodes of vomiting
↓
Vomiting releases endorphins
↓
Behavior is reinforced

- Underlying depression
- Anxiety & panic disorders
- Obsessive-compulsive disorder (OCD)
- Personality disorders
- History of trauma
- Family history of eating disorders
- Poor relationships/connections in early childhood

LAB VALUES:

MEMORY TRICK: **L**ow **L**abs from e**L**ectro**L**ytes **L**eaving the body

- ↓ Potassium levels
- ↓ Sodium levels
- ↑ Serum amylase from parotitis
- ↑ BUN & creatinine

Acute kidney injury is common with this disease

MEDICATIONS:

✓ **Fluoxetine (Prozac)**: the only SSRI approved for these patients

✗ **Bupropion:** contraindicated

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Binge Eating Disorder (BED)

PATHOLOGY:

Person eats amounts exceeding requirements (overeats), or eats past the point of hunger

Most common form of eating disorders

Individual becomes:
- Upset
- Distressed
- Panicked
- Guilty

Cycle of binge eating & emotions

Addiction process may occur (addicted to particular foods), which **immediately & temporarily** makes them feel emotionally better

SIGNS & SYMPTOMS:

BEHAVIORAL SIGNS

- Hiding foods
- Eating late at night or during "off hours"
- Choosing non-nutritious/ indulgent foods
- Feelings of depression & guilt, especially after eating

PHYSICAL SYMPTOMS

- Typically overweight or obese
- GI upset/constipation/diarrhea
- Bloating
- Irregular bowel habits
- Irregular glucose levels
- Development of diabetes
- Hypertension

RISK FACTORS:

- Social anxiety or phobia of social situations
- Drug or alcohol misuse
- Post-traumatic stress disorder (PTSD)
- Depression
- Anxiety
- Family history of eating disorders

TREATMENT:

THERAPIES

- Cognitive behavioral therapy (CBT)
- Dialectical behavioral therapy (DBT)

MEDICATIONS

- Serotonin and norepinephrine reuptake inhibitors (SNRIs)
- Selective serotonin reuptake inhibitors (SSRIs)
- Stimulants

ADHD medications can decrease appetite

NURSING CONSIDERATIONS:

MONITOR: ▪ Strict intake & output ▪ Regular weight checks ▪ Blood pressure ▪ **Activity** (bathroom visits, hiding)

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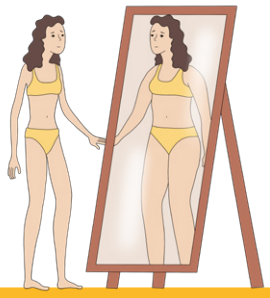
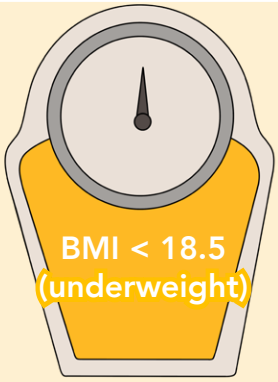
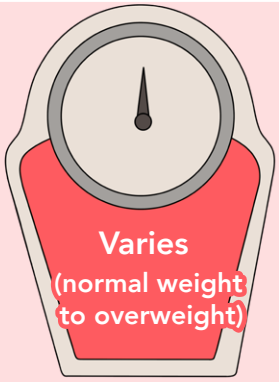
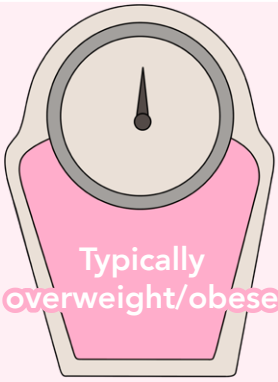
Eating Disorders: Quick Glance



	ANOREXIA NERVOSA	BULIMIA NERVOSA	BINGE EATING
WEIGHT			
CHARACTERISTICS	Starving self or eating less	Eating followed by purging	Overeating large amounts of food in a short amount of time (not followed by purging)
COMPLICATIONS	▪ Amenorrhea (no periods) ▪ Osteoporosis ▪ Infertility ▪ Cardiac issues ▪ Dehydration & electrolyte imbalances ▪ Constipation	▪ Cavities in the teeth ▪ Tooth erosion ▪ Damage to the esophagus ▪ Stomach ulcers ▪ Dehydration	▪ Diabetes mellitus ▪ Hypertension ▪ Hyperlipidemia ▪ Sleep disturbances
CLASSIC SYMPTOMS	▪ Presence of lanugo	▪ Malodorous breath	▪ Eating late at night or during "off hours" ▪ Choosing non-nutritious/ indulgent foods

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Schizophrenia Spectrum Disorder Overview

A serious mental disorder where patients interpret their reality abnormally

POSITIVE SIGNS & SYMPTOMS

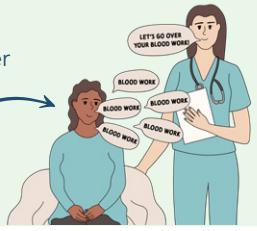
Patients with schizophrenia may show **positive** symptoms. These are symptoms that should not be present.

May also be called *"psychotic symptoms"*

- **Hallucinations**
  False sensory perceptions which can present as *visual*, *tactile* (touch), *olfactory* (smell), *gustatory* (taste) or *auditory* (hearing)

- **Delusions**
  A false belief or judgment
  Example: They feel they are being followed and/or watched by someone

- **Anxiety/agitation**
- **Jumbled speech**
- **Flight of ideas**
  Thoughts that are constantly changing and have no connection to one another
- **Disorganized behavior**
- **Echolalia** MEMORY TRICK: ECHOlalia – think ECHOing
  Repeating a phrase, commonly the last word of the sentence



NEGATIVE SIGNS & SYMPTOMS

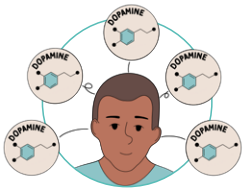
Patients with schizophrenia may show **negative** symptoms. This is the absence of healthy behaviors.

- **Flattened/bland affect**
- **Lack of energy**
- **Reduced speech**

- **Avolition**
  Lack of motivation
- **Anhedonia**
  Not capable of feeling joy or pleasure
- **Lack of social interaction**

RISK FACTORS:

- ↓ Dopamine levels
- Illicit substances (LSD & marijuana)
- Environment (malnutrition, toxins, viruses during pregnancy)
- Genetics (family history)



MEDICATIONS:

- Antipsychotic medications
- Antidepressants
- Mood stabilizers (lithium)
- Benzodiazepines

NURSING CONSIDERATIONS:

How to handle hallucinations:

- **Safety is the #1 priority**
  The patient's auditory hallucinations may be telling them to harm themselves or harm others

- Don't reinforce the hallucinations
  Tell the patient you know that the hallucinations seem real to them but you do not see or hear them

- Don't argue with the patient about the hallucinations

- Distraction helps with auditory hallucinations
  - Watching TV
  - Listening to music through headphones

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Somatic Symptom Disorder (SSD)

PATHOLOGY:

Unexplained physical symptom/illness → Severe anxiety, overwhelming thoughts & obsessions surrounding symptom/illness → Physical symptom/illness is unexplained by any medical diagnosis

SIGNS & SYMPTOMS:

BEHAVIORAL SIGNS

- Frequently scheduling doctor appointments
- Avoiding all appointments & healthcare
- Seeking out new doctors
- Obsessively seeking reassurance
- Researching symptoms online
- Having obsessive thoughts about symptoms

*Patients usually exhibit one **or** the other!*

PHYSICAL SYMPTOMS

- ★ **Pain**
- GI upset
- Chest pain
- Palpitations
- Fidgeting
- Insomnia
- Lack of appetite

RISK FACTORS:

- Could be due to stress
- Existing anxiety & panic disorders
- Major depressive disorder (MDD)
- History of trauma or a traumatic diagnosis

TREATMENT:

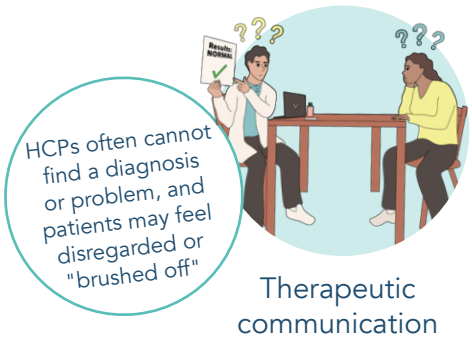
THERAPIES

- Hypnotherapy
- Techniques to ↓ stress

MEDICATIONS

- Antidepressants
  - Selective serotonin reuptake inhibitors (SSRIs)
  - Tricyclic antidepressants (TCAs)

NURSING CONSIDERATIONS:



Therapeutic communication is **KEY**

Limit the time you talk about the diagnosis/symptoms with the patient

Designate specific times for the patient to talk about their physical symptoms

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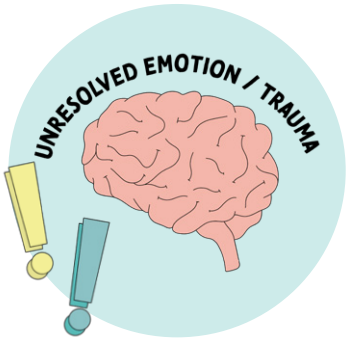
Conversion Disorder

PATHOLOGY:

Also called *functional neurological disorder*

Painful emotions or significant psychological stress → **CONVERTS** into or **MANIFESTS** as physical symptoms or loss of physical function

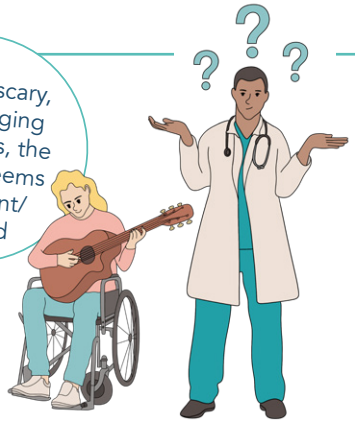


SIGNS & SYMPTOMS:

"La belle indifference"

- Psychogenic non-epileptic seizures
- Loss of mobility
- Loss of consciousness
- Paralysis or weakness
- Tremors
- Episodes of pain
- Lump in throat

Despite scary, life-changing symptoms, the patient seems indifferent/ unfazed



Physical paralysis, pseudo-seizures, and loss of physical function in extremities are MOST COMMON

RISK FACTORS:

- Dissociation & dissociative disorders
- Major depressive disorder (MDD)
- Family history of conversion & dissociative disorders
- Childhood trauma
- Sexual abuse

TREATMENT:

THERAPIES

- Body-oriented psychological awareness (BOPT)
- Dialectical behavioral therapy (DBT)
- Electroconvulsive therapy (ECT)
- Hypnosis
- Physical therapy (PT)

MEDICATIONS

- Antidepressants
- Antipsychotics

NURSING CONSIDERATIONS:

MONITOR for life-threatening changes

- Loss of airway
- Loss of consciousness
- Impaired breathing

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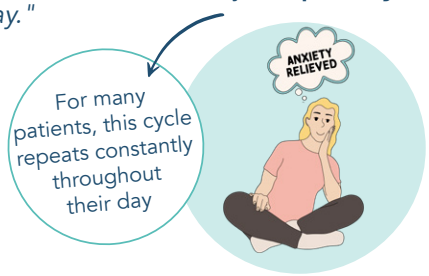
Obsessive-Compulsive Disorder (OCD)

PATHOLOGY:

An intrusive, unwanted thought or picture enters the mind of the patient → Anxiety increases, and the patient's urge to STOP the thought appears → *"I am having an intrusive thoughts about **germs**. If I **clean**, then I can ease my anxiety, and the thoughts will go away."* → Anxiety is eased by performing the **compulsion**, but only **temporarily**



For many patients, this cycle repeats constantly throughout their day

SIGNS & SYMPTOMS:

BEHAVIORAL SIGNS

- **Repetitive behaviors**
- **Trichotillomania** (pulling of one's hair)
- **Excoriation disorder** (picking of the skin)
  - Counting
  - Tapping objects
  - Touching objects
  - Praying
  - Returning to a particular place or object
  - Isolating oneself from friends or work
  - Feeling unable to perform daily tasks

PHYSICAL SYMPTOMS

- Severely dry skin (due to excessive hand washing)
- Broken/open sores or skin
- Lack of hair in patches
- Palpitations
- Chest pain
- Panic (flushing, diaphoresis)
- Fidgeting

RISK FACTORS:

- Childhood trauma
- Childhood abuse
- Family history of obsessive-compulsive disorders
- Depression
- Additional anxiety disorders
- Eating disorders

TREATMENT:

MEDICATIONS

- Antidepressants (SSRIs & SNRIs)
- Antipsychotics

THERAPIES

- Exposure & response

EXPOSURE & RESPONSE THERAPY

Therapy involving exposure to the intrusive thoughts but NOT allowing the patient to perform the compulsive action for relief.

Helps patients relieve anxiety WITHOUT compulsion & regain a healthy lifestyle.

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Types of Depression

MAJOR DEPRESSIVE DISORDER (MDD)

Has **at least 5** of these symptoms every day for **at least 2** weeks:

- **Lowered & depressed mood**
- **Lack of interest/joy**

MUST have 1 of the 2 for MDD diagnosis

- Feelings of:
  - Sadness
  - Emptiness
  - Lack of purpose in life
  - Guilt
  - Emptiness
  - Worthlessness
- Sleep disturbance (too much or too little)
- Episodes of crying or becoming teary-eyed
- Lack of interest in previously loved things

FACTS:

Major depressive disorder (MDD)

- Impairs the patient's normal functioning
- Is not the same depression seen in bipolar disorder
- Is constant, without mood swings
- Is not a normal, expected symptom of aging

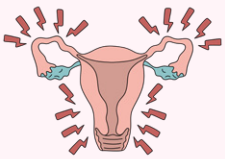
NURSING CONSIDERATIONS:

MONITOR for:

- Worsening of symptoms
- Signs of suicidal ideation

Reckless behavior, talk about death, social withdrawal, feelings of hopelessness

PREMENSTRUAL DYSPHORIC DISORDER (PMDD)



Depression that occurs during the luteal phase of the menstrual cycle

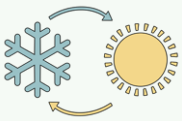
SUBSTANCE-INDUCED DEPRESSIVE DISORDER

Depression associated with withdrawal or the use of alcohol and drugs

PERSISTENT DEPRESSIVE DISORDER (DYSTHYMIA)

A milder form of depression compared to MDD, although it can turn into MDD later in life

SEASONAL AFFECTIVE DISORDER (SAD)



Depression that occurs during the winter months when there is less sunshine

TREATMENT: Light therapy

POSTPARTUM DEPRESSION

Depression that happens after a woman goes through childbirth

SYMPTOMS: Feelings of being disconnected from the world, fear of harming newborn

NURSING CONSIDERATIONS:

SAFETY IS THE #1 PRIORITY

Those struggling with depression have a higher suicide risk.

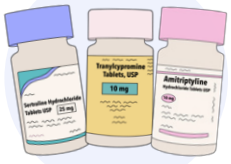
MONITOR FOR:

- Worsening of symptoms
- Signs of suicidal ideation

COPING METHODS

Help the patient identify coping methods, and teach alternatives if needed:

- Exercise
- Self-care
- Meditation
- Healthy relationships
- Individual & group therapy



ANTIDEPRESSANTS

- May take weeks to take effect, so educate the patient on compliance
- INCREASE the risk for suicide

Antidepressants give the patient enough energy to CARRY OUT a suicide plan. BE AWARE of changes, and seek help immediately if safety is a concern!

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Anxiety Disorders

	DEFINITION	CLASSIC SIGNS & SYMPTOMS	NURSING CONSIDERATIONS
GENERALIZED ANXIETY DISORDERS (GAD)	Significant, excessive worry beyond what is "rational"	▪ Persistent worry ▪ Neck & jaw tension ▪ Loss of appetite	Anxiety & elevated stress can indicate a medical issue (pulmonary embolism, myocardial infarction)
SOCIAL ANXIETY DISORDERS	Significant fear of being looked at by other individuals, with underlying fear of judgment & opinion	▪ Tremors ▪ Sweating ▪ Avoiding social events	**EDUCATE** to avoid: ▪ Caffeine ▪ Chocolate ▪ Excess sugar Anxiolytics may be prescribed
SEPARATION ANXIETY	Fear of losing a parent, friend, or significant other, with possible rumination about worst-case scenarios until reunited with the person they miss *(this is normal in late infancy)*	**GI upset (diarrhea, stomachaches, nausea)** ▪ Nightmares ▪ Jaw tension/ grinding of teeth	When a child is in the hospital, keep them in the loop about the plan & maintain routines as much as possible **EXAMPLE:** Visiting hours end at nine o'clock, but you will have your nurse and can call your mother at any time.
PANIC DISORDER	Exacerbations of anxiety ↓ Activation of body's fight-or-flight response, with release of stress hormones (cortisol, adrenaline) ↓ Escalation of PANIC	▪ Hyperventilation ▪ Chest pain/pressure ▪ Tachycardia/ palpitations ▪ Shortness of breath ▪ Ringing in ears	Encourage the patient to share experiences & similar feelings **AVOID:** ✗ Educating during an attack ✗ Generating additional stimulation (Instead, create a soothing environment!) ✗ Telling the patient to "calm down"

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Phobias

PHOBIA DEFINITION:

Constant and irrational fear of something that causes the person to avoid that environment, object, or being.

Types of Phobias

PHOBIA	FEAR OF:
Acrophobia	Heights
Agoraphobia	Open spaces
Arachnophobia	Spiders
Aerophobia	Flying
Claustrophobia	Enclosed spaces
Mysophobia	Germs
Nyctophobia	The dark
Zoophobia	Animals
Hematophobia	Blood
Blood-Injection-Injury (BII) phobia	Blood from injuries & needles

Write in more

MEMORY TRICK

ACROphobia – think of an ACRObat

MEMORY TRICK

AEROphobia – think of an AIRcraft

MEMORY TRICK

ZOOphobia – think of a ZOO full of animals

Most often causes bradycardia, hypotension & loss of consciousness (syncopal episodes/ fainting)

Position patient in supine position prior to any injection/blood draw

Exposure Therapy

Type of behavioral therapy where patients are encouraged to face their fears in a controlled & supportive environment

The hope is that facing their fear & viewing it from a new perspective can help them **overcome** it

TYPE	DESCRIPTION	MEMORY TRICK	EXAMPLE
IMAGINAL EXPOSURE	Patient imagines the situation or object in order to confront the fear	IMAGINAL – they IMAGINE it	A patient with nyctophobia (fear of the dark) will think of a time they were in the dark & describe their feelings & emotions
IN VIVO EXPOSURE	Patient confronts the fear in real life	VIVO means "alive" in Spanish – think of REAL LIFE	A patient with arachnophobia (fear of spiders) will go to a zoo and watch spiders crawl around in their cages
VIRTUAL REALITY EXPOSURE	**Combination of IMAGINAL & IN VIVO** Patient uses a VR headset & audio to simulate the fear	VIRTUAL REALITY Virtual: computer-generated environment Reality: simulation of real life	A patient with aerophobia (fear of flying) will see & hear an airplane experience through a VR headset & headphones

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Personality Disorders

Cluster A

Odd or eccentric

PARANOID

Very suspicious of others. They think everyone wants to harm or trick them.

SCHIZOID

Socially withdrawn. They don't have many expressions & are viewed by others as "strange."

SCHIZOTYPAL

Don't blend in. They are indifferent, seclusive & detached. They do not care for close relationships & experience extreme social anxiety.

Cluster B

Dramatic or emotional

ANTISOCIAL

Have no care for others. Unpredictable, do not follow the rules & can be aggressive.

BORDERLINE

Unstable, manipulative to self & others, afraid of neglect, impulsive & lacking a clear sense of identity.

HISTRIONIC

"Drama queens." Like to be the center of attention through behaviors such as being seductive & flirtatious.

NARCISSISTIC

Egocentric (narcissistic). Are in need of constant applause & have a lack of empathy.

Cluster C

Anxious or insecure

AVOIDANT

Very anxious in social settings & often avoids them. They desire close relationships but fear abandonment & rejection.

DEPENDENT

Extremely dependent on others. Searches urgently to find a new relationship when a previous relationship fails.

OBSESSIVE-COMPULSIVE MOST COMMON

Preoccupied with their perfectionist & control issues. They are stubborn & have minimal emotional expressions.

NURSING CONSIDERATIONS:

- Remember that **safety** is a priority
- Develop a therapeutic relationship
- Respect the patient's needs while still setting limits & consistency
- Give the patient choices to improve their feelings of control

Patients with a personality disorder are at greater risk for violence & self-harm

SAFETY FIRST

TREATMENT:

THERAPIES

- Psychotherapy
- Group therapy
- Cognitive therapy
- Behavioral therapy
- Dialectical therapy

MEDICATIONS

- Antidepressants
- Anxiolytics
- Antipsychotics
- Mood stabilizers

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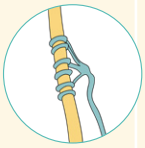
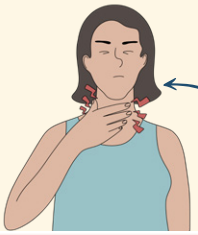
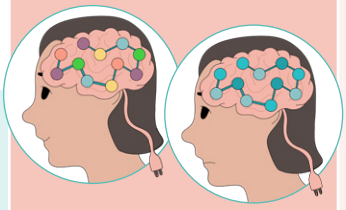
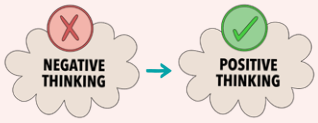
Mental Health Therapies

	DEFINITION	WHY IS IT DONE?	PATIENT EDUCATION
ELECTRO-CONVULSIVE THERAPY (ECT)	**Treatment that causes a quick electrical current to pass through the brain, inducing a seizure** Commonly used on those who are unresponsive to or cannot tolerate medication treatment Works quickly to decrease symptoms, unlike medications which may take weeks or months to take effect	▪ Bipolar disorder ▪ Severe depression ▪ Mania ▪ Catatonia	**EDUCATE** that the patient may have memory loss, confusion, or a headache post-procedure and that this is normal
VAGUS NERVE STIMULATION (VNS)	Electrical stimulation of the vagus nerve (longest cranial nerve)  ↓ Increases neurotransmitters ↓ Improves mood ↓ Decreases depression *(originally used for those with epilepsy but is now used for depression as well)* MEMORY TRICK: Think of VNS as a pacemaker for the brain	▪ Epilepsy ▪ Severe depression 	**EDUCATE** the patient about potential temporary side effects: ▪ Neck pain ▪ Cough ▪ Hoarseness of the voice ▪ Dyspnea Remember: the device is being placed near the larynx
COGNITIVE BEHAVIORAL THERAPY (CBT) 	Therapy that teaches patients how to take their negative thoughts & replace them with healthy & realistic thoughts MEMORY TRICK: Think of CBT as rewiring the brain	▪ Depression ▪ Phobias ▪ Anxiety ▪ Schizophrenia ▪ Bipolar disorder ▪ Obsessive-compulsive disorder (OCD) ▪ Substance use/dependence	**EDUCATE** the patient on the goal of treatment: Shift from negative to positive thinking 
MILIEU THERAPY	Group therapy with leadership & structure, allowing for growth, learning & progress toward improved mental state MEMORY TRICK: Milieu – think of Multiple people	**Schizophrenia** ▪ Can help achieve conflict resolution and decrease violence in this population	**EDUCATE** the patient on the goal of treatment and encourage the patient to share experiences and similar feelings

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Mother Baby

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Tips for Mother Baby Class

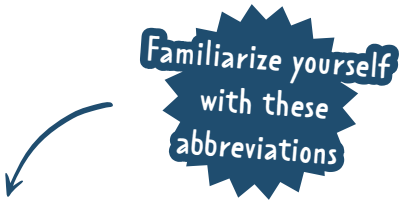
Don't think of real-life scenarios when learning about labor, pregnancy, and newborns

The NCLEX has very specific standards in place. Passing requires knowing and remembering these. Try NOT to relate to real-life scenarios.

Mother baby is like learning a new language. There are a lot of new terms, abbreviations & acronyms

For example:

MED SURG:	MOTHER BABY:
ROM means *range of motion*	ROM means *rupture of membranes*



Abbreviations

IUP	Intrauterine pregnancy
IUFD	Intrauterine fetal demise
SAB	Spontaneous abortion
TAB	Therapeutic abortion
LMP	Last menstrual period
ROM	Rupture of membranes
SROM	Spontaneous rupture of membranes
AROM	Artificial rupture of membranes
PROM	Prolonged rupture of membranes (>24 hours)
PPROM	Preterm premature rupture of membranes
SVD	Spontaneous vaginal delivery
FHR	Fetal heart rate
EFM	Electronic fetal monitoring
US	Ultrasound transducer (detects FHR)
FSE	Fetal scalp electrode (precise reading of FHR)
IUPC	Intrauterine pressure catheter (strength of contractions)
LTV	Long term variability
SVE	Sterile vaginal exam
MLE	Midline episiotomy
NST	Non-stress test
CST	Contraction stress test
BPP	Biophysical profile
VBAC	Vaginal birth after cesarean
AFI	Amniotic fluid index
BUFA	Baby up for adoption
NPNC	No prenatal care
PTL	Preterm labor
BOA	Born on arrival
BTL	Bilateral tubal ligation
D&C / D&E	Dilation & curettage / dilation & evacuation
LPNC	Late prenatal care
TIUP	Term intrauterine pregnancy
VMI / VFI	Viable male infant / viable female infant
EDB	Estimated date of birth
EDC	Estimated date of confinement
EDD	Estimated date of delivery

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GTPAL

Gravidity vs. Parity

Gravidity

A woman who is pregnant / the number of pregnancies

- **NULLIGRAVIDA** — Never been pregnant
- **PRIMIGRAVIDA** — Pregnant for the first time
- **MULTIGRAVIDA** — Pregnant for two or more times

Parity

The number of pregnancies that have reached viability (22-25 weeks of gestation), whether the fetus was born alive or not

- **NULLIPARA** — **0** Zero pregnancies that have reached viability
- **PRIMIPARA** — **1** One pregnancy that has reached viability
- **MULTIPARA** — **2+** Two or more pregnancies that have reached viability

MEMORY TRICK

NULLI = none
PRIMI = one/first
MULTI = multiple

GTPAL is an acronym used to quickly help assess for pregnancy outcomes

G	Gravidity	The number of pregnancies	▪ Includes the present pregnancy ▪ Includes miscarriages/abortions ▪ Twins/triplets count as **one**
T	Term Births	The number born at term > 37th week of gestation	▪ Includes alive or stillborn ▪ Twins/triplets count as **one**
P	Pre-Term Births	The number of pregnancies delivered between the 20th -36 6/7th weeks of gestation	▪ Includes alive or stillborn ▪ Twins/triplets count as **one**
A	Abortions/ Miscarriages	The number of pregnancies delivered before 20 weeks gestation	▪ Counts towards gravidity (the number of pregnancies) ▪ Twins/triplets count as **one**
L	Living Children	The number of current living children	▪ Twin/triplets count **individually**

Think about it: You aren't pregnant twice with twins/triplets

PRACTICE QUESTION 1

You are admitting a client to the mother-baby unit. Two hours ago she delivered a boy on her due date. She gives her obstetric history as follows: she has a three-year-old daughter who was delivered a week past her due date and last year she had a miscarriage at 8 weeks gestation. How would you note this history using the GTPAL system?

A. 2-2-1-0-2
B. 3-2-1-0-1
C. 3-2-1-0-2
D. 3-2-0-1-2

PRACTICE QUESTION 2

A prenatal client's obstetric history indicates that she has been pregnant 3 times previously and that all her children from previous pregnancies are living. One was born at 39 weeks gestation, twins were born at 34 weeks gestation, & another child was born at 38 weeks gestation. She is currently 38 weeks pregnant. What is her gravidity & parity using the GTPAL system?

A. 4-1-3-0-4
B. 4-1-2-0-3
C. 4-2-1-0-4
D. 4-2-2-0-4

ANSWER KEY
Q#1 is (D) 3-2-0-1-2
Q#2 is (C) 4-2-1-0-4

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Pregnancy Duration & Naegele's Rule

Trimesters

FIRST Trimester
1–13 weeks

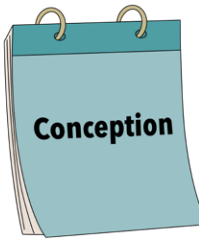
SECOND Trimester
14–26 weeks

THIRD Trimester
27–40 weeks

Pregnancy Duration

40 WEEKS GESTATIONAL AGE
The number of completed weeks counting from the 1st day of the last normal menstrual cycle (LMP).

38 WEEKS FETAL AGE
This refers to the age of the developing baby, counting from the estimated date of conception. The fetal age is usually 2 weeks less than the gestational age.



Naegele's Rule

Used for estimating the expected date of delivery (EDD) based on the (last menstrual period (LMP)

DATE OF LAST MENSTRUAL PERIOD − **3 CALENDAR MONTHS** + **7 DAYS** + **1 YEAR** = **EDD**

EXAMPLE

1st day of last period:	September 2, 2015
Minus 3 calendar months:	June 2, 2015
Plus 7 days:	June 9, 2015
Plus 1 year:	June 9, 2016 → EDD

REMEMBER:
How many days are in each month?

"30 days hath September,
April, June & November.
All the rest have 31, except
February alone (28 days) "

FACTS ABOUT NAEGELE'S RULE:

- Bases calculation on a woman who has a 28-day cycle (most women vary)
- The typical gestation period is 280 days (40 weeks)
- First-time mothers usually have a slightly longer gestation period

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Pregnancy Signs & Symptoms

Presumptive SUBJECTIVE

NOT a definite diagnosis for pregnancy!

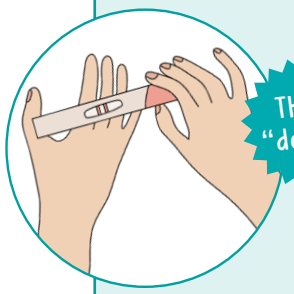
These are changes felt by the woman and are subjective. Can be associated with other things.

- Period absent (amenorrhea)
- Breast enlargement and soreness
- Nausea / vomiting
- Quickening
  - When a mother feels the movement of the fetus in the uterus

Why is quickening not a positive sign?

Quickening can be difficult to distinguish from peristalsis or gas so it cannot be a positive sign.

Probable OBJECTIVE



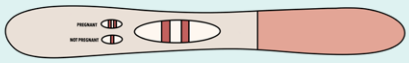
Pregnancy signs that the nurse or doctor can observe

- The **signs**:
  - **Goodell's sign:** a softened cervix
  - **Chadwick's sign:** bluish color of the vulva, vagina, or cervix
  - **Hegar's sign:** lower uterine segment soft
- **Ballottement**
  - When the uterus is pushed with fingers, the fetus goes up and than returns back down to its original position
- **Positive (+) pregnancy test** (high levels of the hormone: hCG)

MEMORY TRICK
Ballottement is like a Bouncy Ball

Why is a positive pregnancy test not a positive sign?

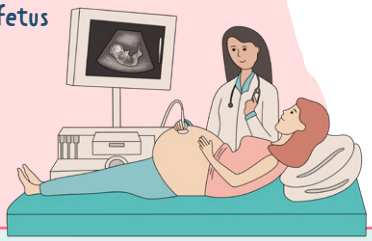
High levels of hCG can be associated with other conditions such as certain medications or hydatidiform mole (molar pregnancy).



Positive OBJECTIVE

DEFINITE diagnosis for pregnancy!

Can only be attributed to a fetus



The baby is:

HEARD	FELT	SEEN
Fetal heart tones	Fetal movement felt by the provider	Visualization by an ultrasound

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Pregnancy Physiology

Integumentary (Skin)

Striae gravidarum
- Stretch marks (abdomen, breasts, hips, etc.)
- May cause itching

Chloasma "mask of pregnancy"
- Brownish hyperpigmentation of the skin (cheeks, nose, & forehead)

Palmar erythema
- Red, mottled, blotchy appearance of the hands

Linea nigra
- Means "black line"
- Presents as vertical line on belly during pregnancy

Montgomery glands / tubercles
- Small rough / nodular / pimple-like appearance of the areola (nipple)

↑ Hair & nail growth

Musculoskeletal

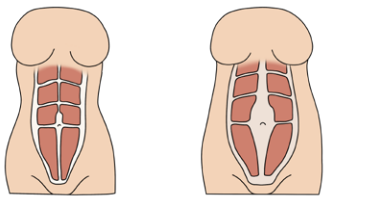
Lordosis
- Center of gravity shifts forward leading to an inward curve of the spine → ↑ risk for falls

Low back pain

Carpal tunnel syndrome
- Edema causes pressure on the peripheral nerves

Calf cramps

Diastasis recti abdominis
- The growing uterus causes *stretching of the abdominal wall*



Non-pregnant | Diastasis recti abdominis during pregnancy

Renal

↑ GFR from ↑ plasma volume

Smooth muscle relaxation of bladder and renal pelvis
- More urine is held in the bladder and becomes stagnant allowing for bacteria to grow
- ↑ Risk for urinary tract infections (UTIs)

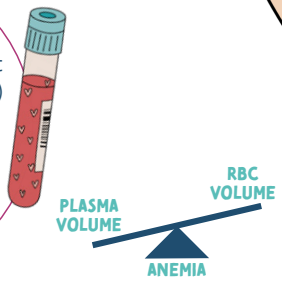
↑ Urgency, frequency & nocturia
- ↑ Progesterone = ↓ tone of bladder, ureter & urethra

PREVENTION OF UTIs
- Drink lots of water to "flush" out the urinary tract
- Cranberry juice can help prevent & control symptoms
- Wipe from front to back
- Void after intercourse
- Avoid bubble baths, perfumes, or sprays
- Wear non-tight cotton underwear

Hematological

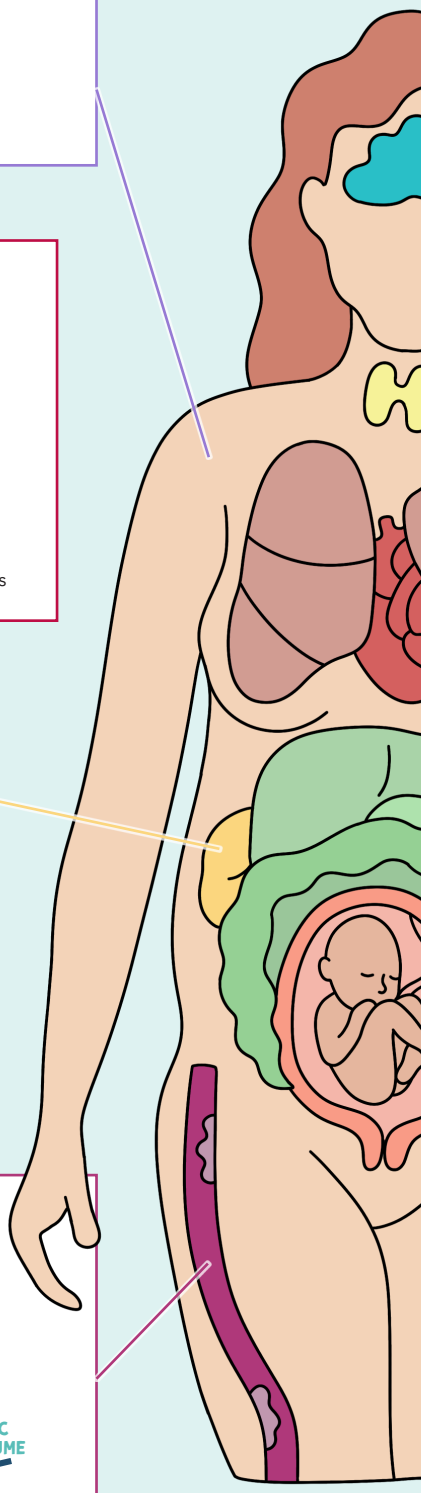
↓ Hemoglobin & Hematocrit
- Due to hemodilution (↑ blood volume is diluting the Hgb & Hct)

Plasma volume is greater than the amount of red blood cells (RBCs) = hemodilution = **physiological ANEMIA**



↑ Fibrinogen
- Pregnant women are hypercoagulable which increases the risk for DVTs

FIBRINOGEN:
non-pregnant levels: 200–400 mg/dL
pregnant levels: up to 600 mg/dL



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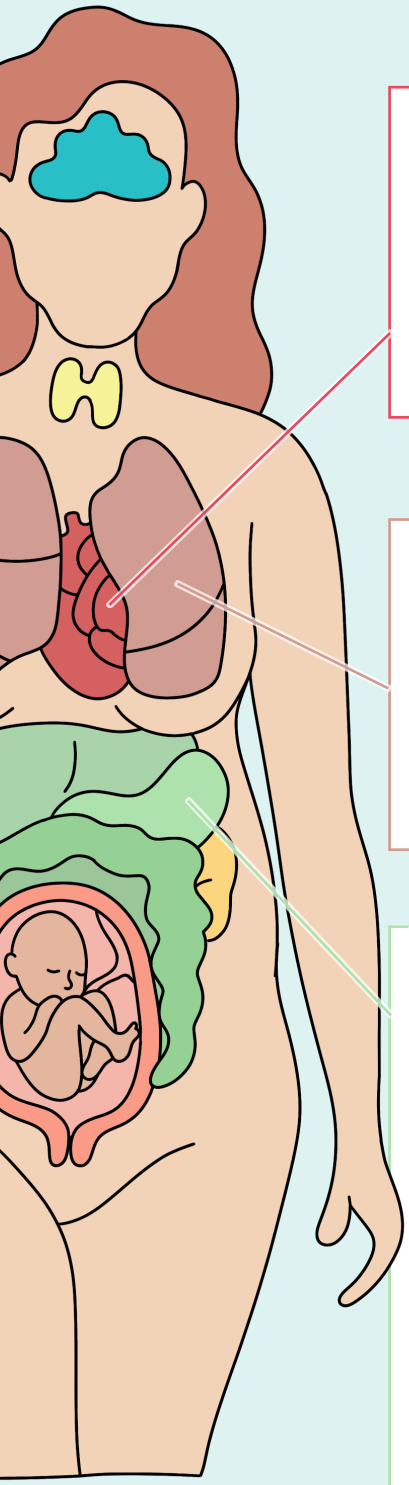
Pregnancy Physiology

HORMONE SPOTLIGHT:

Relaxin & Progesterone	Relaxes smooth muscles
Estrogen	Growth of the uterus, the placenta, growth of new blood vessels, etc.
Human Chorionic Gonadotropin (hCG)	The placenta produces hCG during pregnancy & it prevents menstruation

MEMORY TRICK: Relaxin think Relaxing the body

MEMORY TRICK: Estrogen think Everything is growing



Cardiovascular

↑ Cardiac output
- ↑ heart rate + ↑ stroke volume

Blood pressure
- Stays the same or a slight decrease

↑ in blood volume
- May cause edema

Blood pressure should not be increased! This could indicate preeclampsia

Heart enlarges
- Most develop a systolic murmur

This is not alarming and is not permanent

Cardiac rhythm
- May experience palpitations and cardiac dysrhythmias

Healthy women: no therapy needed

Existing heart condition: needs attention

Hypercoagulable
- ↑ risk for DVTs

FIBRINOGEN:
non-pregnant levels: 200-400 mg/dL
pregnant levels: up to 600 mg/dL

Respiratory

↑ Basal metabolic rate (BMR)

↑ O_2 needs
- Fetus needs oxygen and there is a lot of growing tissue in the uterus & breasts

Mild respiratory alkalosis
- ↑ pH ↓ CO_2

Chest
- ↑ in size to allow for fetal growth and lung expansion

↑ Vascularity of upper respiratory tract
- May develop nose bleeds more frequently, congestion, runny nose, inflammation

Gastrointestinal

Nausea & vomiting
- Commonly called "morning sickness" and is most common during the first trimester

Hyperemesis gravidarum = severe morning sickness & may require hospitalization

Pyrosis (heartburn)
- ↑ Progesterone = lower esophageal sphincter relaxes

Prevention/treatment OF HEARTBURN
- Do not lie down after meals
- Eat small, frequent meals
- Avoid:
  - ✘ Greasy foods
  - ✘ Spicy foods
  - ✘ Trigger foods

Constipation
- ↑ Progesterone = ↓ gut motility

Prevention/treatment OF CONSTIPATION

MEMORY TRICK: Fruits, Fiber and Fluids fill up the toilet!

Hemorrhoids
- Inflamed veins in the anus. Symptoms can include itching, pain, & bleeding with defecation.

Pica
- Craving substances that are not considered food (ice, clay, paper, soil, etc.)

Swollen mouth & gums
- Gums become swollen and easily bleed. May develop epulis (red nodules on the gums)

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What to Avoid During Pregnancy

Teratogenic Drugs

Teratogenic drugs are drugs that can cause birth defects in the developing fetus. These medications should be avoided during pregnancy.

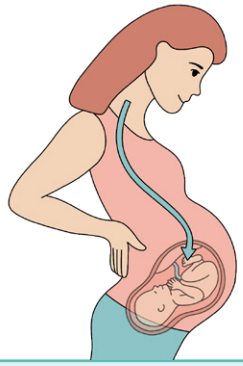
MNEMONIC

"TERA-TOWAS"

- **T** Thalidomide
- **E** Epileptic medications (valproic acid, phenytoin)
- **R** Retinoid (vit A)
- **A** Ace inhibitors, ARBs
- **T** Third element (lithium)
- **O** Oral contraceptives
- **W** Warfarin (coumadin)
- **A** Alcohol
- **S** Sulfonamides & sulfones

"TORCH" Infections

TORCH infections are infections that can cross the placenta. They can harm pregnant women and the developing fetus. Pregnant women should avoid exposure to these infections!



MNEMONIC

"TORCH"

- **T** Toxoplasmosis
- **O** Other infections → Syphilis, Parvovirus B19, Hepatitis
- **R** Rubella
- **C** Cytomegalovirus (CMV)
- **H** Herpes simplex virus (HSV)

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Maternal Nutrition

Consuming a healthy, balanced diet during pregnancy is crucial for proper fetal development and for a safe & healthy pregnancy.

Folic acid	▪ Folic acid intake (600–800 mcg/day) ▪ Important to consume before and in the early stages of pregnancy ★ **Poor folic acid intake increases the fetal risk for neural tube defects (NTDs)**
Protein	▪ Needed for tissue growth ▪ Great sources of protein: lean meats, eggs, cheese, milk, nuts, legumes
Omega-3 fatty acids	▪ DHA intake (300 mcg/day) ▪ Needed for brain development for the growing fetus
Maternal weight gain	▪ **1ST TRIMESTER:** About 2.2–4.4 lbs (1–2 kg) ▪ **2ND & 3RD TRIMESTER:** 0.5–1 lb per week
Iron	▪ Iron intake (25–30 mg/day) ▪ ↑ risk for iron deficiency due to ↑ iron requirements during pregnancy ▪ Adequate intake is needed to supply the fetus with iron
Calcium	▪ Same intake as non-pregnant (1,000 mg/day)
Caloric intake	▪ **Never restrict calories** ▪ **1ST TRIMESTER:** no increase in caloric intake ▪ **2ND & 3RD TRIMESTER:** additional 300–400 calories per day

This varies tremendously based on the woman who is pregnant

FOODS TO AVOID

- ✘ Unpasteurized milk
- ✘ Unwashed fruits & vegetables
- ✘ Deli meat
- ✘ Liver
- ✘ Raw fish/raw meat
- ✘ Fish high in mercury (shark, tilefish, swordfish, etc.)
- ✘ Alcohol

CAFFEINE

Studies are inconclusive whether caffeine has negative effects on pregnancy.

BUT, excess amounts can contribute to intrauterine growth restriction (IUGR).

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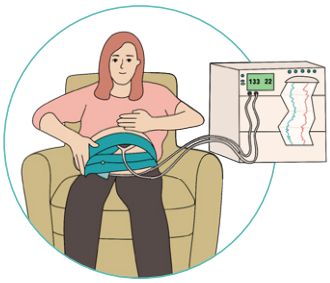
Tests During Pregnancy

NONSTRESS TEST (NST)

TEST & DESCRIPTION

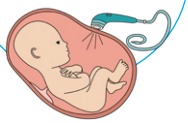


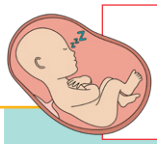
A non-invasive way to assess fetal wellbeing, specifically their heart rate and movements

INTERPRETATION

✓ REACTIVE	2 accelerations in 20 minutes	▪ Indicates fetal well being ▪ Accelerations have been noted and are associated with movement
✗ NONREACTIVE	Does not have at least 2 accelerations in 20 minutes	▪ Requires further evaluation ▪ NST is extended due to the possibility the fetus is sleeping ▪ **Vibroacoustic stimulation** may be used ▪ After 40 minutes of no activity, a biophysical profile (BPP) is needed

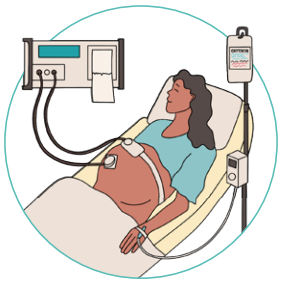
VIBROACOUSTIC STIMULATION
Uses sound & vibration to stimulate the fetus



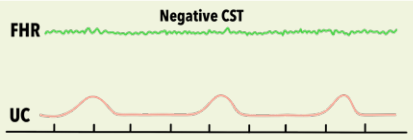
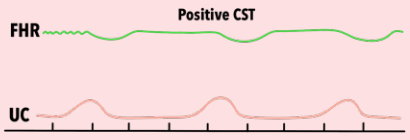
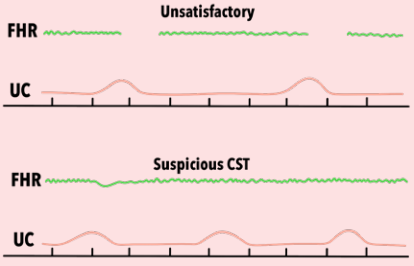


REASONS FOR NO ACCELERATIONS OR MOVEMENT
- The fetus is sleeping (MOST COMMON)
- Certain medications (*Example:* narcotics)
- Maternal smoking
- Fetal malformations

CONTRACTION STRESS TEST (CST)



Also called oxytocin challenge test (OCT)
A test to assess the fetal responses to contractions

✓ NEGATIVE (Negative for decelerations)	3 contractions in 10 minutes	
✗ POSITIVE (Positive for decelerations)	Repetitive decelerations occurring for at least 50% of contractions, EVEN if < 3 occur in 10 min span	
UNSATISFACTORY OR SUSPICIOUS	Test must be repeated within 24 hours	

BIOPHYSICAL PROFILE (BPP)

A noninvasive test that combines a nonstress test (NST) and a fetal ultrasound to assess the following:

	SCORE = 2	SCORE = 0
Fetal movements	> 3 body/limb movement in 30 minutes	< 3 body/limb movement in 30 minutes
Fetal breathing movements (FBM)	> 1 episode of breathing movements of >30 seconds	No breathing movements or < 30 seconds of sustained breathing movements
Muscle tone	> 1 extension/flexion	No movement, extension or flexion
Amniotic fluid index (AFI)	AFI > 5 cm or Pocket of fluid > 2 cm	AFI < 5 cm or No pockets of fluid > 2 cm
Nonstress test	Reactive	Nonreactive

✓ 8–10 = NORMAL ✗ < 8 = FETAL HYPOXIA

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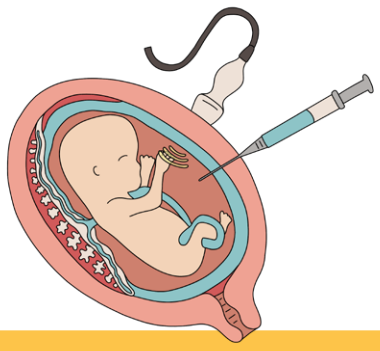
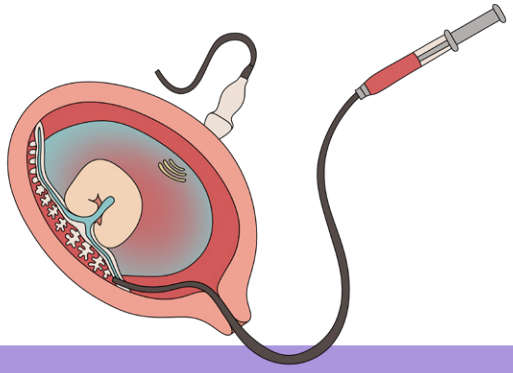
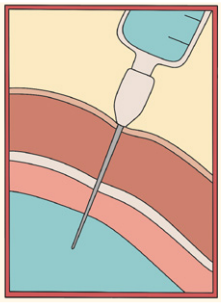
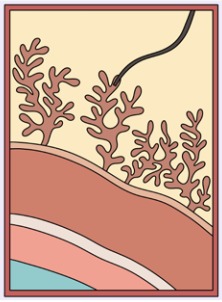
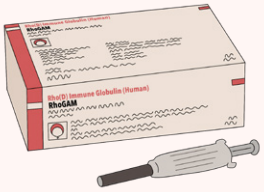
Amniocentesis vs. Chorionic Villus Sampling (CVS)





	AMNIOCENTESIS	CHORIONIC VILLUS SAMPLING (CVS)
WHAT IS IT?	**AMNIOCENTESIS** ↙ Amniotic fluid ↘ Puncture into a body cavity to remove fluid A needle is inserted through the abdomen into the uterus under direct visualization through ultrasonography. Amniotic fluid is withdrawn from the uterus (amniotic fluid contains fetal cells)	A needle is inserted (transcervical or transabdominal) to obtain a tissue sample from the fetus
SAMPLE OF	Amniotic fluid 	Fetal tissue (no amniotic fluid is obtained) 
WHEN IS IT DONE?	After 14 weeks	10–13 weeks
WHY IS IT DONE?	▪ Testing for genetic disorders/congenital anomalies (*Example*: neural tube defects) ▪ Assessment of fetal lung maturity (for amniocentesis)	Both are not routinely done; only done if there is a genetic concern → *Example:* family history of congenital anomalies
COMPLICATIONS	 RhoGAM should be administered to Rh negative mothers (covers ANY chance of blood mixing)	**MOM** ▪ Infection ▪ Hemorrhage ▪ Placental abruption **BABY** ▪ Infection ▪ Injury from the needle

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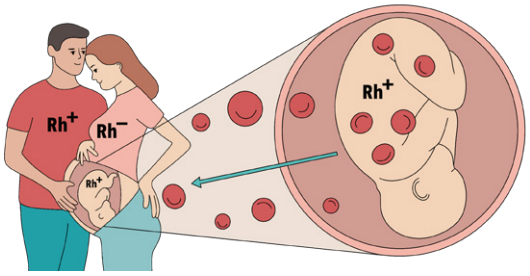
Rh Incompatibility During Pregnancy

PATHOLOGY:

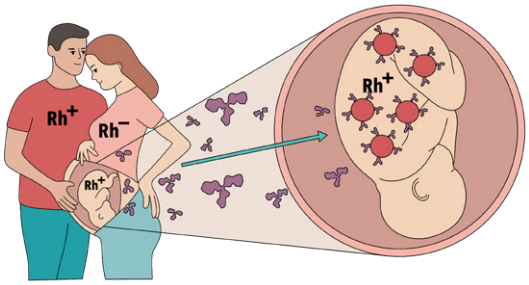
- During pregnancy, the baby's blood does not typically enter the mother's circulatory system and does not mix with her blood
- If a Rh- mother is exposed to Rh+ fetal blood, the mother develops antibodies (this is called maternal sensitization)

PREGNANCY #1

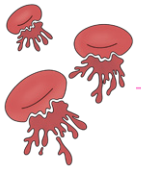


If the Rh- woman developed antibodies during her first pregnancy, it's not usually enough to cause harm to the developing fetus

PREGNANCY #2



In all future pregnancies, there is a greater chance for destruction of fetal blood cells



RISK FACTORS:

Reasons why mom & baby's blood would mix

- Trauma
- Hemorrhage
- Invasive diagnostic testings (amniocentesis)
- Miscarriages
- Induced abortion

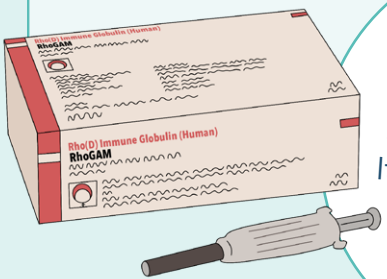
TREATMENT:

Rh immune globulin (RhoGAM) is administered to all Rh- pregnant women:

- 28 weeks gestation
- Within 72 hours after birth
- Anytime there is trauma or blood is at risk for mixing

NURSING CONSIDERATIONS:

- An **indirect Coombs test** screens for Rh incompatibility



RhoGAM does not work if antibodies have already developed. It only helps prevent the development of permanent Rh antibodies.

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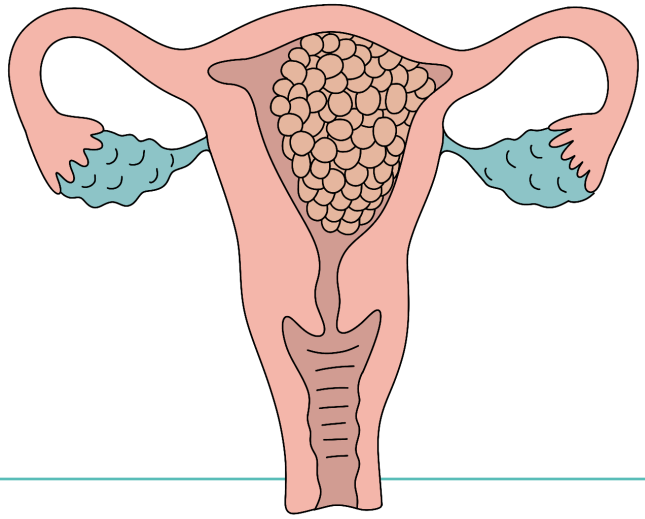
Hydatidiform Mole (Molar Pregnancy)

PATHOLOGY:

A rare condition where there is a benign proliferation of trophoblasts. Trophoblasts are normally what develops into the placenta.

TYPES OF MOLAR PREGNANCIES

COMPLETE MOLE	PARTIAL MOLE
Comes from a fertilized egg, but the nucleus was lost or inactivated	Comes from one normal ovum that is fertilized
Has no fetal parts, amniotic sac, or membranes	Has fetal parts & an amniotic sac



RISK FACTORS:

THE CAUSE IS UNKNOWN!

- History of a molar pregnancy
- Really young or really old
- Nutritional deficiencies

DIAGNOSTIC:

- Transvaginal ultrasound
- ↑ Serum hCG levels

SIGNS & SYMPTOMS:

EARLY SIGNS

- Asymptomatic

LATER SIGNS

- Vaginal bleeding
- Excessive nausea & vomiting
- Anemia (from blood loss)
- Abdominal cramping
- Preeclampsia

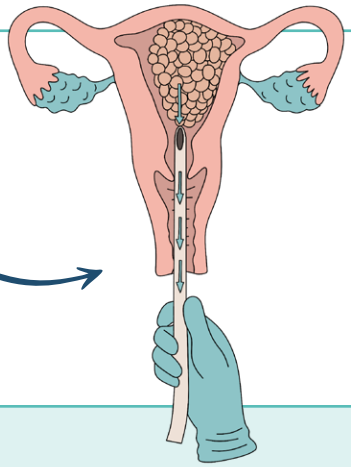
hCG is typically what causes nausea & vomiting, and these patients have ↑ hCG

PATIENT EDUCATION:

- **FOLLOW UP CARE:**
  - Frequent physical exams
  - Pelvic exams
  - Weekly monitoring of hCG levels
- **EDUCATE:**
  - About methods to avoid pregnancy during the follow-up care

TREATMENT:

- Most molar pregnancies will abort spontaneously
- **Suction curettage**
  - Suctioning out the hydatidiform mole



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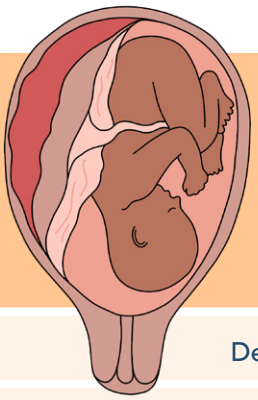
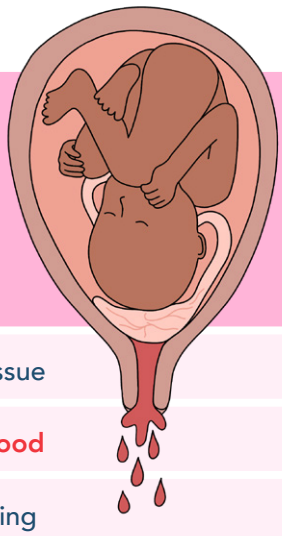
The Placenta

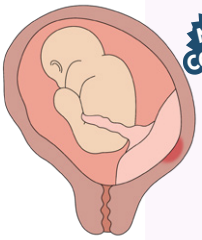
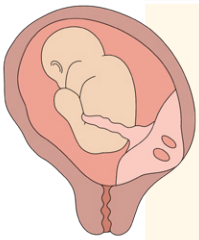
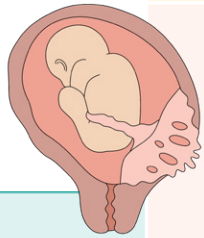


ABRUPTIO PLACENTA	VS.	PLACENTA PREVIA
Detachment issue		Attachment issue
Dark red blood		Bright red blood
May be concealed bleeding		Visible bleeding
Abdominal pain & uterine tenderness		Painless and nontender uterus
Fetal distress (placenta has come off the uterine wall)		Typically will have a normal fetal heart rate



Both are typically seen **AFTER** 20 weeks gestation

Abnormal IMPLANTATION OF THE PLACENTA

	TYPE	DESCRIPTION	MEMORY TRICK	RISK FACTORS FOR ALL
	MOST COMMON **PLACENTA ACCRETA**	Placenta **ATTACHES** too deep into the uterine wall *(A little penetration of the myometrium)*	**A**ccreta think **A**ttaches	History of a cesarean birth
	**PLACENTA INCRETA**	Myometrium is **INVADED** *(Deep penetration of the myometrium)*	**I**ncreta think **I**nvades	History of uterine surgery Placenta previa
	**PLACENTA PERCRETA**	Myometrium is **PENETRATED.** Placenta grows through uterine wall & may attach to anything on the other side (bladder/intestines) *(Perforation of the uterus)*	**P**ercreta think **P**enetrates	Advanced maternal age (AMA)

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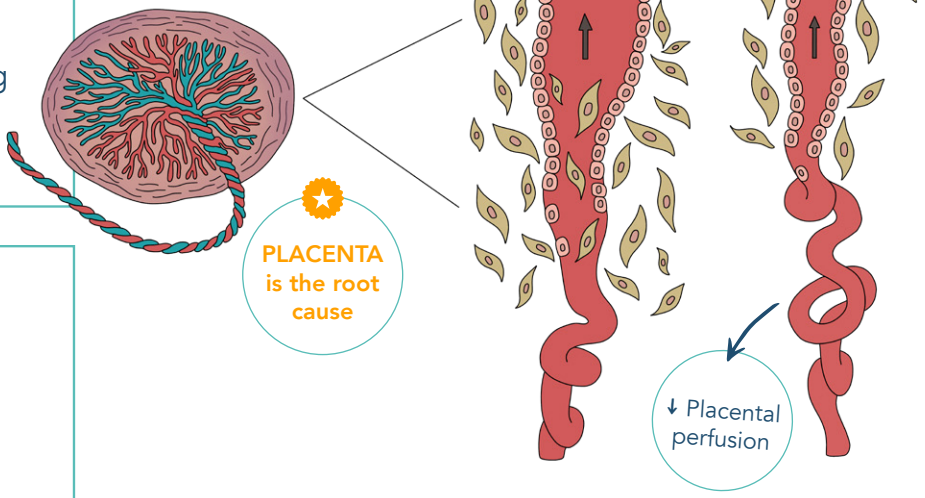
Preeclampsia Overview

PATHOLOGY:

Hypertension & proteinuria after 20 weeks gestation in a pregnant woman who previously had a normal blood pressure.

Pathology is not completely known

- Defective spiral artery remodeling
- Systemic vasoconstriction & endothelial dysfunction



SIGNS & SYMPTOMS:

MEMORY TRICK

"PRE" eclampsia

- **P** Proteinuria
- **R** Rising BP
- **E** Edema

Triad signs

- Severe headache
- RUQ or epigastric pain
- Visual disturbances
- ↓ Urine output
- Clonus (hyperactive reflexes)
- Rapid weight gain

DIAGNOSTIC:

Hypertension + 1 or more of:

- Proteinuria
- Kidney dysfunction
- Low platelet count
- New onset of headaches/visual disturbance
- Impaired liver function
- Pulmonary edema

RISK FACTORS:

- History of:
  - Preeclampsia in previous pregnancies
  - Family that had preeclampsia
- 1st pregnancy
- Obesity
- Younger (< 18) & older (> 35)
- Medical conditions (Chronic HTN, renal disease, diabetes, autoimmune disease)

> 35 = AMA *advanced maternal age*

NURSING CONSIDERATIONS:

GOAL: Prevent eclampsia (seizures)

- **Medication:** magnesium sulfate

MONITOR:

- Deep tendon reflexes
- Neuro status

Have the antidote (calcium gluconate) at the bedside

COMPLICATIONS:

MOM

- Eclampsia (seizures)
- Organ failure/damage
- Cardiovascular disease

BABY

- Intrauterine growth restriction (IUGR)
- Preterm birth
- Placental abruption
- HELLP syndrome

HELLP SYNDROME

LIFE-THREATENING COMPLICATION

Variant of preeclampsia

- **H** Hemolysis (breakdown of red blood cells)
- **EL** Elevated Liver enzymes
- **LP** Low Platelet count

CURE?

Preeclampsia will resolve after the placenta has been expelled

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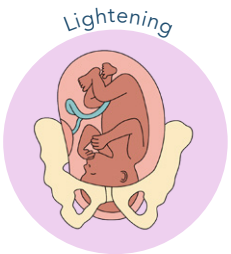
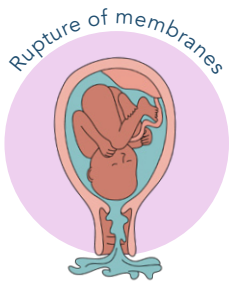
True vs. False Labor

	False LABOR	True LABOR
FREQUENCY OF CONTRACTIONS	Contractions are **IRREGULAR**	Contractions are **REGULAR** They get stronger, longer, & closer together as time goes on
CHANGE WITH MOVEMENT & COMFORT MEASURES	Stops with walking/position change **Stops** with comfort measure	More intense with walking **Continues** despite the use of comfort measures
LOCATION OF PAIN	Felt in the the **abdomen** above the umbilicus	Felt in **lower back**
EFFECTS ON CERVIX	No significant changes in: ▪ Effacement ▪ Dilation	Significant changes in: ▪ Effacement ▪ Dilation ▪ Can cause bloody show
POSITION OF BABY	In **posterior** position: *baby's head facing mom's front of belly*	In **anterior** position: *baby's head facing mom's back*

OTHER SIGNS OF True LABOR





- ✓ Lightening
- ✓ Increased vaginal discharge (bloody show)
- ✓ Return of urinary frequency
- ✓ Rupture of membranes "water breaking"
- ✓ Persistent backache
- ✓ Stronger Braxton Hicks contractions
- ✓ A few days before labor
  - ▪ Surge of energy
  - ▪ Weight loss (1–3.5 pounds) from a fluid shift

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Stages of Labor

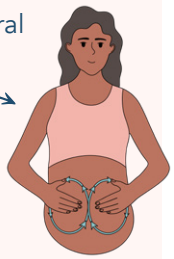
STAGE 1

Cervix DILATES from 0–10 cm

MNEMONIC: **L**abor **A**ctively **T**ransitioning

		CERVIX DILATES	INTENSITY	CONTRACTIONS
L	LATENT (early) — LONGEST STAGE	1–3 cm	Mild	15–30 min
A	ACTIVE	4–7 cm	Moderate	3–5 min (30–60 sec in duration)
T	TRANSITON	8–10 cm	Strong	Every 2–3 min (60–90 sec in duration)

- All about promoting COMFORT
  - Warm shower, massage, or epidural
  - Provide a quiet environment
  - **Effleurage** (light stroking of the abdomen which can be done by the mother or their partner)



- Offer fluids & ice chips
- Encourage voiding every 1–2 hours
- Encourage effective breathing patterns & rest between contractions

STAGE 2

The BABY is delivered

This is the stage where the mother is **PUSHING**

Starts when cervix is fully dilated & effaced

↓

Ends after the baby is delivered

- Provide ice chips & ointment for dry lips
- Provide praise & encouragement to the mother

MONITOR:

- Uterine contractions
- Mothers vital signs
- For signs of birth
  - Perineal bulging
  - Visualization of fetal head

STAGE 3

The PLACENTA is delivered

THE PLACENTA IS EXPELLED
(5–30 min after birth)

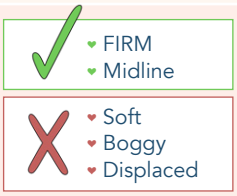
SIGNS OF A PLACENTA DELIVERY

- Lengthening umbilical cord
- Gush of blood
- Uterus changes from oval to globular shape

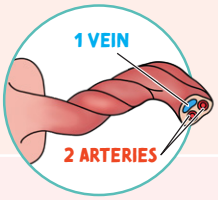
If the placenta stays in the mother for longer than 30 minutes it's called a RETAINED PLACENTA

MONITOR:

- Mothers vital signs
- **Uterine tone (fundal rubs every 15 minutes)**



- Provide warmth to the mother
- Promote mother-baby bonding
- Examine placenta & verify it's intact
  - **Should have 2 arteries & 1 vein**



STAGE 4

RECOVERY!

First 1–4 hours after delivery of the placenta

- Administer IV fluids

MONITOR:

- Uterine tone
- Vital signs
- Signs of bleeding (postpartum hemorrhage)
- Temperature for fever (infection)
- Lochia discharge
- For respiratory depression, vomiting, & aspiration (if anesthesia was used)

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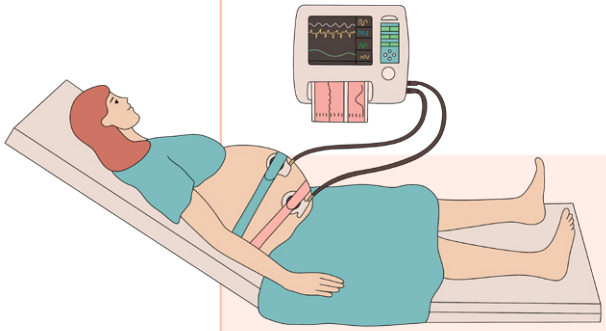
Electronic Fetal Monitoring

Electronic fetal monitoring is commonly used during labor to assess fetal oxygenation, fetal heart rate, and fetal hypoxia. It's a way to continually assess these components.

There are two types: **EXTERNAL** & **INTERNAL**

External

Non-invasive



Monitoring FETAL HEART RATE

Ultrasound transducer uses high-frequency sound waves to record the fetal heart rate (FHR)

Monitoring UTERINE ACTIVITY

Tocotransducer (Tocodynamometer) is placed over the fundus and secured by an elastic belt. It measures uterine activity transabdominally

Monitors: frequency & duration

PROS

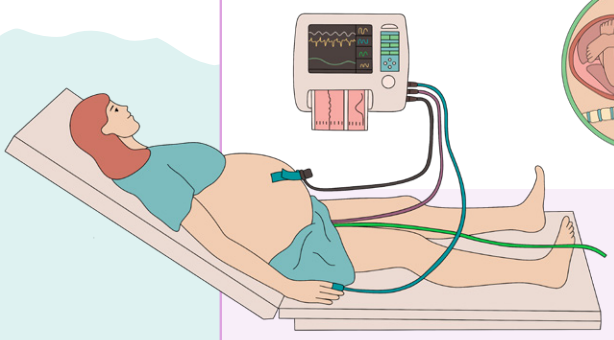
- Noninvasive
- Does not pose any risk to the fetus
- Can be done before rupture of membranes occurs

CONS

- Not as accurate because the transducers can easily move with maternal or fetal movement

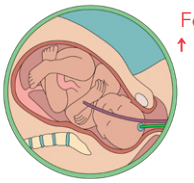
Internal

Invasive: The membranes must be ruptured



Monitoring FETAL HEART RATE

A spiral electrode is placed on the fetal presenting part (typically the head)



Fetal scalp electrode ↑ the risk for infection because a foreign object is being placed into the mother's vagina

Monitoring UTERINE ACTIVITY

An intrauterine pressure catheter (IUPC) is placed into the uterine cavity to measure contractions (measured in mmHg)

Monitors: frequency, duration, & intensity

PROS

- More accurate
- Not affected by movement

CONS

- Risk for infection
- Possible injury to the fetus and/or mother

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Accelerations & Variability

Accelerations

- Accelerations indicate fetal well being
- They are typically caused due to fetal movement or contractions
- They do not require treatment or intervention

If an acceleration lasts for longer than 10 minutes, it's considered a **BASELINE CHANGE**

< 32 WEEKS

10 X 10

Increase in the fetal HR of 10 beats for 10 seconds

> 32 WEEKS

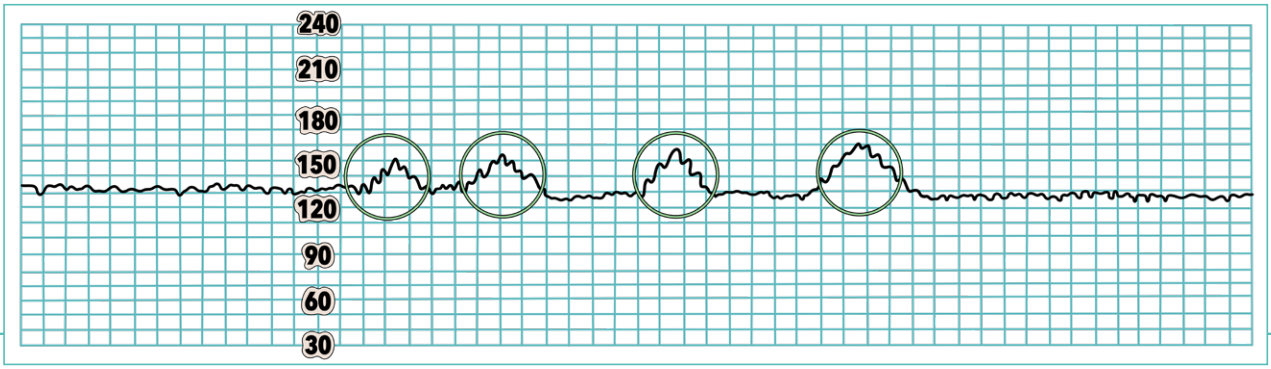
15 X 15

Increase in the fetal HR of 15 beats for 15 seconds

MEMORY TRICK

Accelerations think **A**+

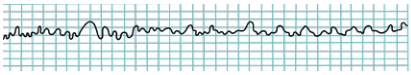
(this is a good thing!)

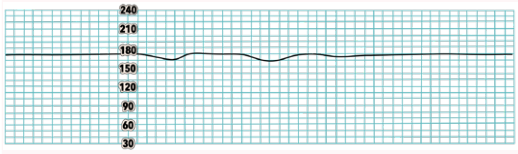
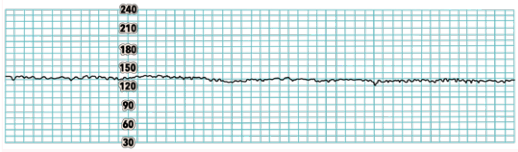
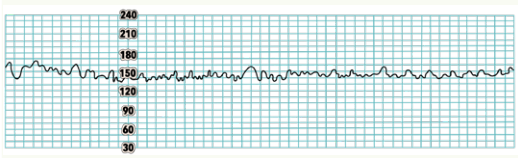
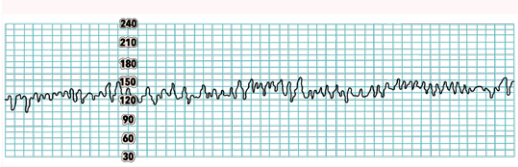
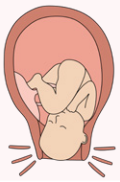


Variability

Irregular fluctuations or waves in the fetal heart rate baseline. A normal fetal heart rate is 110–160 bpm.

looks like chicken scratch



		AMPLITUDE RANGE	INTERPRETATION
ABSENT VARIABILITY		Amplitude range undetectable (flat line)	▪ The fetus is not responding well to contractions, birth process, etc. ▪ Fetal hypoxemia STAT delivery via cesarean birth
MINIMAL VARIABILITY		< 5 bpm	*May be due to:* ▪ The fetus sleeping ▪ Maternal tachycardia ▪ Certain medications ▪ Congenital anomalies
MODERATE VARIABILITY		6–25 bpm	Indicates fetal well-being MEMORY TRICK Moderate = Most desired
MARKED VARIABILITY		> 25 bpm	***Cause not completely known*** *May be due to:* ▪ Hypoxia ▪ Baby stressed while going through the birth canal 

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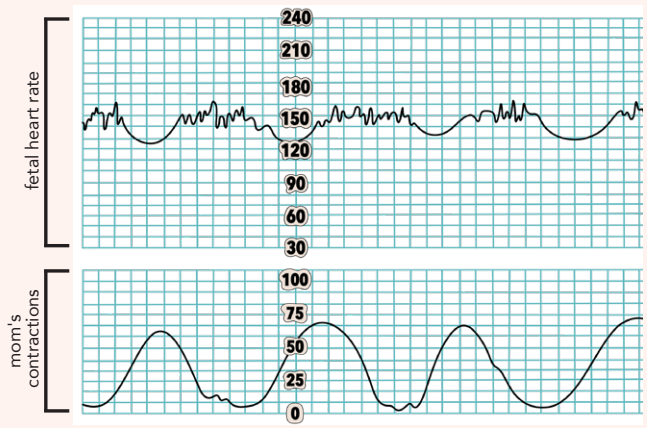
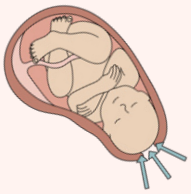
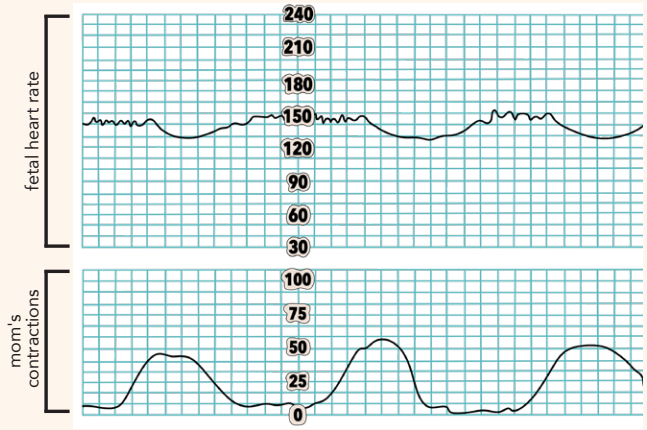
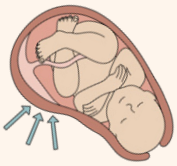
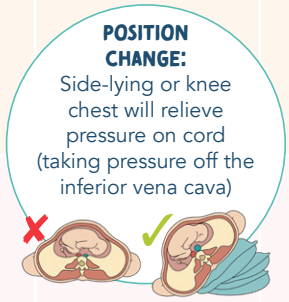
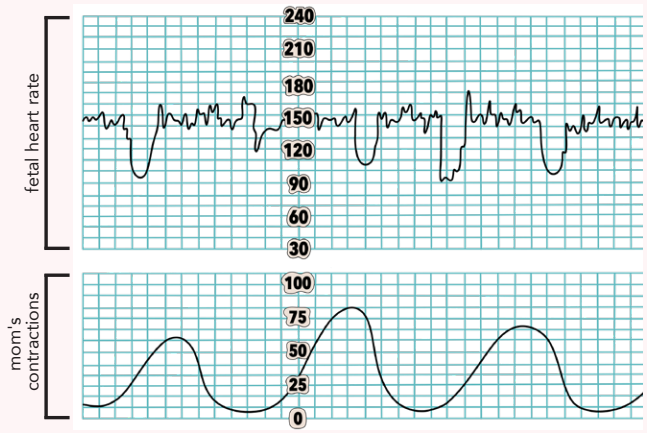
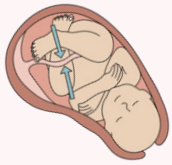
Fetal Heart Tone Decelerations

VEAL CHOP:

A tool to help interpret fetal strips

- **V** Variable Decelerations → **C** Cord Compression
- **E** Early Decelerations → **H** Head Compression
- **A** Accelerations → **O** OK (normal fetal oxygenation)
- **L** Late Decelerations → **P** Placental Insufficiency

		MEMORY TRICK	CAUSE	INTERVENTION
EARLY DECELERATIONS	NORMAL! ✓ 	"Mirror" image of mom's contractions (they don't technically come early)	From head compressions 	Continue to monitor No intervention needed
LATE DECELERATIONS	NON-REASSURING ✗ 	Literally comes late after mom's contraction	Uteroplacental insufficiency  	Discontinue oxytocin Oxygen (non-rebreather) Hydration (IV fluids) Elevate legs to correct the hypotension
VARIABLE DECELERATIONS	NON-REASSURING ✗ 	Variable: Looks "V" shaped	Cord compression 	Discontinue oxytocin Amnioinfusion Oxygen (non-rebreather)

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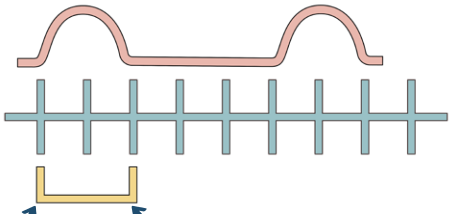
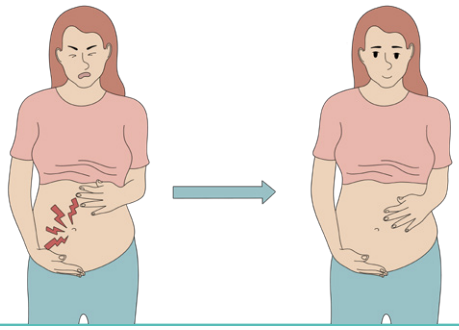
Assessment of Uterine Contractions

Duration



BEGINNING of the contraction

to the **END** of that same contraction

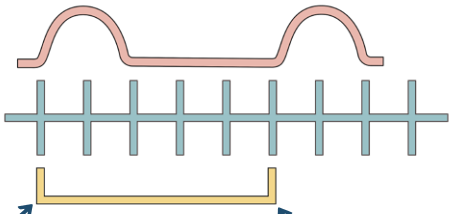


Only measured through external monitoring

INTERPRETATION:

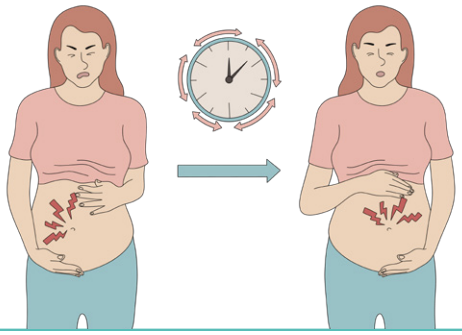
- Lasts 45–80 seconds
- Should not exceed 90 seconds

Frequency



TIME from the START of one contraction

to the **BEGINNING** of the next



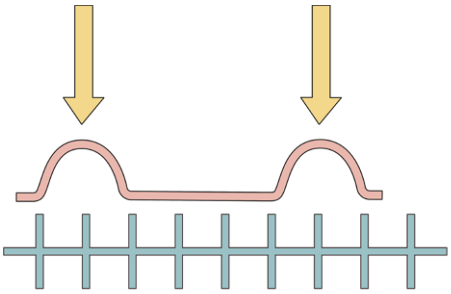
Only measured through external monitoring

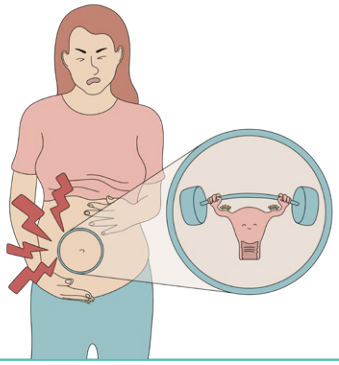
INTERPRETATION:

- 2–5 contractions every 20 minutes
- Should not be more frequent than every 2 minutes

Intensity

Strength of a contraction at its **PEAK**





Can be palpated

INTERPRETATION:

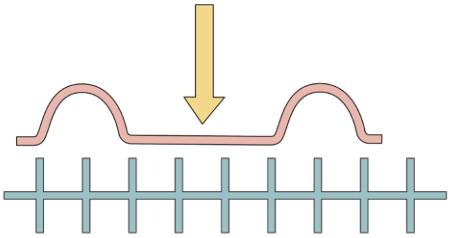
- 25–50 mmHg
- Should not exceed 80 mmHg

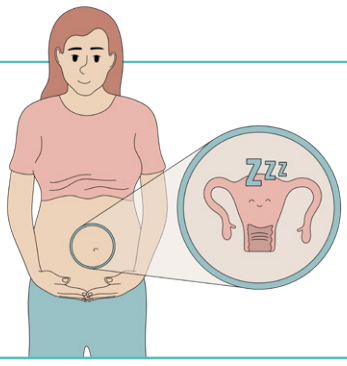
The fundus will feel like:

The nose **(MILD INTENSITY)**
The chin **(MODERATE INTENSITY)**
The forehead **(STRONG INTENSITY)**

Resting Tone

TENSION in the uterine muscle between contractions





Can be palpated

INTERPRETATION:

- Average: 10 mmHg
- Should not exceed 20 mmHg

SOFT = *good*

FIRM = *not resting enough*

✓ **Relaxation of the uterus** = fetal oxygenation between contractions

✗ **No relaxations or not enough** = less oxygen getting to the fetus

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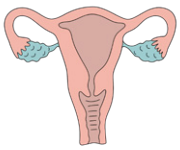
5 Ps That Affect Labor

5 Ps **5 factors that affect the process of labor & birth:**

Passenger	Passageway	Position	Powers	Psychology
FETUS & PLACENTA	THE BIRTH CANAL	POSITION OF THE MOTHER	CONTRACTIONS	EMOTIONAL RESPONSE



Passenger — FETUS & PLACENTA

Fetal Head

FONTANELLES

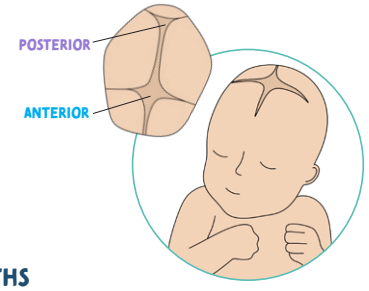
- Space between the bones of the skull allow for molding
- Often referred to as a baby's "soft spots"

ANTERIOR (LARGER)

- Diamond-shaped
- Closes in 12–18 **MONTHS**

POSTERIOR

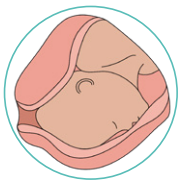
- Triangle shaped
- Closes in 8–12 **WEEKS**



MEMORY TRICK: The *POST* office always *CLOSES EARLY*

MOLDING

- Change in the shape of the fetal skull to "mold" & fit through the birth canal



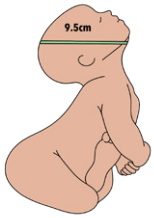
Fetal Attitude

GENERAL FLEXION

- Back of the fetus is rounded so that the chin is flexed on the chest, thighs are flexed on the abdomen, legs are flexed at the knees

BIPARIETAL DIAMETER

- 9.25 cm at term, the largest transverse diameter and an important indicator of fetal head size

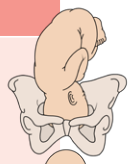
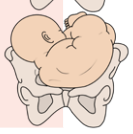


SUBOCCIPITOBREGMATIC DIAMETER

- Most critical & smallest of the anteroposterior diameters

Fetal Presentation

Refers to the part of the fetus that enters the pelvic inlet first through the birth canal during labor

NAME	WHAT'S FIRST?	PRESENTING PART	
MOST COMMON CEPHALIC	Head	Occipital (back of head/skull)	
BREECH	Buttocks, feet, or both	Sacrum	
SHOULDER	Shoulders	Scapula	

Fetal Position

FETAL STATION

Head, foot, buttocks (closest to exit of uterus)

- Where the baby's **PRESENTING PART** is located in the pelvis
- Measured in centimeters (cm)

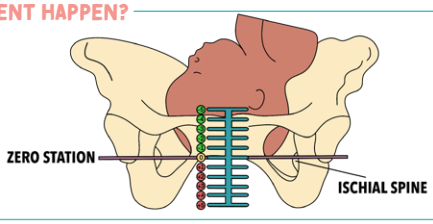
ENGAGEMENT

- Fetal station **ZERO** = baby is "**engaged**"
  - Enganged: Presenting parts have entered down into the pelvic inlet & are at the ischial spine line **(0)**

MEMORY TRICK: I'm (+) that I'm getting this baby out

WHEN DOES ENGAGEMENT HAPPEN?

- **NULLIPARA:** 38 weeks
- **MULTIPARA:** can happen when labor starts



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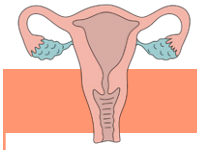
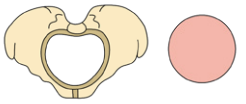
5 Ps That Affect Labor



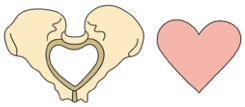
Passageway

THE BIRTH CANAL: Rigid/bony pelvis, soft tissue of cervix, pelvic floor, vagina & introitus

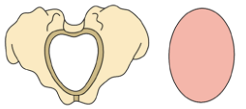
Types of Pelvis



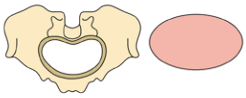
GYNECOID (MOST COMMON)
- Classic female type



ANDROID
- Resembling the male pelvis



ANTHROPOID
- Oval-shaped
- Wider anteroposterior diameter



PLATYPELLOID (LEAST COMMON)
- The flat pelvis

Soft Tissue

LOWER UTERINE SEGMENT
- Stretchy

CERVIX
- Effaces (thins) & dilates (opens)
- After fetus descends into the vagina, the cervix is drawn upward and over the first portion

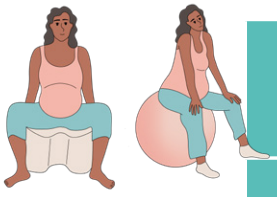
PELVIC FLOOR MUSCLES
- Helps the fetus rotate anteriorly

VAGINA

INTROITUS
- External opening of the vagina

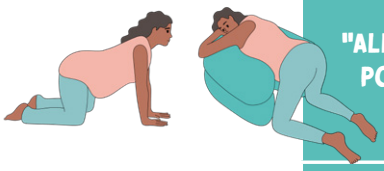
Position

POSITION OF THE MOTHER DURING BIRTH



UPRIGHT POSITION

Sitting on a birthing stool or cushion



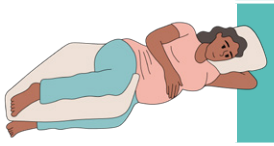
"ALL FOURS" POSITION

On all fours: putting your weight on your hands & feet or commonly on a medicine ball

LITHOTOMY POSITION (MOST COMMON)

Supine position with buttocks on the table

Most common in women who receive epidurals



LATERAL POSITION

Lying on their side

FREQUENT CHANGES IN POSITION HELP WITH:

- ✓ Relieving fatigue
- ✓ Increasing comfort
- ✓ Improving circulation
- ✓ Helps the baby be delivered quicker

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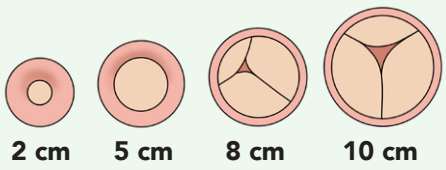
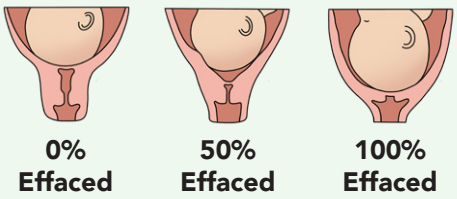
5 Ps That Affect Labor

Powers — CONTRACTIONS: PRIMARY & SECONDARY

INVOLUNTARY & VOLUNTARY powers are used to expel the baby and the placenta from the uterus

PRIMARY POWERS		SECONDARY POWERS
INVOLUNTARY Uterine contractions signal the beginning of labor		**VOLUNTARY** Bearing-down efforts by the woman once the cervix has dilated
DILATION Gradual enlargement or widening of the cervical opening **0** (closed) - **10 cm** (full dilation)	**EFFACEMENT** Shortening & thinning of the cervix during the first stage of labor **0** (Not effaced) - **100%** (Fully effaced)	**FERGUSON REFLEX** When the stretch receptors release oxytocin, it triggers the maternal urge to bear down She uses secondary powers to aid in the expulsion of the baby





Psychology — EMOTIONAL RESPONSE

- Anxiety can increase pain perception & the need for more medications (analgesia & anesthesia).
- Everyone has a unique birthing process based on their social support, past experience, and knowledge. You as the nurse are there to support her in any way she needs. It's important to take into account these factors when caring for a mother in labor and during the postpartum period.

Nursing Considerations

THINGS TO CONSIDER:

- Social support
- Past experience
- Knowledge

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Newborn Assessment

APGAR

7 - 10 supportive care
4 - 6 moderate depression
< 4 aggressive resuscitation

	SCORE	0 POINTS	1 POINT	2 POINTS
A	ACTIVITY (Muscle tone)	Absent	Flexed arms & legs	Active
P	PULSE	0	< 100	> 100
G	GRIMACE (Reflex irritability)	Floppy	Minimal response to stimulation	Prompt response to stimulation
A	APPEARANCE (Skin color)	Blue / pale all over	Pink body Blue extremities (acrocyanosis)	Pink
R	RESPIRATION (Effort)	No breathing	Breathing slow & irregular, weak cry	Vigorous cry

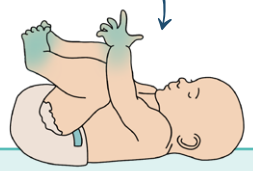
Initial Goals:

1ST PRIORITY = AIRWAY
Suction with bulb syringe / deep suction
*Newborns are obligatory nose breathers

2ND PRIORITY = WARMTH
Dry with a blanket or place in warmer

CIRCULATORY SYSTEM

- Blood flow from umbilical vessels & placenta stop at birth
- **Acrocyanosis**: Blueness of hands & feet *(normal during the first 24 hours of life)*
- Closure of:
  - Ductus arteriosus
  - Foramen ovale
  - Ductus venosus
- Transient murmurs are normal



VITAL SIGNS

BLOOD PRESSURE (BP) (Not done routinely)	SYSTOLIC DIASTOLIC	60 - 80 mmHg 40 - 50 mmHg
HEART RATE (HR)	110 - 160 bpm can be 180 if *crying* can be 100 if *sleeping*	Take apical pulse for 1 full min
RESPIRATORY RATE (RR)	30 - 60 breaths/min	
TEMPERATURE (T) (Axillary)	97.7 – 99.5°F (36.5 - 37.5°C)	
MAP	Equal to the # of weeks gestation or higher	

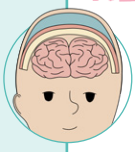
SIGNS OF RESPIRATORY DISTRESS
▪ Retractions ▪ Nasal flaring ▪ Grunting

Breathing pattern is **IRREGULAR.**
Newborns are **ABDOMINAL** breathers.

To count breaths, place your hand on their abdomen.

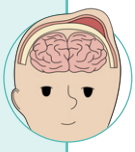
Count for a full minute!

HEAD



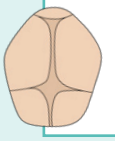
CAPUT SUCCEDANEUM:
- Edema (collection of fluid)
- Crosses the suture lines

MEMORY TRICK: Like a baseball CAP



CEPHALOHEMATOMA:
- Birth trauma (collection of blood)
- Does not cross the suture lines

MOLDING:
Abnormal head shape that results from pressure (*normal*)



FONTANELLES:
Bulging = increase ICP or hydrocephalus
Sunken = dehydration

Fontanelles may be **BULGING** when the newborn cries, vomits, or is lying down. This is normal.

GENERAL CHARACTERISTICS

Length & Weight

EXPECTED LENGTH	44 - 55 cm 17 - 22 in
EXPECTED WEIGHT	2,500 - 4,000 g 5 lb, 8 oz - 8 lb, 14 oz

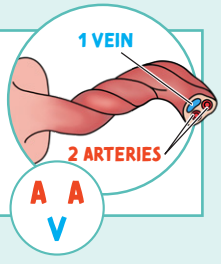
Head & Chest Circumference

HEAD CIRCUMFERENCE	32 - 39 cm 14 - 15 in **measure above eyebrows*
CHEST CIRCUMFERENCE	30 - 36 cm 12 - 14 in **measure above nipple line*

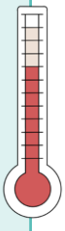
UMBILICAL CORD

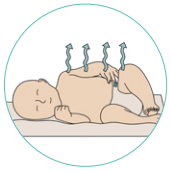
Should have
2 ARTERIES & **1 VEIN**
Should be dry, no odor & no drainage

MEMORY TRICK: looks like a smiley face!

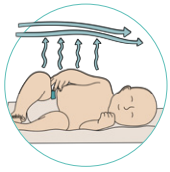


↓ TEMP → HEAT LOSS DUE TO:

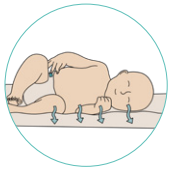




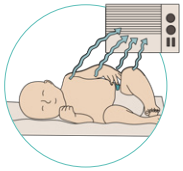
EVAPORATION: Moisture from skin & lungs



CONVECTION: Body heat to cooler air



CONDUCTION: Body heat to a cooler surface in direct contact



RADIATION: Body heat to a cooler object nearby

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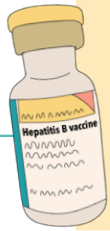
Post-delivery Newborn Medications & Heelstick

Post-Delivery Newborn Medications

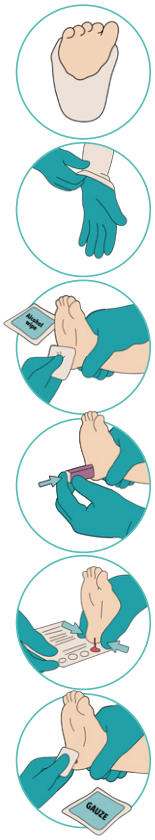
MEDICATION	DESCRIPTION	ROUTE
ERYTHROMYCIN OPHTHALMIC OINTMENT (antibiotic ointment)	▪ This is given to all newborns as a prophylaxis for ophthalmia neonatorum (conjunctivitis of the newborn) ▪ It's required by law in the U.S	Ophthalmic (eye)
VITAMIN K	▪ A newborn's gut is sterile, meaning they do not have any intestinal flora that produce vitamin K until 7 days after birth ▪ Vitamin K is essential for blood clotting ▪ Given to all newborns to prevent hemorrhage & bleeding	Intramuscularly in the vastus lateralis
 **HEPATITIS B VACCINE**	▪ Given to immunize against hepatitis B virus (HBV)	Intramuscularly in the vastus lateralis

Heelstick (HEEL PUNCTURE)

Heelsticks are typically taken in the newborn **24 HOURS AFTER BIRTH** (should be done before discharge!)

WHY IS IT DONE?

To test for medical conditions such as Phenylketonuria (PKU), cystic fibrosis, sickle cell anemia, etc.



Warm the heel before to increase blood flow → *Example:* apply a cloth soaked in warm water

↓

Put on gloves

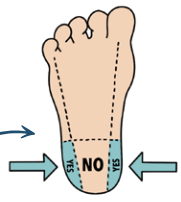
↓

Clean the area with an antiseptic & allow to dry

↓

Hold the infant's foot with the free hand

↓

Puncture the site (side of the heels) 

↓

Collect the sample

↓

Apply gentle pressure with a gauze and cover the site with an adhesive bandage

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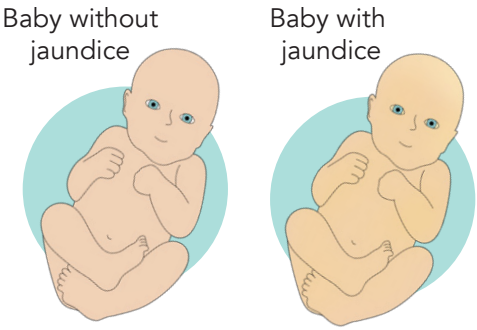
Hyperbilirubinemia

Hyperbilirubinemia (JAUNDICE)

Yellow discoloration of the skin, mucous membranes, and/or sclera caused by ↑ bilirubin levels. Bilirubin is caused by the breakdown of red blood cells (RBCs).



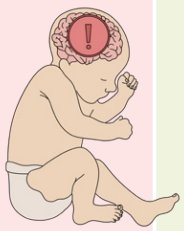
HYPERBILIRUBINEMIA

High levels Bilirubin in the blood

Elevated bilirubin levels in the newborn's blood

MEMORY TRICK: Bilirubin think Breakdown of Red Blood Cells

	PATHOLOGICAL JAUNDICE (ABNORMAL)	PHYSIOLOGICAL JAUNDICE (NORMAL)
TIMELINE	Happens within the **FIRST 24 HOURS** of life	"Nonpathological Jaundice" Happens **AFTER 24 HOURS** of age (begins to see jaundice around day 2–4 of life)
CAUSES	★ Hemolytic disease ▪ RH / ABO incompatibility ▪ Premature infants (the liver is not fully developed) ▪ Failure to pass meconium ▪ Sepsis	▪ Immature liver ▪ ↑ RBCs ▪ A newborn's RBCs have a shorter lifespan ▪ Vacuum Assisted Birth This often causes hematomas to the scalp = ↑ RBCs = more RBCs for the body to breakdown = jaundice
TREATMENT	**PHOTOTHERAPY:** fluorescent light is used to convert the bilirubin to a water-soluble substance so it can be excreted by the body via stool or urine	Typically there is no treatment or complications; it will resolve on it's own *For some cases, phototherapy may be used to help breakdown the red blood cells more quickly*
COMPLICATIONS	**KERNICTERUS** ▪ The excess bilirubin (if untreated) can cause brain damage	



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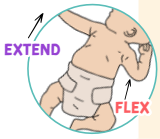
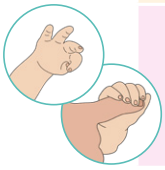
Newborn Reflexes & Types of Heat Loss/Prevention

Newborn Reflexes

	REFLEX	DESCRIPTION	REFLEX DISAPPEARS
	BABINSKI REFLEX	When the bottom of the foot is stroked from the heel upward, the big toe dorsiflexes (bends back) and the other toes spread out Babinski = Big toe fans out	Should disappear after 1 year of age
	ROOTING REFLEX	When the baby's mouth is stroked, the baby will turn its head and open the mouth. This helps the baby find the food source when feeding. *Example:* breastfeeding	Should disappear after 3–4 months (although this can last up to a year)
	MORO REFLEX "startle reflex"	Can be triggered by a sudden loud noise or unexpected movement. The infant will extend the arms with palms up and then move the arms back to the body	Should disappear after 6 months
	**TONIC NECK REFLEX** "fencing"	When an infant is lying on its back and quickly turns its head to one side. The leg and arm on that side will **EXTEND**, while the leg and arm on the opposite side will **FLEX**	Should disappear after 3–4 months
	**PALMAR/ PLANTAR GRASP**	**PALMAR** When a finger is touching the inside of the infant's palm, the hand will close **PLANTAR** When a finger is placed or touching under the toes, the toes will curl	**PALMAR** Response should lessen around 3–4 months **PLANTAR** Response should lessen around 8 months

Types of Heat Loss & Prevention

NEWBORN TEMPERATURE

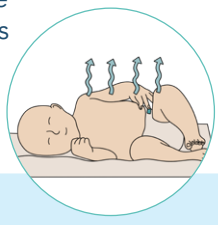
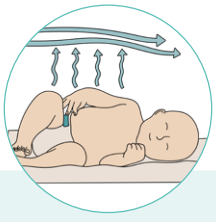
The newborn's temperature at birth is high because they have been snuggled in their mom's uterus which is a warm environment.

The newborn's temperature will immediately **DROP** upon delivery.

There is a balance between heat loss & production.

FACTORS THAT CAUSE HEAT LOSS

- Less subcutaneous fat
- Inability to properly conserve heat
- The temperature of the birthing environment (the operating room is usually cold)

TYPE	EVAPORATION	CONVECTION	CONDUCTION	RADIATION
DEFINITION	Body heat lost due to moisture on skin to cooler air	Body heat lost to cooler air	Body heat lost to a cooler surface in direct contact	Body heat lost to a cooler object nearby
PREVENTION	Dry infant immediately after birth	Keep bed away from open windows	Warm stethoscope & other instruments before use	Keeping infant away from any drafts

MEMORY TRICK: COnvection think COol air

MEMORY TRICK: ConDuction think Direct

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Postpartum Assessment

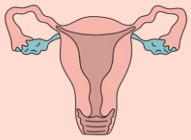
MNEMONIC **"BUBBLES"**

B BREASTS

- May be sore after breastfeeding
- Breastfeed every 2–3 hours (15–20 minutes each breast)
- Position newborn "tummy to mummy"
- Latch should be completely around the areola

MASTITIS
Infection & inflammation of breast tissue
- Continue breastfeeding
- Warm compress
- Hydration
- Rest
- Analgesics
- Wash hands!

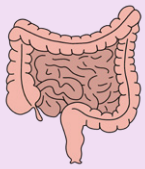
U UTERUS



UTERINE ATONY:

RISK FACTORS	SYMPTOMS	INTERVENTIONS
▪ Retained placenta ▪ Chorioamnionitis (infection) ▪ Uterine fatigue ▪ Full bladder	▪ Enlarged ▪ Soft ▪ Boggy ▪ Not midline ▪ Poorly contracted uterus	▪ Fundal massage ▪ Assist to void or use a straight catheter

B BOWELS



Constipation is common after birth. Increasing **FLUIDS & FIBER** may help!

MEMORY TRICK: Fluids, Fruits & Fiber Fill up the toilet!

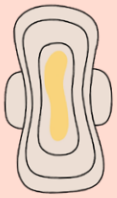
HEMORRHOIDS
- May see blood in the stool
- Should begin to shrink following birth

INTERVENTIONS:
- Tucks/witch hazel
- Ice pack
- Squeeze bottle
- Sitz bath

B BLADDER

- Postpartum urinary retention is common
  - In-and-out catheterization may be needed
  - Bladder distention can cause a displaced & boggy uterus!

L LOCHIA



"Really Sore After"

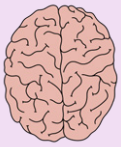
MEMORY TRICK

TYPE	TIMING	DESCRIPTION
Rubra	Really	Birth–4 days	bright or dark red; small clots
Serosa	Sore	4–10 days	pinkish/brown; less or no clotting
Alba	After	10–28 days	whitish/yellow; little to no blood or blood clots

HEMORRHAGE
- Soaking pad in < 1 hour
- Clots larger in size than a nickel

INFECTION
- Foul odor
- Green/yellow purulent discharge
- Fever (> 100.4°F)

E EMOTIONAL STATUS

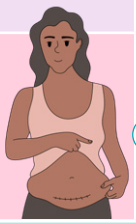


- Postpartum depression (PPD) is common for women following childbirth
- As the nurse ask about feelings of...
  - *depression* ▪ *hopelessness* ▪ *self-harm*
  - *harm to the newborn*

Crying, irritable, sleep disturbances, anxiety, feelings of guilt

S SECTION

C-section incisions/ Episiotomy



- Promote proper wound healing
- **REPORT** to the health care provider:
  - *pain* ▪ *inflammation* ▪ *surrounding skin is warm to touch*

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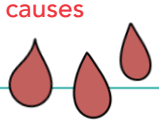
Postpartum Hemorrhage

PATHOLOGY:

#1 cause of maternal death in the U.S.

✓ The uterus is like a basket weave of muscle fibers that crimps vessels, protecting the mother from hemorrhage.

✗ If the uterus is not effectively doing this crimping off, it causes bleeding!



RISK FACTORS:

- Multiple gestations
- Olyhydramnios
- Macrosomic fetus (> 8 lbs 13 oz)
- Multifetal gestation

All these cause an over-distended uterus

Full bladder

- A full bladder can cause distention of the uterus & interfere with contractions

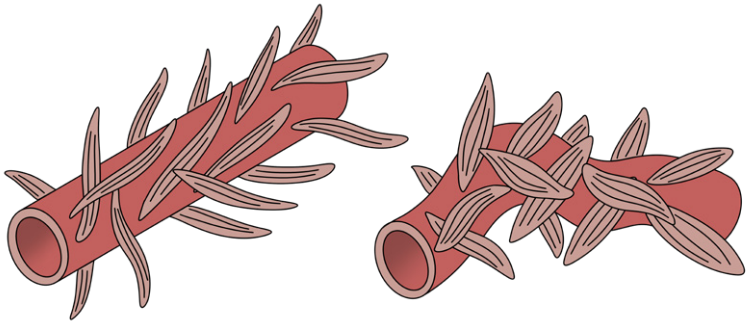
DIAGNOSTIC:

Postpartum hemorrhage is defined as:

VAGINAL & CESAREAN BIRTH

> 1000 mL with signs and symptoms of hypovolemia within 24 hours of birth

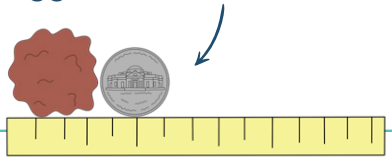
The uterus is often called the **LIVING LIGATURE**



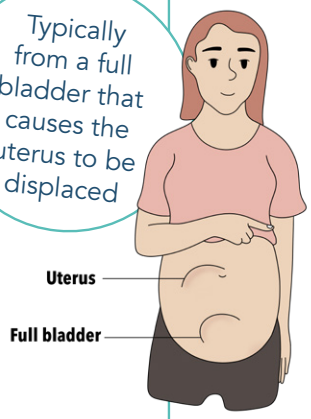
This is showing compression of the arteries by smooth muscle cells in the uterus to stop or prevent postpartum hemorrhage

SIGNS & SYMPTOMS:

- Hypotonia of the uterus
- Uterine atony "a boggy uterus"
- Deviated to the right
- Uncontrolled bleeding
- Saturated perineal pads
- Constant trickling from the vagina
- Blood clots (bigger than a nickel)



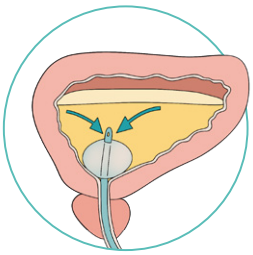
Typically from a full bladder that causes the uterus to be displaced



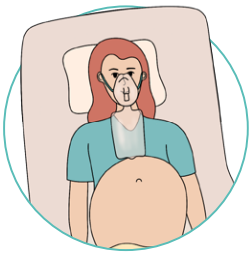
NURSING CONSIDERATIONS:

Firmly massage the fundus

This helps the uterus to contract
(contractions help compress the arteries to stop bleeding)



Assist in emptying the bladder
(possibly a urinary catheter)



Administer oxygen, if needed, through a non-rebreather mask

MEDICATIONS:

MEMORY TRICK: **O**h **M**y **M**aternal **H**emorrhage

- **O**xytocin (Pitocin)
  Most common
- **M**ethergine
  ⚠ Contraindicated in people with hypertension
- **M**isoprostol (Cytotec)
- **H**emabate
  ⚠ Contraindicated in people with asthma

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Breastmilk vs. Formula

DESCRIPTION

Breast Milk

- "Ideal" form of nutrition

COLOSTRUM (BABY'S "SUPERFOOD")

Body stops producing approx. 3 days postpartum
GOLDEN color
Concentrated antibodies, vitamins, nutrient-dense
Minimal amounts to support newborn's nutritional needs

BREAST MILK

Body will begin producing 2–3 days after delivery
WHITE/CREAM color
↑ hand expression & latching, ↑ stimulation to produce
Amount needed is unique to individual

THINK: You are "telling" the body to continue producing

Formula

- Alternative to breast milk, but still contains nutrients necessary for growth
- Comes as a concentrate or a powder
- Follow manufacturer's preparations guidelines (do not overdilute or overconcentrate)

CORE DIFFERENCES

- Digests **FASTER** = more frequent feedings
- Easier to digest for the infant
- Contains antibodies
- Breast milk does not contain Vitamin D (supplements may be needed)

- Digests **SLOWER** = more time between feedings
- May be more difficult to digest
- Lacks antibodies & less "nutrient-complex"

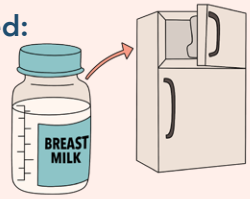
MATERNAL DIET

- Limit caffeine to < 300mg daily
- Limit alcohol (transfers to breast milk)
- Avoid fish high in mercury (tuna, swordfish)

- No restrictions

STORAGE

Freshly pumped or expressed:
- Up to 4 hours at room temp
- Up to 4 days in refrigerator
- Up to 12 months frozen



Once baby has started eating:
use within 2 hours

Once thawed:
use within 1–2 hours **or** refrigerate up to 24 hours
NEVER RE-FREEZE thawed milk

Once prepared:
- Spoils quickly at room temp
- May refrigerate up to 24 hours
- Do not freeze!

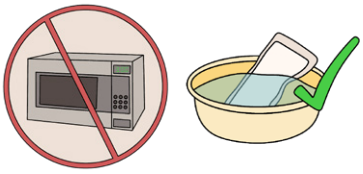
Once prepared:
use within 2 hours

Once baby has started eating:
use within 1 hour

Once powder container is open:
use within 1 month

PATIENT EDUCATION

NEVER microwave; this can cause "hot-spots" & scald baby's mouth
- Run the bottle or bagged breast milk under warm water or place in a cup/bowl of warm water to gently heat
- Always test the temperature with your hand first for safety!



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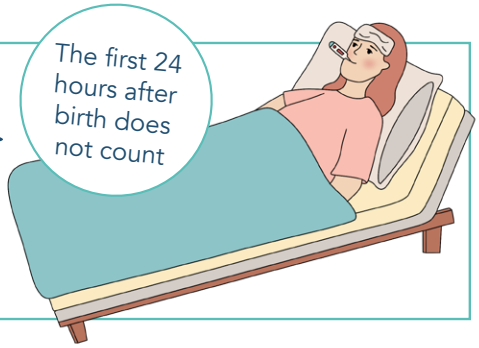
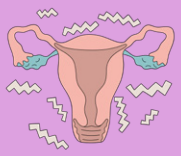
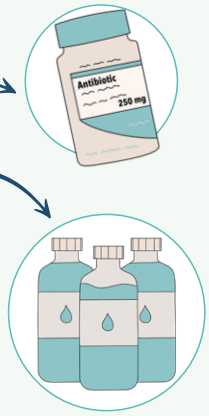
Postpartum Infections

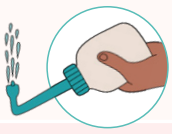
Postpartum Infections

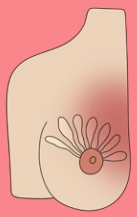
Infection that occurs within 28 days postpartum

DEFINED AS:
Having a fever (> 100.4°F) for 2 or more consecutive days



	PATHOLOGY	SIGNS & SYMPTOMS	TREATMENT
ENDOMETRITIS 	**ENDOMETRITIS** Endometrium + Inflammation Infection of the endometrium (the lining of the uterus)	▪ Fever > 100.4°F (> 38°C) ▪ Tachycardia (↑ HR) ▪ Chills ▪ Fatigue/lethargic ▪ Uterine tenderness ▪ Lochia that is foul-smelling	▪ Antibiotics ▪ Hydration ▪ Rest ▪ Analgesics 
URINARY TRACT INFECTIONS (UTIs)	Infection in any part of the urinary system (bladder, kidneys, urethra)	▪ Low-grade fever ▪ Dysuria (painful or difficult urination) ▪ Frequency & urgency ▪ Hematuria ▪ Flank pain	

WOUND INFECTIONS	Infection of a wound (*Examples:* cesarean incision, episiotomy incision, lacerations, etc.)	▪ Fever > 100.4°F (> 38°C)
		▪ At the site:
		  ▪ Redness
		  ▪ Pain/tenderness
		  ▪ Warmth
			▪ For laceration & episiotomy
			  ▪ Warm or cool compress
			  ▪ Sitz bath
			  ▪ Perineal care (peri bottle) 

MASTITIS 	**MASTITIS** The breast + Inflammation Commonly called "clogged or infected milk duct"	★ **Reddened, hot area/tenderness in one breast** ▪ Temp > 100.4 °F (> 38°C) ▪ One-sided pain to underarm (axillary) region **SUDDEN ONSET OF FLU-LIKE SYMPTOMS** should trigger breast assessment!	▪ Discourage/stop use of underwire bras ▪ **FREQUENT/CONTINUED** breastfeeding on affected side ▪ Encourage pumping and direct feeds to empty infected side ▪ Infection **CANNOT** be transferred to baby MEMORY TRICK **"REST, EXPRESS, & COMPRESS"** **REST** as much as possible **EXPRESS** breast milk frequently to unclog duct **COMPRESS** gently with warm compress

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Pediatrics

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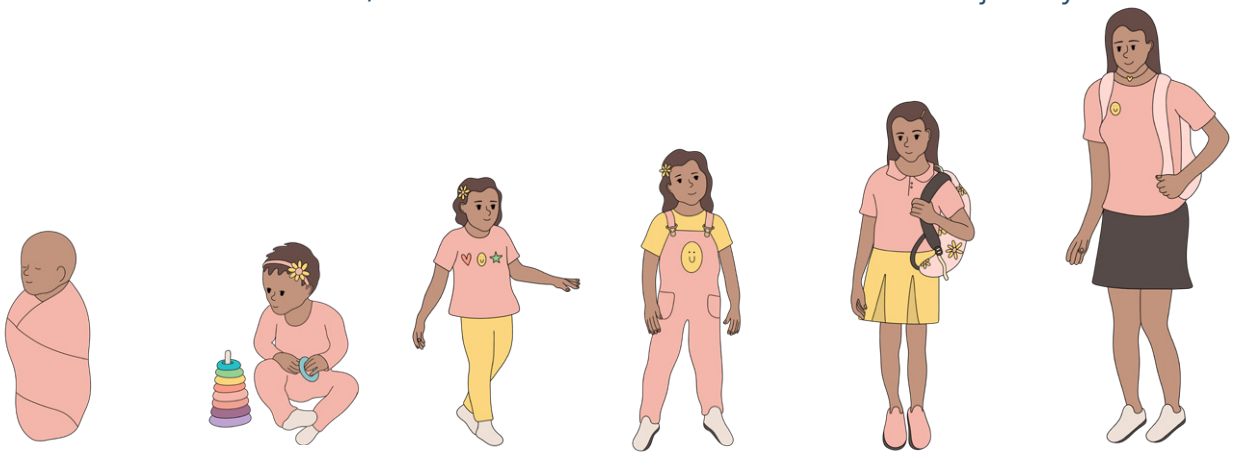
Tips for answering growth & development questions

- Try NOT to relate to real-life scenarios.
- The NCLEX has very specific standards in place.
  Passing requires knowing and remembering these.
- Every milestone is different, just like every child is different!
  In the NCLEX world, the children should follow the outlined trajectory.



NEWBORN/NEONATE
First 4 weeks of life

INFANT
1 month–1 year

TODDLER
1–2 years

PRESCHOOL
3–5 years

SCHOOL AGE
6–12 years

ADOLESCENCE
13–17 years

Children are not just small adults

Many conditions are specific to infants and children; be sure to learn about these conditions & how to intervene.

Pediatric vital signs, blood volume, and presentation will differ from an adult; be sure to learn about these.

Administering medication to children is very critical because many dosages are calculated based on weight and a child's response to the medication can differ significantly from an adult's response.

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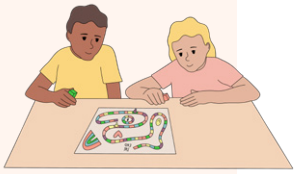
Stages of Play

AGE GROUP	TYPE OF PLAY	KEY POINTS	DESCRIPTION
Infancy (1 month - 1 year)	SOLITARY PLAY	Playing alone	The child is playing all by them self with a toy
2 years	ONLOOKER (spectator)	Watching other children	The child is watching other kids play, but does not want to engage in the play
Toddlers 1 - 2 years	PARALLEL PLAY	Next to, but not interacting	The children will play next to each other, but not WITH each other (not interacting with each other)
Preschool 3 - 5 years	ASSOCIATIVE PLAY	Interacting, but not working together	Children are playing the same game or building the same thing, but not working together or connecting with each other
School Age 6 - 12 years	COOPERATIVE PLAY	Working together	Children are playing the same game or building the same thing and are working together 

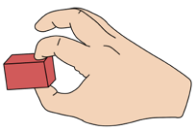
Milestones are broken down into different categories:

GROSS MOTOR SKILLS

Gross motor skills think **LARGER** movements or movement which uses the whole body.

MEMORY TRICK

GROSS = LARGE in medical terms



FINE MOTOR SKILLS

Fine motor skills are more **INTRICATE** and involve one part of the body (often the hands).

Examples:

- Holding
- Touching
- Tapping

MEMORY TRICK

FINe motor skills think FINgers

LANGUAGE

Language includes communication through verbal, crying, and nonverbal expressions.

COGNITIVE

Cognition is how the child understands the world around them. Cognitive function can be shown as expression of emotion, action, or play.

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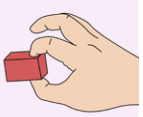
Pediatric Milestones

There are **MANY** pediatric milestones but these are the *most commonly tested on.*

AGE	GROSS MOTOR SKILLS	FINE MOTOR SKILLS	LANGUAGE	COGNITIVE
1 Month	Head lag	Palmar grasp reflex *This is when a baby involuntary wraps their fingers around an adult's fingers when the palm is touched*	Communicates through crying	Watches moving things & tracks with eyes
2 - 3 Months	Able to raise head & chest	Palmar grasp reflex is disappearing/absent	Makes cooing noises	★ **Should be smiling** MEMORY TRICK: **S**econd month think **S**miling
4 - 5 Months	Rolls from prone to supine MEMORY TRICK: Rolls on the FLOOR rhymes with FOUR!	Attempts to grasp objects (*Example:* a stuffed animal) MEMORY TRICK: You grasp something with 5 fingers at 5 months	Babbling ("ba", "ga", etc.)	Remembers faces & cries when left alone
6 - 7 Months	Able to sit in a tripod position MEMORY TRICK: This position kind of makes a 6 (6 months)	Transfers items from one hand to the other	Starts to imitate sounds	★ **Stranger anxiety starts to develop**
8 - 9 Months	**Cruising:** Standing up & stepping while holding onto couch or table	★ **Developing pincer grasp** Uses index finger & thumb to lift or grab something MEMORY TRICK: Pincer grasp think Pinch 	More expressive *(screeching, squealing, giggling)*	★ **Object permanence** Realizing that objects that are out of sight still exist

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Pediatric Milestones

AGE	GROSS MOTOR SKILLS	FINE MOTOR SKILLS	LANGUAGE	COGNITIVE
10 – 12 Months	Takes first steps MEMORY TRICK 12 months: Step 1, step 2	Pincer grasp fully developed	Able to say "ma mu," "da di," & "uh oh"	**Separation anxiety** When the child has anxiety & is distressed over the parent or caretaker leaving them
	The first teeth to show are the **LOWER CENTRAL INCISORS** (usually around 10 months of age)			
15 Months	Able to mostly walk independently	Claps & points at objects	Understands names of people & things	Fussing or "throwing a fit" when unhappy
18 Months	Able to walk on stairs WITH help Able to throw a ball over their head	Colors with marker or crayon (scribbles)	Answers questions with yes/no (head nod, head shake)	Learning independently
2 Years	**Independently walking** Kicks a ball 	Points at pictures in book	Can form 2–3 word sentences like "play outside"	**Parallel play** Playing next to each other but not WITH each other
3 Years	Jumps from a step, walks up and down stairs without putting both feet on one stair (one foot per step)	Uses scissors to cut simple shapes	Asks "WHY" MEMORY TRICK "WHY" has 3 letters in it	**Associative play** Starting to play with each other & learn social skills
4 Years	Plays/rides down slide independently Throws & catches a ball 	Holds pencil appropriately Begins to write name	**Explains** events (answers "what did you do at school?")	**Imaginary play** "Pretend play"–child will make-believe with toys, stuffed animals, or utensils

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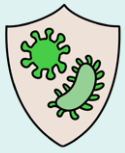
Pediatric Vaccine Schedule

BIRTH	1 MONTH	2 MONTHS	4 MONTHS	6 MONTHS	9 MONTHS	12 MONTHS	15 MONTHS
	Hep B dose #2			Hep B dose #3			
				Inactivated polio (IPV) #3			
				Influenza (every year) ★ Each child should receive a yearly influenza vaccine after 6 months of age			
 Hep B dose #1 MEMORY TRICK B think Birth		Inactivated poliovirus (IPV) #1 Pneumococcal #1 Rotavirus (RV) #1 DTaP #1 Hib #1	Inactivated poliovirus (IPV) #2 Pneumococcal #2 Rotavirus (RV) #2 DTaP #2 Hib #2	Inactivated poliovirus (IPV) #3 Pneumococcal #3 Rotavirus (RV) #3 (depending on brand) DTaP #3 Hib #3 (depending on brand)		MMR #1 Varicella (chickenpox) #1 Pneumococcal #4 Hep A #1 Hep A #2 *(must be 6 months apart from first dose)* Hib #3 or #4 (depending on brand)	DTaP #4

18 MONTHS	19–23 MONTHS	2–3 YEARS	4–6 YEARS	7–10 YEARS	11–12 YEARS	13–15 YEARS	16 YEARS	17–18 YEARS
Hep B dose #3	Yearly **influenza vaccine** recommended for all children ★ Each child should receive a yearly influenza vaccine after 6 months of age							
Hep A dose #2 *(must be 6 months apart from first dose)*		MEMORY TRICK Years two to THREE think vaccine-FREE! (except for annual flu shot & children 2 yrs. and older who have not had 2 doses of Hep A)	MMR #2 Varicella (chickenpox) #2 DTaP #5 Inactivated poliovirus (IPV) #4		Tdap #1 *(switches names from DTaP after age 7)* Meningococcal #1 HPV (human papillomavirus) *optional		Meningococcal #2	
Hep B dose #3 *(if not received in last grouping)* DTaP dose #4 *(if not received in last grouping)*								

DTaP: diphtheria, tetanus, pertussis
Tdap: tetanus, diphtheria, pertussis



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Pediatric CPR (< 12 Months)

Cardiac arrest in infants usually stems from **RESPIRATORY ETIOLOGY**

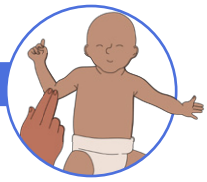
Order of Events:

1 PULSE

- Check pulse for no longer than 10 seconds

INFANT

Check **BRACHIAL** pulse



CHILD

Check **CAROTID** pulse

2 CALL FOR HELP

- Active the emergency response system/shout for nearby **HELP**
- Delegate someone else to call 911/get the AED

3 CHEST COMPRESSIONS

- Perform 2 minutes of CPR before retrieving AED if rescuer is alone
  - Rate of 100–120 compressions/min
  - Use either 2 fingers or 2 thumbs on the sternum
  - Compress to a depth of: **INFANT** About 1.5 inches; **CHILD** 2 inches
  - Allow for full chest recoil between compressions

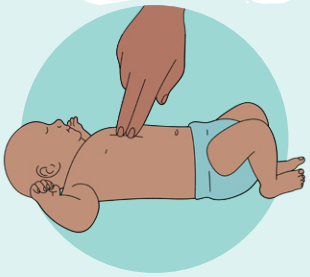
SINGLE RESCUER

30:2 compression-to-breath ratio

TWO RESCUERS

15:2 compression-to-breath ratio

2-FINGER compression technique



2-THUMB encircling hand technique

4 CONTINUE until help arrives or AED becomes available

Pediatric Vital Signs

Age	Heart Rate (HR)	Respiratory Rate (RR)	Systolic Blood Pressure (SBP)	Diastolic Blood Pressure (DBP)	Mean Arterial Blood Pressure (mm Hg)
NEWBORN (first 4 weeks of life)	11 - 205	30 - 60	67 - 84	35 - 53	45 - 60
INFANT (1 month-1 year)	100 - 180	30 - 53	72 - 104	37 - 56	50 - 62
TODDLER (1-2 years)	98 - 140	22 - 37	86 - 106	42 - 63	49 - 62
PRESCHOOLER (3-5 years)	80 - 120	20 - 28	89 - 112	46 - 72	58 - 69
SCHOOL-AGED CHILD (6-12 years)	75 - 118	18 - 25	97 - 115	57 - 76	66 - 72
ADOLESCENTS (13-17 years)	60 - 100	12 - 20	110 - 131	64 - 93	73 - 84

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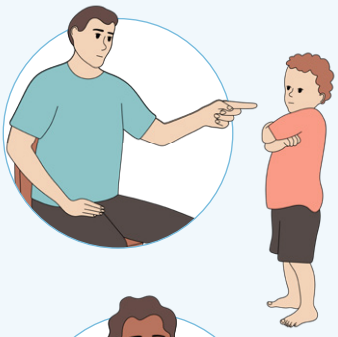
Kohlberg's Theory of Moral Development

Preconventional

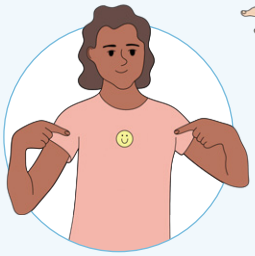
1 PUNISHING VS. OBEYING

A child believes obeying strict rules of a parent or adult will help them to avoid punishments



2 FOCUS ON SELF-INTEREST

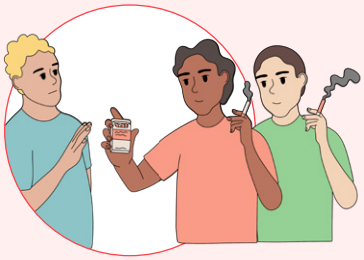
Doing wrong may lead to punishment BUT may benefit the child's self-interests



Conventional

3 SOCIAL ACCEPTANCE & PLEASING OTHERS

The child or adolescent sees social status as important and aims to please friends, while weighing right and wrong



4 RULES ARE FLEXIBLE

Rules and laws exist for order, but the individual realizes that these CAN change and CAN be flexible

Postconventional

5 FOCUS ON INNER DECISIONS & CONSCIENCE

Using past experience and conscience (good vs. bad), the individual is able to make decisions and guide themselves with the HELP of rules, but not ONLY through rules

6 ETHICS & UNIVERSAL PRINCIPLES

An individual's behavior is based on their moral principles

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Piaget's Stages of Cognitive Development

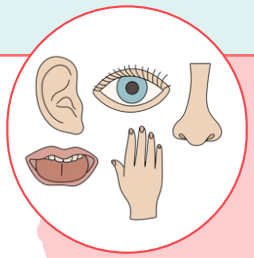
MNEMONIC

Saying Piaget's Cognitive stages is Fun

S Sensorimotor Stage

0–2 YEARS

- Development through the 5 senses
- Development through motor response and reflexes
- **OBJECT PERMANENCE** is developed
  - Realizes that out-of-sight objects still exist



P Preoperational Stage

2–7 YEARS

- Symbolic thinking
- Imagination
  - Magical thinking, animism (thinks objects are alive), plays pretend
- Abstract thinking is still difficult
- Asks a lot of questions (intuition)
- Egocentric
  - Can only see the world from one's own point of view

C Concrete Operational Stage

7–11 YEARS

- Develops concrete cognitive operations
- Sorting blocks in a certain order
- **CONSERVATION** is developed
  - **CONSERVATION:** Understanding that something stays the same in volume even though its shape changes
- Conductive reasoning (mathematical advancements)

F Formal Operational Stage

> 11 YEARS

- More rational, logical, organized, moral, and consistent thinking
- Hypothetical thinking: can think outside the present
- Abstract concepts
  - Love, hate, failures, successes
- Deductive reasoning

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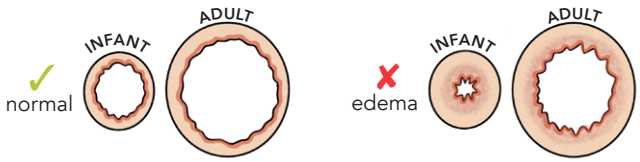
Variations in Pediatric Anatomy

Respiratory

More narrow airways
- Edema in the airways is even more serious in an infant



Larger tongue relative to body size
- Risk for obstruction

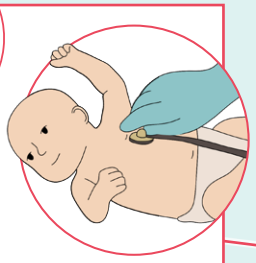
Obligatory nose breathers
- Newborns are unable to breathe through their mouth (orally) so if their noses are blocked or occluded, they can not breathe

Less alveoli than adults
- Thousands of alveoli grow each day for the first few months of life

Cardiovascular

Blood pressure does not fluctuate as much
- Can maintain a safe blood pressure longer than adults (even if they are losing blood)

This is why it's vital to monitor closely. Decline can be **RAPID**



Higher heart rate than adults
- This is due to the infant's immature muscle fibers

Numerous congenital defects can occur
- Most common is ventricular septal defect (VSD) "hole in the heart"

Gastrointestinal

Smaller stomach capacity than adults
- Example:
  - Newborn stomach capacity → 10–20 mL
  - Adult stomach capacity → 2,000–3,000 mL

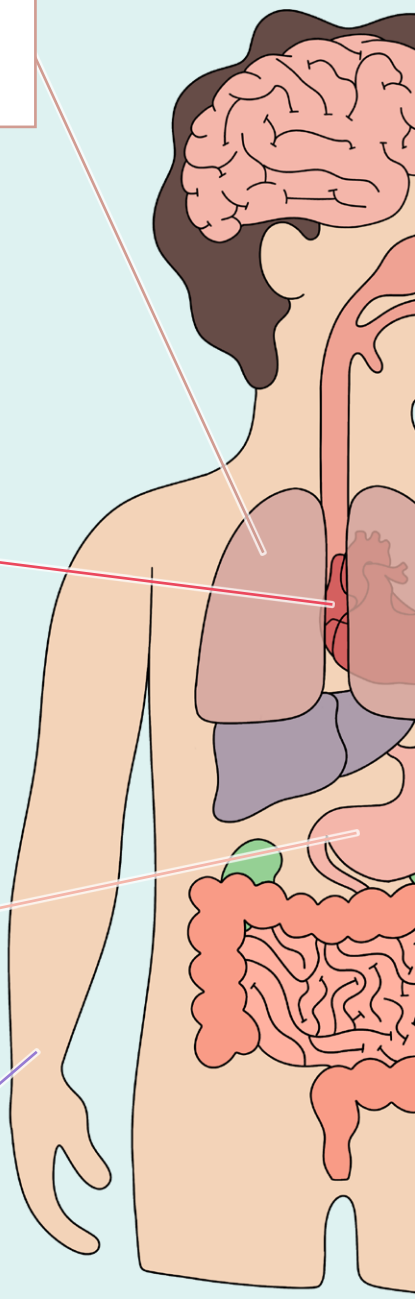
Lower esophageal sphincter (LES) is not fully developed until 1 month
- Can lead to regurgitation & dysphagia

Integumentary (Skin)

The epidermis is thinner & more fragile than adults
- Can lead to tearing with minimal friction

Skin loses water more quickly & dries out more easily

Blood vessels are closer to the surface (lose heat very easily)
- Harder to regulate temperature



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Variations in Pediatric Anatomy

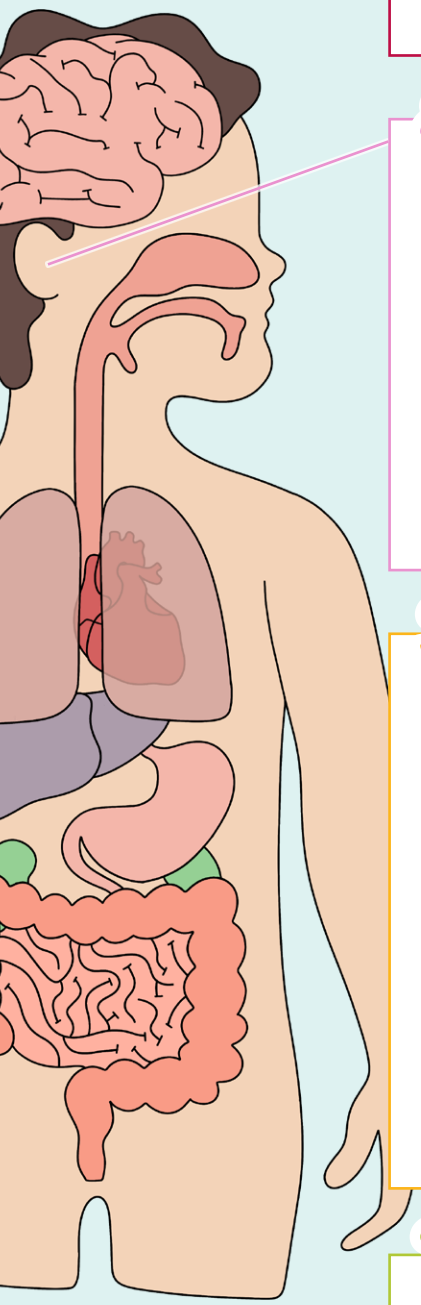
Musculoskeletal

Bones are not ossified yet
- Bones are more flexible/flimsy; they typically bend rather than break

Ligaments & muscles are immature
- Females have more lax ligaments or flexible joints due to female hormones

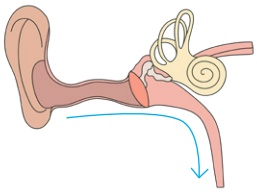
Growth plates sit at the end of a bone
- These are triggered to close during puberty by hormones (estrogen & testosterone); this is when bones will stop growing

Cranial bones not completely fused
- Sutures & fontanelles make the skull flexible and allows for growth of the brain



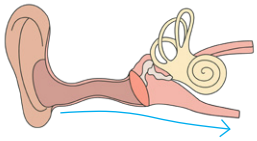
Ears

ADULT EUSTACHIAN TUBES



Adult eustachian tubes are slightly tilted allowing for drainage and are less likely to harbor bacteria

INFANT EUSTACHIAN TUBES



Infant eustachian tubes are short, wide, & flat

Proper drainage is difficult → Accumulation of secretions occurs → harboring of microorganisms → ↑ Risk for ear Infection

Nervous System

Infants are "top heavy" & neck muscles are not fully developed
- ↑ Risk for falls that lead to head injuries

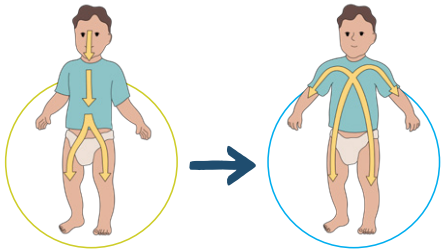
Less cerebrospinal fluid (CSF) to cushion the brain

The brain is highly vascular
- ↑ Risk for hemorrhage

The spine is very mobile
- ↑ Risk for cervical spine injury

Myelination is incomplete at birth and continues until around 2 years of age

How myelination happens in **CEPHALOCAUDAL** to **PROXIMODISTAL** fashion:



CEPHALOCAUDAL DIRECTION:
Develops head to tail

Example:
Head control happens before walking

PROXIMODISTAL DIRECTION:
Develops inward to outward

Example:
Infants use their arms before being able to effectively use their hands/fingers

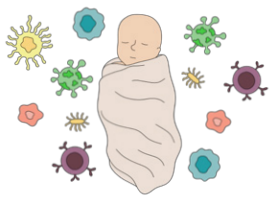
Immune System

Immature immune system

↓ Inflammatory response

Limited exposure to diseases (loses immunity from maternal antibodies)

↑ Risk for Infection



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Fever Management

Having a fever is not an illness; it's simply a symptom of an illness/infection

Signs & Symptoms

- Sweating
- Dehydration
- Chills & muscle aches
- Feeling weak

FEBRILE SEIZURE:

A seizure that is caused by a fever, NOT by any issues with the brain such as a CNS infection. Most require no treatment.

If the seizure lasts > 5 minutes, find immediate help.

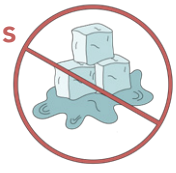
Treatment

- Encourage fluid consumption
- Apply a moist, cool compress
- Keep the room at a cool temperature
- Don't over-bundle the child
- Sponge bath/tepid bath

A child with a high fever is at risk for dehydration & electrolyte imbalances

✓ **Use lukewarm water to help lower the body temperature**

✗ **Do not use cold water as this can cause body temperature to lower too quickly**



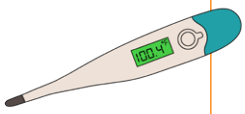
Diagnostic

- Taking the child's temperature

Most accurate: orally & rectally

MEASUREMENT METHOD	NORMAL TEMPERATURE RANGE
RECTAL	97.9°F–100.4°F (36.6°C–38°C)
EAR	96.4°F–100.4°F (35.8°C–38°C)
ORAL	95.9°F–99.5°F (35.5°C–37.5°C)
AXILLARY	94.5°F–99.1°F (34.7°C–37.3°C)

Temperatures above normal ranges should be evaluated by a HCP as this could indicate an illness requiring medical management.



Risk Factors

- Viral infections (influenza, RSV)
- Bacterial infections
- Post-vaccination periods (24–48 hour immune response)
- Exposure to illness/infection
- Children who are unvaccinated
- Children who are immunocompromised

Medications

Antipyretics

Acetaminophen (Tylenol) — NSAIDs such as ibuprofen (Advil)

Do not give aspirin due to the risk for Reye's syndrome!

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Sudden Infant Death Syndrome (SIDS)

Pathology

Sudden, unexplained death of a previously healthy infant younger than 1 year of age

ABCs OF SAFE SLEEPING

MEMORY TRICK

Alone
on their **B**ack
in a **C**rib

Risk Factors

THE EXACT CAUSE IS UNKNOWN!

- Age: birth–6 months (↑ risk)
- Preterm
- Sleep position
- Sibling death
- Nicotine exposure
- Socioeconomic status
- Lack of prenatal care
- Genetics
- Bedding (can be smothered)

Signs & Symptoms

THERE ARE NO SIGNS OR SYMPTOMS!

Parent Education & Prevention

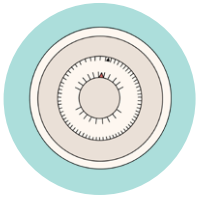
AVOID over-bundling or overdressing the infant

Have baby sleep in a supine position (on their backs)

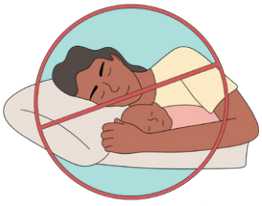
AVOID smoking and vaping during and after pregnancy
Second hand smoke
↑ the risk for SIDS

Encourage pacifier use
A pacifier might
↓ the risk for SIDS

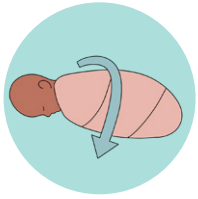
Bedding
- Firm mattress
- No sleeping with toys, blankets, pillows, or stuffed animals



Normal room temp
(ideal temperature is 68°F–72°F)
Overheating may
↑ the risk for SIDS



No co-sleeping
Infant should sleep separate from the parents



STOP swaddling once the baby shows signs of rolling over
Swaddling when a baby rolls over can ↑ risk for SIDS & suffocation

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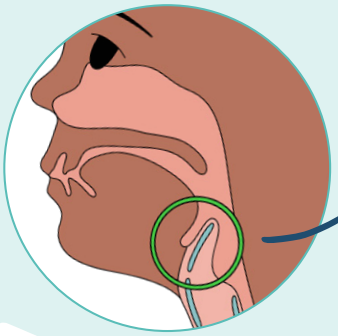
Epiglottitis

Pathology

Inflammation of the epiglottis leading to an upper airway obstruction

Most common cause:
Haemophilus influenzae type b (Hib) bacteria



The epiglottis prevents water & food from entering the windpipe

This is considered a medical emergency

Signs & Symptoms

Abrupt onset of symptoms

- Sore throat
- Dysphagia
- High fever
- Anxious/apprehensive/agitated
- Difficulty speaking
- Stridor (frog-like croak on inspiration)
- **Drooling/dysphagia**
- **Tripod position**
- Retractions (chest)
- Nasal flaring
- Absent cough
- Tachycardia

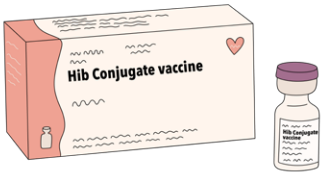
Sitting forward with the neck extended & mouth open to breathe

Risk Factors

- Unvaccinated for Hib
- Upper respiratory/throat infection

Prevention

- Hib conjugate vaccine



Nursing Considerations

DON'Ts

- **Do not visualize the throat with a tongue blade; take oral temperature or obtain a throat culture**
- Do not place in supine position because it may make breathing harder
- Do not leave the child

Why? It can cause **reflex laryngospasms** (cutting off the airway)

DOs

- Assess oxygen status (may need emergency intubation)
- Create a calm environment
  - Allow child to stay with parents
  - Don't restrain the child
  - Help to avoid crying
  - Place in most comfortable position (usually tripod position)

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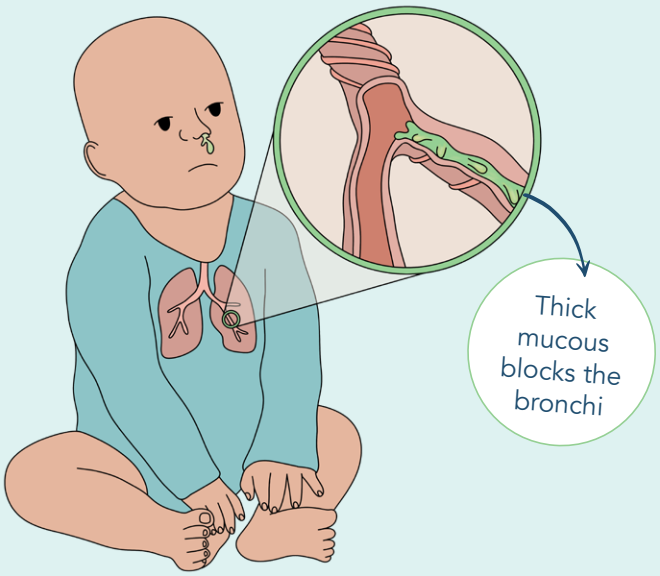
Respiratory Syncytial Virus (RSV) "Bronchiolitis"

Pathology

BRONCHIOLITIS

small airways in the lungs → inflammation

- Viral illness usually caused by **respiratory syncytial virus (RSV)**
- **Very contagious**



Risk Factors

- Time of year (winter season)
- Age: newborn–2 years old
- Underlying asthma
- Exposure to secondhand smoke

Signs & Symptoms

Starts as an upper respiratory infection & moves into the chest

INITIAL →

- Upper respiratory symptoms:
  - Nasal congestion
  - Runny nose
  - Cough
  - Sneezing
- Fever

CONTINUED →

- Lower respiratory tract symptoms:
  - Tachypnea
  - Cough
  - Wheezing

EMERGENT

- ⚠ Grunting
- ⚠ Nasal flaring
- ⚠ Cyanosis
- ⚠ Hypoxia
- Respiratory failure
- Apneic episodes

⚠ Hospitalization may be needed for severe cases

Treatment

Most children can be managed at home

- Viral supportive measures
  - Hydration
  - Fever management
  - Humidifier
  - Suction bulb use
- Nebulizers/steroids
- Cool humidified air & oxygen

Suctioning the nares with a bulb syringe helps to remove the excess secretions before feeding or at bedtime

✗ **Avoid using cough suppressants:**

- Cough suppressants stop secretions from coming out and we want secretions to be cleared from the airways!

Parent Education

- Encourage hand hygiene and masks around newborns
- Do not smoke around babies
- Know **EMERGENT** signs & when to call 911

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Scarlet Fever

Pathology

A complication of **group A streptococcal infection "strep throat"**
(not all children who have strep will develop scarlet fever)

MEMORY TRICK

Scarlet Fever think Strep!

Signs & Symptoms

ABRUPT onset!

- **Sandpaper-like rash**
- **Strawberry tongue**
- Fever, body aches, chills
- Tender lymph nodes
- Tonsils are red & exudate may be present

Begins on the **NECK & CHEST** and spreads outward to **THE EXTREMITIES**

Rash is usually not seen on the palms & soles of the feet

MEMORY TRICK

S's of SCARLET FEVER:
Strawberry tongue
Sandpaper rash

Transmission

- Droplets & respiratory tract secretions

Transmission happens in close contact such as schools & daycare settings

Treatment

- Fluids & soft foods
- Provide comfort
- Cool mist humidifier

Soups, teas, popsicles, slushies

Medications

- Antibiotics
  - Penicillin or amoxicillin

Finish the entire prescription even if the child appears to be better!

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Laryngotracheobronchitis "Croup"

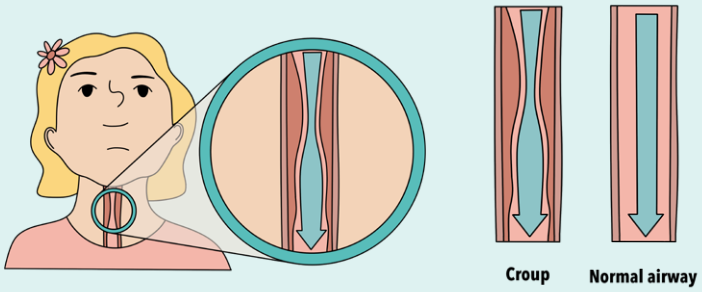
Pathology

Inflammation of the larynx, trachea, & bronchi occur as a result of a viral infection

Most commonly caused by the parainfluenza virus

LARYNGO TRACHEO BRONCHI ITIS

Larynx ↓ Trachea ↓ Bronchi ↓ Inflammation



CROUP vs. EPIGLOTTITIS

	CROUP	EPIGLOTTITIS
ONSET	Sudden (at night)	Rapid (within hours)
FEVER	Fluctuating	High
COUGH	Yes	No
DYSPHAGIA	No	Yes
CAUSE	Viral	Bacterial
EMERGENCY	Not typically	⚠ Yes

Signs & Symptoms

THE 4 Ss

- **S**tridor
- **S**ubglottic swelling (causes hoarseness in the voice)
- **S**eal-bark cough (sometimes described as a "brassy" cough)
- **S**leeping (symptoms typically occur at night)

Risk Factors

- Young age (6 months–3 years have the greatest risk)

Treatment

Often self-limiting (usually resolves on its own)

CARE IN THE HOME

- Cool humidified air (go outside, open the fridge, go in a cool basement)
- Encourage rest & fluid intake
- Calm environment for the child

CARE IN THE HOSPITAL

- Corticosteroids (↓ inflammation)
- Racemic epinephrine
- Humidified oxygen with mist

Medical emergency when the child is showing signs of RESPIRATORY DISTRESS:

- Child is confused/restless
- Blue lips/nails
- ↑ Respiratory rate (breathing faster, but less air is going in)
- Retractions
- Nasal flaring
- Drooling/can't swallow

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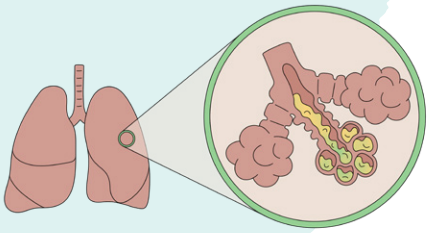
Cystic Fibrosis (CF)

Pathology

A gene mutation (CFTR) which prevents **EXOCRINE GLANDS** from properly functioning. This causes a multi-system disorder where there is an increased production of thick mucus.

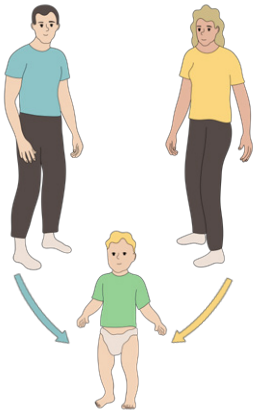
EXOCRINE GLANDS:
Produce & transfer secretions (mucus, tears, sweat, & enzymes) through ducts

↑ Viscosity of mucus
↓
↑ Resistance to ciliary action
↓
Slowing the flow rate of mucus
↓
Mucus plugging



Risk Factors

CF is an **AUTOSOMAL RECESSIVE** genetic disorder:



Dad is a carrier of CF gene

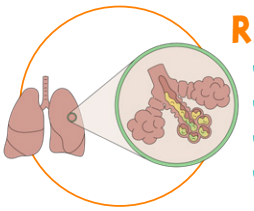
Mom is a carrier of CF gene

2 mutated CF genes = **CYSTIC FIBROSIS**

Diagnostic

- Ambry test
- (+) Sweat sodium chloride test
- Genetic screen

Signs & Symptoms and Patient Education



RESPIRATORY

- Pulmonary hypertension
- Chronic coughing
- Wheezing
- Recurrent respiratory infections (pneumonia)

Thick mucus creates an environment for bacterial growth

CHEST PHYSIOTHERAPY (CPT)

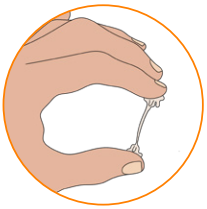
NOT done right before or after meals!

- Causes vibrations & percussions to break apart the mucus
- Done multiple times a day in 1–2 hour increments

INTEGUMENTARY

- Sweat glands produce ↑ chloride = **salty skin**
- Salty sweat & salty tears

Can lead to dehydration & electrolyte imbalance

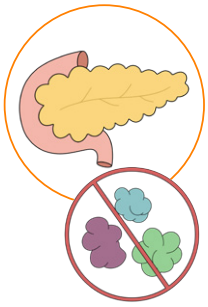


REPRODUCTIVE

BOYS: Thick mucus blocks the vas deferens

GIRLS: Thick cervical mucus blocks sperm from penetrating

Infertility



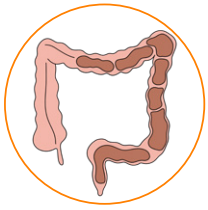
PANCREAS & LIVER

- Deficient in pancreatic enzymes: (protease, amylase, lipase)
  - Causes weight loss & inadequate protein absorption
- Hyperglycemia
- CF-related diabetes
- Bile duct blockage from thick mucous causes gallstones

DIET MODIFICATIONS:

- ↑ Protein, ↑ fat, ↑ calorie intake
- ↑ Fat soluble vitamin supplementation (A, K, E, D)
- **Pancreatic enzymes:** pancrelipase or pancreatin

Can swallow capsules or sprinkle enzymes on foods that are acidic such as apple sauce!



GASTROINTESTINAL

- Fecal impaction
- Rectal prolapse
- Constipation/bowel obstruction
- Meconium ileus in infants
- Steatorrhea
  - Frothy (bulky), fatty, foul-smelling stools

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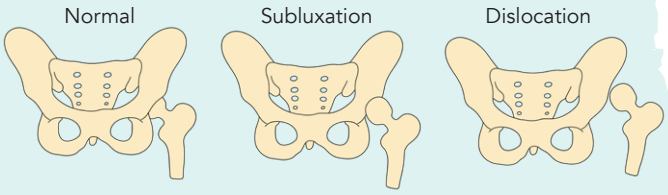
Developmental Dysplasia of the Hips (DDH)

Pathology

- Abnormal development of the hip joint
- A baby's bones are not ossified yet so they can dislocate & relocate easily

TYPES:



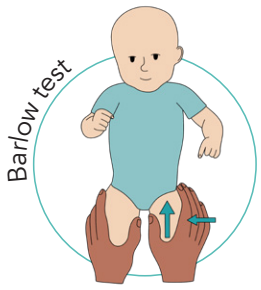
Signs & Symptoms

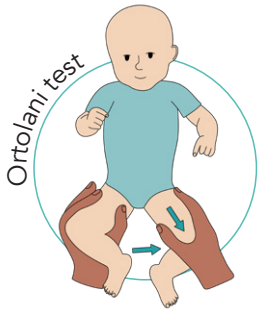
- Uneven leg length
- Appears to "waddle" when walking
- Lordosis (lower back curves in exaggeratedly)

Diagnostic

- Ultrasound while in utero
- X-ray for those older than 6 months

Barlow & Ortolani tests





Listen for any noises during the exam. There should be no "clicks" or "clunks" heard or felt.

If "clicks" or "clunks" are heard or felt
=
a positive sign for DDH
& the child will get an X-ray or ultrasound of the hip

Risk Factors

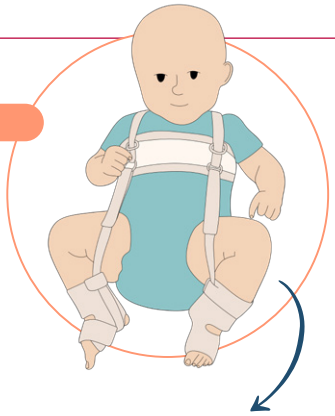
- FEMALE → more lax ligaments from maternal hormones
- Breech positioning
- Oligohydramnios

Treatment

< 6 MONTHS OLD

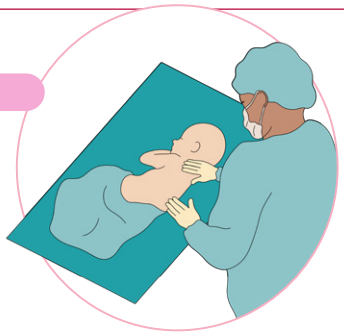
Pavlik harness worn for 6–12 weeks

MEMORY TRICK: HIP think Pavlik Harness



> 6 MONTHS OLD

- Surgery
- Physical therapy
- Gait training



DOs & DON'Ts for Pavlik harness:

DOs

- ✓ Wear the harness at all times (24 hours a day)
- ✓ **MONITOR** for redness, irritation or skin breakdown 2–3 times per day
- ✓ Place baby on their back to sleep
- ✓ Place knee-long socks, undershirt, diaper, & clothing **UNDER** straps to prevent rubbing of the harness
- ✓ Gently massage the skin under the strap to promote circulation

DON'Ts

- ✗ Adjust the straps or remove harness until instructed by the HCP
- ✗ Use lotions under harness (this causes excess moisture and skin breakdown)

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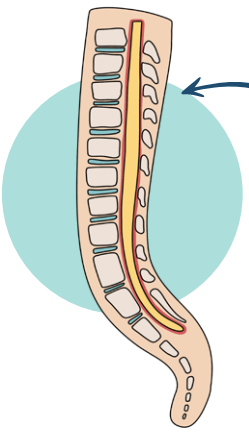
Neural Tube Defects



NORMAL SPINE

The neural tube closes during the 3rd–4th week of gestation

Spina Bifida

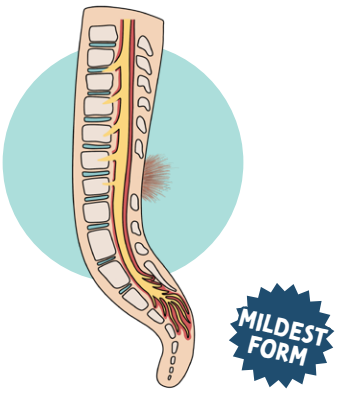
is a general term for a birth defect typically diagnosed during pregnancy where the spinal column fails to close.

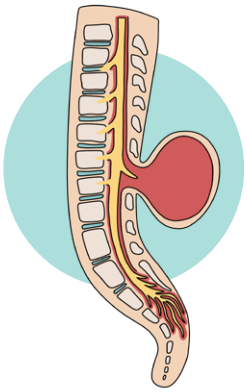
Spina bifida means "split spine"

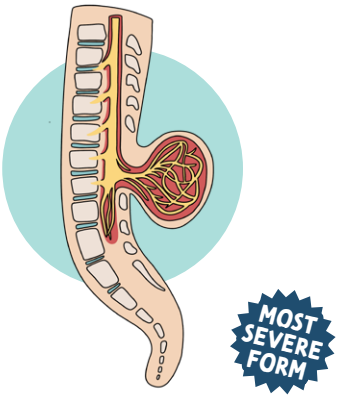
CAUSES:

Definitive cause is unknown.
Maternal predisposing factors thought to increase the risk such as:

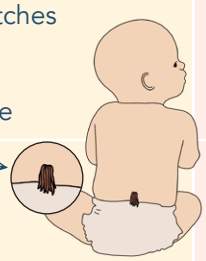
- ★ Low folic acid intake
- Maternal obesity
- Gestational diabetes
- Malnutrition
- Certain drugs
- Chemical or radiation exposure







	SPINA BIFIDA OCCULTA	MENINGOCELE	MYELOMENINGOCELE
PATHO	Defect of the vertebral body **WITHOUT** protrusion of the spinal cord or meninges	▪ Sac protruding from the spinal area ▪ Most are covered with skin	▪ Protrusion of the meninges, cerebrospinal fluid, and spine ▪ Skin may be exposed as well
SIGNS & SYMPTOMS	▪ Typically asymptomatic ▪ May have dimpling, abnormal patches of hair, or discoloration near the spine	▪ Meninges herniate through a defect in the vertebrae ▪ Usually minor or no neurological deficits	The spinal cord often ends at the point of the defect = Absent motor & sensory function beyond that point
TREATMENT	▪ Does not need immediate medical care if asymptomatic ▪ If symptoms are present, the patient may get an MRI	Surgical correction of the lesion	▪ Surgical correction of the lesion (requires multiple surgical procedures) ▪ **Can cause:** ▪ Paralysis ▪ Bladder/bowel incontinence ▪ Neurogenic bladder ▪ Meningitis (infection) ▪ Hypoxia ▪ Hemorrhage ▪ **Frequent catheterization causes:** ▪ Latex allergy ▪ Urinary tract infection (UTIs) ▪ Renal damage Until surgery can be performed: cover the area with a wet sterile dressing ★ Risk for infection



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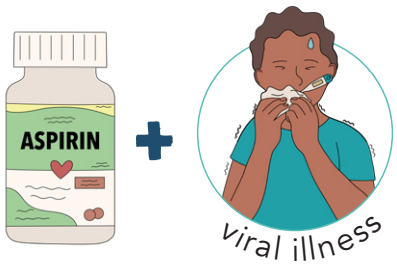
Reye's Syndrome

Pathology

Rare disease affecting young children recovering from a viral illness (flu or chicken pox)

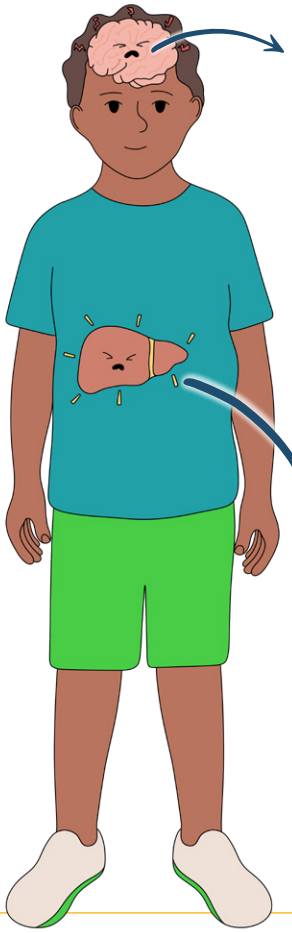
Risk Factors

Triggered due to the intake of salicylates or salicylate-containing products such as **aspirin** to treat a viral illness



↑ RISK FOR REYE'S SYNDROME

Signs & Symptoms



SWELLING OF THE BRAIN

Encephalopathy/cerebral edema

MNEMONIC **"CHILD"**

- **C** Confusion
- **H** Hurl (vomiting)
- **I** Increased ICP
- **L** Lethargy & irritability
- **D** Deadly complications
  - Seizures
  - Coma

SWELLING OF THE LIVER

Liver failure

LABS:

- ↑ Ammonia
- ↑ Liver enzymes (↑ AST & ↑ ALT)

Treatment

- Stay hydrated & balance electrolyte levels
- Administer diuretics to help ↓ ICP
- Strict I&O
- Prevent bleeding
  - Administration of vitamin K or platelets

Damage to the liver may cause bleeding

Prevention

- Administer acetaminophen or ibuprofen instead of aspirin when a child is sick
- **EDUCATE** on salicylate-containing products & to never administer to a child with a current or recent viral infection

SALICYLATE-CONTAINING PRODUCTS:

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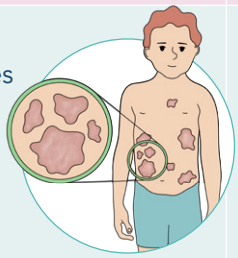
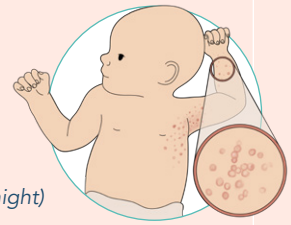
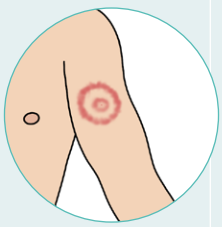
Pediatric Skin Conditions

CONDITION	PATHO	CLASSIC SYMPTOMS	TREATMENTS	CONTAGIOUS?
IMPETIGO	**Bacterial** skin infection that can affect any part of the body, but most commonly affects the face *Typically caused by Staphylococcus aureus (staph) or group A Streptococcus (GAS)*	★ **Honey-colored crusted lesions**	Topical or oral antibiotics	✓
MOLLUSCUM CONTAGIOSUM	A **viral** skin infection caused by pox virus	★ **Lesions will be umbilicated (small donut-like indent inside)**	Chemical agents (example: tretinoin) & cryotherapy	✓
PSORIASIS	Chronic inflammatory **autoimmune** skin disease that causes rapid turnover of the epidermal cell	Itchy red patches covered with ★ **silvery plaques** MEMORY TRICK: Psoriasis think Plaques 	Topical corticosteroids or immunosuppressants	✗
SCABIES	**"Mites" (tiny bugs)** invade the skin creating tunnels and burrows where they can lay eggs and cause skin irritation	Itchy rash caused by the mites burrowing in the skin *(more severe at night)* 	Permethrin topical ointment (FULL BODY application)	✓
PEDICULOSIS CAPITIS "Head Lice"	Infestation & irritation of the scalp and hair follicles by a small **parasite** called the "head louse"	★ **Visualization of "nits" (eggs) in the hair** ★ **Itchy scalp**	Pediculicides (example: permethrin) Anti-lice shampoo & thorough combing of hair	✓
LYME DISEASE	Lyme disease is a **vector-borne bacterial disease** caused by a bite from a deer tick. *The tick does not cause the disease. Rather, it's the disease that the tick is carrying.*	★ **RED "bulls-eye" at the bite site** 	Doxycycline	✗

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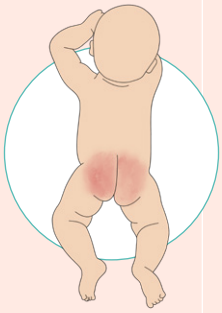
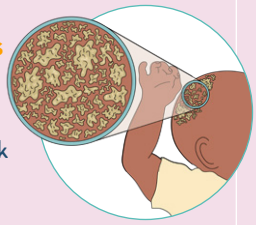
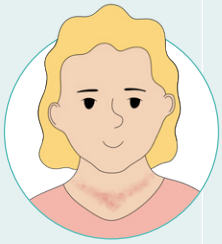
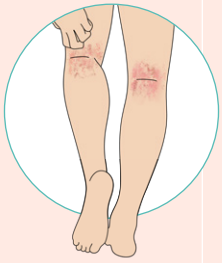
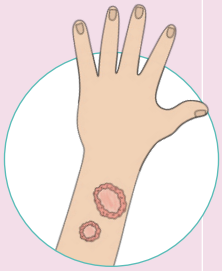
Pediatric Skin Conditions

CONDITION	PATHO	CLASSIC SYMPTOMS	TREATMENTS	CONTAGIOUS?
DIAPER CANDIDIASIS	**Fungal infection, or yeast,** which grows in the perineal and "diaper" region of infants and toddlers	▪ Redness in the groin or buttocks ▪ Cries with diaper change 	Dimethicone & zinc oxide ▪ A&D ointment ▪ Desitin ▪ Triple paste ▪ Nystatin cream	✗
SEBORRHEIC DERMATITIS "Cradle cap"	Excessive production of sebum (oil) in the body which disturbs the skin's normal process of shedding	**Yellow crusting/scales of the scalp** MEMORY TRICK: SEBorrheic think SEBum makes a Scaly Scalp 	Massage with cleanser, and brush with a soft-bristled brush **Do not remove skin or pick at the scalp**	✗
CONTACT DERMATITIS	General description for inflammation of the skin after contact with an irritant	Rash (generalized or localized), pruritus (itching), redness, & inflammation 	▪ Corticosteroids ▪ Gentle, fragrance-free cleansers ▪ Antihistamines	✗
ATOPIC DERMATITIS "Eczema"	Chronic inflammatory skin condition which causes dry, flaky, & itchy skin	Red, dry, & itchy skin 	▪ Corticosteroids (topical or oral) ▪ Antihistamines	✗
DERMATOPHYTOSIS TINEA "Ringworm"	**Fungal** infection which can occur anywhere on the body (It's not actually a worm; the infection appears like a worm under a microscope)	**Circular patches of red skin** MEMORY TRICK: CIRCULAR think RINGS 	Topical anti-fungal cream (ends in "-azole") MEMORY TRICK: AZOLE rhymes with FUNGAL	✓

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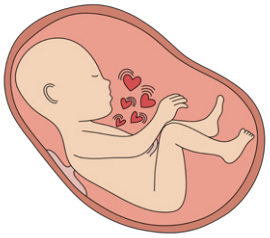
Three Shunts of Fetal Circulation



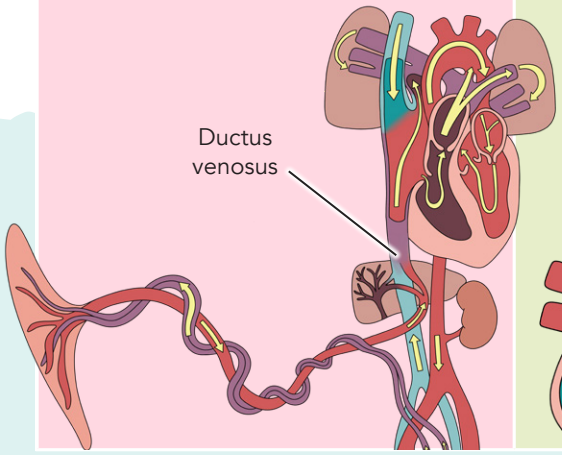
These **3 SHUNTS** are the main steps in healthy & typical **FETAL CIRCULATION**

DUCTUS VENOSUS

- Venous blood returning from the placenta
- Oxygenated blood in umbilical vein goes straight to the **RIGHT** atrium

MEMORY TRICK

Ductus Venosus think a Venous Delivery from the placenta



FORAMEN OVALE

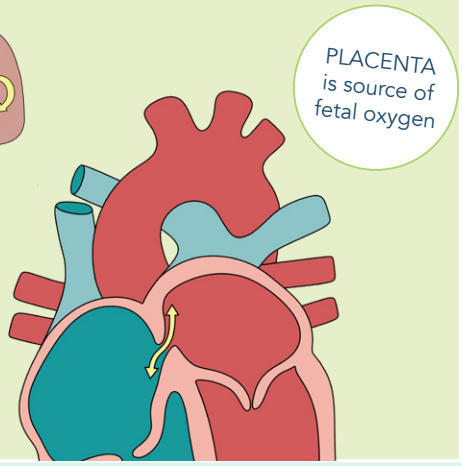
- Opening between TOP CHAMBERS (left and right atria)
- Bypasses the lungs

The lungs do not supply oxygen yet. Blood flows from **HIGH** resistance to **LOW** resistance.

Lungs are high resistance from all the fluid. So the blood does not want to go in the lungs!

MEMORY TRICK

Ovale think Opening

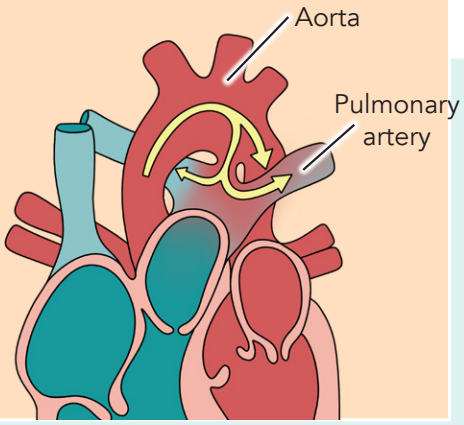


DUCTUS ARTERIOSUS

- Shunts blood from the pulmonary artery into the aorta
- This detours blood away from the lungs (since the lungs are not used in the womb)

MEMORY TRICK

Ductus Arteriosus think Deoxygenated blood carried Away



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Fetal Circulation in Utero

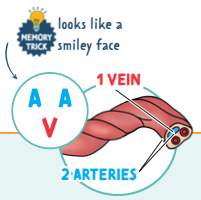
THE PLACENTA

2 UMBILICAL ARTERIES | 1 UMBILICAL VEIN

A think Away

Takes deoxygenated blood & waste AWAY from the baby and back to the placenta

looks like a smiley face



START

THE PLACENTA
is the "lifeline" between mother & fetus

The placenta acts as *"temporary lungs"* for the fetus while in utero

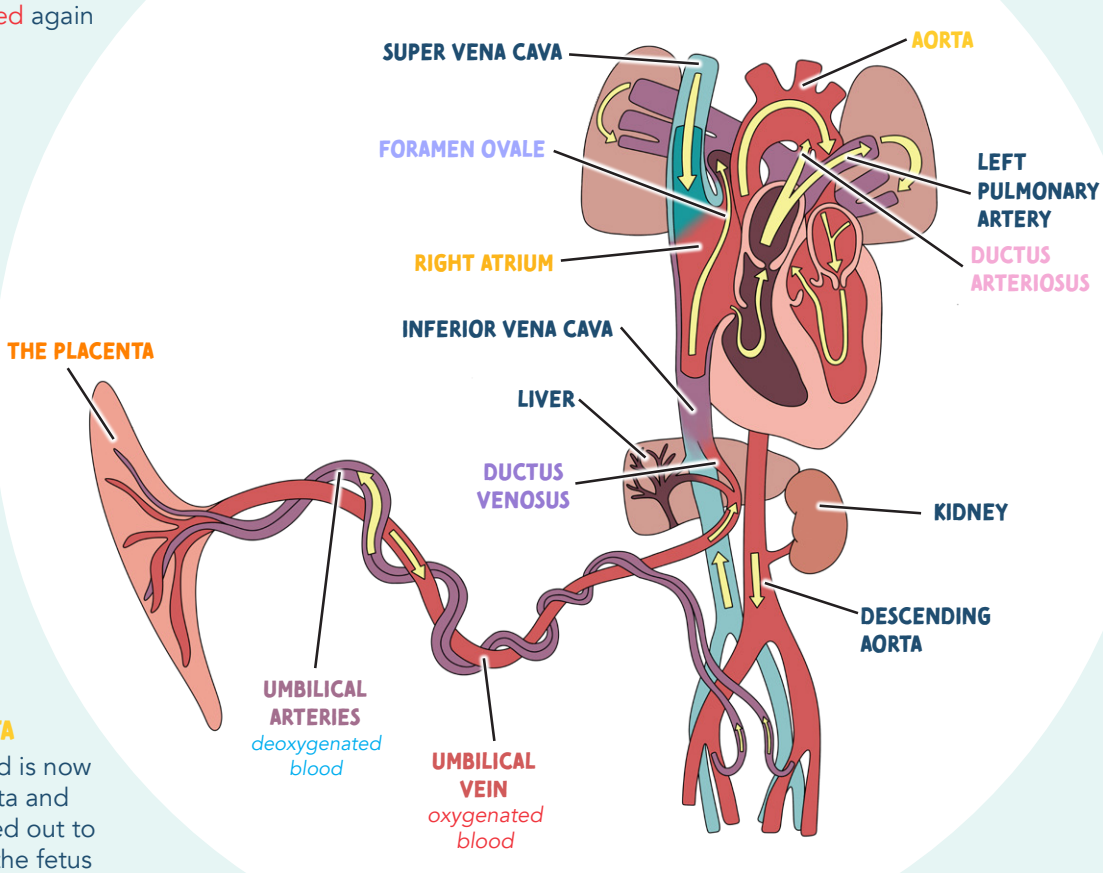
DUCTUS VENOSUS
Umbilical vein is carrying oxygenated blood from the placenta. It passes the **LIVER** (some blood will go to the liver) but most will be **SHUNTED** to the **INFERIOR VENA CAVA** by the **DUCTUS VENOSUS**

Liver not fully functioning yet

RIGHT ATRIUM
Blood goes from the inferior vena cava to the right atrium as well as some deoxygenated blood coming from the **SUPERIOR VENA CAVA**

So the blood is now MIXED (oxygen-rich & oxygen-poor blood)

Blood goes back to the **PLACENTA** to get oxygenated again



AORTA
Mixed blood is now in the aorta and being pushed out to oxygenate the fetus

FORAMEN OVALE
Blood is **SHUNTED** from the right atrium to the left atrium by the **FORAMEN OVALE**

Blood bypasses the lungs because it's already oxygenated from the placenta (mom)

How can blood be shunted from the right atrium to the left atrium?

PRESSURE DIFFERENCE!
Blood flows from *high resistance* to *low resistance*

Lungs: high resistance from all the fluid, so the blood does not want to flow in the lungs

DUCTUS ARTERIOSUS
Blood is **SHUNTED** from the pulmonary artery into the aorta by the **DUCTUS ARTERIOSUS**

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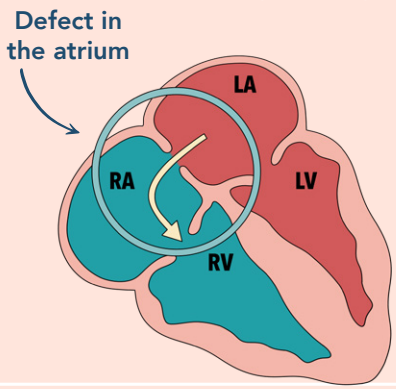
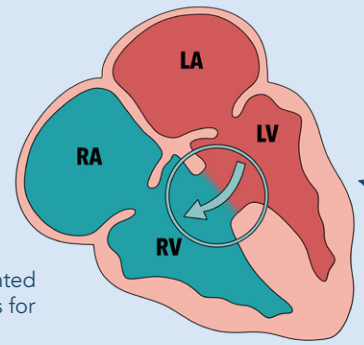
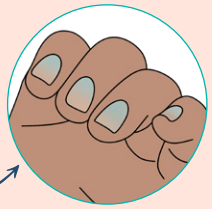
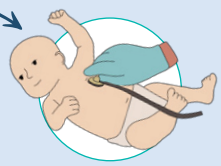
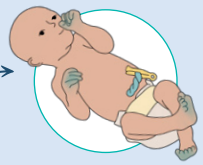
ASD vs. VSD

	ATRIAL SEPTAL DEFECT (ASD)	VENTRICULAR SEPTAL DEFECT (VSD)
PATHOLOGY	 Septal defects (holes between chambers) allow for the blood to mix! **LEFT SIDE of heart:** pumps oxygenated blood to the rest of the body **RIGHT SIDE of heart:** pumps deoxygenated blood to the lungs for oxygenation 	
MEMORY TRICK	Think defect in the ATRIUM	Think defect in the VENTRICLE
PREVALENCE	**Less** common diagnosis	**More** common diagnosis
PATHOLOGY	Septum (division) between the left and right atria does not fully form and close prior to birth, leaving a small opening for blood to travel through	Septum (division) between the left and right ventricle does not fully form and close prior to birth, leaving a small opening for blood to travel through
SIGNS & SYMPTOMS	▪ Frequent illness ▪ Failure to gain weight ▪ Poor appetite & feeding ▪ Tachycardia ▪ Tachypnea ▪ Blue-tinged fingernails (oxygen disturbance) 	★ **Whooshing or murmur**  ▪ Signs of congestive heart failure (CHF) & fluid excess ▪ Inability to gain weight ▪ Poor feeding/appetite ▪ Frequent illness ▪ Tachycardia ▪ Tachypnea ▪ Cyanosis (fingernails, toes) 
TREATMENT/MEDICATIONS	**SMALL ASDs** Will close on their own without intervention **LARGE ASDs** Need surgical intervention	Most cases will resolve on their own without surgery (typically by age 1)

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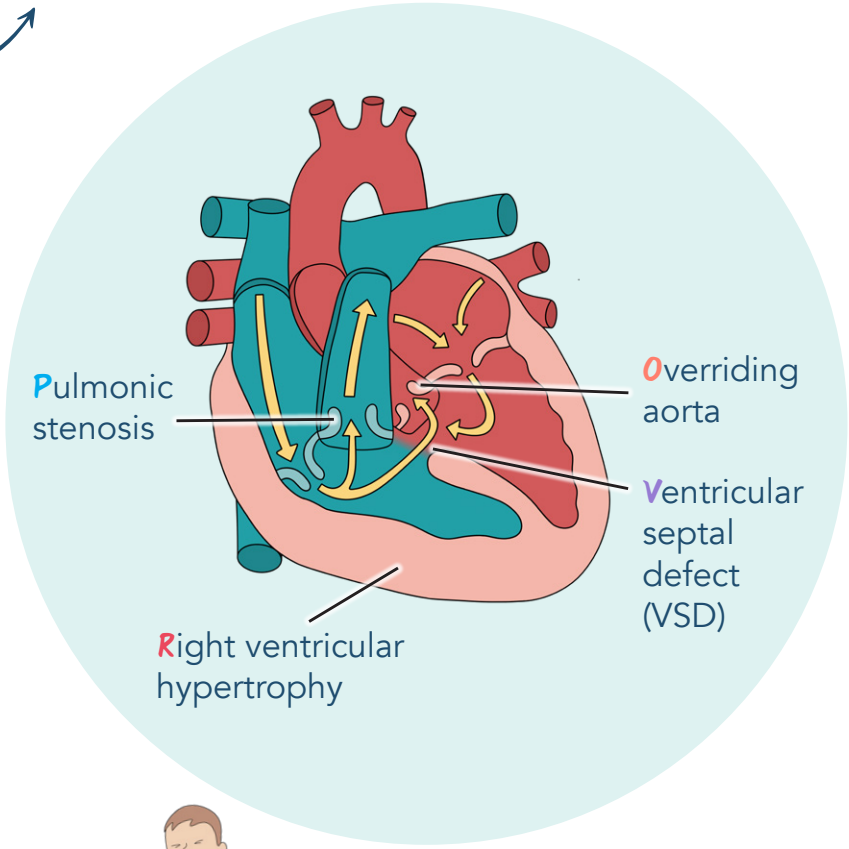
Tetralogy of Fallot

Pathology

A rare congenital heart defect which consists of **4** heart abnormalities:

- **P**ulmonic stenosis
- **R**ight ventricular hypertrophy
- **O**verriding aorta
- **V**entricular septal defect (VSD)

MEMORY TRICK — Need to **PROVe** they have tetralogy of Fallot



Diagnostic

- Echocardiography or EKG
- Chest X-ray
- Pulse oximetry → Oxygen saturation will be LOW

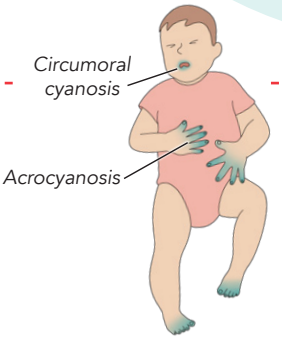
Treatment

- Surgery
  (typically within 1 year of birth)

⚠"TET" SPELLS⚠

Also called *"hypercyanotic spells"*

- Caused by a lack of oxygen which causes the child to become hypoxic



CAUSES

Anything that causes an ↑ in oxygen demand

- STRESS or PAIN:
  - Crying, stimuli, hospital visit, hunger, being scared or startled

PREVENTION

- Small frequent meals
- Create a calm, non-stressful environment
- Provide a pacifier when crying or agitated

CARE DURING A SPELL

- Adm. supplemental oxygen
- Adm. morphine
- Change position as follows:

NEWBORN: Place the knees on their chest

CHILD: Have them sit in a squatting position

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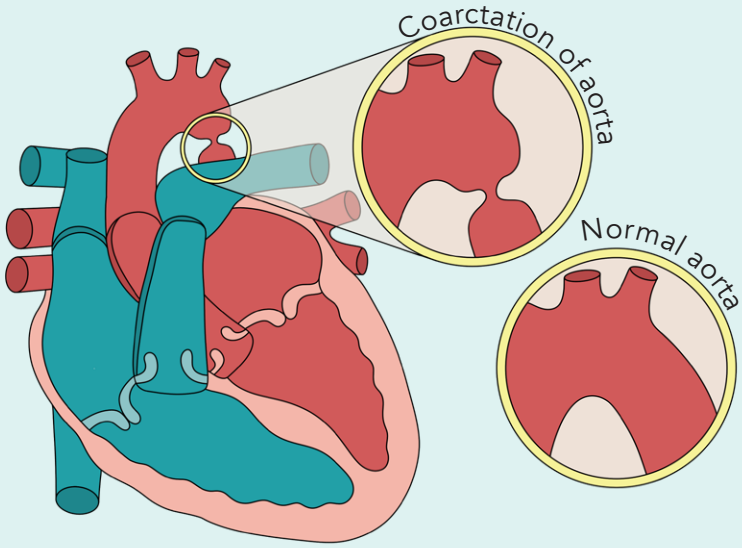
Coarctation of the Aorta

Pathology

Birth defect which causes the aorta to be narrowed

The aorta is what brings oxygenated blood to the body



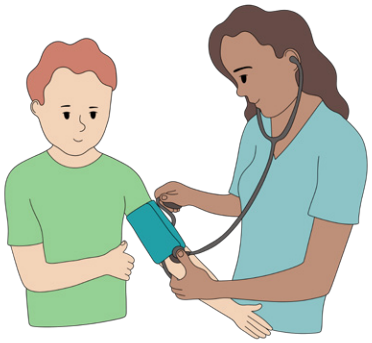
MEMORY TRICK

COArctation
think
Cut Off Aorta

Diagnostic

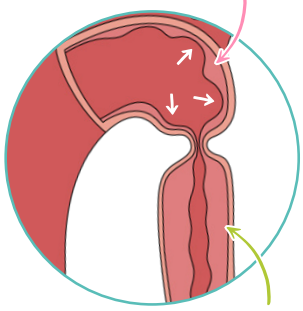
Take blood pressure reading of all 4 extremities & compare

- If there is a drastic difference, it may indicate this condition



Signs & Symptoms

Above the constriction: pressure is **HIGH**



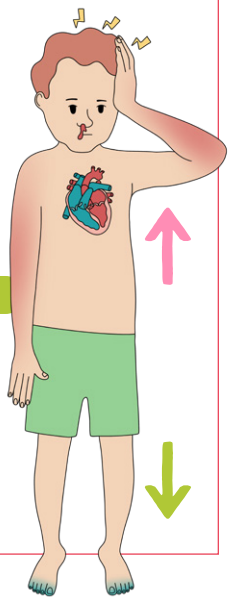
Below the constriction: pressure is **LOW**

UPPER EXTREMITIES, HEAD, & NECK

- ↑ Blood pressure
- Strong pulse (bounding)
- Nose bleeds
- Headaches
- Stroke
- Heart failure

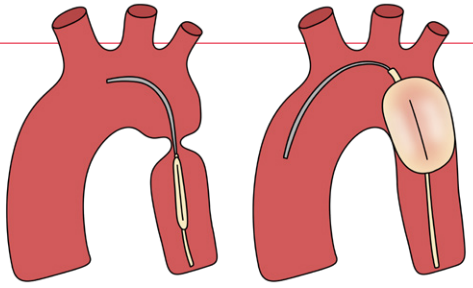
LOWER EXTREMITIES

- ↓ Blood pressure
- Weak pulse
- Legs/feet are cool to the touch



Treatment

- Surgical repair
- Balloon angioplasty (widening the narrowed area) — MOST COMMON
- Bypass graft (rerouting the area that is narrowed)



Balloon angioplasty

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Transposition of the Great Arteries (TGA)

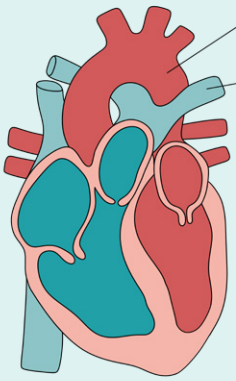
Pathology

Also referred to as ***transposition of the great vessels (TGV)***

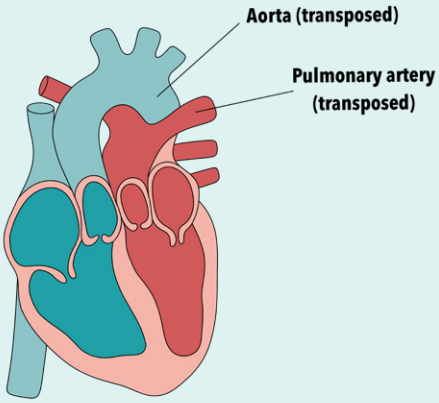
This is a congenital heart defect where the **pulmonary artery** and the **aorta** are transposed or REVERSED!

The only way they survived in utero was due to the shunts

NORMAL	TGA
PULMONARY ARTERY carries deoxygenated blood to the lungs **THE AORTA** carries oxygenated blood to the tissues/body	In TGA, these are swapped & there is no communication between the systemic system & the pulmonary system



Normal heart anatomy



Transposition of the great arteries

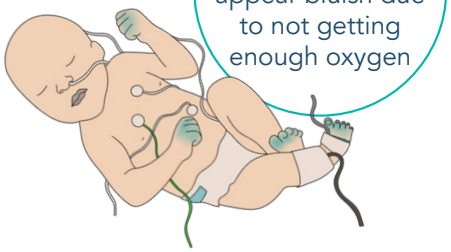
MEMORY TRICK

TGA think everything is TRANSPOSED (switched places)

Signs & Symptoms

Cyanosis

- Cold extremities
- Poor feeding
- ↑ Breathing (labored)
- ↑ Heart rate (but weak)
- ↓ Oxygen saturation



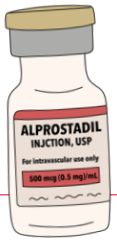
Skin, lips, or mucous membranes appear bluish due to not getting enough oxygen

Medications

PREOPERATIVELY

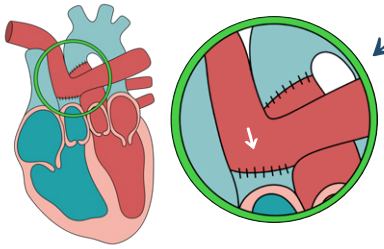
- IV prostaglandins

Example: alprostadil



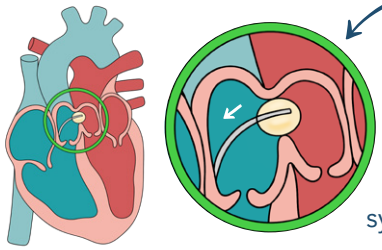
This keeps the ductus arteriosus open (PDA) which allows for systemic blood flow while the child waits for surgery

Treatment



JATENE PROCEDURE

Redirecting the blood flow through the heart



BALLOON ATRIAL SEPTOSTOMY

Balloon is used to enlarge the foramen ovale, allowing systemic and pulmonary blood to mix

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Intussusception

Pathology

Ileum telescopes into the cecum

↓

Obstruction & compression of blood vessels

↓

Pain & decreased blood flow

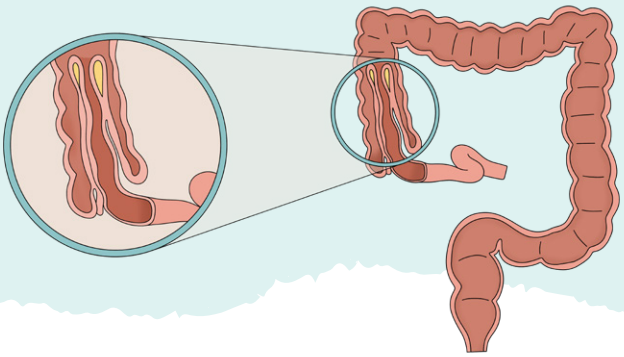
↓

The bowel begins to die

↓

Rectal bleeding

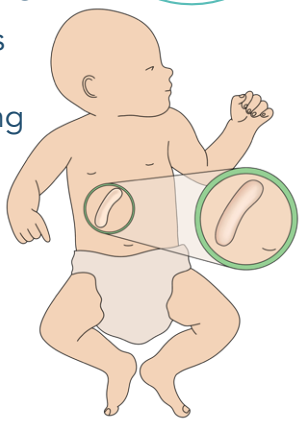
PEDIATRIC EMERGENCY



Signs & Symptoms

- **Currant-jelly stools (bloody)**
- **Sausage-shaped mass in the upper mid-abdomen**
  - Lethargy
  - Intermittent pain/cramping
  - Child draws up their legs toward the abdomen in severe pain while crying
  - Vomiting & diarrhea

This is because telescoping is intermittent

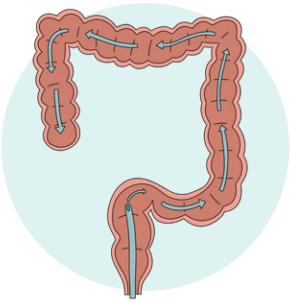


Causes

- **Not completely known**
- May be due to a virus that causes swelling

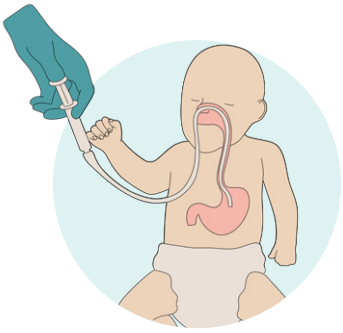
Treatment

MAY RESOLVE ON ITS OWN



AIR OR LIQUID BARIUM ENEMA

Works to diagnose & also helps reduce the intussusception. The air or fluid helps to push the telescoping area back to normal.



DECOMPRESSION VIA NG TUBE

Helps to get rid of excess air that is trapped in the bowel.

IV FLUIDS & ANTIBIOTICS

Adm. because a hole may develop in the bowel leading to dehydration or an infection.

MONITOR for signs of perforation & shock

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Hypertrophic Pyloric Stenosis

Pathology

A hypertrophied pyloric muscle causes narrowing of the pyloric canal

↓

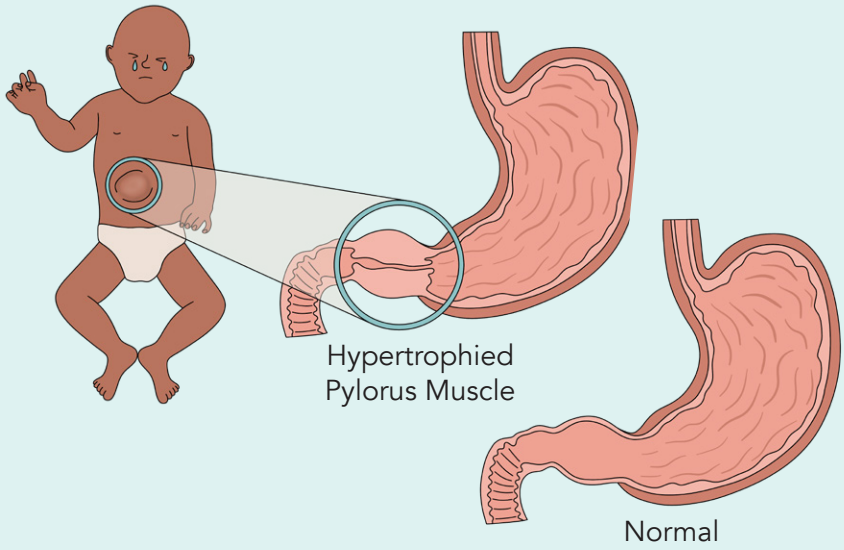
Thickness creates a narrow stomach outlet

HYPERTROPHIC ↓ increase in size

PYLORIC ↓ pylorus

STENOSIS ↓ narrowing

(The opening from the stomach into the small intestines)

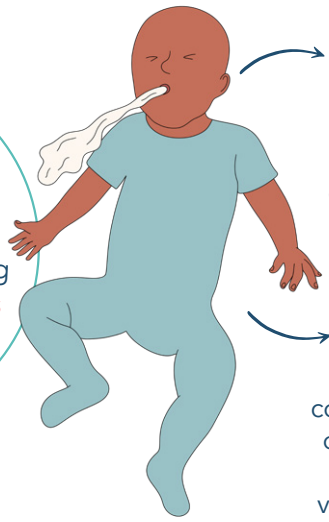


Signs & Symptoms

Symptoms usually start within the first weeks of life!

- **Projectile vomiting**
- **Non-bilious emesis**
- **Olive-shaped mass palpable in the right upper quadrant**
- Infant will be hungry & fussy regardless of regular feedings
- Weight loss or failure to thrive
- Dehydration
  - **Hemoconcentration causes:** ↑ Hematocrit from ↑ BUN

Stomach contains acid which becomes depleted when vomiting, leading to **metabolic alkalosis** (↑ pH & ↑ HCO_3)



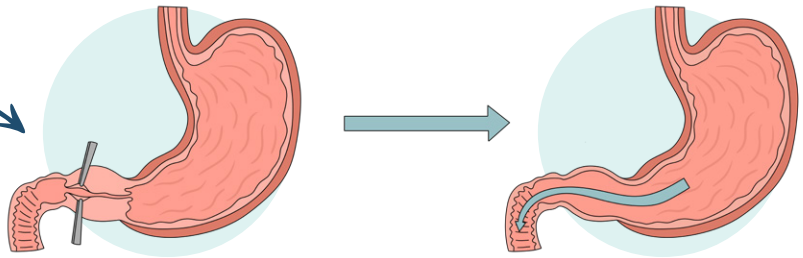
Projectile vomiting typically occurs after a feeding

After vomiting the child is usually hungry and irritable

Non-bilious emesis, (This vomit is not the color of bile because the obstruction is proximal to the bile duct. This vomit is typically white.)

Treatment

- Pyloromyotomy
- Replace fluid & electrolytes
- Stomach decompression



Cut the muscle of the pylorus

Relieves the gastric outlet obstruction

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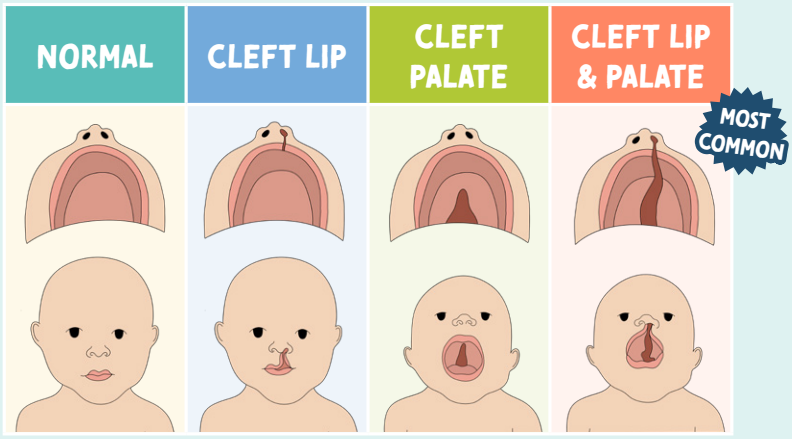
Cleft Lip & Palate

Pathology

- A birth condition where the mouth and/or lip does not form properly
- There is no known cause

Most common congenital anomaly in the Untied States



Can be unilateral or bilateral

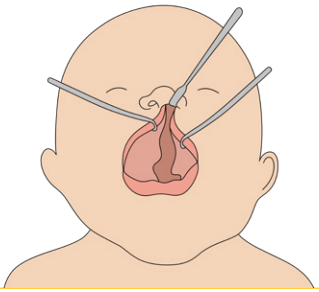
Complications

- Issues with:
  - Feeding
  - Swallowing
  - Speaking
- Impaired bonding between mom & baby

Babies with this condition have trouble suctioning during feeding because air leaks out through the cleft

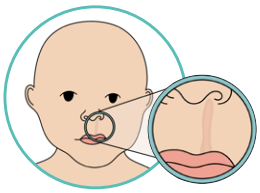
Treatment

- Reconstructive surgical repair
  - Fisher repair
  - Millard rotation-advancement technique



Surgery typically resolves all the complications and the child is just left with a small scar

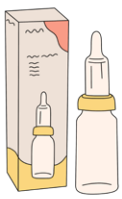
Parent Education (POST-OP):



SURGICAL SITE

- Proper wound care (prevent crush formation)

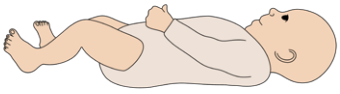
MONITOR airway (swelling may occur around surgical site)



FEEDING

- Special feeders/bottles
- Give feedings slowly
- Give small, frequent meals

MONITOR for choking/aspiration



Lying on the back

POSITIONING

Child should lie **supine** when not eating/drinking to lower risk of damaging sutures

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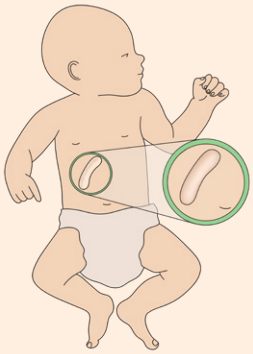
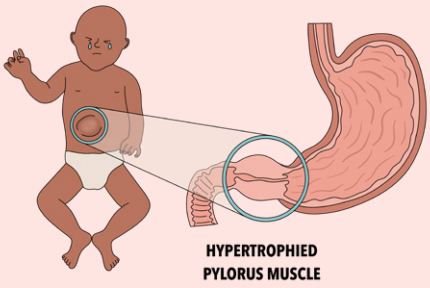
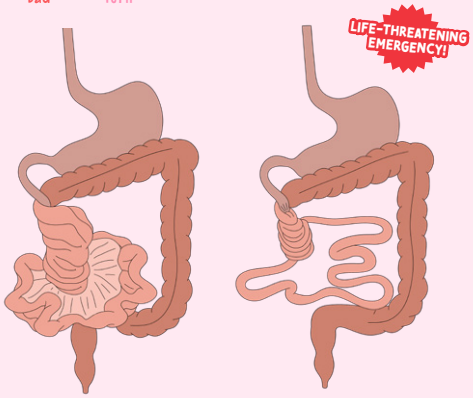
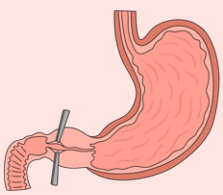
Pediatric Gastrointestinal Conditions

	INTUSSUSCEPTION	HYPERTROPHIC PYLORIC STENOSIS (HPS)	MALROTATION & VOLVULUS
PATHOLOGY	Ileum **telescopes** into the cecum ↓ Obstruction & compression of blood vessels ↓ Pain & decreased blood flow ↓ The bowel begins to die ↓ Rectal bleeding 	The pyloric sphincter muscle becomes thickened causing narrowing of the pyloric canal ↓ Thickness creates a narrow stomach outlet **HYPERTROPHIC PYLORIC STENOSIS** ↓ Increase in size ↓ Pylorus ↓ Narrowing  HYPERTROPHIED PYLORUS MUSCLE	**MALROTATION** → **VOLVULUS** Intestines fail to move in healthy way → Complication of malrotation MAL (Bad) ROTATION (Turn) Intestines are **TWISTED** causing a **BLOCKAGE**!  LIFE-THREATENING EMERGENCY!
CLASSIC SYMPTOMS	★ **Currant-jelly stools (bloody)** ★ **Sausage-shaped mass in the upper mid-abdomen**	★ **Projectile vomiting** ▪ Can lead to metabolic alkalosis ★ **Non-bilious emesis** ★ **Olive-shaped mass palpable in the right upper quadrant**	▪ Abdominal pain ▪ Diarrhea ▪ Nausea & vomiting ▪ Bloody stool
TREATMENT	May resolve on its own (as evidenced by passage of normal, brown stools)	Pyloromyotomy (cutting the muscle of the pylorus) 	▪ IV antibiotics & IV Fluids ▪ Volvulus requires surgery to untwist bowel

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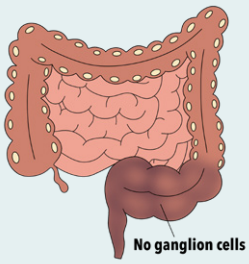
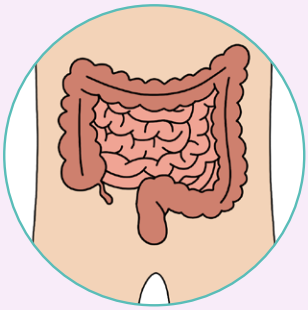
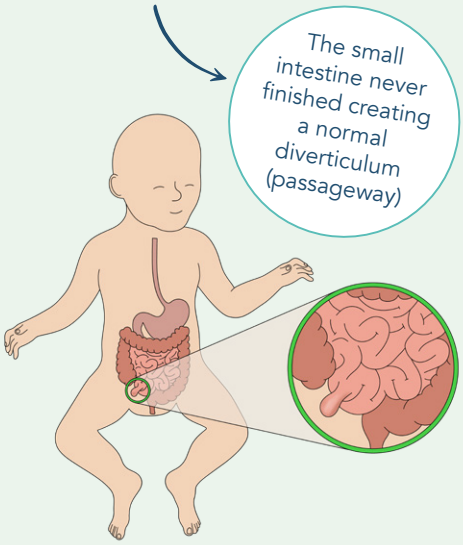
Pediatric Gastrointestinal Conditions

	HIRSCHSPRUNG'S DISEASE	IMPERFORATE ANUS	MECKEL'S DIVERTICULUM
PATHOLOGY	A congenital anomaly (present from birth) which causes a lack of mobility/peristalsis in the intestine and can lead to an obstruction of the intestine **Ganglion cells are missing (called the aganglionic portion) in the affected area which causes loss of function**  	The pyloric sphincter muscle becomes thickened which causes narrowing of the pyloric canal ↓ Thickness creates a narrow stomach outlet 	▪ Congenital defect of the bowel ▪ Malformation of small intestine ▪ Part of umbilical cord remains & makes a pouch 
CLASSIC SYMPTOMS	Delayed passage of meconium	Newborn's first bowel movement (meconium) should happen within the first 48 hours of birth	▪ Bloody stools — MOST COMMON SIGN ▪ Abdominal pain ▪ Stool with mucous and "currant jelly stool"
TREATMENT	Surgical procedure (cutting out/removing the affected area)	▪ **Posterior Sagittal Anorectoplasty (PSARP)** Creating an anal opening where one was missing ▪ **Anoplasty** If the rectum is connected to the genitals, this involves moving the opening to the correct location	▪ Surgical removal of diverticulum (pouch) — Colon resection involving SMALL intestine ▪ ***Bleeding/blood loss management:*** blood transfusion & IV fluid volume replacement

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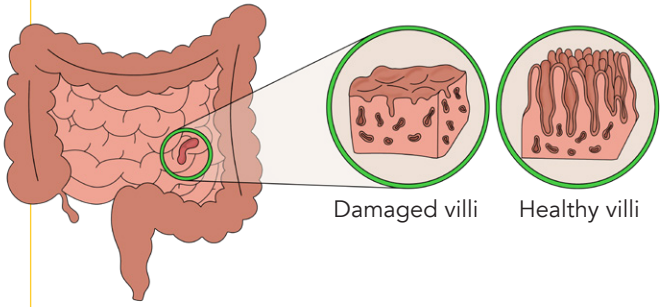
Celiac Disease & Lactose Intolerance

Celiac Disease

PATHOLOGY:

- Autoimmune disease caused by the intake of gluten
- The protein gluten causes villi in the small intestine to be damaged. This damage to the villi causes a decrease in surface area and causes malabsorption



SIGNS & SYMPTOMS:

- Steatorrhea
- Chronic diarrhea
- Abdominal distention
- Fatigue
- Failure to thrive
- Canker sores

Frothy (bulky), fatty, foul-smelling stools

TREATMENT:

EDUCATE to read ALL food labels!

DIET MODIFICATIONS

Follow a gluten-free diet for life

- Likely will need to follow a dairy-free diet as well because the villi are damaged, so the intestine is more sensitive to lactose

✗ GLUTEN-CONTAINING FOODS

MEMORY TRICK: **NO BROW**

- ✗ Barley
- ✗ Rye
- ✗ Oat
- ✗ Wheat

Mainly in wheat & rye (small amounts in barley & oats)

✓ APPROVED FOODS

- ✓ Rice
- ✓ Corn
- ✓ Soy
- ✓ Potatoes
- ✓ Millet
- ✓ Quinoa
- ✓ Nuts & legumes

Lactose Intolerance

PATHOLOGY:

- The body is unable to break down lactose due to a **lack of lactase enzyme** from the small intestine
- Inability to break down lactose leads to discomfort and GI upset

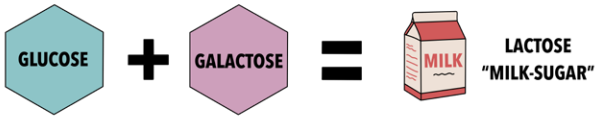
MEMORY TRICK: LACtose INTolerance think LACk of LACtase from INTestine

WHAT IS LACTOSE?

Natural sugar in dairy & cheese

WHAT IS LACTASE?

Breaks down lactose into smaller molecules (glucose & galactose) to be absorbed in the intestines



SIGNS & SYMPTOMS:

- Gas
- Cramping
- Nausea & vomiting
- Diarrhea
- Bloating & discomfort

Usually after consuming milk or dairy

TREATMENT:

DIET MODIFICATIONS:

- Eliminate dairy & lactose from the diet
- Switch to soy-based formulas
- Use artificial lactase supplements when consuming dairy
- Take calcium & vitamin D supplements

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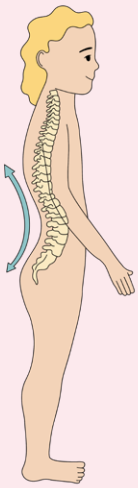
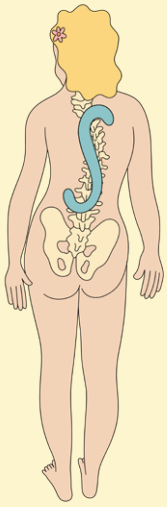
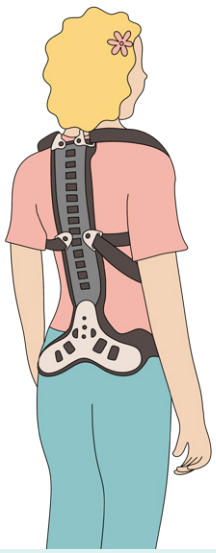
Abnormal Spinal Curvatures

	KYPHOSIS	LORDOSIS	SCOLIOSIS
DESCRIPTION	Also called *"hunchback"* Upper back curves at the top, creating a "hunch" below the neck 	Also called *"swayback"* Lower back swoops IN, creating a C-shape of the lower spine 	Spine curves irregularly and sideways, creating misalignment of shoulders, hips, and ribs 
MEMORY TRICK	When you drop your KEYS (Kyphosis) you have to lean down & pick them up which makes you have a spinal curvature	LORDOSIS think standing tall with chest puffed out like a LORD	Scoliosis think Swaying Sideways
SIGNS & SYMPTOMS	Observation of a hunch in the upper back	▪ Observation of exaggerated inward curve of lower back ▪ Lower back pain	▪ Hip height & shoulders appear uneven ▪ Ribs are asymmetric when bending forward to touch toes ▪ Poor-fitting clothing (especially pants)
TREATMENT	**MONITOR** & maintain comfort ▪ Prevent injury ▪ Braces for support ▪ Strengthen physical therapy ▪ Surgery is rare Pain is more commonly associated with lordosis! (lower back pain) 		**TREATMENT DEPENDS ON THE DEGREE OF THE SCOLIOSIS** ▪ Braces worn up to 23 hours per day ▪ Strengthens the back & prevents worsening ▪ Exercise & physical therapy ▪ Spinal fusion, rods, or bone graft if degree of the scoliosis is severe (typically > 45 degrees)

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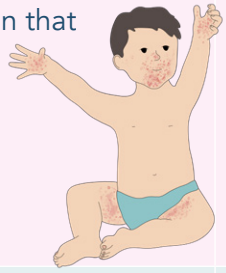
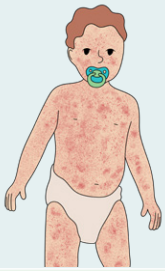
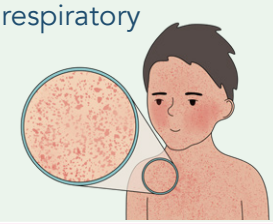
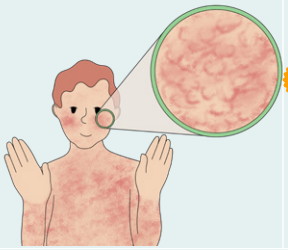
Quick Overview of Pediatric Infectious Diseases

CONDITION	PATHO	CLASSIC SYMPTOMS	MICROBE
THE FLU "Influenza"	Viral infection with multiple strains that mainly target the respiratory system 	▪ Fever ▪ Body aches ▪ Nausea/vomiting ▪ Cough & congestion	VIRAL
HAND-FOOT -MOUTH DISEASE	**HAND**, **FOOT**, **MOUTH** is a mild but contagious viral infection that causes blisters/sores; spread by direct contact with saliva & mucus ★ **Primarily caused by coxsackievirus** 	▪ Blisters & reddened sores on **HANDS**, **FEET**, and inner thighs ▪ **MOUTH** sores	VIRAL
MEASLES "Rubeola"	▪ Virus which infects the respiratory system and nasopharyngeal tissue ▪ Spreads to lymph nodes ★ **Highly contagious** 	★ **Koplik's spots** (tiny white spots appearing on mucous membranes of inner cheek) ▪ Red, blotchy rash (a later sign)	VIRAL
RUBELLA "German Measles"	▪ Virus does not target respiratory system like measles ▪ Targets lymph nodes, skin, and mucous membranes 	★ **Forchheimer spots** ▪ Rash, usually originating on head/face ▪ Low-grade fever ▪ Headache ▪ Conjunctivitis Small red spots on the top of the mouth (the soft palate)	VIRAL
SCARLET FEVER	▪ Complication of group A streptococcal infection (strep throat) ▪ Not all children who have strep will develop scarlet fever 	MEMORY TRICK **S's of Scarlet fever** ★ **Sandpaper-like rash** ★ **Strawberry tongue**	BACTERIAL
ERYTHEMA INFECTIOSUM "Fifth Disease"	Infection caused by human parvovirus B19 	▪ Flu-like symptoms ★ **Rash on cheeks (slapped cheek appearance)** ▪ Rash on arms, legs & chest	VIRAL

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Notes

If we're **GROWING**, we're always going to be out of our **COMFORT ZONE**.

– JOHN MAXWELL

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Med-Surg

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Tips for Med Surg Class

Synthesize your notes

I had the most success in remembering content when synthesizing my notes. Here's what I did: after each class, I synthesized (combining into one place) my notes from the PowerPoint and the notes I wrote down from the lecture. I rewrote them in one organized document. Then, I would print out those new notes and place them in my class binder for test review later on!

Understand how the body works

You will learn diseases & conditions in every body system in your med surg class (respiratory, cardiac, renal, etc.)

Understanding how the body is supposed to function can help you understand what to do when things go wrong or a patient becomes ill.

Teach it out loud!

Do you really know the content? Passively listening or passively reading the content usually does not produce the same benefits as actively speaking the material out loud. Don't sit around listening to your friends or teachers talk through the material; actively engage with it.

SOLUTION:
Teach the material OUT LOUD to yourself, your friends, your family, or even your pet. Try not to use your notes. When you can teach the content without hesitation, you really know the content. It's a perfect test of how prepared you actually are for the next exam.

Comparison charts

Most of med surg class consists of comparing different diseases.
For example:
hypocalcemia vs. hypercalcemia,
Cushing's disease vs. Addison's disease.

So, making condensed comparison charts and understanding the core differences can help a lot!

GOOD NEWS:
This book has a ton of comparison charts already made for you! :)

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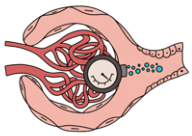
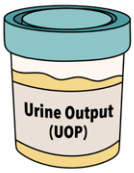
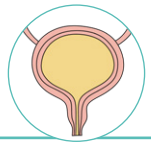
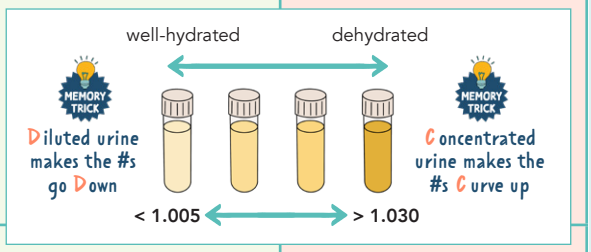
Lab Values Related to the Kidneys

	DESCRIPTION	EXPECTED RANGE	POSSIBLE CAUSES ↓	POSSIBLE CAUSES ↑
GLOMERULAR FILTRATION RATE (GFR) 	Rate of blood flow through the kidneys	**90 – 120** mL/min	Kidney dysfunction *(such as chronic kidney disease)*	Pregnancy
CREATININE	End product of muscle metabolism; solely filtered from the blood via glomerulus	**0.6 – 1.2** mg/dL MEMORY TRICK **RHYME**: Creatinine over 1.3 = a bad kidney	▪ Low muscle mass ▪ Hyperthyroidism ▪ Starvation ▪ Liver disease	**Acute or chronic kidney disease** ▪ Congestive heart failure ▪ Dehydration ▪ Certain drugs *Creatinine is a better indicator of kidney function than BUN*
BLOOD UREA NITROGEN	Normal waste product resulting from the breakdown of proteins; high levels can indicate a kidney problem & be toxic in the body	**7 – 20** mg/dL MEMORY TRICK Think of hamburger BUNs – hamburgers can cost anywhere from $7-$20	▪ Liver damage ▪ Malabsorption ▪ Poor diet ▪ Low-nitrogen diet	**Acute or chronic kidney disease**
URINE SPECIFIC GRAVITY	A measure of the kidney's ability to excrete or conserve water	**1.005 – 1.030**	▪ Adequate or excessive fluid intake ▪ Diabetes insipidus	▪ Dehydration ▪ Syndrome of inappropriate antidiuretic hormone secretion (SIADH)
URINE OUTPUT 	The amount of urine a person excretes from their bladder via the urethra	Urine output: **AT LEAST 30** mL/hr (The average adult will void about **1500** mL/day) 	▪ Shock ▪ Hypotension ▪ Trauma ▪ Infection ▪ Chronic kidney disease (CKD)	▪ Diabetes mellitus ▪ Diabetes insipidus ▪ Too many diuretics
NORMAL FINDINGS	Urine free from glucose, ketones, blood, protein, bilirubin, nitrates, or leukocyte esterase			



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Kidney Overview

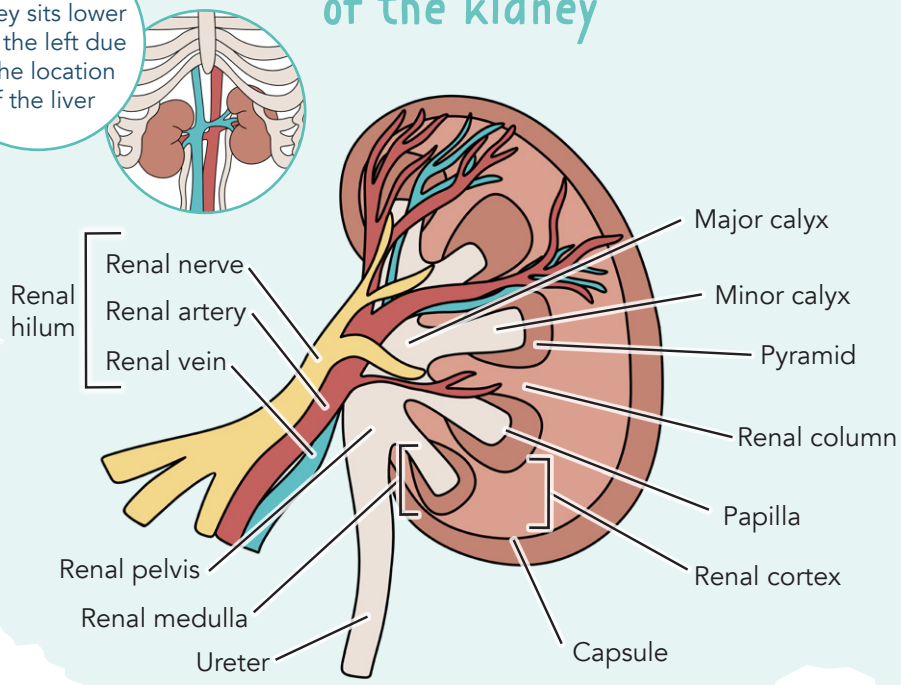
Functions of the kidney

MNEMONIC **"A WET BED"**

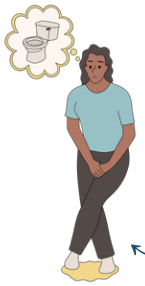
- **A** **A**cid-base balance
- **W** **W**ater balance
- **E** **E**lectrolyte balance
- **T** **T**oxin removal
- **B** **B**lood pressure control
- **E** **E**rythropoietin
- Vitamin **D** metabolism

FUN FACT: The right kidney sits lower than the left due to the location of the liver

Anatomy of the kidney



Terms to Know



- **DYSURIA** ..... Pain while urinating
- **NOCTURIA** ..... Excessive urination at night
- **HEMATURIA** ..... Bloody urine
- **FREQUENCY** ..... Voiding more than every 3 hours
- **URGENCY** ..... Strong desire to void
- **INCONTINENCE** ..... Involuntary voiding
- **ENURESIS** ..... Involuntary voiding during sleep
- **PROTEINURIA** ..... Abnormal amounts of protein in the urine
- **OLIGURIA** ..... Urine output: < 400 mL/day
- **ANURIA** ..... Urine output: < 100 mL/day
- **MICTURITION** ..... Voiding

URINE FORMATION

1 GLOMERULAR FILTRATION

Blood flows into the kidneys: **120 mL/min**

Filters water, electrolytes & small molecules **into the glomerulus** (large molecules stay in the bloodstream)

2 TUBULAR REABSORPTION

Fluid moves **from the renal tubules into the capillaries**, which reabsorb fluid into the venous circulation

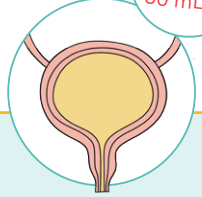
3 TUBULAR SECRETION

Fluid moves **from the capillaries into the renal tubules** to get eliminated/excreted

4 URINE EXCRETION

Adults should void 1-2 L/day

No less than 30 mL/hr



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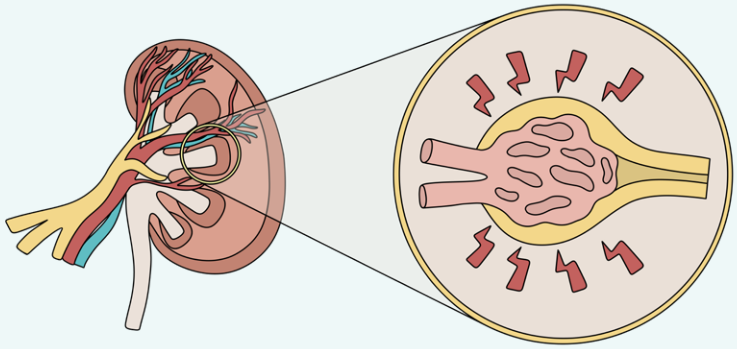
Acute Glomerulonephritis (AGN)

PATHOLOGY

Also referred to as ***Post-streptococcal***



Glomerulonephritis
- glomerulus
- kidneys
- inflammation

✗ It's not the strep that causes the inflammation of the kidneys

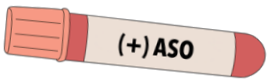
✓ It's the antigen-antibody complexes that form ***due to*** the strep

Untreated or recent streptococcal infection → Immune system responds by creating **antigen-antibody complexes** (14 days after infection) → Antibodies get "lodged" in the glomeruli → Inflammation & scarring → ↓ Glomerular filtration rate (GFR) → **Fluid volume overload & edema**

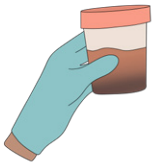
SIGNS & SYMPTOMS

MNEMONIC **"HAD STREP"**

- **H** **H**ypertension (↑ blood pressure due to sodium retention)
- **A** Positive **A**ntistreptolysin O (**A**SO) test results
- **D** **D**ecreased GFR



- **S** **S**welling in the face/eyes (edema)
- **T** **T**ea-colored or cola-colored urine (due to azotemia: build-up of waste products)
- **R** **R**ecent strep infection
- **E** **E**levated labs: azotemia (↑ BUN & creatinine levels)
- **P** mild **P**roteinuria (protein in the urine)



TREATMENT

Treat the underlying cause

MEDICATIONS

- Antibiotics
  - *If infection is present*
- Antihypertensives
  - *To control blood pressure*
- Diuretics
  - *To decrease fluid retention & edema*
- Corticosteroids
  - *To help with inflammation*

FLUID STATUS MONITORING

- Strict I&Os
- Assessing urine color
- Daily weights → Best indicator of fluid status
- Assessing for edema
- Auscultating heart and lung sounds → Checking for fluid in the lungs

DIET MODIFICATIONS

- Fluid & sodium restriction
- ↓ Protein
- ↑ Carbohydrates → Carbohydrates provide energy & stop the breakdown of protein

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Nephrotic Syndrome

PATHOLOGY

Inflammatory response in the glomerulus

↓

Damage to membrane

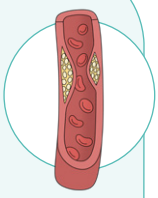
↓

Loss of protein (albumin)
Albumin regulates oncotic pressure

↓

HYPOALBUMINEMIA
Low albumin levels

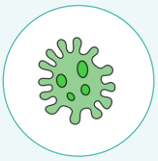
- Synthesis of cholesterol & triglycerides → Hyperlipidemia



- Fluid shift → Generalized edema

- Decreased albumin, which normally prevents clot formation → Possible blood clots (thrombosis)

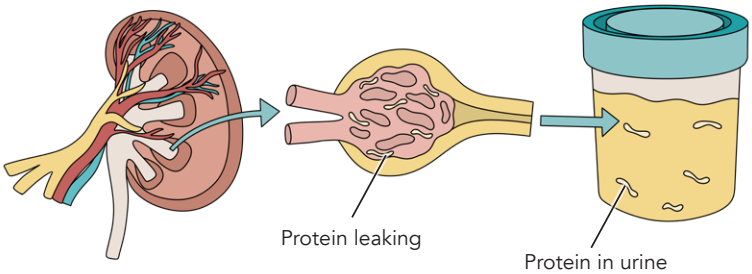
- Possible loss of protein that helps fight infections (immunoglobulins) → Risk for infection



SIGNS & SYMPTOMS

Hypoalbuminemia
- Edema
- Fatigue & loss of appetite
- Hyperlipidemia

Proteinuria (> 3 g/day)
- Large amounts of protein in the urine



TREATMENT

Treat the underlying cause

MEDICATIONS

- Diuretics
- Statins → Lipid-lowering drugs
- Corticosteroids (example: prednisone) → ↓Inflammation
- Antineoplastic agents
- Immunosuppressants

MONITOR FLUID STATUS

- Daily weights → Best indicator of fluid status
- I&Os
- Assessing for swelling & abdominal girth

DIET MODIFICATIONS

- ↓ Cholesterol & saturated fats
- ↓ Na+ intake
- Moderate protein intake

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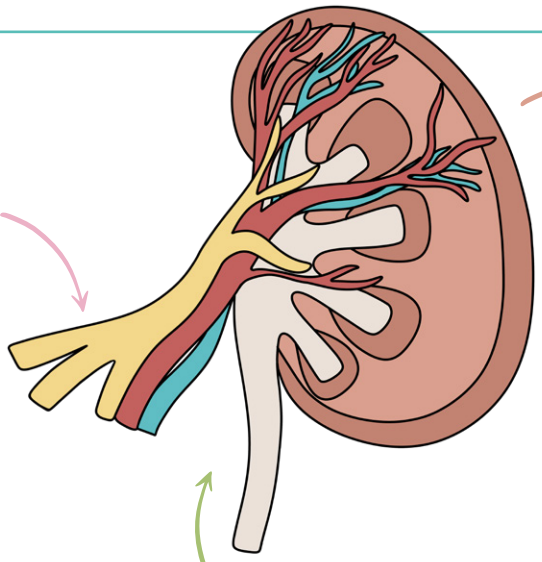
Acute Kidney Injury (AKI)

WHAT IS IT?

SUDDEN RENAL DAMAGE!

Abrupt damage that causes a buildup of waste, fluid, and electrolyte imbalance. Can be reversible. Formerly called *Acute Renal Failure*.



Prerenal FAILURE:

Damage **BEFORE** the kidneys

↓ VOLUME/PERFUSION TO THE KIDNEYS

- Cardiac damage
  - Decreased or impaired cardiac output
  - *Example:* myocardial infarction (MI)
- Vasodilation
- Hemorrhage (hypovolemia)
- Burns
- GI losses (vomiting/diarrhea)

Intrarenal FAILURE:

Damage **IN** the kidneys

PROLONGED ISCHEMIA

- Myoglobinuria
- Hemoglobinuria
- Rhabdomyolysis
- Nephrotoxic drugs
  - *Examples:* NSAIDs, antibiotics (aminoglycosides), chemo drugs, contrast dyes
- Infections
  - *Example:* Glomerulonephritis

Postrenal FAILURE:

Damage **AFTER** the kidneys

OBSTRUCTION/BLOCKAGE IN THE URINARY TRACT

▪ Renal calculi (stones) ▪ Blood clots ▪ Retroperitoneal issues
▪ Benign prostatic hyperplasia (BPH) ▪ Tumors ▪ Neurological damage (stroke)

Phases

MEMORY TRICK "OH OH DARN RENAL"

	OH ONSET/INITIATION	OH OLIGURIA	DARN DIURETIC	RENAL RECOVERY
LENGTH	When the injury occurred	1–7 days	1–3 weeks	3–12 months
DESCRIPTION	Triggering event (Prerenal, intrarenal, or postrenal failure)	Glomerulus ↓ the ability to filter blood = **↓ urine output**	▪ Cause of AKI is corrected ▪ Gradual ↑ in urinary output Large amount of diluted urine with electrolytes	↑ in kidney function
TREATMENT	Treat the underlying cause to prevent long-term damage	▪ Strict I&Os & daily weights ▪ ↑ *BUN & creatinine* ▪ Dialysis may be needed until kidney function returns **DIET MODIFICATIONS:** ▪ Low-protein diet ▪ Limited fluid intake **MONITOR** EKG & labs ▪ *Watch for hyperkalemia (K^+ > 5.0)*	**MONITOR** the patient for dehydration & *hypokalemia*	Some patients may develop chronic kidney disease (CKD) ▪ Dialysis ▪ Kidney transplant

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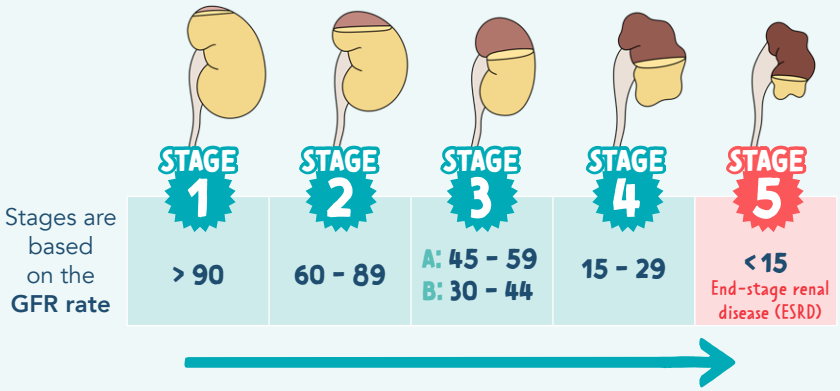
Chronic Kidney Disease (CKD)

PATHOLOGY

Gradual & irreversible loss of kidney function that occurs over a long period of time

Stages



CAUSES

- ★ Diabetes mellitus
- ★ Hypertension
- Untreated acute kidney injury (AKI)
- Recurrent infections
- Autoimmune disorders

High blood sugar over a long period of time can damage the vessels in the kidneys

TREATMENT

- Dialysis
- Kidney transplant

SIGNS & SYMPTOMS

★ In the end stages of CKD, almost every body system is negatively affected

CENTRAL NERVOUS SYSTEM

- Lethargy
- Weakness
- Altered LOC
- Confusion
- Seizures

RENAL SYSTEM

- ↓ Urinary output (UOP)
- Oliguria = < 400 mL/day
- Anuria = < 100 mL/day
- Proteinuria & hematuria

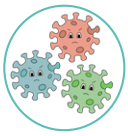
CARDIAC SYSTEM

- ★ Fluid volume excess (hypervolemia)
- Hypertension
- Heart failure

GASTROINTESTINAL SYSTEM

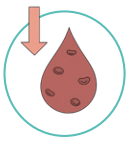
- ★ Uremic fetor (ammonia breath or a metallic taste)
- Anorexia
- Nausea/vomiting

Due to ↑ ammonia levels



IMMUNE SYSTEM

Impaired immune system
=
↑ Risk for infection



HEMATOLOGICAL SYSTEM

- Anemia
  - ↓ Erythropoietin [EPO]
- ↑ Risk for bleeding

REPRODUCTIVE SYSTEM

- Amenorrhea
- Erectile dysfunction
- ↓ Libido

INTEGUMENTARY SYSTEM

- ★ Uremic frost
- Pruritus

Due to crystallized urea deposits

LABS

- ↑ BUN
- ↑ Creatinine
- ★ ↑ Potassium
- ↑ Magnesium
- ↓ Calcium
- ↑ Phosphate

Monitor for EKG changes!

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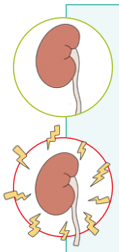
Types of Dialysis: Hemodialysis

Hemodialysis

MOST COMMON METHOD OF DIALYSIS

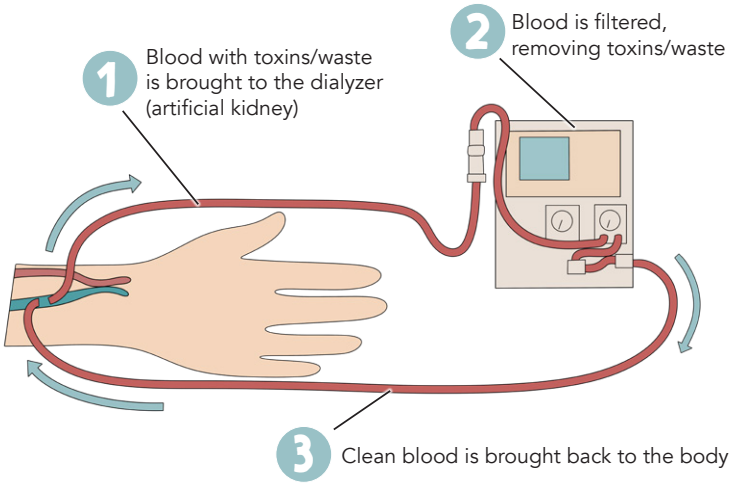
Uses a dialyzer (an artificial kidney) to remove excess fluids and toxins



In a healthy body, the kidneys are able to filter waste products.

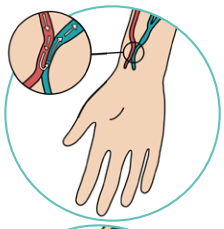
Kidneys that are not functioning properly (injured) need help removing excess waste from the blood. Otherwise, waste accumulates and becomes toxic and harmful to the body.

THE PROCESS

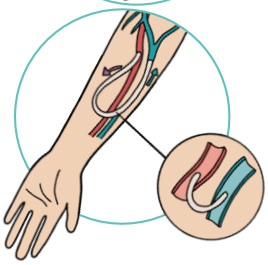


ACCESS

FISTULA
Joining an artery to a vein



GRAFT
Inserting synthetic graft material between an artery and a vein



★ Increased risk for infection due to the synthetic material insertion

COMPLICATIONS

- Hypotension
- Hemorrhage
- Air embolus
- Electrolyte imbalances
- ★ **Dialysis disequilibrium syndrome**
  - Rare complication caused by rapid removal of urea during the filtration process
    - Neurological symptoms
    - Nausea/vomiting
    - ↓ Level of consciousness (LOC)
    - Restlessness
    - Seizures

PATIENT EDUCATION

On the arm that has vascular access, the patient needs to avoid:

- ✗ Compression
- ✗ Blood draws
- ✗ Blood pressure readings
- ✗ Tight clothing
- ✗ Pressure from carrying bags
- ✗ Sleep on that arm

NURSING CONSIDERATIONS

HOLD these medications before dialysis:

- Water-soluble vitamins
- Antibiotics
- Antihypertensives
- Digoxin

MEMORY TRICK

Vitamin B complex & Vitamin C – think of "WATER at the BeaCh"

EVALUATE patency of vascular access

- ✓ **Feel the thrill** (palpating the fistula)
- ✓ **Hear the bruit** (heard during auscultation)

REPORT if these are not noted over the site

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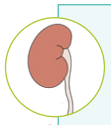
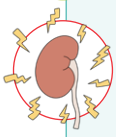
Types of Dialysis: Peritoneal Dialysis

Peritoneal Dialysis

Drains of the peritoneum to remove excess fluids and toxins

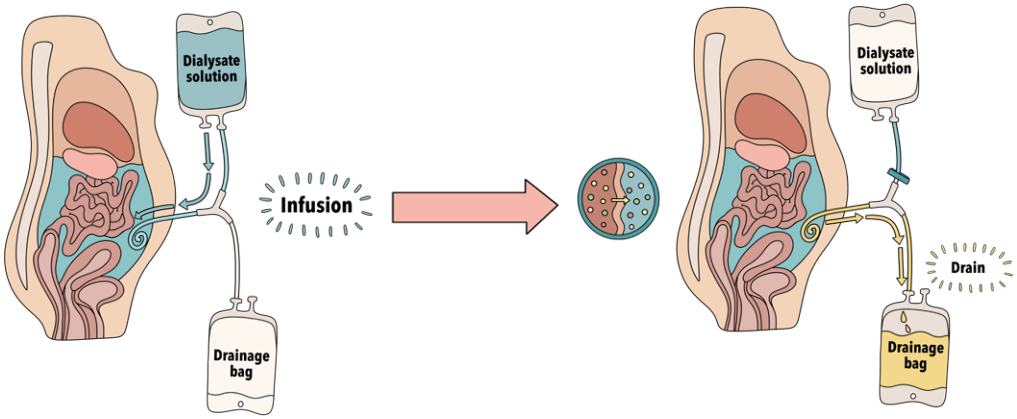


In a healthy body, the kidneys are able to filter waste products.



Kidneys that are not functioning properly (injured) need help removing excess waste from the blood. Otherwise, waste accumulates and becomes toxic and harmful to the body.

THE PROCESS



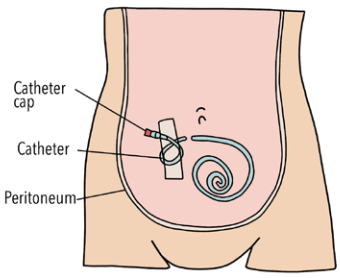
Dialysate is infused into the peritoneal cavity by gravity
↓
The clamp is closed on the infusion line
↓
Dialysate dwells for a set amount of time (this is called the dwell time)

The drainage tube is unclamped
↓
Fluid drains from the peritoneal cavity by gravity
↓
A new container of dialysate is infused as soon as drainage is complete
↓
REPEAT!

ACCESS

PERITONEAL CATHETER

Procedure is performed at the bedside or in the operating room



COMPLICATIONS

- Hyperglycemia → The dialysate infusion contains glucose
- **Peritonitis**
  - This procedure is commonly done at home and has an increased risk for infection in the peritoneum
  - Signs & symptoms
    - Cloudy or bloody drainage
    - Fever > 100.4°F (38°C)
    - Abdominal pain
    - Malaise

PATIENT EDUCATION

How to AVOID infections:

- ✓ Practice good hand hygiene before and after dialysis
- ✓ Clean catheter site daily
- ✓ Keep supplies in a clean, dry place

Warm the solution prior to administration:

✗ Do not use a microwave

✓ Use an incubator or heating cabinet

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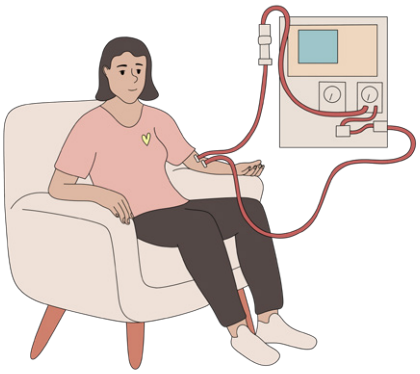
Dialysis Quick Comparison



Hemodialysis

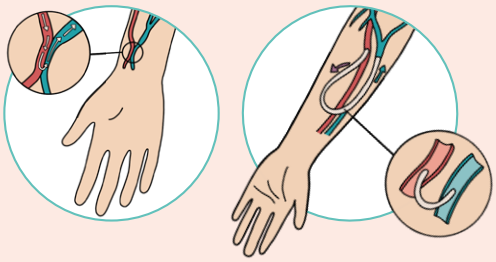
MOST COMMON METHOD OF DIALYSIS

Outside the body
(blood goes to a machine)

Done in the **hospital** or a **dialysis clinic**

Performed 3–5x per week

ACCESS:
Fistula or graft



COMPLICATIONS:

- Fistula infection or thrombosis
- Hypotension
- Disequilibrium syndrome
- Hemorrhage

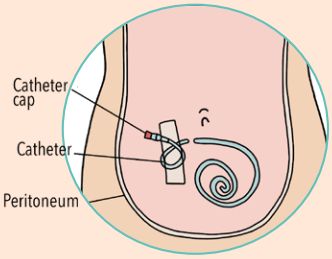
Peritoneal Dialysis

Inside the body
(fluid goes through the abdominal wall)

Can be done at home

Performed daily (7x per week)

ACCESS:
Peritoneal catheter



COMPLICATIONS:

- Peritonitis
- Hyperglycemia

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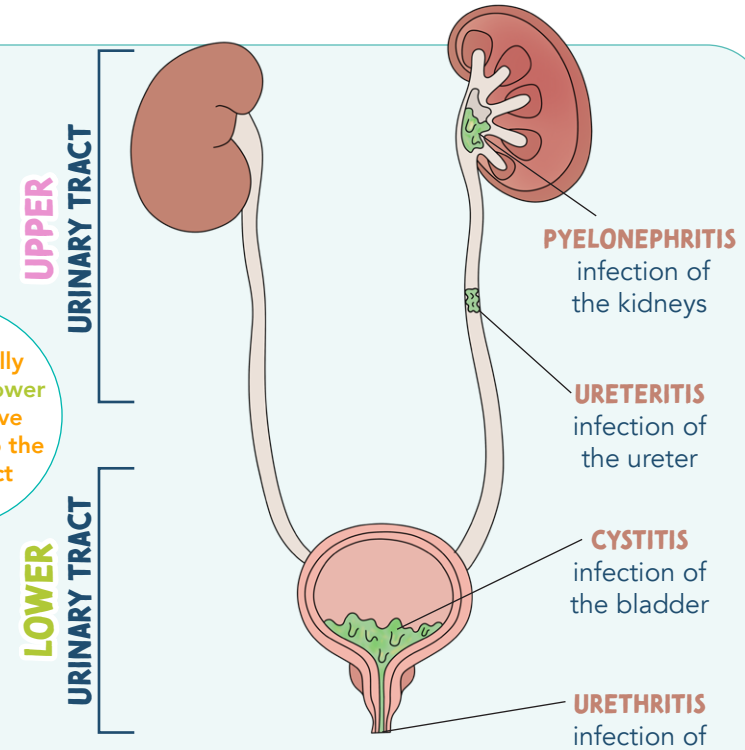
Urinary Tract Infection (UTI)

PATHOLOGY

Infection within the urinary system caused by a **bacteria**, **virus**, or **fungus**.

BACTERIA IS THE MOST COMMON (*E. coli*)

UTIs typically start in the lower tract & move upwards into the upper tract



RISK FACTORS

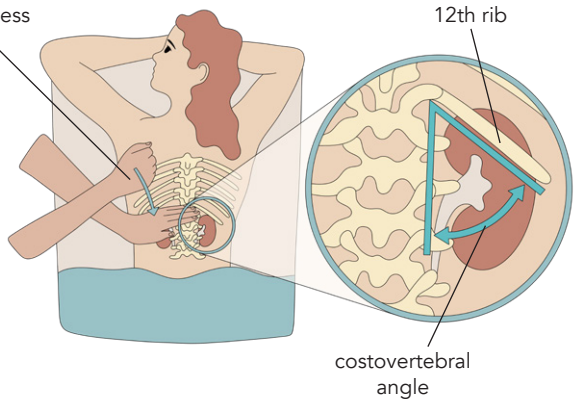
- **Most common in women**
  - **Shorter urethra & urethra close to the rectum**
- **Indwelling catheters**
- Overuse of antibiotics
- Hormone changes (pregnancy)
- Diabetes
- Lifestyle (baths, scented tampons, perfumes)

SIGNS & SYMPTOMS

- **Pain/burning on urination (dysuria)**
- **Costovertebral angle (CVA) tenderness**
- Foul-smelling urine
- Chills & fever
- Headache/malaise
- Frequency & urgency
- Nocturia
- WBCs & RBCs in the urine

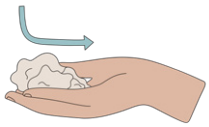
Elderly patients may show atypical symptoms:

- Change in mental status/confusion
- Lethargy
- New incontinence



PREVENTION & PATIENT EDUCATION

- **Wipe from front to back**
- Wear loose cotton underwear
- Void after intercourse, which helps flush out bacteria from the urethra
- Avoid bubble baths, perfumes, or sprays
- Finish entire antibiotics course
- Remove any catheters ASAP (per HCP orders)
- Avoid caffeine & alcohol during an active infection



- Maintain fluid status
  - 2–3 L per day

"Flushes" out the urinary tract

MEDICATIONS

- Antibiotics

Take urine culture BEFORE giving first dose of antibiotics

- Analgesics to manage pain
  - Example: Phenazopyridine (Pyridium)

May turn urine orange

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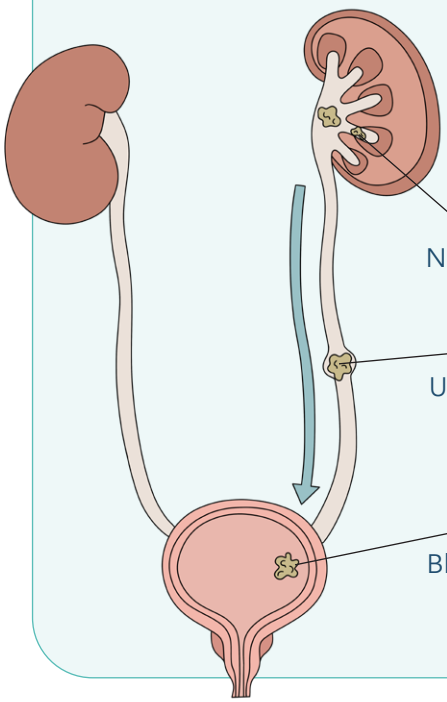
Renal Calculi

PATHOLOGY

Stones (calculi) found in the urinary tract & kidney that eventually pass through the ureter & bladder

Stones can be very large or very small



SIGNS & SYMPTOMS

- **Pain**
- Discomfort
- Hematuria (RBCs in the urine)
- Pyuria (WBCs in the urine)
- Nausea & vomiting

DIAGNOSTICS

- KUB: X-ray of kidneys, ureters, bladder
- Intravenous pyelogram (IVP)
- Ultrasound or CT scan
- Urine test

TREATMENT

- Medications to control the pain
  - NSAIDs → **↓ Pain & inflammation** *(makes the stone easier to pass)*
  - Opioid analgesics
- **Strain the urine**
  - Keep any stones & send them to the lab for evaluation
- Get the patient moving or frequently reposition them
- ↑ Fluids
- **DIET MODIFICATIONS:**
  - Limit protein, sodium & calcium
- Procedures:

NONINVASIVE **Extracorporeal shock wave lithotripsy (ESWL):** Sends shock waves to break up the stone

INVASIVE

Percutaneous nephrolithotomy: Removes stone through an incision made on the back where the kidneys are located

Ureteral Stent Placement: Stent is inserted to allow urine and/or stones to pass from the bladder to the ureters and out of the body

Most commonly, the stone will pass on its own

Push stone forward & out to ↓ risk of infection

Uric acid is a waste product of the breakdown of **purines**

What is URIC ACID?

	MOST COMMON CALCIUM	URIC ACID	STRUVITE	CYSTINE
FORMATION	Forms due to excess calcium & oxalate in the urine	Forms due to excess uric acid in the urine (acidic urine)	Forms due to a **bacteria** that causes ammonia-rich urine and an alkaline environment	Forms due to excess cystine in the urine
CAUSES	**Dehydration** ▪ Hypercalcemia ▪ Hypercalciuria ▪ Hyperparathyroidism ▪ ↑ Intake of sodium ▪ GI disorders ▪ ↑ Intake of calcium supplements with vitamin D	▪ Gout ▪ Foods high in purine or animal proteins ▪ Dehydration ▪ Metabolic issues (diabetes)	▪ Chronic urinary tract infections (UTIs) ▪ Foreign bodies ▪ Neurogenic bladder	▪ Rare genetic disorder that affects renal absorption of cystine

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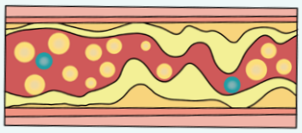
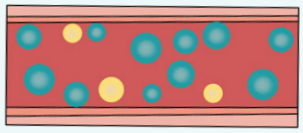
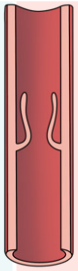
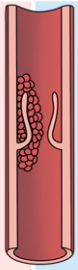
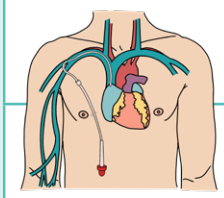
Lab Values Related to the Cardiac System

	EXPECTED RANGE	DESCRIPTION	↓	↑
TOTAL CHOLESTEROL	< 200 mg/dL	Measurement of the total amount of cholesterol in the blood	Lower risk for heart disease and stroke	Higher risk for heart disease and stroke
TRIGLYCERIDES	< 150 mg/dL	Most common type of fat in the body: takes food and stores it as excess energy		
LOW DENSITY LIPOPROTEINS (LDL)	< 100 mg/dL	 **LDL** bad MEMORY TRICK: LDL= we want Low levels because it's a "bad" fat		
HIGH DENSITY LIPOPROTEINS (HDL)	F > 40 mg/dL M > 55 mg/dL	 **HDL** good MEMORY TRICK: HDL = we want High levels because it's a Happy cholesterol	Higher risk for heart disease and stroke	Lower risk for heart disease and stroke
D-DIMER	< 0.5 mcg/mL	Fragments of fibrin that are in the blood when a clot dissolves or is broken down **D-dimers help to determine if a clot is present somewhere in the body**	 Normal/low levels ▪ Blood clot is ruled out	 Elevated/high levels (positive result) **POSSIBLE CAUSES:** ▪ Blood clot present in the body ▪ Disseminated intravascular coagulation (DIC)
BNP	< 100 pg/mL	Peptide that is released when the ventricle stretches from being filled with too much fluid	Helps to rule out heart failure	May indicate congestive heart failure (CHF)

Hemodynamic Parameters

CARDIAC OUTPUT (CO)	4 – 8 L/min	Total volume pumped per minute
 **CARDIAC INDEX (CI)**	2.5 – 4.0 L/min/m^2	Cardiac output per body surface area $CI = \frac{CO}{\text{surface area}}$
CENTRAL VENOUS PRESSURE (CVP)	2 – 8 mmHg	Pressure in the superior vena cava: shows how much pressure from the blood is returned to the right atrium from the superior vena cava
MEAN ARTERIAL PRESSURE (MAP)	70 – 100 mmHg At least 60 mmHg is required to adequately perfuse the vital organs	Average pressure in the systemic circulation (entire body) through the cardiac cycle
SYSTEMIC VASCULAR RESISTANCE (SVR)	800 – 1200 dynes/sec/cm	**Resistance** it takes to push blood through the circulatory system to create blood flow

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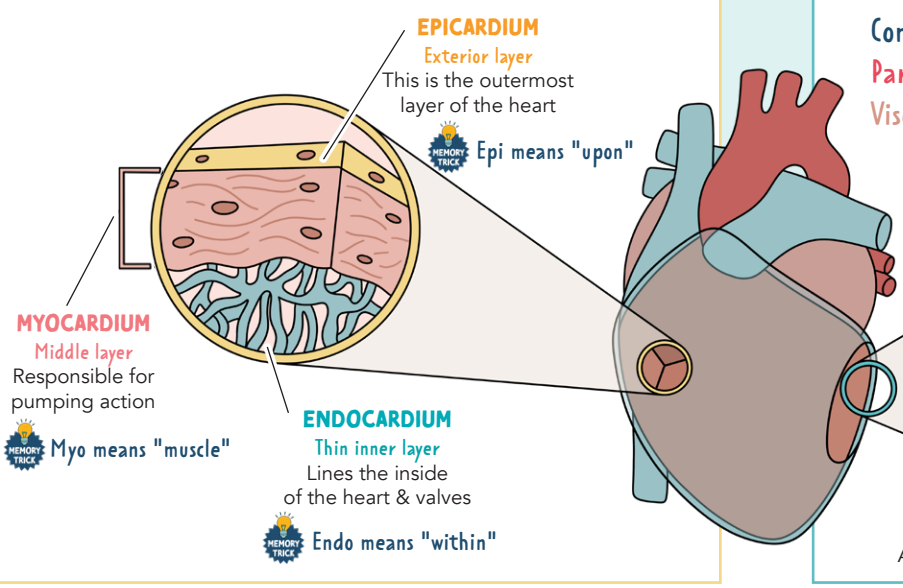
Cardiac Overview

Layers of the Heart

There are three layers of the heart:
EPICARDIUM, **MYOCARDIUM**, and **ENDOCARDIUM**



EPICARDIUM
Exterior layer
This is the outermost layer of the heart
Epi means "upon"

MYOCARDIUM
Middle layer
Responsible for pumping action
Myo means "muscle"

ENDOCARDIUM
Thin inner layer
Lines the inside of the heart & valves
Endo means "within"

Pericardium

Thin sac that encases the heart

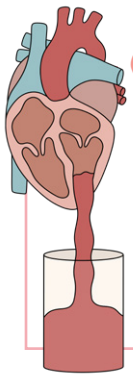
Composed of two layers:
Parietal pericardium
Visceral pericardium

Parietal pericardium
The outer layer of the pericardium. Thin sac of tissue which surrounds the heart.

Visceral pericardium
Adheres to the epicardium

Pericardial fluid
Lubricates the surface of the heart & reduces friction

Cardiac Terms



CARDIAC OUTPUT (CO)

This is the total volume of blood ejected (pumped) by the heart per minute.

It's the amount of blood reaching the tissues.

FORMULA

HR x SV = CO

Heart Rate | Stroke Volume | Cardiac Output

HR = The # of times the heart contracts each minute (normal: **60-100** bpm)

SV = The amount of blood ejected from the left ventricle with each heartbeat

NORMAL CO:
4 - 8 L/min

INTERPRETATION

↓ **CO = LESS** volume
(↓ perfusion to the vital organs)

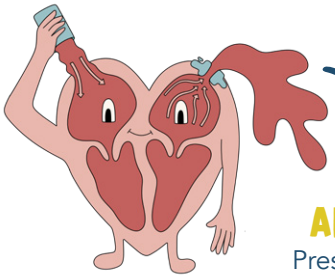
↑ **CO = MORE** volume
(possible causes include hypervolemia)

CONTRACTILITY

Force/strength of contraction of the heart muscle

PRELOAD

Amount of blood returned to the right side of the heart at the end of diastole



EJECTION FRACTION (EF)

% of blood expelled from the left ventricle with every contraction

NORMAL EF:
50 - 70%

EXAMPLE:
If the EF is 55%, the heart is pumping out 55% of what's inside the left ventricle

AFTERLOAD

Pressure that the left ventricle has to pump against (the resistance it must overcome to circulate blood)

Clinically measured by systolic blood pressure!

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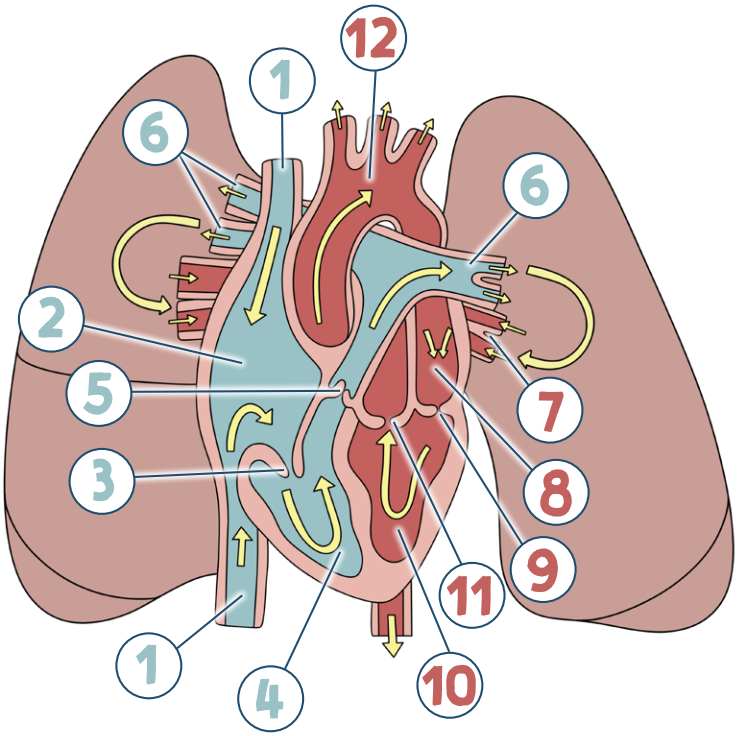
Blood Flow Through the Heart



Right

Deoxygenated blood

1. Superior Vena Cava/ Inferior Vena Cava
2. Right Atrium
3. Tricuspid Valve
4. Right Ventricle
5. Pulmonary Valve
6. Pulmonary Artery*

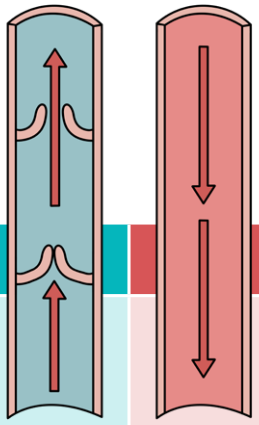
carries DEOXYGENATED blood to the LUNGS

Left

Oxygenated blood

7. Pulmonary Vein*
8. Left Atrium
9. Bicuspid/Mitral Valve
10. Left Ventricle
11. Aortic Valve
12. Aorta

carries OXYGENATED blood to the TISSUES/BODY



Vein

Carries **DEOXYGENATED** blood ***back to the heart***

Artery

Carries **OXYGENATED** blood ***away from the heart***

MEMORY TRICK Arteries = Away from the heart

Vein	Artery
Flows smoothly	Flows in pulses
Thin walls	Thick walls with muscle tissue
Has valves to prevent backflow	Has no valves
Under low pressure	Under high pressure

***EXCEPTIONS:**

The only exceptions to this are the **PULMONARY ARTERY** and the **PULMONARY VEIN**

- **PULMONARY ARTERY** ↓ carries deoxygenated blood from the heart to the lungs
- **PULMONARY VEIN** ↓ carries oxygenated blood from the lungs to the heart

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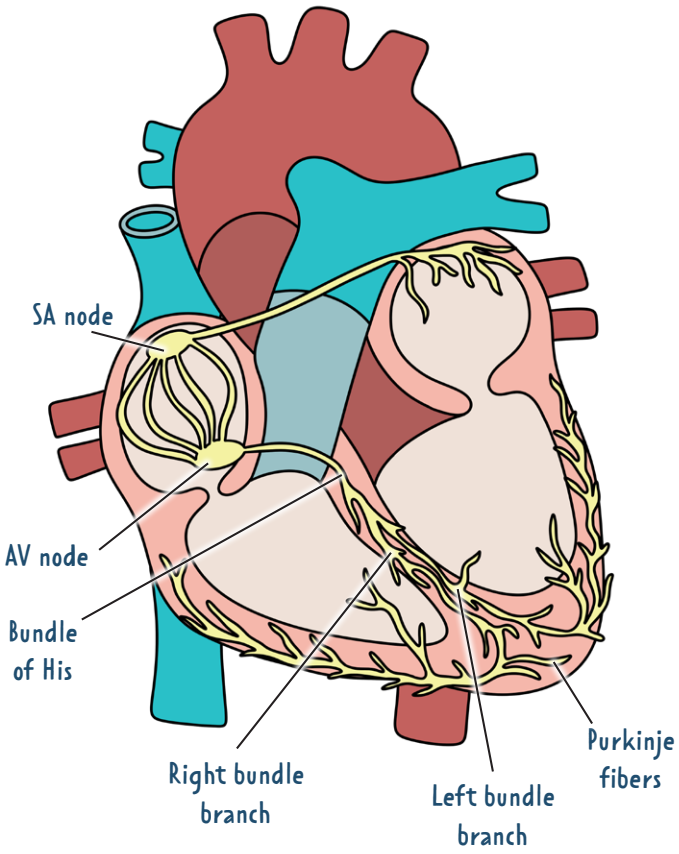
Electrical Condition of the Heart

Generates & transmits

ELECTRICAL IMPULSES

that stimulate contractions of the atria and then the ventricles



Steps in the Heart's Conduction System

MNEMONIC		
Send	**SA** node (**S**ino**A**trial node)	Primary pacemaker of the heart Creates electrical impulses of **60–100 bpm**
A	**AV** node (**A**trio**V**entricular node)	Secondary pacemaker of the heart, or "backup pacemaker" Takes over at a rate of **40–60 bpm** if the SA node malfunctions
Big	**B**undle of His	Branch into the right bundle branch & left bundle branch
Bounding	**B**undle branches	
Pulse	**P**urkinje fibers	Fire at a rate of **30–40 bpm** if the SA & AV nodes fail

This is a normal heart rate

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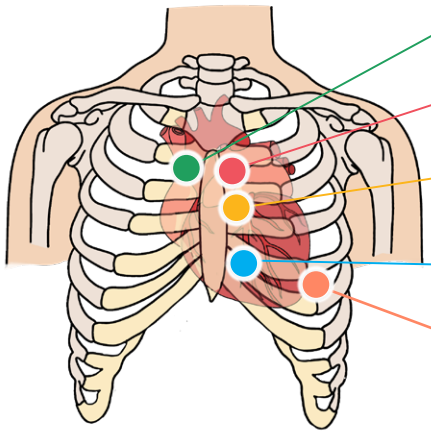
Auscultating Heart Sounds

5 AREAS FOR LISTENING TO THE HEART

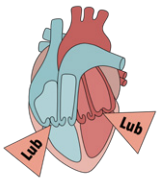
MNEMONIC **All People Enjoy Time Magazine**



- **Aortic** Right 2nd intercostal space
- **Pulmonic** Left 2nd intercostal space
- **Erb's point** (S1, S2) Left 3rd intercostal space
- **Tricuspid** Lower left sternal border 4th intercostal space
- **Mitral** Left 5th intercostal space, medial to midclavicular line

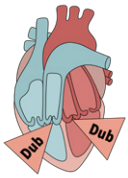
MEMORY TRICK **M for Midclavicular**

NORMAL



S1 *LUB* — Tricuspid & mitral valve closure

↓



S2 *DUB* — Aortic & pulmonic valve closure

CLOSING OF THE VALVES
Valve opening does not normally produce a sound

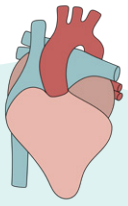
ABNORMAL

S3 EARLY DIASTOLE in rapid ventricle filling

↓

S4 LATE DIASTOLE & high atrial pressure (forcing blood into a stiff ventricle)

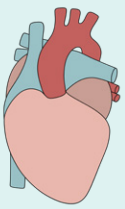
ABNORMAL VENTRICULAR FILLING
Extra heart sounds



CONTRACTED

Systole

Ventricle pump/ejection = LUB (S1)



RELAXED

Diastole

Ventricle relax/fill = DUB (S2)

LUB (S1) ⇄ DUB (S2)

MEMORY TRICK

"COZY RED"

CO (*contract*) **ZY** (*systole*)
RE (*relax*) **D** (*diastole*)

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Congestive Heart Failure (CHF)

Pathology

- Cardiac disorder that impairs the ability of the ventricles to fill or eject properly
- **The heart muscle can't pump enough blood to meet the body's needs**

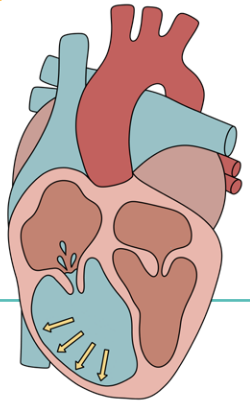
Risk Factors

- Uncontrolled hypertension
- Congenital heart defect
- Arrhythmias
- Coronary artery disease
- Faulty heart valves
- Damage or inflammation of the heart muscle

Diagnostic

- ↑ **B-type natriuretic peptides (BNP)**

BNP is a peptide released when the ventricle stretches from being filled with too much fluid



BNP	< 100 pg/mL	Expected range
BNP	100 – 300 pg/mL	HF is suspected
BNP	> 300 pg/mL	Mild HF
BNP	> 600 pg/mL	Moderate HF
BNP	> 900 pg/mL	Severe HF

- Chest X-ray
  - Will show an enlarged heart & pulmonary infiltrate
- Echocardiogram
  - Measures ejection fraction
- Cardiac stress test

Patient Education

- Report S&S of fluid retention
- Elevate the head of the bed (HOB)
- Balance periods of activity & rest
- Monitor daily weight

Edema, weight gain

Daily weight is the best way to monitor HF

Weight gain of 2–3 lbs over a 24-hour period would be alarming

DIET MODIFICATIONS

- ↓ Sodium
- ↓ Fat
- ↓ Cholesterol

Space out fluid intake throughout the day & use sugar-free hard candy to ↓ thirst

Medications

- Antihypertensives
  - ACE inhibitors & beta blockers
- Digoxin
- **Diuretics**

MEMORY TRICK

Diuretics cause the body to excrete urine (Diurese), which causes the body to be Dry inside

POTASSIUM-WASTING DIURETICS	POTASSIUM-SPARING DIURETICS
Furosemide (Lasix) Hydrochlorothiazide	Spironolactone

MEMORY TRICK

Spironolactone = Sparing

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Congestive Heart Failure (CHF)

A PATIENT CAN HAVE BOTH!

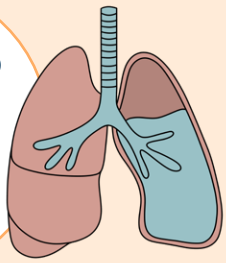
Left-Sided HEART FAILURE

MOST COMMON

Also called
left ventricular (LV) heart failure

	Description	Ejection fraction
SYSTOLIC HF	**Weakened heart muscle** The ventricle does not **EJECT (squeeze)** properly	**Ejection fraction reduced** Also called heart failure with reduced ejection fraction (HFrEF)
DIASTOLIC HF	**Stiff & non-compliant heart muscle** The ventricle does not **FILL** properly	**Normal ejection fraction** Also called heart failure with preserved ejection fraction (HFpEF)

Fluid is backing up into the **LUNGS** = **PULMONARY SYMPTOMS**



MEMORY TRICK

Left side = Lungs

Right-Sided HEART FAILURE

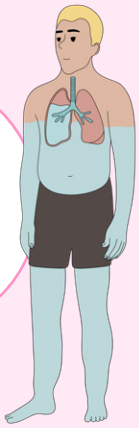
Also called
right ventricular (RV) heart failure

Typically occurs as a result of LEFT-SIDED HF

When the **LEFT** ventricle fails, pressure from fluid builds up and causes a backflow of fluids into the **RIGHT** side of the heart

This causes damage to the RIGHT side of the heart

Fluid is backing up into the **VENOUS SYSTEM**



MEMORY TRICK

Right side = Rest of the body

Signs & Symptoms

PULMONARY SYMPTOMS

- Trouble breathing (dyspnea)
- Trouble breathing while lying down (orthopnea)
- Shortness of breath while sleeping (nocturnal paroxysmal dyspnea)
- Rales (crackles) heard in the lungs
- Constant cough (frothy, blood-tinged sputum)

CHRONIC HF may show BOTH types of signs & symptoms

VENOUS SYMPTOMS

- Swelling of the legs & hands
- Weight gain (2–3 lbs. per day)
- Edema (pitting)
- Large neck veins
  - Jugular vein distension (JVD)
- ↑ Abdominal girth (ascites)
- Enlarged liver from buildup of fluid (hepatomegaly)

This may cause nausea, anorexia & bloating

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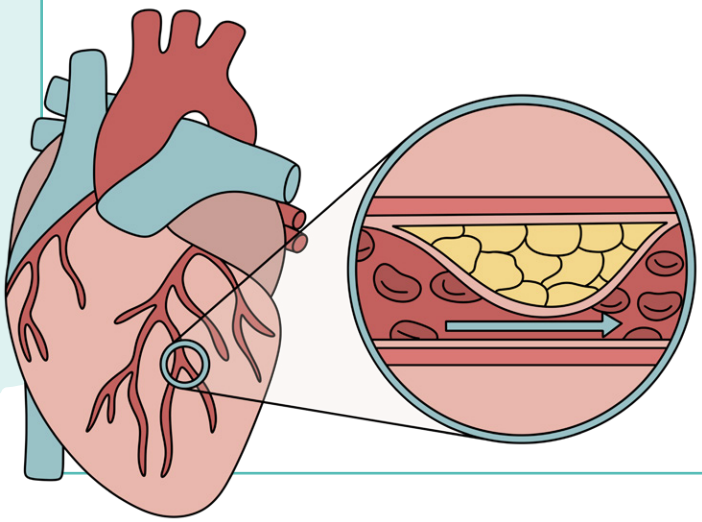
Coronary Artery Disease (CAD)

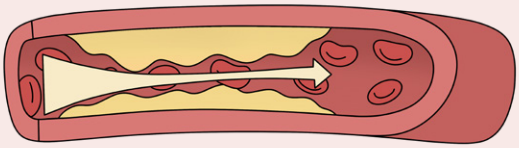
Pathology

Damage in the coronary arteries due to **ATHEROSCLEROSIS**



ATHEROSCLEROSIS

is plaque buildup that happens over time on the blood vessel walls. It causes narrowing of the vessels and limits blood supply to the heart.



The plaque may rupture, causing thrombi (clots), and may obstruct blood flow, leading to an acute MI.

Risk Factors

NON-MODIFIABLE

- Age
- Gender
- Race
- Family history

MODIFIABLE

- Diabetes
- Hypertension
- Smoking
- Obesity
- Physical inactivity
- High cholesterol
- Metabolic syndrome

Diagnostic

- Blood tests:
  - ↑ LDL, total cholesterol, triglycerides
  - ↓ HDL
- EKG: assess for changes in ST segments
- Stress test
- Cardiac catheterization

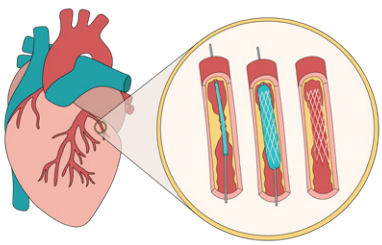
"BAD" CHOLESTEROL	"GOOD" CHOLESTEROL
↓ LDL < 100 mg/dL ↓ Triglycerides < 150 mg/dL ↓ Total cholesterol < 200 mg/dL	↑ HDL > 60 mg/dL
MEMORY TRICK: LDL= we want Low levels because it's a "bad" fat	MEMORY TRICK: HDL = we want High levels because it's a Happy cholesterol

Signs & Symptoms

- Usually asymptomatic
- Chest pain (Angina) may occur

Treatment

- Percutaneous coronary intervention (PCI)



Medications

- Antiplatelets
- Medications to normalize cholesterol levels
  - **Statins**
  - Bile acid sequestrants

Patient Education

Heart-healthy diet

- ↓ Saturated fats
- ↑ Fiber

PREVENTABLE MEASURES

- Check cholesterol levels
- Control hypertension & diabetes
- Stop smoking
- Increase physical activity
- Lose weight if needed

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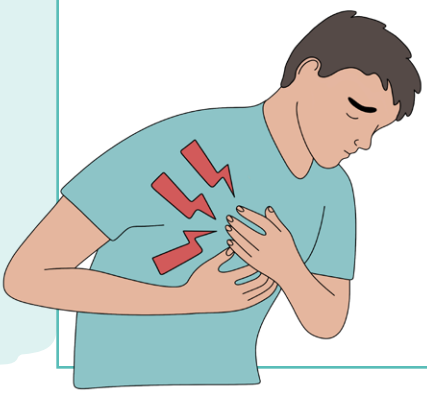
Angina Pectoris

Pathology

Angina is **CHEST PAIN** associated with ischemia. It's due to narrowing of at least one major coronary artery.



TYPES OF ANGINA

STABLE	Predictable	Occurs with **EXERTION** EXAMPLE Exercise or strenuous activity
UNSTABLE	Preinfarction	Occurs at **REST** More common than stable or Prinzmetal's anginas
PRINZMETAL'S/ VARIANT	Coronary artery vasospasm	Occurs with pain at **REST** with reversible **ST ELEVATION**

Signs & Symptoms

- Chest pain **RELIEVED** by **rest** & **nitroglycerin**

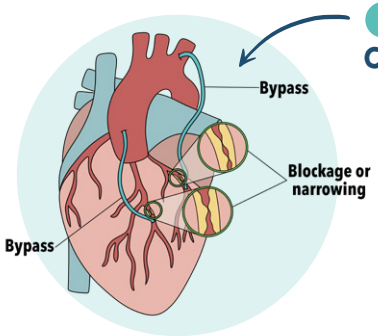
Interventions

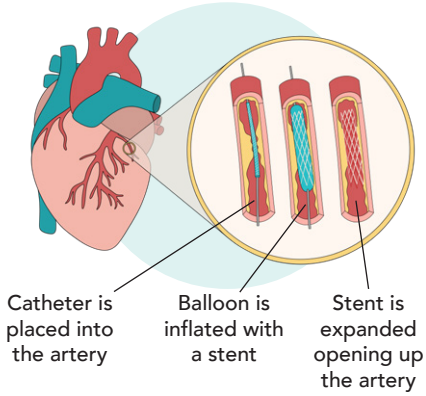
- Reperfusion procedures

GOAL: ↓ Oxygen demand

CABG — Coronary Artery Bypass Graft

PCI — Percutaneous Coronary Interventions





Medications

NITRATES

Nitroglycerin

- Vasodilators
- ↓ Ischemia = ↓ pain
- Usually sublingual administration
- Keep in original container (dark, glass bottle) in a dry, cool place
- Do not swallow or chew these tablets

CALCIUM CHANNEL BLOCKERS

- Relax blood vessels
- ↑ Oxygen supply to the heart
- ↓ Workload of heart

ANTIPLATELETS/ ANTICOAGULANTS

- Prevent platelet aggregation & thrombosis

BETA BLOCKERS

- ↓ Myocardial oxygen consumption

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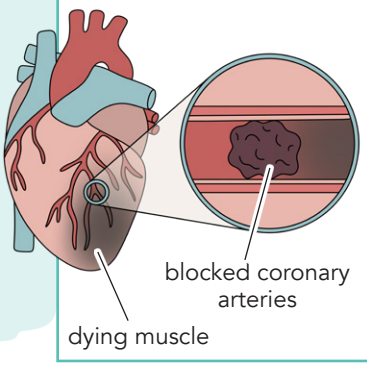
Myocardial Infarction (MI)

Pathology

Complete blockage in one or more arteries of the heart

EMERGENCY!



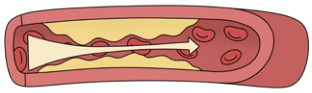
ATHEROSCLEROSIS
Coronary arteries become narrow due to plaque buildup

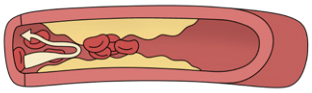
→

ANGINA
Chest pain due to ischemia (low O_2)

→

MYOCARDIAL INFARCTION (MI)
Plaque rupture becomes a blood clot that blocks arteries of the heart





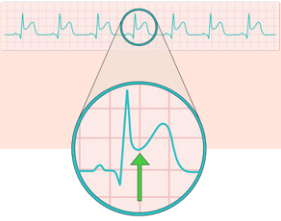
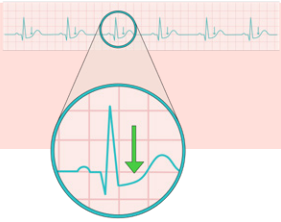
Signs & Symptoms

- **Sudden, crushing, radiating chest pain that continues despite rest & medications**
- Shortness of breath
- Nausea & vomiting
- Sweating
- Pale & dusky skin
- Pain radiating to the:
  - Left arm or shoulder
  - Jaw
  - Back
  - Epigastric area (i.e. heart burn)

Women may present with different/abnormal symptoms

Diagnosis

- Troponin
- Stress tests (chemical & exercise)
- ECG/EKG: assess the ST segment

STEMI	NSTEMI
ST Elevation **M**yocardial **I**nfarction	**N**on-**ST E**levation **M**yocardial **I**nfarction
Complete block	Partial block
ST segment is ELEVATED	ST segment is NOT ELEVATED ST depression or T inversion
	

Treatment

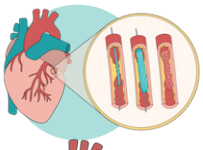
IMMEDIATE

- **M** **Morphine**
  ↓ Workload of the heart & ↓ pain
- **O** **Oxygen**
  ↑ O_2 to the heart
- **N** **Nitroglycerin**
  Opens up the vessels
- **A** **Antithrombotic medications**
  - **Aspirin** (prevents platelets from sticking together)
  - **Heparin** (anticoagulant)
  - **Thrombolytics** (bust apart the clot)
    - These end in the suffix "-teplase" like Ateplase

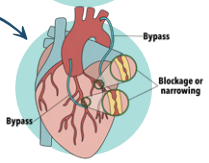
Any time you give a thrombolytic, watch for signs of bleeding!

SURGERY

- Percutaneous coronary intervention (PCI)



- Coronary artery bypass graft (CABG)
- Endarterectomy
  - Cuts out the blockage



PREVENTION & REST

- Prevent/stabilize clot (IV heparin)

PATIENT EDUCATION:

- Engage in only light activity

Patients tend to ask: When can I resume sexual activity?

You can answer: When you can do physical activity without symptoms.

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Cardiac Biomarkers

TROPONIN

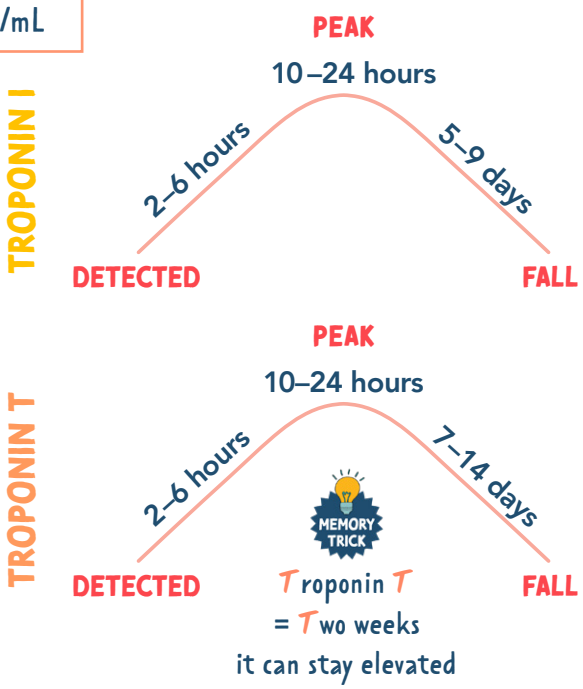
EXPECTED RANGE
TROPONIN I < 0.03 ng/mL
TROPONIN T < 0.1 ng/mL

BEST indicator of an acute MI

- Protein released into the bloodstream when the heart muscle is damaged
- Three isomers of troponin:

TROPONIN C:
Binds calcium to activate muscle contraction

TROPONIN I & T:
Specific for cardiac muscle



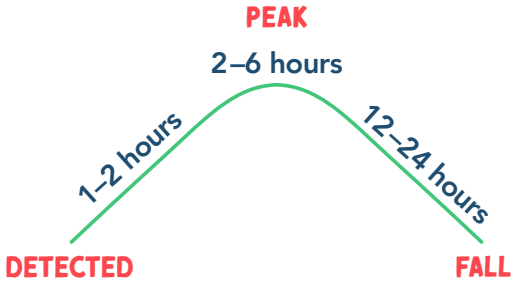
MYOGLOBIN

EXPECTED RANGE
5 - 70 ng/mL

- Found in cardiac & skeletal muscle

MEMORY TRICK: Myoglobin = Muscle

- NOT a specific indicator of an acute MI, but a (-) sign is good for ruling out an acute MI

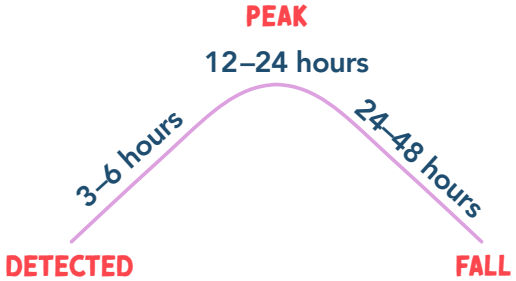


CK-MB

EXPECTED RANGE
0 - 5 ng/mL

CREATINE KINASE-MB

- Cardiac-specific isoenzyme
  BUT less reliable than troponin
- An enzyme released into the bloodstream when the heart, muscles, or brain is damaged



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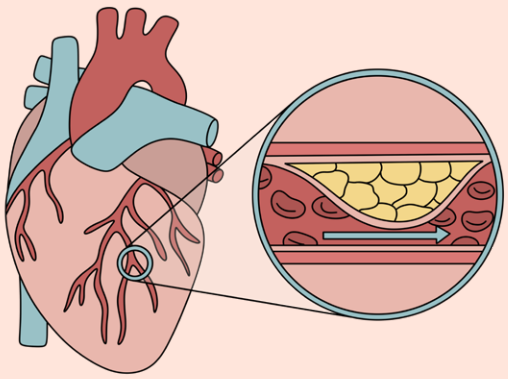
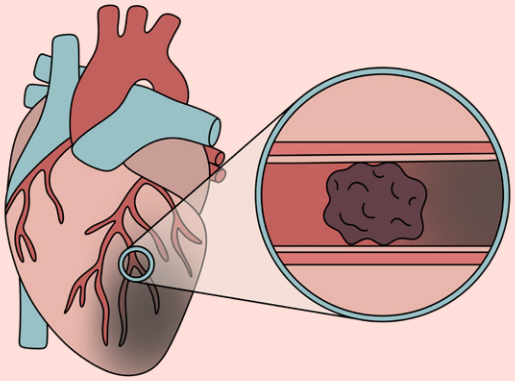
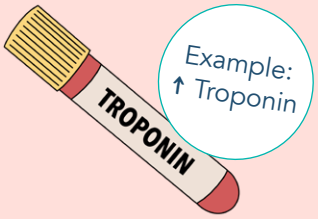
Angina vs. Myocardial Infarction

	Angina	Myocardial Infarction
	LOW oxygen from narrowed arteries 	**NO** oxygen from a **SUDDEN** blockage 
CAUSE	Precipitated by exertion, exercise, or stress	Can occur without cause, typically in the morning
SYMPTOMS	Typically only lasts 1–3 minutes	**CHEST PAIN** Lasts longer than 20 minutes
	RELIEVED by rest and/or nitroglycerin	**NOT RELIEVED** by rest or nitroglycerin
SERUM BIOMARKERS	Not elevated	Elevated 

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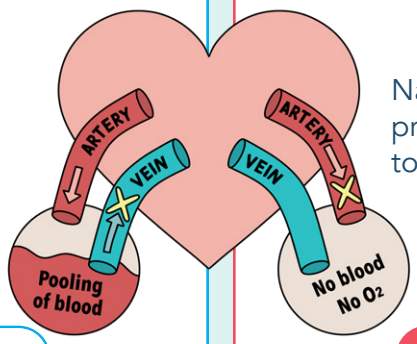
Peripheral Vascular Disease

PERIPHERAL VENOUS DISEASE (PVD)

Deoxygenated blood can't get back to the heart.

Oxygenated blood pools in the extremities.



PAIN ?	✓	Dull, constant, achy pain!
PULSE ?	✓	May not be palpable due to edema
EDEMA ?	✓	Blood **POOLING** in the legs
TEMP ?		Warm legs (blood is warm)
COLOR ?		Stasis dermatitis (brown/yellow)
WOUNDS ?		Venous **STASIS** ulcers, shallow & irregular shaped wounds
GANGRENE ?	✗	We have too much blood! Gangrene is caused by insufficient amounts of blood.
POSITIONING ?		Elevate Veins. Positions that make it worse: dangling, sitting/standing for long periods of time

PERIPHERAL ARTERIAL DISEASE (PAD)

MEMORY TRICK: PAD = "BAD"

Narrow arteries (atherosclerosis) prevent oxygenated blood from getting to the distal extremities (hands & feet).

Ischemia & necrosis of the extremities

PAIN ?	✓	Sharp pain that gets worse at night ("rest pain"), intermittent claudication
PULSE ?	✓	Very poor or nearly absent
EDEMA ?	✗	No blood in the extremities
TEMP ?		Cool. No blood = cool leg (blood is warm)
COLOR ?		Pale, hairless, dry, scaly, thin skin due to lack of nutrients (↓ O_2)
WOUNDS ?		Red sores with a regular shape and a round, "punched-out" appearance
GANGRENE ?	✓	Tissue death caused by a lack of blood supply
POSITIONING ?		Dangle arteries

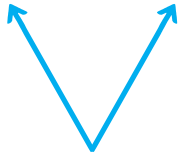
CAUSES OF BOTH: Smoking • Diabetes • High cholesterol • Hypertension

DIAGNOSIS FOR BOTH: Doppler ultrasound or ankle brachial index (ABI)

TREATMENT

Keep vein open!

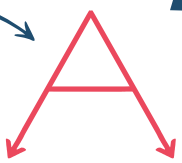
- **E**le**V**ate **V**eins
- **Medications**
  - Aspirin or clopidogrel
  - Cholesterol-lowering drugs or "statins"
- **Surgery**
  - Angioplasty
  - Bypass (CABG)
  - Endarterectomy



TREATMENT

Get blood moving!

- D**A**ngle **A**rteries (Dependent position)
- **Perform daily skin care using moisturizers**
- **Stop smoking**
- **Avoid tight clothing** (vasoconstriction)
- **Avoid heating pads**
- **Medications**
  - Vasodilators
  - Antiplatelets



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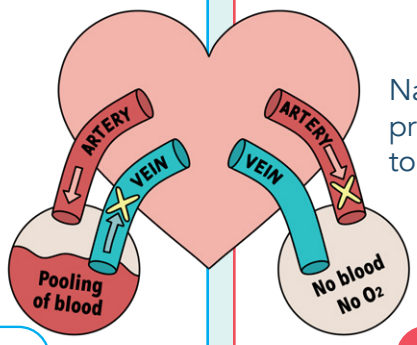
Peripheral Vascular Disease

PERIPHERAL VENOUS DISEASE (PVD)

Deoxygenated blood can't get back to the heart.

Oxygenated blood pools in the extremities.



- PAIN ?
- PULSE ?
- EDEMA ?
- TEMP ?
- COLOR ?
- WOUNDS ?
- GANGRENE ?
- POSITIONING ?

PERIPHERAL ARTERIAL DISEASE (PAD)

MEMORY TRICK: PAD = "BAD"

Narrow arteries (atherosclerosis) prevent oxygenated blood from getting to the distal extremities (hands & feet).

Ischemia & necrosis of the extremities

- PAIN ?
- PULSE ?
- EDEMA ?
- TEMP ?
- COLOR ?
- WOUNDS ?
- GANGRENE ?
- POSITIONING ?

CAUSES OF BOTH:

DIAGNOSIS FOR BOTH:

TREATMENT

- Position:
- Medications
  -
  -
- Surgery
  -
  -
  -

TREATMENT

- Position:
- Perform
- Stop
- Avoid
- Avoid
- Medications
  -
  -

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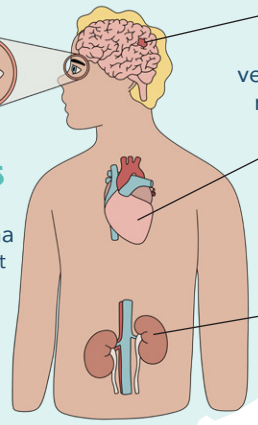
Hypertension (HTN)

HYPERtension = **HIGH** BP

Most accurate diagnosis for HTN

CATEGORIES	SYSTOLIC (SQUEEZE)	DIASTOLIC (DECOMPRESS)
HYPOTENSION	< 100	< 60
NORMAL	< 120 mmHg	< 80 mmHg
PRE-HTN	120–129 mmHg	< 80 mmHg
STAGE 1 HTN	130–139 mmHg	80–89 mmHg
STAGE 2 HTN	≥ 140 mmHg	≥ 90 mmHg
⚠ HTN CRISIS	> 180 mmHg	> 120 mmHg

Affected Organs



HEMORRHAGIC STROKE
Weak & narrow vessels could lead to rupture of vessels

VISUAL CHANGES
Damages blood vessels in the retina (blurred vision, can't focus on objects)

CONGESTIVE HEART FAILURE (CHF)
Overworking of the heart muscle (ventricle enlarges)

RENAL FAILURE
Too much blood flowing to the kidneys at a fast rate & high pressure

Risk Factors

PRIMARY HTN (MOST COMMON)

Also called **ESSENTIAL** or **IDIOPATHIC HTN**

- Cause is unknown
- Not curable, only controllable

- **R** **R**ace (African Americans)
- **I** **I**ntake of Na/ETOH
- **S** **S**moking
- **K** Low **K**⁺ & vitamin D levels

- **F** **F**amily history
- **A** **A**dvanced age
- **C** ↑ **C**holesterol
- **T** **T**oo much caffeine
- **O** **O**besity
- **R** **R**estricted activity
- **S** **S**leep apnea

SECONDARY HTN

Has a direct cause/preexisting condition

- Chronic kidney disease
- Diabetes
- Hypo/Hyperthyroidism
- Cushing's disease
- Pregnancy
- Certain drugs (oral contraceptives)

Signs & Symptoms

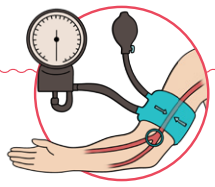
- Usually ***asymptomatic***
- **Symptoms *(if seen)*:**
  - Blurred vision
  - Headaches
  - Chest pain
  - Nosebleeds

Commonly called the "SILENT KILLER"

Patient Education

- Limit sodium intake
- Limit alcohol intake
- Stop smoking
- Measure BP & keep a record
- Participate in exercise programs for weight loss if needed

CHECKING BLOOD PRESSURE

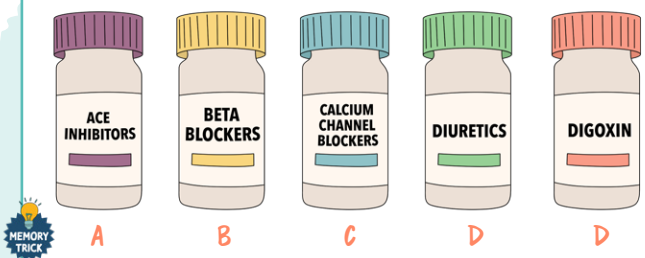


- Place stethoscope over brachial artery
- Patients should not smoke or exercise within 30 minutes of having their BP checked *(could lead to inflated BP)*
- Instruct the patient to:
  - Sit in a chair with legs uncrossed
  - Keep arm at heart level
- Use correct size cuff
- Do not assess BP in affected arm of patients with:
  - Mastectomy
  - History of AV shunt
  - Blood clots
  - PICC lines/central lines

Too small = false high BP

Too large = false low BP

ANTIHYPERTENSIVE MEDICATION OVERVIEW



SUFFIXES

- **A** ACE inhibitors — -pril
- **B** BETA blockers — -olol
- **C** Calcium channel blockers — -pine -amil
- **D** Diuretics
- **D** Digoxin

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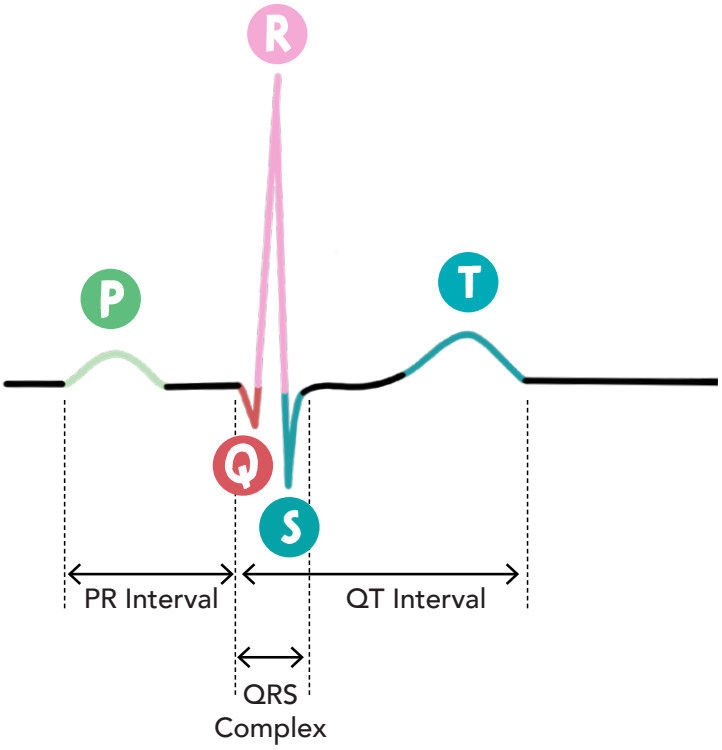
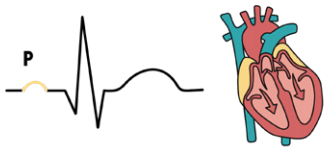
EKG Waveforms



P-WAVE

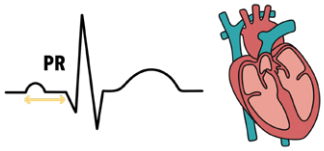


ATRIAL CONTRACTING

DEpolarization = DEcompressing

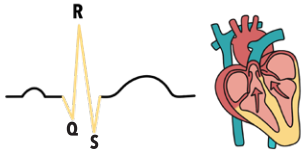
MEMORY TRICK

PR INTERVAL



The time between atrial depolarization & ventricular depolarization

QRS INTERVAL

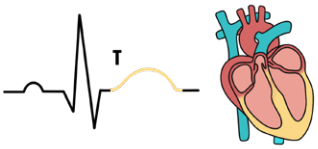


VENTRICLE CONTRACTING

DEpolarization = DEcompressing

MEMORY TRICK

T-WAVE



VENTRICLE RELAXING

REpolarization = RElaxing & REfilling with blood

MEMORY TRICK

Basic Rhythms

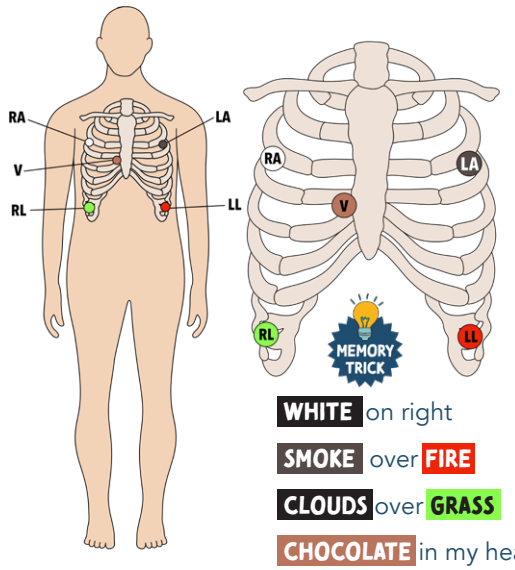
NORMAL SINUS	60 - 100 bpm
SINUS TACHYCARDIA	> 100 bpm
SINUS BRADYCARDIA	< 60 bpm

DEpolarization = DEcompressing

MEMORY TRICK

REpolarization =
- RElaxing
- REpolarizing
- REfilling with blood

5-LEAD EKG PLACEMENT



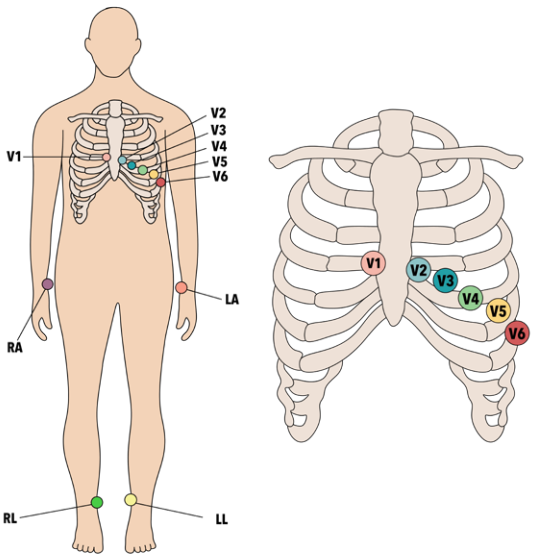
WHITE on right

SMOKE over **FIRE**

CLOUDS over **GRASS**

CHOCOLATE in my heart

12-LEAD EKG PLACEMENT



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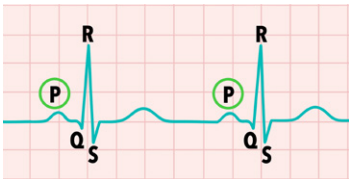
Steps to Interpreting EKGs

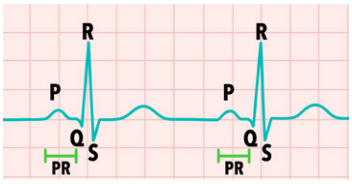
Step 1 — Identify the P-Wave



Evaluate the P-wave

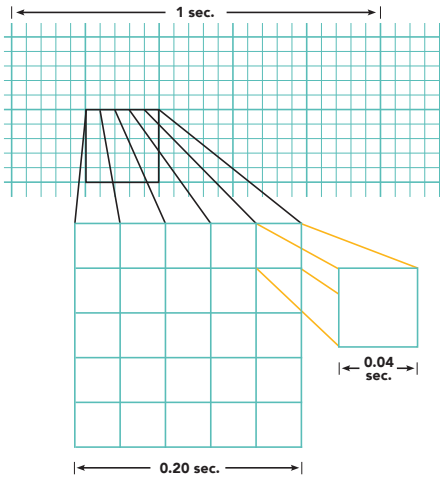
- Should be present & upright
- Comes before QRS Complex
- One P-wave for every QRS Complex

Step 2 — Identify the PR Interval



Calculate the PR interval

NORMAL PR INTERVAL
0.12 - 0.20 seconds

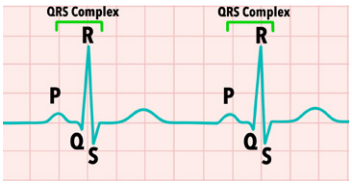


1 small box = 0.04 seconds
1 large box = 0.20 seconds
5 large boxes = 1 second

Heart Rhythm Measurements

PR INTERVAL	QRS COMPLEX	QT INTERVAL
0.12–0.20	0.06–0.12	< 0.40

Step 3 — Identify the QRS Complex



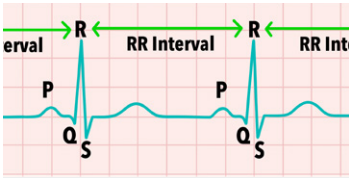
Evaluate the QRS complex

- Check whether every P-wave is followed by a QRS complex
- Should not be *widened or shortened*—this may indicate a problem

Widened is often seen in PVCs, electrolyte imbalances & drug toxicity!

NORMAL QRS COMPLEX
0.06 - 0.12 seconds

Step 4 — Identify the R-R Interval



Are the R-R intervals consistent?

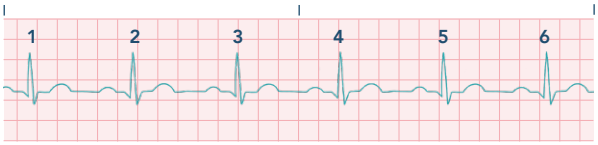
- Check whether they are regular or irregular

Step 5 — Determine the Heart Rate

6 SECOND METHOD

Count the number of Rs between the 6 second strips & multiply by 10

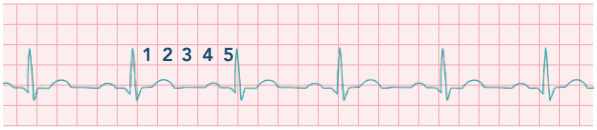
Be sure to check that the strip is 6 seconds! Count the boxes.



6 Rs x 10 = 60 beats per minute

BIG BOX METHOD

Divide 300 by the number of big boxes between 2 Rs



300 ÷ 5 = 60 beats per minute

Step 6 — Identify the EKG Finding!

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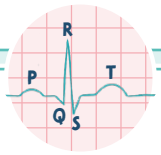
Normal Sinus Rhythm, Sinus Brady, Sinus Tachy

Normal Sinus Rhythm



RATE	60 - 100 bpm
RHYTHM	Regular
P-WAVE	Upright & uniform before each QRS
PR INTERVAL	Normal
QRS COMPLEX	Normal

Sinus Bradycardia

The sinus node creates an impulse at a ***slower***-than-normal rate

RATE	< 60 bpm
RHYTHM	Regular
P-WAVE	Upright & uniform before each QRS
PR INTERVAL	Normal
QRS COMPLEX	Normal

CAUSES

- Lower metabolic needs
  - Sleep
  - Athletic training
  - Hypothyroidism
- Vagal stimulation
- Certain medications
  - Calcium channel blockers
  - Beta blockers
  - Amiodarone

THIS IS NORMAL: Athletes have a low RESTING heart rate. This is because the heart is strong and pumps more efficiently than in non-athletes

TREATMENT

- Correct the underlying cause
- ↑ Heart rate to normal

Sinus Tachycardia

The sinus node creates an impulse at a ***faster***-than-normal rate

RATE	> 100 bpm
RHYTHM	Regular
P-WAVE	Upright & uniform before each QRS
PR INTERVAL	Normal
QRS COMPLEX	Normal

CAUSES

- Physiological or psychological stress
  - Blood loss, fever, exercise, dehydration, infection, sepsis
- Certain medications
  - Stimulants: *caffeine, nicotine*
  - Illicit drugs: *cocaine, amphetamines*
  - Drugs that stimulate sympathetic response: *epinephrine*
  - Beta-2 agonists
- Heart failure
- Cardiac tamponade
- Hyperthyroidism

TREATMENT

- Correct the underlying cause
- ↓ Heart rate to normal

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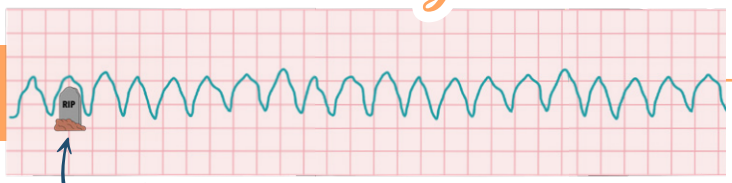
Ventricular Tachycardia

Ventricular Tachycardia (VT)



MEMORY TRICK
Look like a tombstone

Waveforms are irregular, coarse, and of different shapes. The ventricles are quivering, and there is **no contraction or cardiac output**, which may be **fatal**!

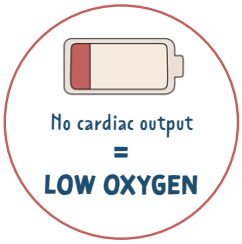
RATE	100 - 250 bpm
RHYTHM	Regular
P-WAVE	Not visible
PR INTERVAL	None
QRS COMPLEX	Wide (like tombstones) > 0.12 seconds

CAUSES

- Myocardial ischemia/infarction
- Electrolyte imbalances
- Digoxin toxicity
- Stimulants: caffeine & methamphetamine

MANIFESTATIONS

- Usually awake (unlike V-fib)
- Chest pain
- Lethargy
- Anxiety
- Syncope
- Palpitations



TREATMENT

STABLE PATIENT WITH A PULSE

- Oxygen
- Antiarrhythmics (stabilizes the rhythm)
  - Amiodarone
- Synchronized cardioversion
  - Synchronized administration of shock (delivery in sync with the QRS wave)
  - Cardioversion is NOT defibrillation! (defibrillation is only given with deadly rhythms)

UNSTABLE PATIENT WITHOUT A PULSE

Also called **PULSELESS V-TACH**

- CPR
- ACLS protocol for defibrillation
- Possible intubation
- Drug therapy
  - Epinephrine, vasopressin, amiodarone

SHOCK!

UNTREATED VT which can lead to → **VENTRICULAR FIBRILLATION** → **DEATH**

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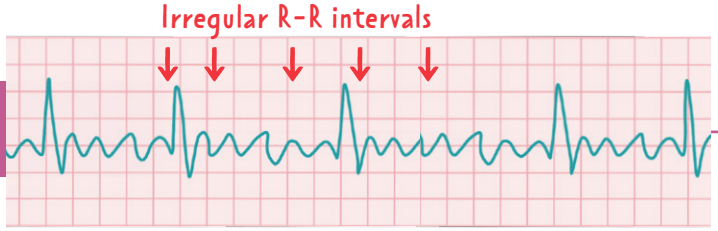
Atrial Fibrillation

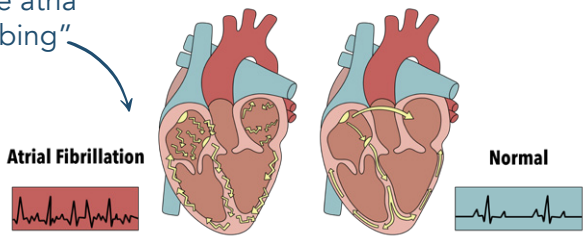
Atrial Fibrillation (A-FIB)



THE ATRIA ARE QUIVERING!

RATE	Usually > 100 bpm
RHYTHM	Irregular
P-WAVE	None (fibrillatory waves may exist, but these are not P-waves)
PR INTERVAL	Visible
QRS COMPLEX	Narrow & irregular

Uncoordinated electrical activity in the atria that causes rapid & disorganized "fibbing" of the muscles in the atrium



CAUSES

- Open heart surgery
- Heart failure
- COPD
- Hypertension
- Ischemic heart disease

MANIFESTATIONS

- Most commonly asymptomatic
- Fatigue
- Malaise
- Dizziness
- Shortness of breath
- Tachycardia
- Anxiety
- Palpitations

All due to low O_2

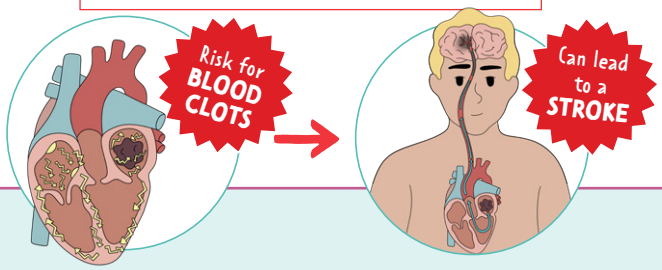
TREATMENT

STABLE PATIENT

- Oxygen
- Drug therapy
  - Beta blockers
  - Calcium channel blockers
  - Digoxin
  - Amiodarone
  - Anticoagulant therapy to prevent clots

RISK FOR CLOTS

The atrial quiver causes pooling of blood in the heart, which increases the risk for clots
= ↑ risk for MI, PE, CVA & DVTs



UNSTABLE PATIENT

- Oxygen
- Cardioversion
  - Synchronized administration of shock (delivery in sync with the QRS wave)
  - Cardioversion is NOT defibrillation!

DEFIBRILLATION

Defibrillation is only given with deadly rhythms

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Premature Ventricular Contractions (PVCs) & Asystole

Premature Ventricular Contractions (PVC)

Early or premature conduction of a QRS complex

RATE	Depends on the underlying rhythm
RHYTHM	Regular but interrupted due to early P-waves
P-WAVE	Visible but depends on the timing of the PVC (may be hidden)
PR INTERVAL	Slower than normal but still 0.12–0.20 seconds
QRS COMPLEX	Sharp, bizarre, and abnormal during the PVC

CAUSES

- Heart failure
- Cardiomyopathy
- Electrolyte imbalances
- Myocardial ischemia/infarction
- Drug toxicity
- Caffeine, tobacco, alcohol
- Stress or pain
- ↑ Workload on the heart
  - Exercise
  - Fever
  - Hypervolemia
  - Heart failure
  - Tachycardia

BIGEMINY: every **other** beat
TRIGEMINY: every **3rd** beat
QUADRIGEMINY: every **4th** beat

R-ON-T PHENOMENON: PVC arises spontaneously from the repolarization gradient (T-wave) & may precipitate V-fib

TREATMENT

Treatment is based on underlying cause

- Administer oxygen
- ↓ Caffeine intake
- Correct electrolyte imbalances
- D/C or adjust the drug causing toxicity
- ↓ Stress or pain

Treatment may not be needed if the patient has a healthy heart

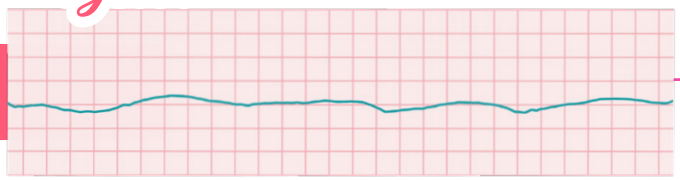
MANIFESTATIONS

- May be asymptomatic
- Feels like heart:
  - skipped a beat
  - is pounding
- Chest pain

CHEST PAIN

Notify the healthcare provider if the patient complains of chest pain, if the PVC increases in frequency, or if the PVC occurs on the T-wave (R-on-T phenomenon)

Asystole



~~RATE~~
~~RHYTHM~~
~~P-WAVE~~
~~PR INTERVAL~~
~~QRS COMPLEX~~

FLATLINE

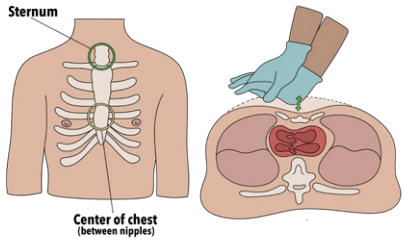
CAUSES

- Myocardial ischemia/infarction
- Heart failure
- Electrolyte imbalances (common: hypo/hyperkalemia)
- Severe acidosis
- Cardiac tamponade
- Illicit drug overdose

TREATMENT

Asystole is a non-shockable rhythm & defibrillators cannot be used

HIGH-QUALITY CPR



- Heel of hand on center of chest
- Arms straight
- Shoulders aligned over hands
- Compress at 2–2.4 inches at a rate of 100–120 per minute
- 30 compressions to 2 rescue breaths
- Minimal interruptions

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Atrial Flutter & Ventricular Fibrillation (V-Fib)

Atrial Flutter

RATE	75–150 bpm
RHYTHM	Usually regular
P-WAVE	"**Sawtooth**"-shaped flutter waves
PR INTERVAL	Unable to measure
QRS COMPLEX	Usually normal & upright

Similar to A-fib, but the heart's electrical signals spread through the atria (heart's upper chambers), which are beating too quickly though at a regular rhythm

CAUSES

- Coronary artery disease (CAD)
- Hypertension
- Heart failure
- Valvular disease
- Hyperthyroidism
- Chronic lung disease
- Pulmonary embolism
- Cardiomyopathy

MANIFESTATIONS

- May be asymptomatic
- Fatigue/syncope
- Chest pain
- Shortness of breath
- Low blood pressure
- Palpitations
- Dizziness

TREATMENT

STABLE PATIENT

- Drug therapy
  - Calcium channel blockers
  - Antiarrhythmics
  - Anticoagulants

RISK FOR CLOTS
Atrial flutter causes pooling of blood in the atria = risk for clots

UNSTABLE PATIENT

- Cardioversion
  - Synchronized administration of shock (delivery in sync with the QRS wave)
  - Cardioversion is NOT defibrillation!

DEFIBRILLATION
Defibrillation is only given with deadly rhythms

Ventricular Fibrillation (V-Fib)

RATE	Unknown
RHYTHM	Chaotic & irregular
P-WAVE	Not visible
PR INTERVAL	Not visible
QRS COMPLEX	Not visible

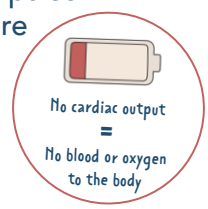
Rapid, disorganized pattern of electrical activity in the ventricle in which electrical impulses arise from many different foci

CAUSES

- Cardiac injury
- Medication toxicity
- Electrolyte imbalances
- Untreated ventricular tachycardia

MANIFESTATIONS

- Loss of consciousness
- May not have a pulse or blood pressure
- Respirations may stop
- Cardiac arrest



TREATMENT

- CPR
- Oxygen
- Defib
  (follow ACLS protocol for defibrillation)

MEMORY TRICK "Defib the V-fib"

- Possible intubation
- Drug therapy
  - Epinephrine (causes vasoconstriction)
  - Antiarrhythmics: Amiodarone, lidocaine
  - Possibly magnesium

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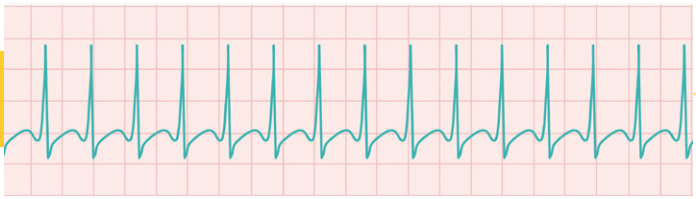
Supraventricular Tachycardia (SVT)

Supraventricular Tachycardia (SVT)



RATE	> 100 bpm (often 180–220 bpm)
RHYTHM	Regular
P-WAVE	None
PR INTERVAL	Unable to identify (no P-wave)
QRS COMPLEX	Narrow (< 0.12 seconds)

"SUPRA" = ABOVE
Occurs as extra & abnormal electrical activity **ABOVE** the ventricles & AV node

CAUSES

- "Accessory" (or extra) electrical pathway → Can be *Wolff-Parkinson-White syndrome*
- Reentering of electrical signals FROM ventricles to atria
- High levels of stress (physical or mental)
- Electrolyte imbalances

This is NOT supposed to happen! **IMAGINE:** that the impulses are turning around on a one-way street and re-entering the atria.

MANIFESTATIONS

- Hypotension (low BP)
- Shortness of breath
- Dizziness
- Chest discomfort

All due to ↓ perfusion

PAROXYSMAL SVT
Rhythm occurs intermittently with normal sinus rhythm in between

TREATMENT

STABLE PATIENT

- Valsalva maneuver
  - Teach patient to hold their breath & bear down as if having a bowel movement
- Carotid massage
  - Vagal maneuvers (activate parasympathetic nervous system & RELAX the heart)
- Cold water or ice on face
- If non-pharmacological treatment does not work...

MEMORY TRICK
SVT:
Start with
Vagal
Treatment

MEMORY TRICK
ABCDs of **SVT**s:
Adenosine
Beta-Blockers
Cardiac Ablation (ablate or burn away extra pathway)
Digoxin

UNSTABLE PATIENT

- Synchronized cardioversion

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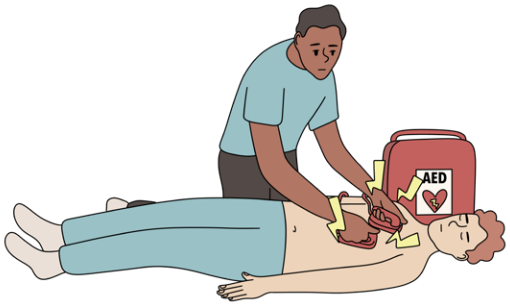
Cardioversion vs. Defibrillation



Cardioversion		Defibrillation
SYNCHRONIZED SHOCK Synced shock delivered only during the R-wave of the QRS complex Synchronizer switch must be turned on! If the shock is accidentally delivered during the **T-wave**, it can cause **R-on-T phenomenon**	SHOCK	**ASYNCHRONOUS SHOCK** Delivered with an automated external defibrillator (AED)
Lower number of joules (energy) used	JOULES	**Higher** number of joules (energy) used
✗ Not combined with CPR	CPR	✓ Resume CPR after shock
Stable patients who have a pulse (must have a QRS complex)	USES	**Unstable patients** or patients who need to be resuscitated back to life
Atrial fibrillation (A-fib)	EXAMPLES	Pulseless ventricular tachycardia (VT) Ventricular fibrillation (VF)
Cardioversions are Carefully planned in a Controlled environment *(Patients are sedated for procedure & may not always require hospitalization)*		Defibrillation is for Deadly rhythms

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Lab Values Related to the Endocrine System

Thyroid Panel

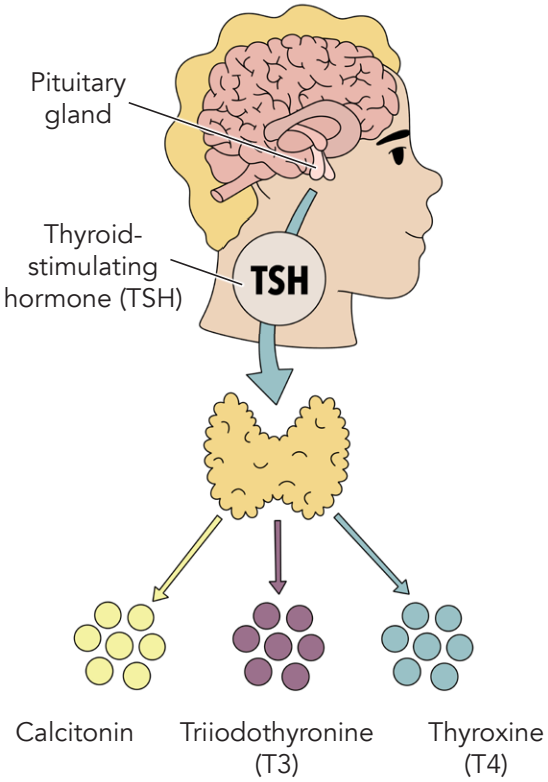
	EXPECTED RANGE
T3 (Triiodothyronine)	**80 – 220** ng/dL
T4 (Thyroxine)	**4 – 12** mcg/dL
THYROID-STIMULATING HORMONE (TSH)	**0.5 – 5** mU/L

T3 & T4 levels
always move in the opposite direction of
TSH levels (negative feedback mechanism)

↓

HYPERTHYROIDISM:
↑ T3 & T4 ↓ TSH

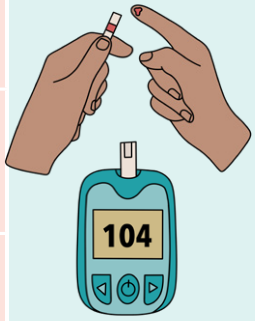
HYPOTHYROIDISM:
↓ T3 & T4 ↑ TSH



Blood Glucose

	EXPECTED RANGE	DESCRIPTION
CASUAL/RANDOM GLUCOSE TEST	**70 – 110** mg/dL	Target for any time of day (doesn't matter when the last meal was)
FASTING BLOOD SUGAR (FBS)	**< 100** mg/dL	Level after no caloric intake for at least 8 hours
ORAL GLUCOSE TOLERANCE TEST (OGTT)	**< 140** mg/dL	Test requiring ingestion of 75g of glucose dissolved in water, then measuring blood sugar levels every hour for up to 3 hours
HBA1C	**< 5.7%**	Blood test that measures the average blood glucose (sugar) levels for the last 2–3 months

A finger stick blood sugar test is the most common way people with diabetes check their blood glucose levels



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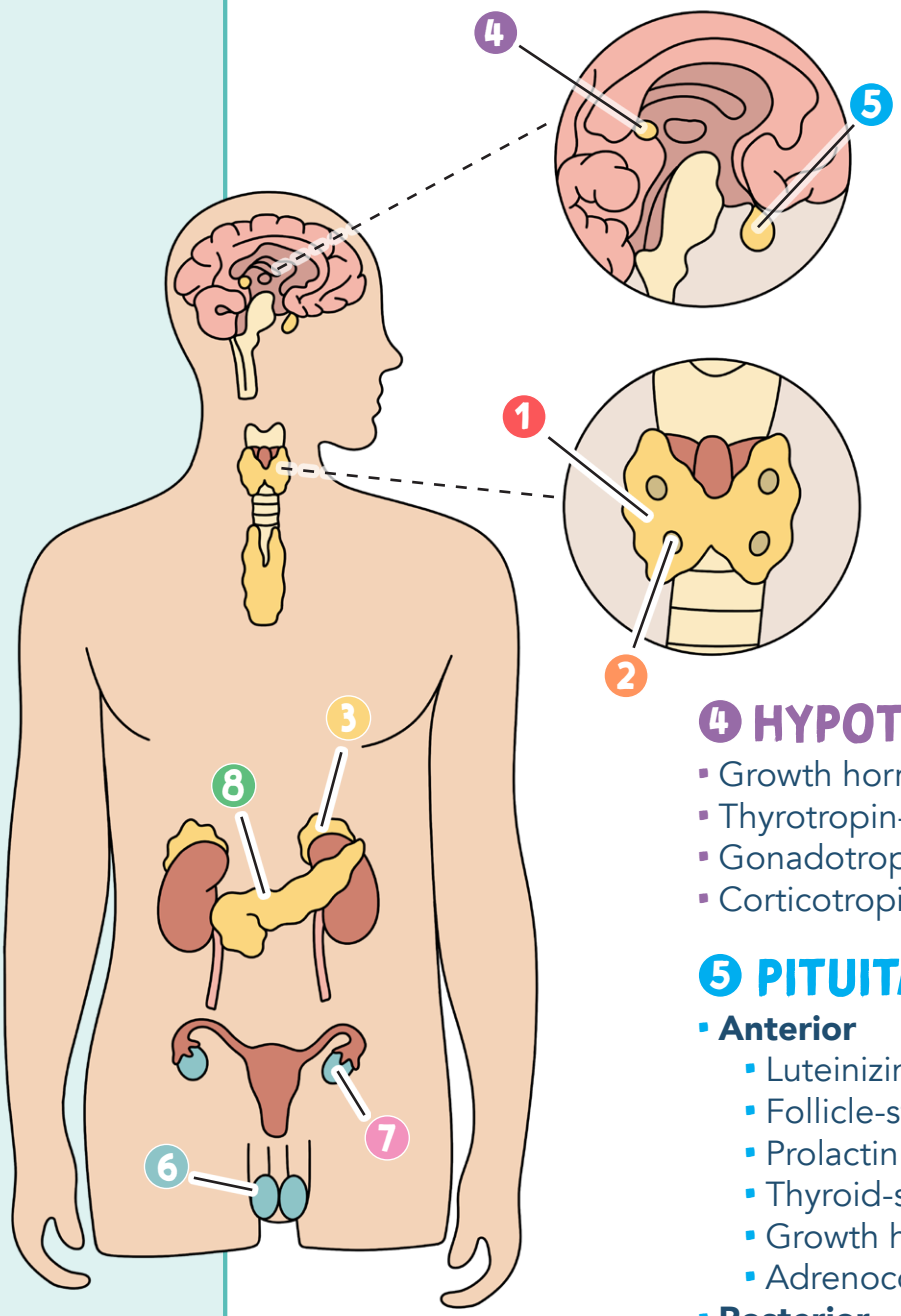
Endocrine System Overview

Function of the ENDOCRINE SYSTEM: The endocrine system is made up of glands & organs that release hormones (chemical messengers). These chemical messengers carry information & instructions from one cell to another.

HORMONES RELEASED by the endocrine organs/glands



1 THYROID GLAND
- Thyroxine (T4)
- Triiodothyronine (T3)
- Calcitonin

2 PARATHYROID GLAND
- Parathyroid hormone (PTH)

3 ADRENAL GLAND
- **Adrenal cortex**
  - Aldosterone
  - Cortisol
- **Adrenal medulla**
  - Epinephrine
  - Norepinephrine

4 HYPOTHALAMUS
- Growth hormone-releasing hormone (GHRH)
- Thyrotropin-releasing hormone (TRH)
- Gonadotropin-releasing hormone (GnRH)
- Corticotropin-releasing hormone (CRH)

5 PITUITARY GLAND
- **Anterior**
  - Luteinizing hormone (LH)
  - Follicle-stimulating hormone (FSH)
  - Prolactin
  - Thyroid-stimulating hormone (TSH)
  - Growth hormone (GH)
  - Adrenocorticotropic hormone (ACTH)
- **Posterior**
  - Antidiuretic hormone (ADH) (Vasopressin)
  - Oxytocin

6 TESTES
- Testosterone

7 OVARIES
- Estrogen
- Progesterone

8 PANCREAS
- Insulin
- Glucagon

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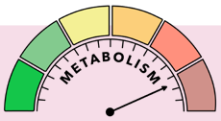
Endocrine Hormones

Thyroxine (T4) Triiodothyronine (T3)	Created and stored in the thyroid Maintain body metabolism in a steady state 
Calcitonin	Secreted by the thyroid Regulates **calcium** in the body CALCitonin – think CALCium
Thyroid-Stimulating Hormone (TSH)	Stimulates the thyroid, causing T3 & T4 to be released
Oxytocin	Causes muscle contractions to help expel the baby during childbirth or to contract the uterus after childbirth (prevention of hemorrhage)
Prolactin	Stimulates milk production after childbirth
Insulin	Works to **DECREASE** blood glucose levels Puts sugar & potassium into the cells to be used later as energy
Glucagon	Works to **INCREASE** blood glucose levels Breaks down stored glucose (glycogen) in the liver
Epinephrine & Norepinephrine	Catecholamines (stress hormones) that are released when blood pressure drops Help in times of **ACUTE** stress
Cortisol	Glucocorticoid (stress hormone) which helps regulate metabolism, ↑ blood glucose levels, & reduces inflammation Helps in times of **CHRONIC** stress
Antidiuretic Hormone (ADH)	Helps regulate the amount of water in the body
Aldosterone	Mineralocorticoid that helps balance fluids
Parathyroid Hormone (PTH)	Helps to increase serum calcium in the blood
Estrogen	Helps to regulate the menstrual cycle, stimulates uterus growth during pregnancy, maintains the pregnancy, and supports the fetus as it grows
Progesterone	Helps to regulate the menstrual cycle, stimulates growth of maternal tissues & fetal organs during pregnancy Progesterone – think Pregnancy hormone
Testosterone	Helps in the development of male sex organs & reproductive tissue, plays a vital role in sperm production, promotes secondary sex characteristics (↑ bone mass, ↑ muscle mass, ↑ growth of body hair) TESTosterone – think TESTes

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Negative vs. Positive Feedback Loop

How do I counteract this?

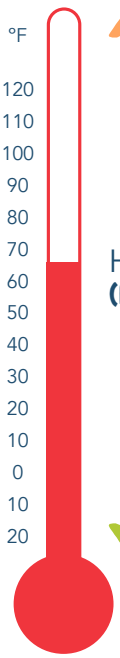
Negative FEEDBACK LOOP

Hormone increases in the body:
Additional production of that hormone is inhibited

Hormone decreases in the body:
Production of that hormone increases

Example:

THERMOSTAT ANALOGY



Heat turns off

Once the room reaches above 65°F, the **heat turns off.**

Heat set at 65°F **(HOMEOSTASIS)**

Let's say you set the heat in a room to 65°F. We'll call this temperature **HOMEOSTASIS**.

Heat turns on

When the temperature falls below 65°F again, the heat **turns back on.**

Hormones in the endocrine system work the same way!

More, More, More!

Positive FEEDBACK LOOP

Almost all endocrine hormones are regulated by a negative feedback loop, but some hormones are regulated by a positive feedback loop

Increase of a **HORMONE** causes another **VARIABLE** to increase, which causes the hormone to increase even more

Example:

OXYTOCIN

Variable — Childbirth

↓

Hormone — Pressure on the cervix causes a release of **OXYTOCIN**

↓

Variable — This triggers contractions

↓

Hormone — This causes the release of more **OXYTOCIN**

↓

Variable — This causes more contractions & pressure

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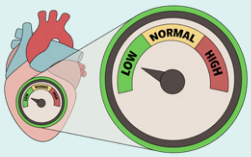
Renin Angiotensin Aldosterone System (RAAS)

Purpose:

The RAAS systems help to regulate blood pressure, systemic vascular resistance, and electrolyte balances.

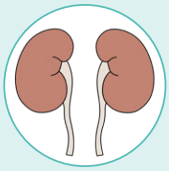
Pathway:

START:
Blood pressure drops



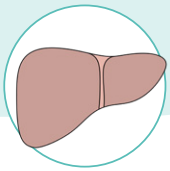
Sympathetic nervous system is stimulated

Kidneys (juxtaglomerular apparatus [JGA] region) release **RENIN**



RENIN is the first part of the RAAS system

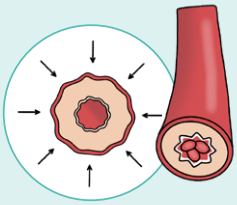
Activates **ANGIOTENSINOGEN** (created and released by the liver)



Creates **ANGIOTENSIN 1**

ACE (angiotensin-converting enzymes) converts **ANGIOTENSIN 1** ↓ **ANGIOTENSIN 2**

ANGIOTENSIN 2 acts on the smooth muscle, which **constricts** blood vessels and increases blood volume



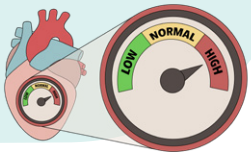
HOW DOES THIS HAPPEN?

KIDNEYS
Hold on to Na+ & H_2O

PITUITARY GLAND
Secretes antidiuretic hormone (ADH), which keeps more H_2O in the vessels

ADRENAL CORTEX
Stimulates **ALDOSTERONE**, which causes the kidneys to reabsorb Na+ & H_2O and excrete potassium

END RESULT:
Increased blood pressure



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Diabetes: Type 1 & Type 2

	TYPE 1 DIABETES MELLITUS (T1DM)	TYPE 2 DIABETES MELLITUS (T2DM)
PATHOLOGY	**PRODUCES NO INSULIN** MEMORY TRICK: Type *ONE* – we have n*ONE* ▪ Caused by an autoimmune response ▪ The cells are starved of glucose since there is no insulin to bring glucose into the cells ▪ The cells break down protein & fat into energy, causing ketones to build up = **ACIDOSIS**!	**DOES NOT PRODUCE ENOUGH INSULIN, OR PRODUCES BAD INSULIN THAT DOES NOT WORK PROPERLY** MEMORY TRICK: *T*ype *T*wo think *T*echnical difficulty ▪ Caused by insulin resistance ▪ Insulin receptors are worn out & not working properly!
ONSET	**ABRUPT** Usually diagnosed in **CHILDHOOD**	**GRADUAL** Usually diagnosed in **ADULTHOOD**
RISK FACTORS	▪ Genetics ▪ Family history	▪ High blood sugar ▪ Hypertension ▪ Obesity ▪ Inactivity ▪ High cholesterol ▪ Family history ▪ Smoking
SIGNS & SYMPTOMS	3 Ps **Polyphagia**: excessive hunger	 **Polydipsia**: excessive thirst **Polyuria**: excessive urination
TREATMENT	Only has **1** treatment: **INSULIN** Oral hypoglycemic agents will not work for type 1 **Insulin-dependent for life!** 	Has **2+** treatments: 1. **DIET & EXERCISE** 2. **ORAL HYPOGLYCEMIC AGENTS** *Example:* Metformin 3. **POSSIBLY INSULIN** ▪ Insulin may or may not be given routinely ▪ Insulin may be given initially or be increased during times of stress, surgery, or sickness

MEMORY TRICK: **CHILDHOOD** comes 1ST in life, and **ADULTHOOD** comes 2ND

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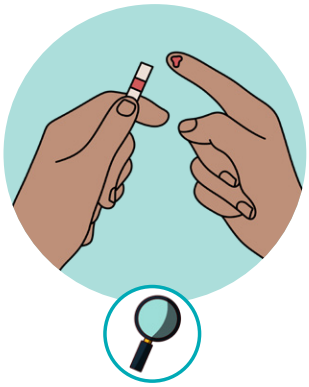
Diabetes: Sick Day Management & Diagnostic

Sick Day Management

When the body experiences sickness, it causes the release of hormones that increase blood sugar levels. This makes it hard to manage diabetes when a patient is sick or experiencing an illness.

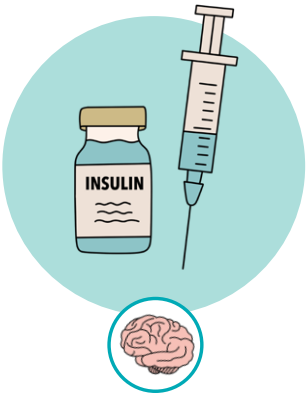


MONITOR:

- Blood glucose
- Temperature
- Urine for ketones

REPORT:

- If ketones are present in urine
- If blood sugar is
  - > 250 mg/dL
  - < 70 mg/dL
- If temperature is > 101° F



PATIENT EDUCATION:

- Stay hydrated (avoid dehydration)
- **Do not skip insulin or oral hypoglycemic agents when feeling sick (take them as usual)**

Diagnostic Criteria For Diabetes

CASUAL/RANDOM GLUCOSE TEST

Target for any time of the day (doesn't matter when the last meal was)

> 200 mg/dL

FASTING BLOOD SUGAR (FBS)

Level after no caloric intake for at least 8 hours

> 126 mg/dL

ORAL GLUCOSE TOLERANCE TEST (OGTT)

Drink 75g of glucose dissolved in water, then measure blood sugar levels every hour for up to 3 hours

> 200 mg/dL

HBA1C

Blood test that measures the average blood glucose (sugar) levels for the last 2-3 months

> 6.5%

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Diabetes: Complications & Foot Care

Diabetic Complications

Diabetes can negatively affect almost every organ system.
This is because high levels of sugar in the blood damage the blood vessel walls and the nerves.

ORGAN AFFECTED	KIDNEYS	NERVES	EYES	HEART	BRAIN
	NEPHROPATHY Kidney damage Excessive blood glucose can damage the tiny blood vessels in the filtering system (glomeruli).	**PERIPHERAL NEUROPATHY** Excessive blood glucose can damage the nerves outside the brain & spinal cord.	**DIABETIC RETINOPATHY** Eye damage Excessive blood glucose can damage the blood vessels of the retina.	**CARDIOVASCULAR DISEASE** Damage to the heart & major coronary arteries Excessive blood glucose can damage the blood vessels & nerves controlling the heart.	**STROKE** Excessive blood glucose can damage the blood vessels and makes them stiff – it can also cause a buildup of fatty deposits.
COMPLICATIONS	**This may cause kidney failure & even end-stage kidney disease.**	**This may cause tingling, numbness & eventually loss of sensation.** Nerve damage in the foot can cause serious complications such as major infections in cuts & blisters. **All this sugar in the blood also causes delayed wound healing = risk for infection**	**This may cause blindness, cataracts & glaucoma.**	**This may cause coronary artery disease, hypertension & atherosclerosis.**	**This may cause a blood clot that travels to the brain, causing a stroke.**

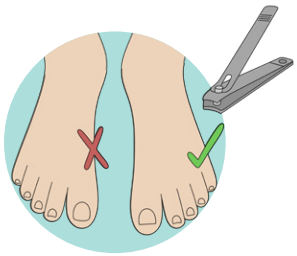
Diabetic Foot Care

Wash feet daily
Use warm water, not hot water (test temperature beforehand)

Gently pat feet completely dry
(dry between toes as well)

Inspect feet daily with a mirror
(check for any cuts, blisters, swelling, or sores)

Avoid over-the-counter products
(callus remover, alcohol, etc.)



Cut toenails straight across & file the edges

Improve blood flow
- Do not cross legs
- Elevate feet when sitting

Use lotion to keep skin soft, but avoid putting lotion between toes

Always wear shoes & socks
Never walk barefoot

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Hyperglycemia vs. Hypoglycemia

	HYPERGLYCEMIA	HYPOGLYCEMIA
PATHOLOGY	↑ **BLOOD SUGAR** > 200 mg/dL	↓ **BLOOD SUGAR** < 70 mg/dL
ONSET	Gradual (hours to days)	Sudden
RISK FACTORS	MEMORY TRICK: 4 Ss that cause an increase in Sugar - **S**teroids - **S**epsis (infection) - **S**tress - **S**kipping insulin or oral hypoglycemic agents - Not eating a diabetic diet	- Exercise - Swimming, cycling, college athletics - Alcohol - More likely to occur during peak time of insulin - Skipping a meal after taking certain insulins **RAPID-ACTING INSULIN** has the highest risk for hypoglycemia
SIGNS & SYMPTOMS	MEMORY TRICK: The 3 Ps **P**olyuria: Excessive urination **P**olydipsia: Excessive thirst **P**olyphagia: Excessive hunger - Dry mouth/dehydration - Fruity breath - Deep, rapid breaths (air hunger) - Numbness & tingling - Slow wound healing - Vision changes	- Palpitations - Sweating (diaphoresis) - Shakiness - Headaches - Fatigue & weakness - Confusion - Inability to arouse from sleep - Can lead to coma These are all neurological symptoms – the brain needs glucose to function
SKIN	Hot & dry skin — MEMORY TRICK: Hot & DRY – Sugar's HIGH	Cold & clammy skin — MEMORY TRICK: Cool & Clammy – need some Candy
TREATMENT	- Administer insulin as needed - Test urine for ketones - Follow a diabetic diet **DIABETIC DIET** also called a *consistent carb diet* ✓ Complex carbohydrates ✓ Fiber-rich foods ✓ Heart-healthy fish ✓ "Good fats" ✓ Sugar-free fluids ✗ Saturated fats ✗ Trans fats ✗ Cholesterol ✗ Sodium	**CONSCIOUS PATIENTS** 15 X 15 X 15 Oral intake of **15 GRAMS** of carbohydrates Recheck blood glucose in **15 MINUTES** Give another **15 GRAMS** of carbohydrates if needed **UNCONSCIOUS PATIENTS** ⚠ **EMERGENCY**: call a rapid response Administer IV 50% dextrose (D50) **OR** glucagon (IM, IV, subQ) Do not put anything in an unconscious patient's mouth—they can **ASPIRATE**!

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DKA vs. HHNS

	DIABETIC KETOACIDOSIS (DKA)	HYPERGLYCEMIC HYPEROSMOLAR NONKETOTIC SYNDROME (HHNS)
PATHOLOGY	Happens mostly in **Type 1** diabetic patients Not enough insulin ↓ Body can't allow blood sugar into the cells for energy ↓ Blood sugar becomes VERY high ↓ Cells break down protein & fat into energy ↓ **KETONES** build up = **ACIDOSIS!** Ketones are a byproduct of metabolism	Happens mostly in **Type 2** diabetic patients NO acidosis present Simply high amounts of glucose in the blood ACIDOSIS ✗ KETONES ✗
RISK FACTORS	4 Ss ▪ **S**tress (surgery) ▪ **S**epsis (infection) ▪ **S**kipping insulin ▪ **S**tomach (stomach virus: nausea/vomiting) ▪ Undiagnosed diabetes	▪ Inadequate fluid intake ▪ ↓ Kidney function ▪ Infection ▪ Stress ▪ Older age
SIGNS & SYMPTOMS — **GLUCOSE LEVELS**	300–500 mg/dL	HYPERGLYCEMIA > 600 mg/dL
SIGNS & SYMPTOMS — **ONSET**	**ABRUPT**	**GRADUAL**
SIGNS & SYMPTOMS — **METABOLIC ACIDOSIS**	✓ Metabolic acidosis	✗ No metabolic acidosis
SIGNS & SYMPTOMS	▪ Kussmaul respirations (trying to blow off CO_2) ★ Acidic breath or "fruity breath" 	▪ 3 **P**s (**P**olyuria, **P**olydipsia, **P**olyphagia) ▪ Neurovascular changes (confusion, ↓ LOC, headache)
SIGNS & SYMPTOMS	Dehydration (hypovolemia)	
TREATMENT	★ IV insulin with potassium (K^+) ▪ Fluid replacement (IV fluids) ▪ Correction of electrolyte imbalance ▪ Administration of bicarbonate for metabolic acidosis (only for DKA)	MEMORY TRICK: *IN*sulin causes sugar & K+ to go *IN* the cells, causing hypokalemia unless we administer K^+ with IV insulin MEMORY TRICK: For DKA, remember to monitor K^+ levels MEMORY TRICK: Regular goes Right into the vein Regular insulin is the only insulin given via IV

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Cushing's Syndrome vs. Addison's Disease

Disorder of the **adrenal cortex,** which produces...

CUSHING'S

PATHOLOGY

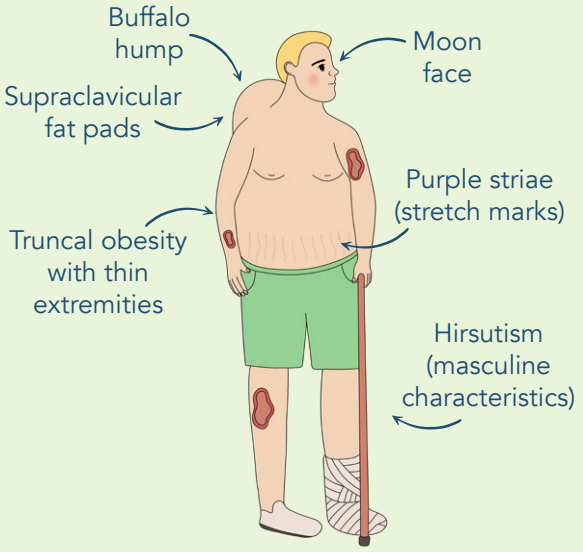
TOO MANY steroids

They have a CUSHION

RISK FACTORS

- Overuse of cortisol medications
- More prevalent in females
- Tumor in the adrenal gland that secretes cortisol

SIGNS & SYMPTOMS



VITAL SIGNS & LAB VALUES

CUSHINGs is PUSHING levels up

↑	BLOOD PRESSURE
↑	FLUID VOLUME
↑	WEIGHT
↑	BLOOD SUGAR
↑	SODIUM & WATER
↓	POTASSIUM

TREATMENT

- Adrenalectomy
  - Requires lifelong glucocorticoid replacement after surgery
- Chemotherapeutic agents (if adrenal tumor is present)

PATIENT EDUCATION

EDUCATE
- Avoid stress

ADDISON'S

PATHOLOGY

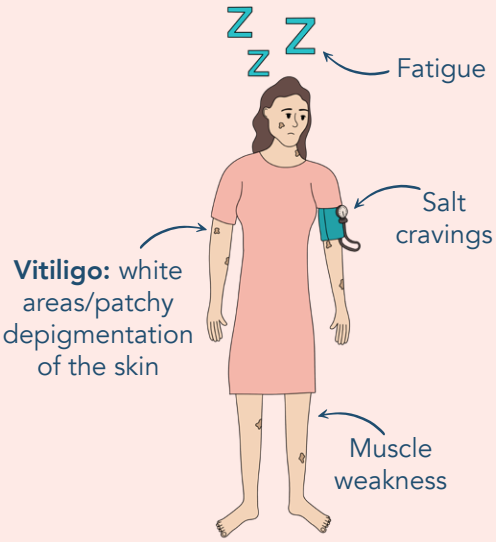
NOT ENOUGH steroids

We need to ADD some

RISK FACTORS

- Surgical removal of one or both adrenal glands
- Infection of the adrenal glands
- Certain bacterial infections
- Tuberculosis (TB)
- Cytomegalovirus

SIGNS & SYMPTOMS



VITAL SIGNS & LAB VALUES

ADdison's = Arrows Down

TREATMENT

- Oral glucocorticoids (steroids)

SAVED BY THE SUFFIX: End in "**-sone**"; *hydrocorti**sone**, predni**sone***

PATIENT EDUCATION

DIET MODIFICATIONS
- ↑ Protein & carbohydrates

EDUCATE
- May need to increase steroid dose during stress

REPORT signs of an infection (corticosteroid use can cause immunosuppression)

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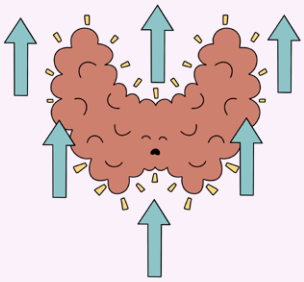
Hyperthyroidism vs. Hypothyroidism

HYPERTHYROIDISM

PATHOLOGY



EXCESSIVE production of thyroid hormone

TOO MUCH energy

RISK FACTORS

Graves' disease

- Too much iodine
- Toxic nodular goiter
- Thyroid replacement medication (Example: Levothyroxine toxicity)

TREATMENT

MEDICATIONS

- Iodine solution (Lugol's solution)
- Beta blockers (↓ HR & BP)
- Antithyroid medications
  - *Examples:* Methimazole, propylthiouracil (PTU)

Methimazole, Propylthiouracil (PTU)

MEMORY TRICK: Prevents Thyroid from being Up

TREATMENT

- Radioactive iodine therapy
- **THYROIDECTOMY**
  - thyroid
  - surgical removal of

(removal of all or part of the thyroid)

COMPLICATIONS

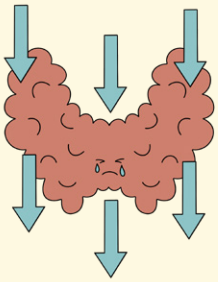
Thyroid storm

HYPOTHYROIDISM

PATHOLOGY

LOW production of thyroid hormone

NOT ENOUGH energy



RISK FACTORS

Iodine helps makes T3 & T4

Hashimoto's disease

- Not enough iodine
- Thyroidectomy
- Antithyroid medications

Affects women more often than men

TREATMENT

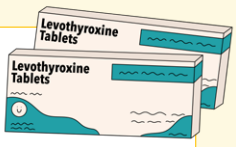
Hormone replacement

GENERIC	TRADE NAME
levothyroxine	Synthroid

MEMORY TRICK: levoTHYroxine think THYroid

MEMORY TRICK: SYNthetic THYroid

MEMORY TRICK: Levothyroxine is a LifeLong therapy



COMPLICATIONS

Myxedema coma

LIFE-THREATENING!

SIGNS & SYMPTOMS

(see next page)

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Hyperthyroidism vs. Hypothyroidism

Signs & Symptoms

	HYPERTHYROIDISM HYPERthyroidism = HIGH S&Ss		HYPOTHYROIDISM HYPOthyroidism = LOW S&Ss	
	SIGNS & SYMPTOMS	MEMORY TRICK	SIGNS & SYMPTOMS	MEMORY TRICK
THYROID HORMONES	↑ T3 & T4 ↓ TSH	HYPER = HIGH thyroid hormones (T3 & T4)	↓ T3 & T4 ↑ TSH	HYPOthyroidism = LOW thyroid hormones (T3 & T4)
MOOD	▪ Hyperexcitable ▪ Nervous (tremors) ▪ Irritable	TOO MUCH energy	▪ Depressed ▪ Fatigued	NO energy
TEMPERATURE	Heat intolerance	HYPER = HOT	Cold intolerance	HYPO = COLD
SKIN	▪ Sweaty skin ▪ Smooth, soft skin	HYPER = HIGH sweat production	Dry, brittle skin	HYPO = LOW sweat production
HAIR	Soft hair	HYPER = HIGH amounts of soft hair	Hair loss	HYPO = LOW amounts of hair
APPETITE	↑ Appetite	HYPER = HIGH appetite	↓ Appetite	HYPO = LOW appetite
WEIGHT	Weight loss	HYPER metabolism – think HUGE calories burned = weight loss	Weight gain	HYPO metabolism – think NO calories burned = weight gain
EYES	Exophthalmos	Exophthalmos = Excess thyroid Bulging eyes due to fluid accumulation behind the eyes	Myxedema	Generalized puffiness & edema
GI FUNCTION	Diarrhea	HYPER GI = HYPER bowel movements	Constipation	HYPO GI = NO bowel movements
VITAL SIGNS	↑ Blood pressure ↑ Heart rate	HYPER = HIGH vital signs	↓ Blood pressure ↓ Heart rate	HYPO = LOW vital signs
GOITER?	Yes	A goiter is an enlarged thyroid gland	Not typical	–

Other symptoms: *amenorrhea & low blood glucose*

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Hyperparathyroidism vs. Hypoparathyroidism

HYPERPARATHYROIDISM

PATHOLOGY

HIGH secretion of PTH by the parathyroid gland

↑ PTH
=
↑ Calcium
+
↓ Phosphorus

Calcium & phosphorus have an inverse relationship

RISK FACTORS

PRIMARY CAUSE
Tumor or hyperplasia of the parathyroid

SECONDARY CAUSE
Chronic kidney failure

SIGNS & SYMPTOMS

MEMORY TRICK: STONES, BONES, GROANS & MOANS

- **STONES:** Kidney stones (↑ calcium)
- **BONES:**
  - Skeletal pain
  - Pathological fractures from bone deformities

- Abdominal **GROANS**
  - Nausea, vomiting & abdominal pain
  - Weight loss/anorexia
  - Constipation

- Psychiatric **MOANS**
  - Mental irritability
  - Confusion

TREATMENT

- Parathyroidectomy

- Administration of:
  - Phosphates, calcitonin, & IV or oral bisphosphonates

DIET MODIFICATIONS

↑ Fiber & moderate calcium

HYPOPARATHYROIDISM

PATHOLOGY

LOW secretion of PTH by the parathyroid gland

↓ PTH
=
↓ Calcium
+
↑ Phosphorus

RISK FACTORS

Can occur due to the removal of the parathyroid gland (accidental or intentional removal)

- Thyroidectomy, parathyroidectomy, or radical neck dissection
- Exposure to radiation

The parathyroid gland is as small as a grain of rice

SIGNS & SYMPTOMS

- Numbness & tingling
- Muscle cramps
- Tetany
- Hypotension
- Anxiety, irritability & depression

Same S&S as hypocalcemia!

POSITIVE TROUSSEAU SIGN:
Carpal spasm caused by inflating a blood pressure cuff

CHVOSTEK'S SIGN:
Contraction of facial muscles with light tap over the facial nerve

MEMORY TRICK: Think "C" for Cheesy smile

TREATMENT

- Administration of:
  - IV calcium & phosphorus-binding drugs

DIET MODIFICATIONS

↑ Calcium ↓ Phosphorus

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SIADH vs. DI

SYNDROME OF INAPPROPRIATE ANTIDIURETIC HORMONE (SIADH)

DIABETES INSIPIDUS (DI)

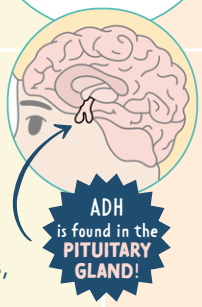
PATHOLOGY

SIADH: Too much **ADH**

RETAINS water

MEMORY TRICK: SIADH = Soaked Inside

ADH regulates & balances the amount of water in your blood



DI: Not enough **ADH**

LOSES water

MEMORY TRICK: DI = Dry Inside

RISK FACTORS

SIADH is often of non-endocrine origin

- **Damage to the central nervous system**
  - Head injury
  - Brain tumor
  - Infection of the brain
- Certain medications such as antidepressants, anticonvulsants, vincristine, etc.)

- Head trauma
- Brain tumor
- Infections of the central nervous system: Meningitis, encephalitis, or TB
- Manipulation of the pituitary gland
- Surgical ablation, craniotomy, sinus surgery, or hypophysectomy

SIGNS & SYMPTOMS

	SIADH	DI
URINE OUTPUT	Low urinary output of concentrated urine	High urinary output of diluted urine
VOLUME	CAUSES: Weight gain, Water intoxication — Fluid volume overload	Fluid volume deficit (dehydration) — CAUSES: Dry mucous membranes, ↓ Skin turgor, Polydipsia (↑ thirst)
VITAL SIGNS	↑ BP & ↓ HR	↓ BP & ↑ HR
URINE SPECIFIC GRAVITY	**High** urine specific gravity (> 1.030) — MEMORY TRICK: Concentrated urine makes the #s Curve up	**Low** urine specific gravity (< 1.005) — MEMORY TRICK: Diluted urine makes the #s go Down
SODIUM	Hyponatremia (**LOW** serum sodium)	Hypernatremia (**HIGH** serum sodium)
MISCELLANEOUS	▪ Muscle weakness ▪ Seizures	▪ Muscle pain & weakness ▪ Fatigue ▪ Headache

TREATMENT

DIET MODIFICATIONS:
- Restrict fluid intake
- Increase sodium intake
- Implement seizure precautions

Fix the underlying cause!

- Administer IV therapy (hypotonic solution)
- Provide safety precautions due to postural hypotension

MEDICATIONS

- Loop diuretics
- Vasopressin antagonists

- Administration of:
  - ADH replacement (vasopressin or desmopressin)

Replace the missing hormone!

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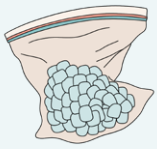
⚠ Endocrine Disorder Emergencies ⚠

Addisonian Crisis

PATHOLOGY	SIGNS & SYMPTOMS	TREATMENT
Acute adrenal insufficiency, which causes low levels of adrenal hormone	▪ Confusion ▪ Hyponatremia ▪ Hypokalemia ▪ Hypoglycemia ▪ Weak, rapid pulse ★ **Hypotension (leading to SHOCK!)**	Administer IV fluids & high-dose hydrocortisone

Thyroid Storm (THYROTOXICOSIS)

PATHOLOGY	SIGNS & SYMPTOMS	TREATMENT

Life-threatening complication of hyperthyroidism
 	
Excessively high levels of thyroid hormone
	Signs and symptoms are the same as those seen in hyperthyroidism, but even **more severe**
	★ **High fever**
	★ **Rapid heart rate**
		▪ ↓ Temperature:
		  ▪ Cooling blankets
		  ▪ Ice packs 
		▪ Administer antipyretics
		▪ Administer antihypertensives
		▪ Maintain airway
		▪ Prepare for a potential thyroidectomy

Myxedema Coma

PATHOLOGY	SIGNS & SYMPTOMS	TREATMENT

Coma due to continous low levels of thyroid hormone (severe hypothyroidism)	↓ Energy (lethargy)
	↓ Breathing (hypoventilation)
	↓ Mental status
	↓ Temperature (hypothermia)
	★ **Coma**
		▪ Administer IV levothyroxine
		▪ Provide respiratory support
		  ▪ Intubation
		  ▪ Ventilation

Hypoglycemia

PATHOLOGY	SIGNS & SYMPTOMS	TREATMENT
Life-threatening complication where a patient's **blood sugar gets dangerously low** Without treatment, it can cause seizures, coma, or brain damage	▪ Cold & clammy skin ▪ Sweating (diaphoresis) ▪ Shakiness ▪ Confusion ▪ Inability to arouse from sleep	Administer IV 50% dextrose (D50) or glucagon (IM, IV, SubQ) *depends on facility protocol*

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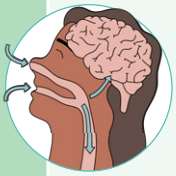
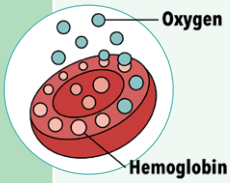
Lab Values Related to the Respiratory System

ABGs

	DEFINITION	EXPECTED RANGE	INTERPRETATION
PH	Measurement of how acidic or alkalotic your blood is	**7.35 - 7.45**	7.35 — 7.40 — 7.45 Acidosis ← Absolute Normal → Alkalosis
$PaCO_2$	Measurement of carbon dioxide in the blood MEMORY TRICK: CO_2 think aCid	**35 - 45**	CO_2 **> 45** = Acidosis CO_2 **< 35** = Alkalosis
HCO_3	Measurement of bicarbonate in the blood MEMORY TRICK: Bicarbonate think Base	**22 - 26**	HCO_3 **> 26** = Alkalosis HCO_3 **< 22** = Acidosis
PaO_2	Measurement of oxygen in the blood	**80 - 100**	PaO_2 **< 80** = Hypoxemia the patient is not getting enough oxygen
SaO_2	Percentage (%) of hemoglobin that is bound to oxygen in the blood	**95 - 100%**	SaO_2 **< 95** = Hypoxemia the patient is not getting enough oxygen COPD patients may have lower than average O_2 levels (as low as 88%) at baseline

OXYGEN LEVELS EXPLAINED

	DEFINITION	EXPECTED RANGE	INTERPRETATION
FiO_2 	FiO_2 Fraction of inspired Oxygen (the air you breathe in)	Room air has **21%** oxygen	–
PaO_2	Partial pressure of oxygen in the arterial blood MEMORY TRICK: PaO_2 = arterial	**80 - 100** mmHg	**HYPOXEMIA** low oxygen in the blood Decreased oxygen in the **BLOOD** **HYPOXEMIA** usually leads to **HYPOXIA** MEMORY TRICK: HypoxEmia think Earlier
SaO_2 	Percentage of hemoglobin that is bound to oxygen in the blood (hemoglobin saturation) MEMORY TRICK: SaO_2 = Saturation (%)	**95 - 100%** (measured with a pulse oximeter)	**HYPOXIA** low oxygenation Decreased oxygen supply to the **TISSUES**

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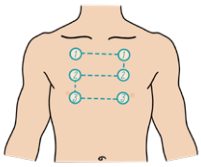
Auscultating Lung Sounds

Tips for Listening

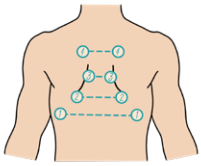
- Listen directly on the skin with the diaphragm of a stethoscope
- Listen at the **IN**tercostal spaces (***IN*** *between the ribs)*
- Listen to the anterior & posterior chest
- Have the patient sit upright (High Fowler's), arms resting across the lap
- Instruct the patient to take deep breaths
- Listen from top to bottom (comparing sides)

Listen for a
FULL INHALATION TO EXPIRATION
on each spot

ANTERIOR
Will hear
UPPER lobes well



POSTERIOR
Will hear
LOWER lobes well



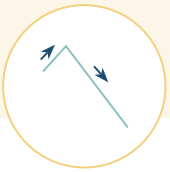
Normal Sounds

BRONCHIAL (TRACHEAL)

DESCRIPTION
High, loud, hollow tubular sound

LOCATION HEARD
Anteriorly only
(heard over trachea & larynx)

DURATION
Inspiration < expiration

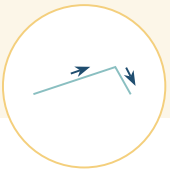


VESICULAR

DESCRIPTION
Soft, low-pitched, breezy/rushing sound

LOCATION HEARD
Heard anteriorly & posteriorly

DURATION
Inspiration > expiration

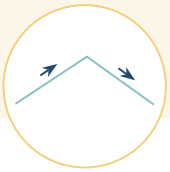


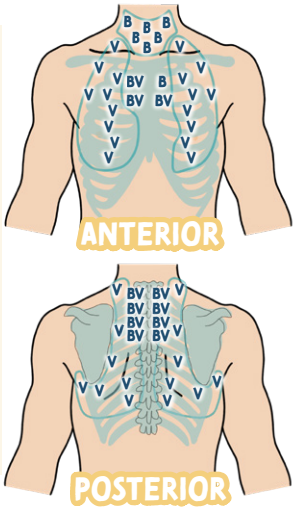
BRONCHOVESICULAR

DESCRIPTION
Medium-pitched, hollow sound

LOCATION HEARD
Heard anteriorly & posteriorly

DURATION
Inspiration = expiration





Abnormal (ADVENTITIOUS) Sounds

DISCONTINUOUS SOUNDS

DISCRETE CRACKLING SOUNDS

FINE CRACKLES (RALES)

DESCRIPTION: High-pitched crackling sounds (like fire crackling or Velcro coming apart)
DUE TO: Previously deflated airways that are popping back open
EXAMPLE: Pulmonary edema, asthma, obstructive diseases

COARSE CRACKLES (RALES)

DESCRIPTION: Low-pitched, wet bubbling sounds
DUE TO: Inhaled air colliding with secretions in the trachea or large bronchi
EXAMPLE: Pulmonary edema, pneumonia, depressed cough reflex

PLEURAL FRICTION RUB

DESCRIPTION: Low-pitched, harsh grating sounds
DUE TO: Pleura being inflamed and losing its lubricant fluid; The surfaces are actually rubbing together during respirations
EXAMPLE: Pleuritis

CONTINUOUS SOUNDS

CONNECTED MUSICAL SOUNDS

WHEEZES

DESCRIPTION: High-pitched musical instrument with more than one type of sound quality (polyphonic)
DUE TO: Air moving through a narrow airway
EXAMPLE: Asthma, bronchitis, chronic emphysema

STRIDOR

DESCRIPTION: High-pitched whistling or gasping with harsh sound quality
DUE TO: Disturbed airflow in larynx or trachea
EXAMPLE: Croup, epiglottitis, any airway obstruction

REQUIRES MEDICAL ATTENTION

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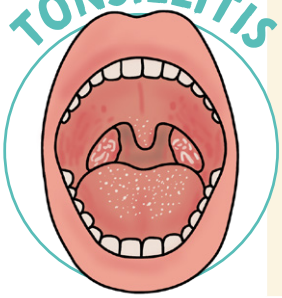
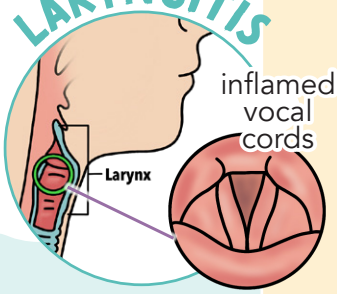
Upper Respiratory Tract Disorders

	PATHOLOGY	SIGNS & SYMPTOMS	TREATMENT
RHINITIS	Inflammation of the **mucous membrane in the nose** *Can be nonallergic or allergic*	▪ Runny nose ▪ Nasal congestion ▪ Nasal discharge ▪ Sneezing ▪ Headache	▪ Saline or steroid nasal sprays ▪ Antihistamines ▪ Decongestants
SINUSITIS	Inflammation of the tissue lining the **sinuses** (Sinus infection)	▪ Runny & stuffy nose ▪ Pressure & pain in the face ▪ Headache ▪ Post-nasal drip ▪ Mucus dripping down the throat ▪ Sore throat	▪ **VIRAL:** supportive measures ▪ **BACTERIAL:** antibiotics ▪ Nasal saline irrigation ▪ Corticosteroids ▪ Antihistamines
TONSILLITIS 	Inflammation of the **tonsils**	▪ Sore throat ▪ Fever ▪ Snoring ▪ Difficulty swallowing ▪ Tonsil stones	▪ Fluids ▪ Salt water gargles ▪ Rest ▪ Humidified air ▪ Tonsillectomy (surgical removal of the tonsils)
LARYNGITIS 	Inflammation of the **larynx**	▪ Hoarse voice ▪ Aphonia (loss of voice) ▪ Cough ▪ Dry sore throat ▪ Symptoms worsening with cold air or cold liquid	▪ Rest voice ▪ Avoid smoking & alcohol ▪ Avoid whispering & clearing throat (can irritate vocal cords) ▪ Humidified air & adequate hydration
PHARYNGITIS	Inflammation of the **pharynx** (Strep throat)	▪ Sore throat ▪ Red & swollen pharyngeal membrane & tonsils ▪ Swollen lymph nodes ▪ White exudate ▪ Fever	▪ **VIRAL:** supportive measures ▪ **BACTERIAL:** antibiotics ▪ Rest ▪ Salt water gargles

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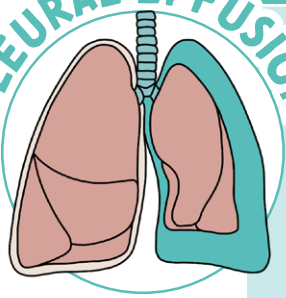
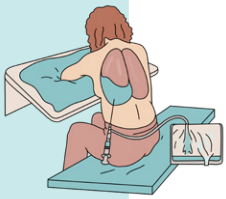
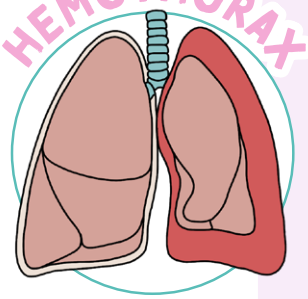
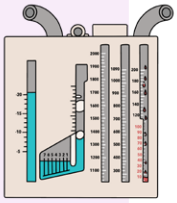
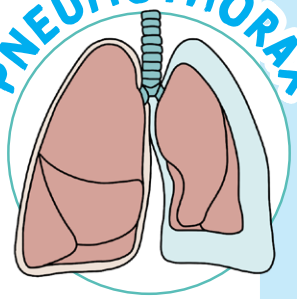
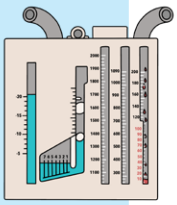
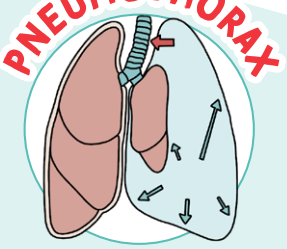
Hemothorax, Pleural Effusion, Pneumothorax, Tension Pneumothorax

	PATHOLOGY	RISK FACTORS	TREATMENT
PLEURAL EFFUSION 	Lung collapse due to a collection of **FLUID** in the pleural space	▪ Trauma ▪ Infection (pneumonia)	▪ Thoracentesis 
HEMOTHORAX 	Lung collapse due to a collection of **BLOOD** in the pleural space "Hemo" means blood	▪ A pneumothorax is often followed by a hemothorax	▪ Chest tube 
PNEUMOTHORAX 	Lung collapse due to a collection of **AIR** in the pleural space	▪ Trauma (blunt or penetrating) ▪ Medical procedure (central line placement) ▪ Gun shot or stab wound	▪ Chest tube 

TENSION PNEUMOTHORAX 

MEDICAL EMERGENCY DUE TO:

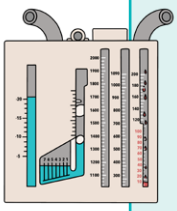
- Complications from a **PNEUMOTHORAX**, which occurs when the opening to the pleural space creates a one-way valve, then air or blood collects in the lungs and can't escape (pressure builds up).
- Inappropriate mechanical ventilation settings

Signs & symptoms:

- Jugular vein distention (JVD)
- Compression on the heart (tachycardia, hypotension, chest pain)
- Compression on other lung (tachypnea, hypoxia)
- Tracheal shift

Treatment:

- Needle decompression (aspirate the air)
- Chest tube



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Chronic Obstructive Pulmonary Disease (COPD)

Pathology

Progressive pulmonary disease that causes chronic airflow obstruction

Diagnostic

- Arterial blood gases (ABGs)
  - ↑ CO
  - ↓ O_2
- Chest X-ray
- Pulmonary function test
  - Spirometry

Obstructive lung disease
FEV1/**FVC** ratio of less than 70%

FEV1 = FORCED EXPIRATORY VOLUME	FVC = FORCED VITAL CAPACITY

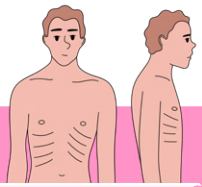
Risk Factors

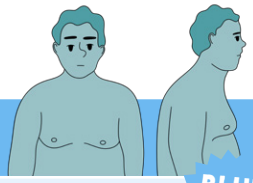
Smoking

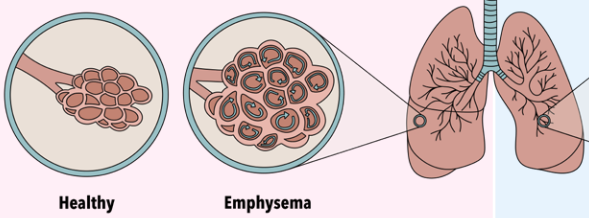
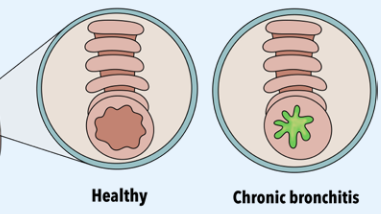
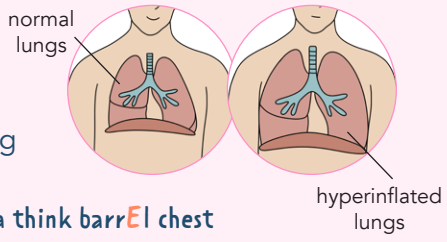
- Breathing in harmful irritants
- Occupation exposure
- Exposure to "secondhand" smoke
- Infection

- Air pollution
- Genetic abnormalities
- Asthma
- Severe respiratory infection in childhood

Deficiency of Alpha-1 antitrypsin (protects the lining of the lungs)





	EMPHYSEMA (PINK PUFFERS)	CHRONIC BRONCHITIS (BLUE BLOATERS)
PATHOLOGY	• The alveoli are damaged & enlarged, which causes loss of lung elasticity • Results in loss of lung tissue recoil & air trapping MEMORY TRICK: EmphysemA think Entrapped Air  	**Chronic productive cough & sputum production for > 3 months (in each of 2 consecutive years)** • Mucus secretion • Airway obstruction (inflammation)  MEMORY TRICK: Bronchitis think Boogers (lots of mucus)
SIGNS & SYMPTOMS	Weight loss (appears very thin)	Overweight or obese Peripheral edema
	Hyperinflation of the lungs (barrel chest) from air trapping  MEMORY TRICK: Emphysema think barrEl chest	Cyanosis (blue) from hypoxemia MEMORY TRICK: Bronchitis think Blue appearance
	Shortness of breath & severe dyspnea	Chronic cough, rhonchi & wheezing

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Chronic Obstructive Pulmonary Disease (COPD)

Nursing Considerations & Patient Education

OXYGEN THERAPY

THOSE WITHOUT COPD

Healthy patients are stimulated to breathe due to ↑ CO_2

COPD PATIENTS

COPD patients are stimulated to breathe due to ↓ O_2 (if you give too much O_2, they lose their "drive to breathe")

Give oxygen with caution

PREVENTING INFECTION

Stay up to date on vaccines

- Influenza vaccine
- Pneumococcal vaccine

↓ Chances of pneumonia

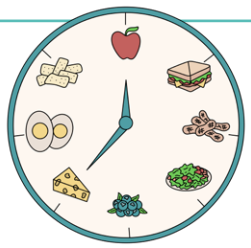
DIET MODIFICATIONS

- ↑ Calories for those with emphysema
- Stay hydrated but drink in between meals (not during) to prevent stomach distention

Fluids help to thin mucous secretions

Patients with emphysema use a lot of their energy to breathe, thereby burning a lot of calories, which causes weight loss!

A full stomach will increase pressure on the diaphragm

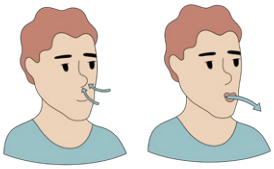


Eat small, frequent meals rich in protein

PROPER BREATHING TECHNIQUES

PURSED LIPS

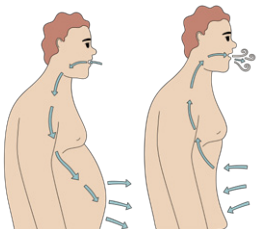
Promotes carbon dioxide elimination & prevents collapse of the airways



Breathe in (inhale) through your nose and then breathe out (exhale) with "pursed lips" (like you are going to whistle) for 4 seconds

DIAPHRAGMATIC BREATHING

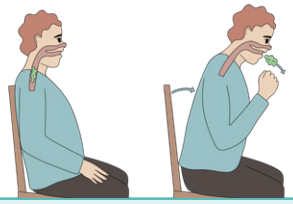
Uses the DIAPHRAGM rather than the accessory muscles to breathe



Slowly breathe in (inhale) and let your abdomen expand, then breathe out (exhale) and let your abdomen go back inward

HUFF COUGHING

Helps to get rid of excess mucus in the lungs/airway



Slowly inhale, hold for a few seconds, and **forcefully** exhale; repeat a few times

MEDICATIONS

Bronchodilators:
Relaxes smooth muscle of lung airways
=
better airflow

Corticosteroids:
↓ Inflammation
(oral, IV, inhaled)

SUFFIXES: "-sone" "-ide"

EDUCATE on taking a bronchodilator & a corticosteroid:

1. Use Bronchodilator first to help open the airways
2. WAIT 5 minutes
3. Then administer Corticosteroid after airways are **open enough to allow it through**

MEMORY TRICK: B comes before C in the alphabet

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Pneumonia

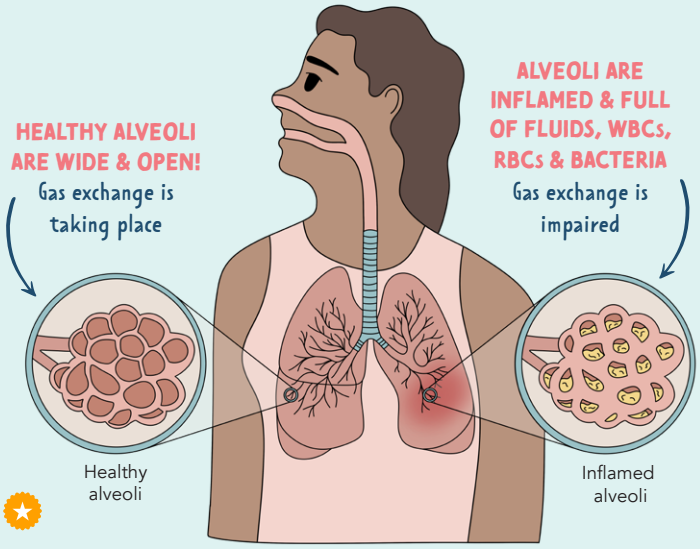
Pathology

Lower respiratory tract infection that causes inflammation of **ALVEOLI SACS**!

TYPES

- **Community-acquired pneumonia (CAP)**
- **Hospital-acquired pneumonia (HAP)**
- **Healthcare-associated pneumonia (HCAP)**
- **Ventilator-associated pneumonia (VAP)**
- **Aspiration pneumonia**

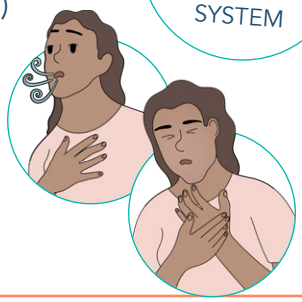
REMEMBER Gas exchange takes place in the alveoli... so pneumonia causes **impaired gas exchange**



Signs & Symptoms

- Productive cough (purulent sputum)
- Trouble breathing (dyspnea)
- ↑ Respiratory rate
- ↑ Heart rate
- Crackles heard in the lungs
- Chest pain
- Fever (> 100.4°F)
- Sweating and/or chills

Most of these are related to the RESPIRATORY SYSTEM



Risk Factors

Can be **community-acquired** or **hospital-acquired!**

- Prior infection
- Immunocompromised
  - HIV, young/old, autoimmune infections
- Immobility
- Aspiration risk
- Smoking
- Lung diseases
  - COPD
- Recent surgery

Diagnostic

- **↑ White blood cells**
- Chest X-ray → Shows pulmonary infiltrates or pleural effusions
- Sputum culture → Can be **bacterial**, **viral**, or **fungal**

Nursing Considerations

MONITOR

- Respiratory status
- Oxygen saturation
- Color, consistency & amount of sputum

DIET MODIFICATIONS

- ↑ Calories
- ↑ Protein
- ↑ Fluids (oral or IV)
- Small, frequent meals

Thins secretions & compensates for dehydration from fever

- Medications
  - Antipyretics
  - **Antibiotics** *(only for bacteria)*
  - Antivirals
  - Bronchodilators
  - Cough suppressants
  - Mucolytic agents

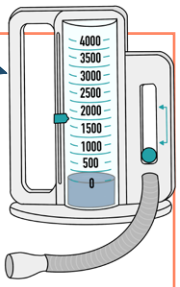
Always take blood and sputum cultures before starting antibiotics if possible

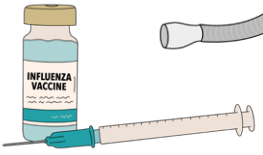
- Semi-Fowler's or High Fowler's position

Helps lung expansion

Patient Education

- Use of **incentive spirometer**
  - Helps to pop open the alveoli sacs & get the air moving
- Up-to-date vaccines
  - Annual flu shot
  - Pneumococcal vaccine
- Smoking cessation
- Hand-washing & avoiding sick people





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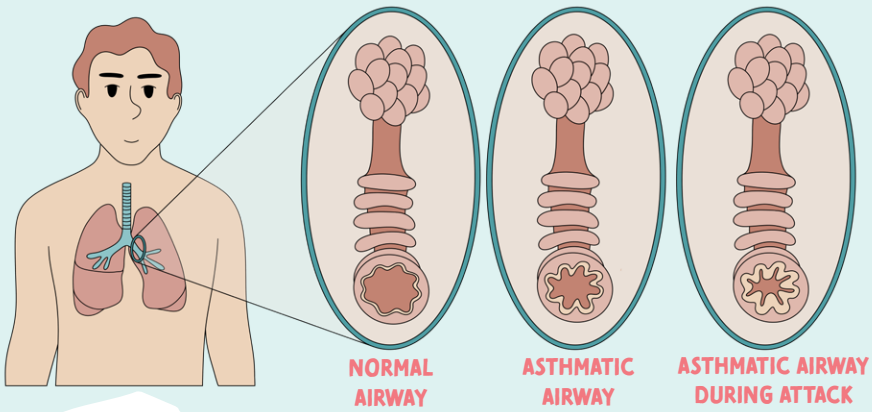
Asthma

Pathology

Chronic lung disease that causes an inflamed, narrow & swollen airway (bronchi & bronchioles)



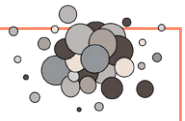
Classifications

Based on symptoms

- **MILD INTERMITTENT**: < 2 exacerbations per week
- **MILD PERSISTENT**: > 2 exacerbations per week, but not daily
- **MODERATE PERSISTENT**: Daily symptoms with 2 exacerbations per week
- **SEVERE PERSISTENT**: Continuous symptoms with frequent exacerbations

Causes

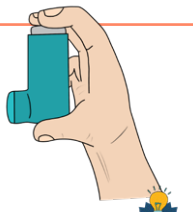
NOT COMPLETELY KNOWN!



- Genetic
- Environmental
  - Smoke, pollen, perfumes, dust mites, pet dander, cold or dry air
- GERD
- Exercise (exercise-induced asthma)
- Certain drugs
  - NSAIDs, aspirin

Medications

- Bronchodilators (These **dilate** the airways)
  - Short-acting (Albuterol) Provides rapid relief
  - Long-acting (Salmeterol) Prevents asthma attack
  - Methylxanthines (Theophylline)
- Corticosteroids Anti-inflammatory agents
  - Suffixes **-SONE** & **-IDE**
  - *Ex: dexametha**sone**, predni**sone***
- Leukotriene modifiers
- Anticholinergics



MNEMONIC

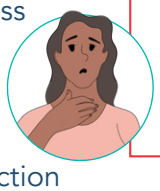
Certain medications are known to cause bronchospasms in patients with asthma. We want to "**BAN**" these medications from asthma patients.

- **B** Beta blockers
- **A** Aspirin
- **N** NSAIDs

Signs & Symptoms

CHARACTERIZED BY FLARE-UPS
(it comes & goes)

- Dyspnea (shortness of breath)
- Tachypnea (fast respiratory rate)
- Chest tightness
- Anxiety
- Wheezing
- Coughing
- Mucus production
- Use of accessory muscles
- **Air trapping**



Air trapping causes the patient to retain CO_2 which is **ACIDIC** = **RESPIRATORY ACIDOSIS**

⚠ STATUS ASTHMATICUS ⚠

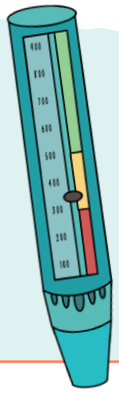
Medical emergency
Life-threatening asthma episode

oxygen
↓
hydration
↓
nebulization
↓
systemic corticosteroid

Nursing Considerations

- Assess patient's airway
- Place in High Fowler's position
- Provide frequent rest periods
- Adm. oxygen therapy
  - Goal: keep the O_2 at 95–100%
- Maintain a calm environment to ↓ stress
- Assess **PEAK FLOW METER** reading
- Assess for cyanosis & retractions

PEAK FLOW METER



- Used to determine how controlled the asthma is & if it's getting worse
- Establish a baseline by performing a "personal best" reading
- Patient will exhale as hard as they can & get a reading

GREEN = GOOD
YELLOW = NOT TOO GOOD
RED = BAD

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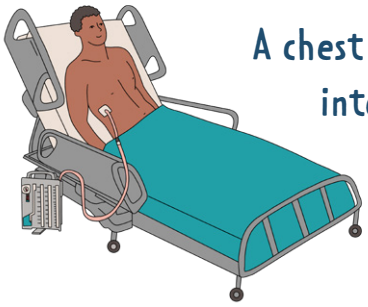
Chest Tubes



A chest tube is a tube that is inserted into the pleural space to remove excess air, blood, or fluid. This helps re-expand the lungs.

WHY IS IT USED?

- After thoracic surgery
- During cardiac surgery (drain fluid from around the heart)
- Spontaneous pneumothorax
- Pneumothorax
- Hemothorax
- Pleural effusion
- Empyema (infection)

3 CHAMBERS:

DRAINAGE CHAMBER

This is where the fluid is collected from the patient

WATER-SEAL CHAMBER

Allows air to be removed from the pleural space WITHOUT outside air entering the lungs

Tidaling (water rises & falls with each breath) = **GOOD**

Excessive continuous bubbling in the water seal chamber = **BAD**

If the water stops fluctuating, this could mean:
1. The lung has re-expanded
2. The tubing is kinked

SUCTION-CONTROL CHAMBER

Two types: **WET SUCTION** & **DRY SUCTION**

If the tube becomes DISLODGED:
Cover the insertion site with a sterile dressing

If the chamber becomes DAMAGED:
Place the tubing in sterile water while waiting for a new system

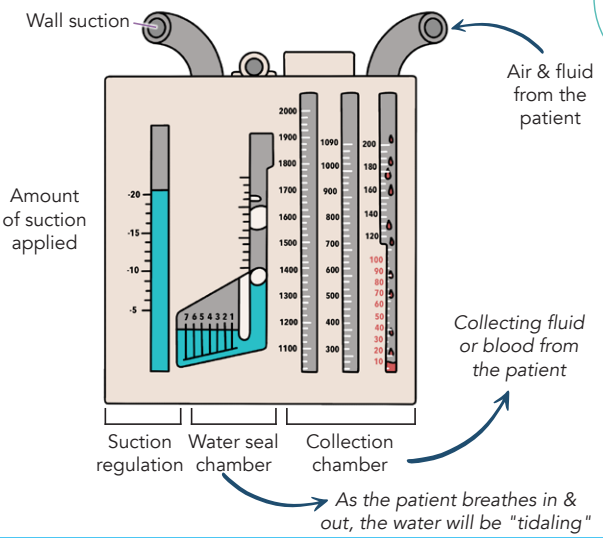
NURSING CONSIDERATIONS

- Always keep the drainage system BELOW the patient's chest
- **NEVER** strip or "milk" the tubing
- **NEVER** clamp the tubing
- **EDUCATE** the patient to do the **Valsalva maneuver** when the HCP is removing the chest tube
  - *Take a deep breath, exhale, and bear down*
- **MONITOR:**
  - Color & quantity of the drainage in the drainage collection chamber every hour
  - Lung sounds
  - Insertion site
  - Occlusive dressing integrity & evidence of moisture
- **REPORT** bright red blood (dark red is expected)
  - *Bright red blood indicates an active bleed*

WET SUCTION

Uses **WATER** to control the level of suction
(actually filling the suction control chamber with water)

WILL HAVE GENTLE BUBBLING

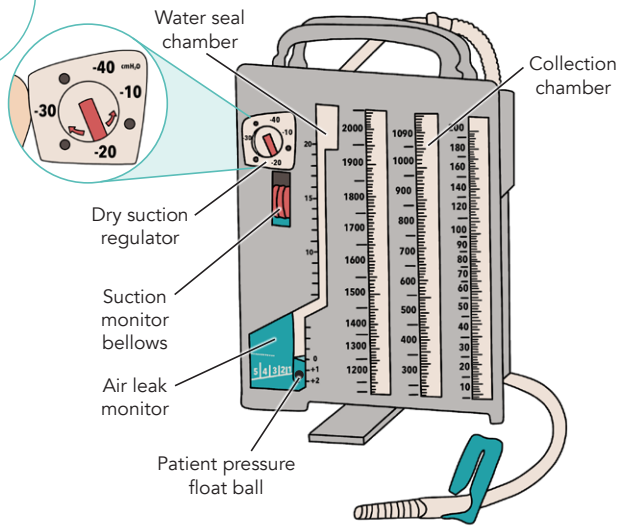


Both have a **collection chamber** and an **air leak monitor**

DRY SUCTION

There is no water column (it's DRY)—the suction is controlled by a suction monitor bellows that balances wall suction

THERE WILL BE NO BUBBLING



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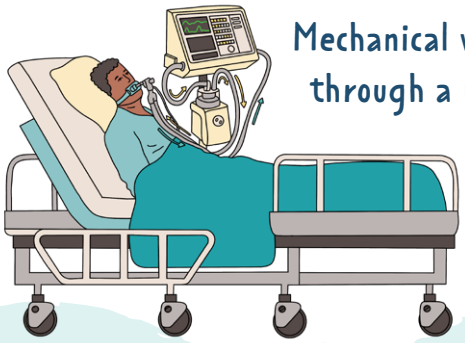
Mechanical Ventilation



Mechanical ventilation is performed through a machine that helps a person breathe by pumping air into the lungs

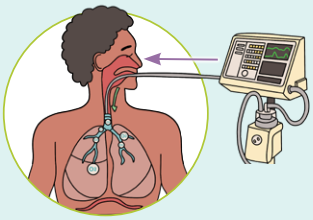
WHY IS IT USED?

- Control breathing during surgery
- Rest the respiratory muscles
- Help a patient who is unable to breathe on their own (respiratory failure such as ARDS)

POSITIVE PRESSURE VENTILATION

THE AIR IS PUSHED INTO THE LUNGS

This forceful air entering into the lungs can cause **barotrauma**

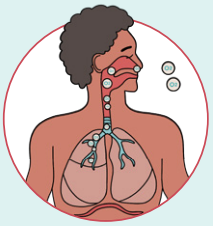


NEGATIVE PRESSURE VENTILATION

NORMAL BREATHING

The diaphragm uses negative pressure to bring in oxygen

MEMORY TRICK *N*egative think *N*ormal breathing



VENTILATOR SETTINGS

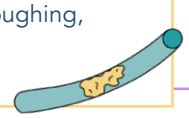
TIDAL VOLUME (VT)	Volume of gas delivered with each breath **500–800** mL
RESPIRATORY RATE	Number of breaths delivered to the patient **12–20** breaths per min
FiO_2	Fraction of inspired oxygen (O_2 concentration of the air being delivered to the patient) **21%–100%**
POSITIVE END EXPIRATORY PRESSURE (PEEP)	Amount of pressure in the lungs after expiration (prevents collapse of the alveoli)

UNDERSTANDING ALARMS

HIGH PRESSURE ALARMS

MEMORY TRICK *H*igh think *H*igh blockage of airflow

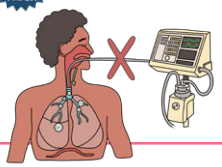
CAUSES:
Excessive mucus or secretions, kinks, coughing, pulmonary edema, pneumothorax, a patient "fighting" the ventilator



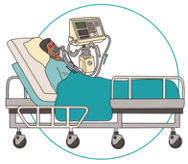
LOW PRESSURE ALARMS

MEMORY TRICK *L*ow think *L*eaks

CAUSES:
Disconnection, cuff leak, tube displacement

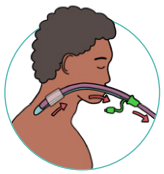


NURSING CONSIDERATIONS



MONITOR:

- Level of consciousness
- Vital signs
- Lung sounds
- Arterial blood gases
- Symptoms of ventilator-associated pneumonia
- Replace the gastrointestinal system with bowel sounds
- Nutritional status



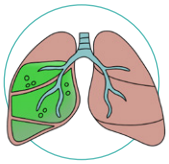
SUCTIONING

Suction secretions only when needed

- Never suction when inserting a catheter into the airway
- Never suction for longer than 10 seconds
- Administer 100% oxygen before suctioning (hyperoxygenate)

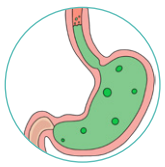
ORAL CARE

- Clean the mouth with chlorhexidine every 2 hours



MOBILIZE SECRETIONS

- Turn/reposition the patient every 2 hours
- Keep the head of the bed > 30°



GASTROINTESTINAL SYSTEM

- Administer PPIs & H2 blockers to prevent stress ulcers and decrease acid

Proton pump inhibitors (PPIs) end in -prazole

Histamine H2 antagonists (H2 blockers) end in -tidine

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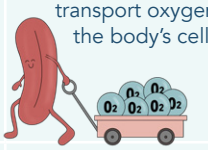
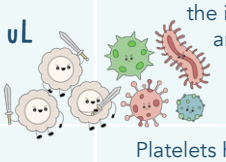
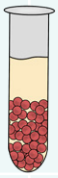
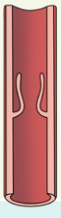
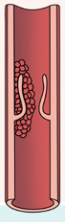
Lab Values Related to the Hematological System

	EXPECTED RANGE	DESCRIPTION	↓	↑
COMPLETE BLOOD COUNT (CBC)				
RED BLOOD CELLS (RBCs)	F 4.2 – 5.2 X 10^6 /uL M 4.7 – 6.1 X 10^6 /uL	Red blood cells transport oxygen to the body's cells. 	▪ Fluid volume overload ▪ Hemorrhage ▪ Anemia ▪ Renal disease (lack of erythropoietin production) More volume dilutes the RBCs	▪ Dehydration/ fluid volume deficit Less volume concentrates the RBCs ▪ Hyperactivity of the bone marrow (polycythemia vera)
WHITE BLOOD CELLS (WBCs)	4,500 – 11,000 / uL 	The white blood cells are a part of the immune system and help to fight infections and diseases.	**LEUKOPENIA** WBCs < 4,500 /uL ▪ Immunosuppression	**LEUKOCYTOSIS** WBCs > 11,000 /uL ▪ Current or recent infection & inflammation ▪ Glucocorticoid therapy
PLATELETS (PLT)	150,000 – 450,000 /uL	Platelets help clot the blood. Platelet aggregation is the clumping together of platelets that form a plug at the site of the injury.	**THROMBOCYTOPENIA** PLTs < 150,000 /uL ↓ Platelets think **BLEEDING**	**THROMBOCYTOSIS** PLTs > 450,000 /uL ▪ Certain cancers ▪ Infection
HEMOGLOBIN (HGB)	F 12 – 16 g/dL M 13 – 18 g/dL	Hemoglobin is an iron containing protein found in red blood cells. It transports oxygen from the lungs to the tissues. It also returns CO_2 from the tissues back to the lungs.	▪ Fluid retention (hemodilution) ▪ Anemia ▪ Hemorrhage	▪ Dehydration (hemoconcentration)
HEMATOCRIT (HCT)	F 36% – 48% M 39% – 54% 	The percent of blood that is made up of red blood cells (expressed as a %).	▪ Fluid retention (hemodilution) ▪ Anemia ▪ Hemorrhage	▪ Dehydration (hemoconcentration) ▪ Low oxygen availability (smoking, pulmonary diseases (COPD), high altitudes)
TYPES OF COAGULATION TESTS				
ACTIVATED PARTIAL THROMBOPLASTIN TIME (aPTT)	NORMAL (not on anticoagulants) 30 – 40 seconds ON HEPARIN THERAPY 1.5 – 2.0 x the normal value	aPTT measures how long it takes for a blood clot to form. It's also used to monitor the effectiveness of the anticoagulant: **HEPARIN**.	Numbers are LOW = Clots will GROW	▪ Heparin therapy MEMORY TRICK: Numbers are too HIGH = Patient will DIE (from increased bleeding)
PROTHROMBIN TIME (PT)	NORMAL (not on anticoagulants) 10 – 12 seconds ON HEPARIN THERAPY 1.5 – 2.0 x the normal value	Prothrombin time measures the amount of time needed to form a clot. It's also used to monitor the effectiveness of the anticoagulant: **WARFARIN**.		▪ Deficiency in vitamin K ▪ Deficiency in clotting factor ▪ Liver disease
INTERNATIONAL NORMALIZED RATIO (INR)	NORMAL (not on anticoagulants) < 1 ON HEPARIN THERAPY INR 2.0 – 3.0 INR 2.5 – 3.5 (heart valve replacement)	 INR is calculated from the prothrombin time and is used to monitor oral anticoagulants such as **WARFARIN**.		▪ Warfarin therapy MEMORY TRICK: Numbers are too HIGH = Patient will DIE (from increased bleeding)
D-DIMER	< 0.5 mcg/mL	D-dimers are fragments of fibrin that are in the blood when a clot dissolves or is broken down. D-dimer helps to determine if a clot is present somewhere in the body	▪ Blood clot is ruled out 	▪ Additional tests are needed to confirm and determine a specific diagnosis ▪ Blood clot may be present in the body 

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Iron Deficiency Anemia

Pathology

TYPE OF ANEMIA CAUSED BY ↓ IRON LEVELS

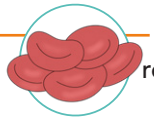
Insufficient levels of iron in the body

Your body needs iron to produce **hemoglobin**

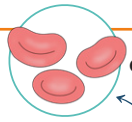
↓ iron = ↓ hemoglobin

↓ oxygen to the tissues and the body

Hemoglobin is the part of the RBC that makes the blood red & allows oxygen to be transported throughout the body



Normal red blood cells

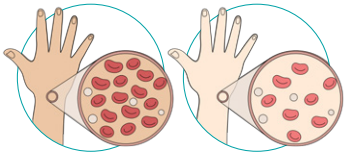


Iron deficiency anemia

Smaller in size & paler in color
(because hemoglobin makes the blood bright red)

Signs & Symptoms

- Pallor
- Weakness & fatigue
- Shortness of breath (from lack of oxygen)
- Tachycardia
- Microcytic (small) red blood cells



Normal Anemia

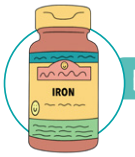
Specific to iron deficiency anemia

SEVERE SYMPTOMS

- **Glossitis:** Inflammation & burning of the tongue
- **Cheilitis**: Inflammation of the lips
- Brittle & ridged nails

Treatment / Medications

TREAT THE CAUSE: Discontinue any drugs causing the anemia



IRON SUPPLEMENTS (oral or liquid) — MOST COMMON TREATMENT

Examples: ferrous sulfate, ferrous gluconate, ferrous fumarate

IV ADMINISTRATION OF IRON

If oral iron is poorly absorbed or poorly tolerated

Risk Factors

- Lack of iron (vegetarian diet)
- Blood loss (excessive menstruation, surgery, or trauma)
- Pregnancy
- Iron malabsorption (due to bariatric surgery or Celiac disease)

Diagnostic

- Complete blood count (CBC)
  - ↓ **hemoglobin** & ↓ **hematocrit**
- Bone marrow aspiration
- Stool sample, colonoscopy, endoscopy (checking for blood)

NORMAL VALUES:

Hemoglobin (Hgb)
Female: 12–16 g/dL *Male:* 13–18 g/dL

Hematocrit (HCT)
Female: 36%–48% *Male:* 39%–54%

Patient Education

EDUCATE on administering iron supplements:

Side effects of iron supplements: Black stool, Constipation, Foul aftertaste

↑ ABSORPTION

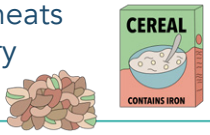
Vitamin C:
Take iron with fruit juice & multivitamin
Take on an empty stomach

↓ ABSORPTION

Calcium:
Do not take iron with milk or antacids

EDUCATE on foods high in IRON: MNEMONIC **"EAT LOTS OF IRON"**

Egg yolks
Apricots
Tofu
Legumes
Oysters
Tuna
Seeds
P**O**tatoes
Fish
Iron-fortified cereals
Red meats
P**O**ultry
Nuts



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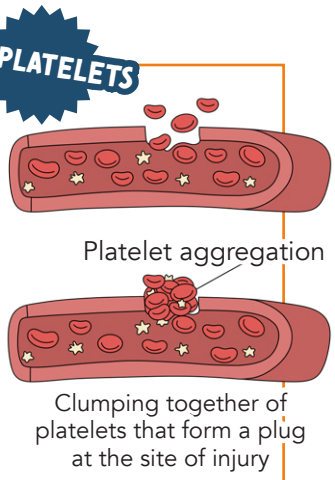
Thrombocytopenia

Pathology

Condition in which the platelet count in the blood is too low

Platelets help clot the blood by platelet aggregation

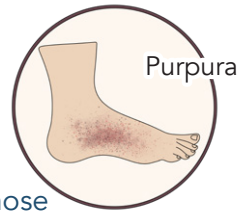
↓ Platelets
=
Bleeding

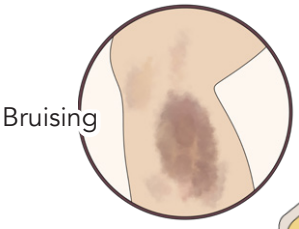


NORMAL PLATELET COUNT	THROMBOCYTOPENIA
150,000–450,000 platelets/μL	< 150,000 platelets/μL

Signs & Symptoms

- Weakness, dizziness, tachycardia, hypotension
- Prolonged bleeding time
- Petechiae (pinpoint bleeding)
- Purpura
- Bruising
- Bleeding from the gums & nose
- Heavy menstrual cycles
- Blood in the stool or urine





Risk Factors

MNEMONIC

- **P** Platelet disorders
- **L** Leukemia
- **A** Anemia
- **T** Trauma
- **E** Enlarged spleen
- **L** Liver disease
- **E** Ethanol (alcohol-induced)
- **T** Toxins (drug-induced)
- **S** Sepsis

Treatment

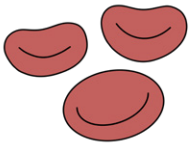
- Platelet transfusion →

- Bone marrow transplant
  - Platelets are made in the bone marrow
- Splenectomy →
  - Removal of the spleen for those unresponsive to other therapy

The spleen destroys platelets, so removal will increase platelet levels

Diagnostic

- ↑ Bleeding time
- ↑ INR & ↑ PT/PTT
- ↓ Hgb & HCT
- Bone marrow aspiration & biopsy



Patient Education

EDUCATE patients to follow **BLEEDING PRECAUTIONS**:

- Use electric razors
- Use small needle gauges
- AVOID aspirin
- Decrease needle sticks
- Protect from injury

IMMUNE THROMBOCYTOPENIC PURPURA (ITP)

Type of thrombocytopenia, formerly called **idiopathic thrombocytopenia purpura**

"Purpura" is in the name because the body bruises easily, and petechiae may occur in the trunk and extremities

PATHOLOGY

Autoimmune disease in which the body produces antibodies against its own thrombocytes (platelets)

ITP

< 100,000 platelets/μL

RISK FACTORS

- Children after viral illness
- Females (ages 20–40)
- Pregnancy

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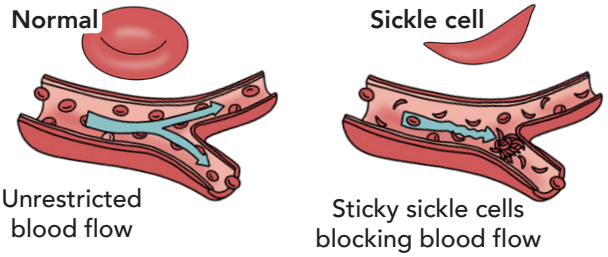
Sickle Cell Anemia

Pathology

An inherited disease that causes the body's hemoglobin molecules to be defective, resulting in weak RBCs that die earlier than healthy RBCs



Hemoglobin S is sensitive to low amounts of oxygen in the body!

Low oxygen
↓
Causes RBCs to change their shape
↓
RBCs that are sickle-shaped, sticky & stiff
↓
Causes clumping, which blocks blood flow to the tissues
↓
⚠ SICKLE CELL CRISIS ⚠

3 CELL TYPES OF SICKLE CELL CRISIS:

Acute vaso-occlusive crisis	RBCs stick in vessels = hypoxia *(this is very painful!)* MOST COMMON
Aplastic crisis	The body stops producing enough RBCs *(bone marrow can't keep up)*
Sequestration crisis	The spleen stops working & becomes flooded with sickle cells

Risk Factors

A patient is born with this genetic blood disorder. It's an autosomal recessive disorder in which the sickle hemoglobin (HbS) gene is inherited. It's commonly recognized early in life after maternal iron stores have been depleted.

Signs & Symptoms

- Anemia symptoms (fatigue, tachycardia, pallor)
- **Pain**
- Dactylitis (swelling of the hands & feet)
- Stroke
- Acute chest syndrome (tachypnea, wheezing, fever, cough)

Diagnostic

- Blood sample
- Test before birth (testing the amniotic fluid)

AUTOSOMAL RECESSIVE

Either parent can have the sickle cell trait without their child having sickle cell anemia. BOTH parents must pass down the sickle hemoglobin (HbS) gene for the condition to occur.

Medications

- Analgesics & opioids (to help with the pain)

Treatment & Nursing Considerations

- **IV fluids** (stops the clumping of RBCs)
- Oxygen therapy
- RBC transfusions
- Stem cell transplant

Patient Education

EDUCATE on how to **prevent** sickle cell crisis:

- ✓ **Drink lots of water** (stay hydrated)
- ✓ Keep vaccines up to date
- ✓ Prevent infection (hand hygiene, avoid big crowds)
- ✓ Limit stress
- ✓ Avoid high altitudes
- ✓ Stop smoking
- ✓ Avoid overexertion

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Disseminated Intravascular Coagulation (DIC)

Pathology

Causative factor (underlying disease)

↓

Inflammatory response causes inflammation & coagulation in the vasculature

↓

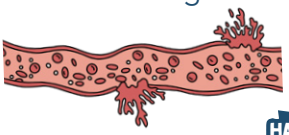
Fibrinolytic system is halted

↓

Causes lots of small clots & platelets to clump

↓

Lots of small clots are using all the blood's clotting factors, which leaves other parts of the body with no means to stop any bleeding



TOO LITTLE CLOTTING
(BLEEDING)

+

HAPPENING AT THE SAME TIME

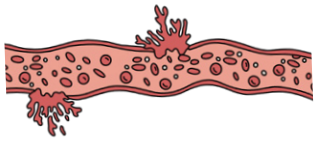
Excessive clotting causes blockage of the blood vessels

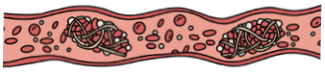
TOO MUCH CLOTTING

↓

Can lead to organ ischemia (because organs are not getting blood supply)

Signs & Symptoms





BLEEDING & **BLOOD CLOTS**

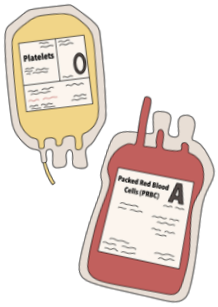
Bleeding can be minimal up to the point of widespread hemorrhaging

- Petechiae & purpura
- Hematuria
- Melena (black, tarry stools)
- Nosebleeds

Can cause a troke, heart attack, deep vein thrombosis, or a pulmonary embolism

Treatment

- **Treat the underlying cause!**
- Transfusion
  - Packed RBCs
  - Fresh frozen plasma (FFP)
  - Platelets



Medications

- **Vasopressors**
  Cause vasoconstriction, which ↑ blood flow & ↑ perfusion to the organs
- **Heparin infusion**
  Stops the clotting, which ↑ blood flow to the organs
- **Cryoprecipitate**
  Replaces fibrinogen

Risk Factors

DIC is not a disease. Rather, DIC occurs due to an underlying condition or disease:

- Infection/sepsis
- Malignancy
- Allergic reactions
- Obstetric complications
- Trauma
- Shock
- Toxins

Diagnostic

- Lab tests
  - ↓ Platelet & ↓ fibrinogen levels
  - Prolonged clotting time (↑ PT & aPTT)
  - ↑ D-dimer (indicates there is a clot somewhere in the body)

Nursing Considerations

- Administer oxygen
- Administer IV fluids
- Correct electrolyte imbalances

MONITOR:

- For signs of bleeding
- Vital signs
- Lab values

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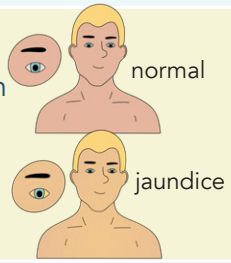
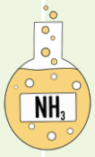
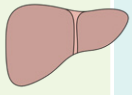
Lab Values Related to the Gastrointestinal System

	EXPECTED RANGE	INTERPRETATION	DESCRIPTION
AMYLASE Pancreatic enzyme	**30 – 110** U/L	↑ levels could indicate pancreatitis	**LIPASE** is a better indicator of pancreatitis than **AMYLASE** because serum lipase remains elevated for a longer period of time. MEMORY TRICK Lipase think Longer
LIPASE Pancreatic enzyme	**< 200** U/L		
BILIRUBIN Produced by the liver	Total **0.2 – 1.2** mg/dL	↑ levels could indicate liver dysfunction	**JAUNDICE** is a yellow discoloration of the skin due to high levels of bilirubin. It is visible when serum bilirubin is > 2 mg/dL.  normal jaundice
ALBUMIN	**3.5 – 5.5** g/dL	↑ levels could indicate dehydration	**ALBUMIN** helps keep fluid in the bloodstream.
PREALBUMIN	**15 – 36** mg/dL	↓ levels could indicate malnutrition	**PREALBUMIN** is great for assessing **NUTRITIONAL STATUS**.
AST Liver enzyme	**0 – 35** U/L	↑ levels could indicate liver dysfunction	**AST** must be taken with **ALT**. AST is a less specific marker of liver function than the enzyme ALT.
ALT Liver enzyme	**0 – 48** U/L		
AMMONIA	**10 – 80** mcg/dL	↑ levels could indicate liver dysfunction	**AMMONIA** (NH_3) is produced by cells throughout the body and is used by the **LIVER** to make urea. If the liver stops working, ammonia increases in the body. Too much ammonia is very toxic, especially to the brain.

Part of the liver function test (LFT)





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Gastrointestinal System Overview

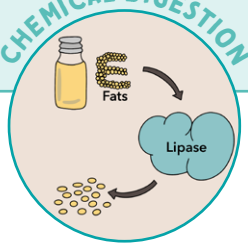
ORAL CAVITY COMPONENTS

MECHANICAL DIGESTION

When food is broken down into smaller pieces

EXAMPLES:
- Chewing
- Churning of the stomach

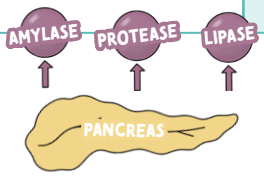
CHEMICAL DIGESTION



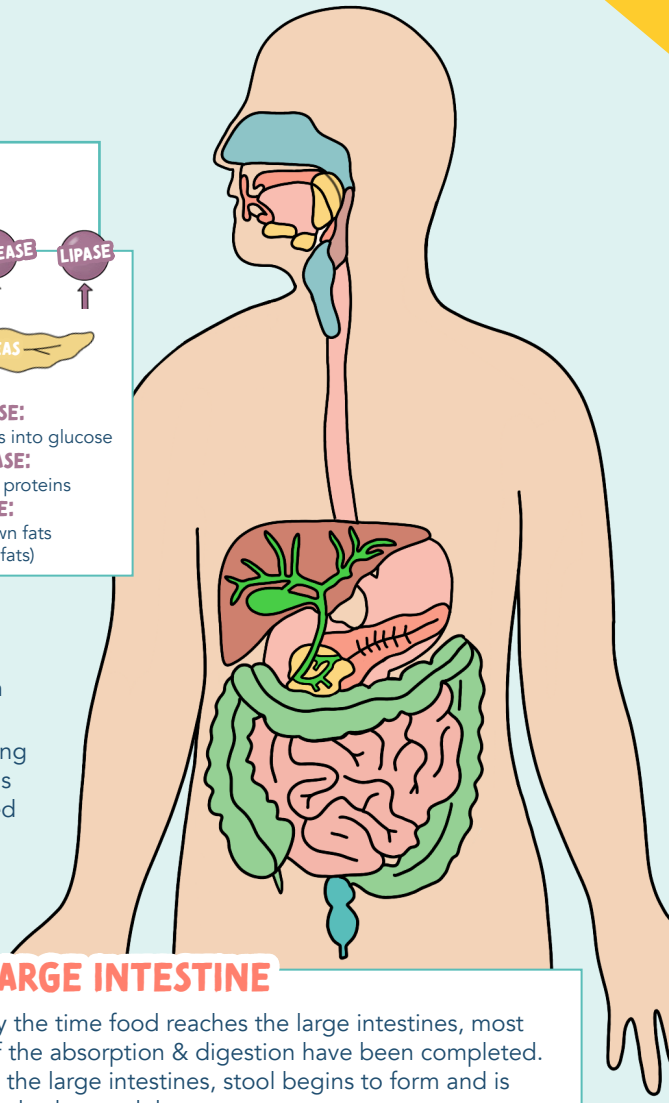
When food is broken down by **ENZYMES** and digestive juices

TIP: Enzymes end in "**-ASE**"

MEMORY TRICK: Protease think Proteins
Lipase think Lipids (fat)



AMYLASE:
breaks down carbs into glucose
PROTEASE:
breaks down proteins
LIPASE:
breaks down fats
(lipids = fats)



ESOPHAGUS

A hollow muscular tube that carries food & liquid from the mouth to the stomach using peristalsis

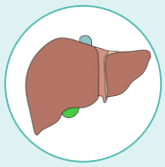
STOMACH

A hollow muscular organ

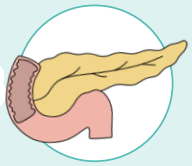
FUNCTIONS:
- Stores food during eating
- Secretes digestive fluids
- Moves partially digested food (chyme) into the small intestine

LIVER

FUNCTIONS:
- Filters the blood
- Metabolizes sugar, protein & fat
- Synthesizes lipoproteins (VLDL & HDL)
- Makes vitamin D
- Detoxifies/excretes bilirubin and other toxins
- Forms bile
- Metabolizes drugs
- Helps in blood clotting
- Synthesizes proteins such as albumin & coagulation factors



PANCREAS

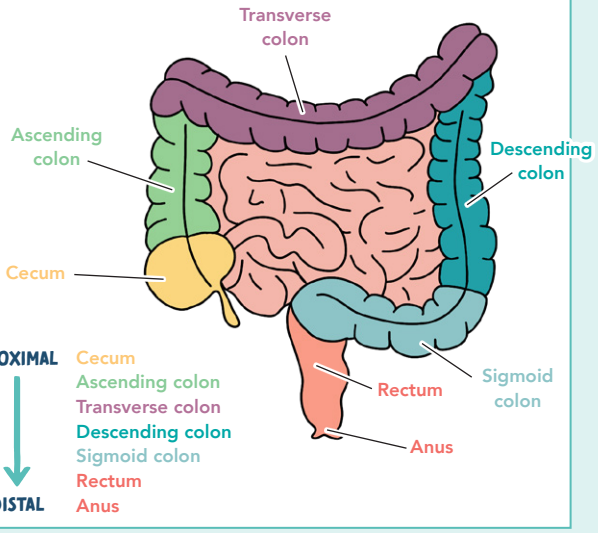


Helps make pancreatic juice (enzymes), which breaks down sugar, fat & starch. The pancreas has both exocrine & endocrine functions.

LARGE INTESTINE

By the time food reaches the large intestines, most of the absorption & digestion have been completed. In the large intestines, stool begins to form and is pushed toward the rectum.

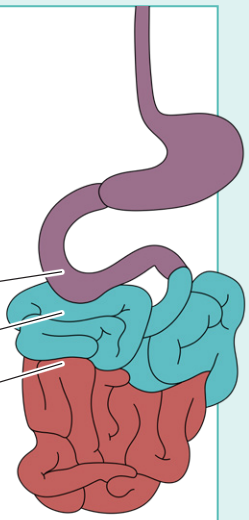
FUNCTIONS:
- **ABSORPTION** of water and electrolytes from food that has not been digested yet
- **DEFECATION** rids the body of any waste left over from food & removes it through the rectum & anus

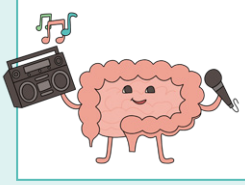


SMALL INTESTINE

The longest portion of the GI tract (longer than the large intestine)

FUNCTIONS:
- **DIGESTION** of food from the stomach
- **ABSORPTION** of nutrients, fats, carbohydrates, vitamins, minerals & water from food into the bloodstream to be used by the body





MEMORY TRICK: To remember the order of Proximal to Distal think DJ Ileum in the club!

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Acute vs. Chronic Pancreatitis

Pathology

Pancreatitis is **AUTODIGESTION** of the pancreas by its own digestive enzymes that are released improperly in the pancreas. This causes the pancreatic enzymes to destroy its own tissue, leading to inflammation.

If the pancreas isn't working, the **ENZYMES** aren't working properly either.

DIGESTIVE ENZYMES (exocrine)

AMYLASE:
- Breaks down carbs to glucose

PROTEASE:
- Breaks down proteins

LIPASE:
- Breaks down fats

Labs

- ↑ Amylase
- ↑ Lipase
- ↑ WBCs
- ↑ Bilirubin
- ↑ Glucose
- ↓ Platelets
- ↓ Ca^+ & Mg

ACUTE vs. CHRONIC

ACUTE: Sudden inflammation that is **REVERSIBLE** with prompt recognition and treatment

CHRONIC: Chronic inflammation that is **IRREVERSIBLE**

CAUSES

ACUTE
- **Gallstones**
  - Block the bile duct
- **Alcohol (ETOH)**
  - Damages the cells of the pancreas
- **Infection**
- **Medications**
- **Tumor**
- **Trauma**

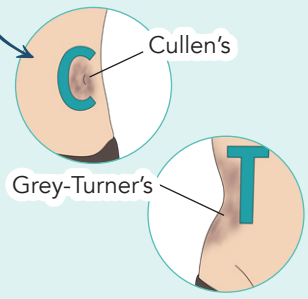
CHRONIC
- **Repeated episodes of acute pancreatitis**
- **Excessive & prolonged consumption of alcohol (ETOH)**
  - Recurrent damage to the cells of the pancreas
- **Cystic fibrosis**

SIGNS & SYMPTOMS

In **ACUTE**, there will still be working functions of the pancreas

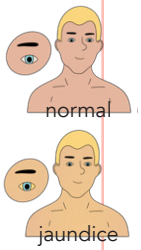
- **Sudden, severe PAIN**
  - Mid-epigastric pain LUQ
- **Nausea & vomiting**
- **Fever**
- **↑ HR & ↓ BP**
- **↑ Glucose**
- **Mental confusion & agitation**
- **Abdominal guarding**
- **Rigid/board-like abdomen**
- **Grey Turner's sign**
  - Bluish discoloration at the flanks
- **Cullen's sign**
  - Bluish discoloration of the umbilicus

MEMORY TRICK: **C**ullen's = **C**ircle belly button



In **CHRONIC**, you will see different S&S due to the prolonged damage & loss of function

- **Chronic epigastric pain or no pain**
- **Pain increases after drinking alcohol or eating a fatty meal**
- **Steatorrhea or "fatty stools"**
  - Oily/greasy, frothy stool
- **Weight loss**
  - Can't digest food properly
- **Jaundice**
  - Yellowish color of the skin from buildup of bile



- **Diabetes mellitus**
  - Damage to the islet of Langerhans
- **Dark urine**
  - From excess bile in the body

Nursing Considerations

- Rest the pancreas!
  - NPO *(we don't want stimulation of the enzymes)*
- Administer IV fluids
- Manage pain
- Position the patient:
  - ✓ Side lying → Fetal position
  - ✗ NOT supine
- Insert NG tube
  - Remove stomach contents

MONITOR:
- Glucose
- Blood pressure
- Intake & output (I&O)
- Laboratory values
- Stools

Medications

- Opioid analgesics
- Antibiotics
- Pancreatic enzymes
- Insulin
- Proton pump inhibitors (PPIs), H2 antagonists, antacids

Patient Education

DIET MODIFICATIONS
- Avoid alcohol

- ↑ Protein
- ↑ Complex carbohydrates (fruits, vegetables, grains)
- ↓ Fat (no greasy, fatty foods)
- Limit sugars

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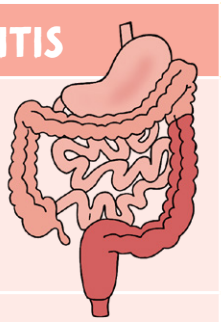
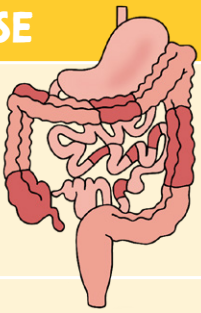
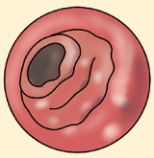
Ulcerative Colitis vs. Crohn's Disease

Types of INFLAMMATORY BOWEL DISEASE (IBD)

This is not the same thing as irritable bowel syndrome (IBS)

	ULCERATIVE COLITIS (MOST COMMON)	CROHN'S DISEASE
DESCRIPTION	Chronic ulceration & inflammation of the rectum & colon 	Inflammation of the gastrointestinal tract wall at ANY point through ALL layers 
LOCATION	Affects the large intestine & rectum only	Can affect anywhere in the GI tract (mouth to anus)
THICKNESS	Inflammation affects the submucosa or mucosa	Inflammation is transmural (occurring across the entire wall)
APPEARANCE	Inflamed areas are continuous with no patches showing the appearance of ulcers MEMORY TRICK: ULCERative colitis think bloody ULCERs	Patches of inflammation are present throughout the bowel. This makes a cobblestone appearance  MEMORY TRICK: Crohn's think Cobblestone
CURE	✗ **NO!** but surgical intervention (colectomy) can help with symptoms	✗ **NO** cure, but surgery can help with symptoms
COMPLICATIONS	Toxic megacolon, rupture of bowel, dehydration. Increased risk for hemorrhage/shock	Abscess, fistulas. Increased risk for infection (sepsis)
CLASSIC SYMPTOMS	▪ Diarrhea ▪ Abdominal pain ▪ Weight loss ▪ Nutritional deficiencies ▪ Fatigue	**ULCERATIVE COLITIS**: Can have mucus, pus, or blood in the stool **CROHN'S DISEASE**: Steatorrhea (fat in the stool)
DIAGNOSTIC	Colonoscopy	
SIMILARITIES	**BOTH:** ▪ Are a form of inflammatory bowel disease (IBD) ▪ Have causes that are not completely known ▪ Increase the risk for colon cancer ▪ Cause inflammation & ulcers	**DIET MODIFICATIONS FOR BOTH:** ↓ fiber ↑ protein ↑ calories ↑ fluids Small, frequent meals

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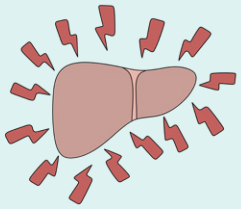
Types of Hepatitis

HEPATITIS

liver → inflammation

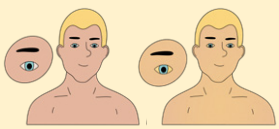
"Inflammation of the liver"



CAUSED BY:
- Virus (A, B, C, D, E) — MOST COMMON
- Excessive use of alcohol
- Hepatotoxic medications

	TRANSMISSION	SIGNS & SYMPTOMS	DIAGNOSTIC	TREATMENT	VACCINE
HAV ACUTE ONLY	**Fecal & oral** Food & water	GI symptoms (N&V, stomach pain, anorexia) Dark-colored urine Clay-colored stool Vomiting Flu-like symptoms Jaundice	Anti-HAV IgM = Active infection IgG = Recovered (It's Gone)	Supportive therapy & REST	✓
HBV ACUTE & CHRONIC	B think Body fluids *(blood, semen, saliva)* ▪ Childbirth ▪ Blood ▪ Sex ▪ IV drugs		HBsAg = Active infection Anti-HBs = Immune/recovered	**ACUTE** Supportive therapy & rest **CHRONIC** Antivirals	✓
HCV ACUTE & CHRONIC	**Body fluids** Most common: IV drug users		Anti-HCV No post-exposure immunoglobulin	**ACUTE** Supportive therapy & rest **CHRONIC** ▪ Antivirals ▪ Interferon	✗
HDV ACUTE & CHRONIC	Depends on B B & D = BuDs Hep D only occurs with Hep B		HDAg Anti-HDV	**ACUTE** Supportive therapy & rest **CHRONIC** ▪ Antivirals ▪ Interferon	✗
HEV ACUTE ONLY	**Fecal & oral** Food & water *(uncooked meats, developing countries)*		Anti-HEV	Supportive therapy & REST	✗

YELLOW DISCOLORATION of the skin from the buildup of bilirubin



normal jaundice

Patient Education for all types of hepatitis

- Rest
- Practice proper hand hygiene
- Do not share personal hygiene products
- Avoid sex until hepatitis antibodies are negative
- Avoid hepatotoxic substances:
  - Alcohol, acetaminophen, aspirin, sedatives

DIET MODIFICATIONS:
- ↑ Carbohydrates
- ↑ Calories
- ↓ Protein & fat

Small, frequent meals

Labs

All will be elevated with hepatitis

Liver enzymes:
ALT: 0–48 U/L
AST: 0–35 U/L
Bilirubin: 0.2–1.2 mg/dL
Ammonia: 10–80 mcg/dL

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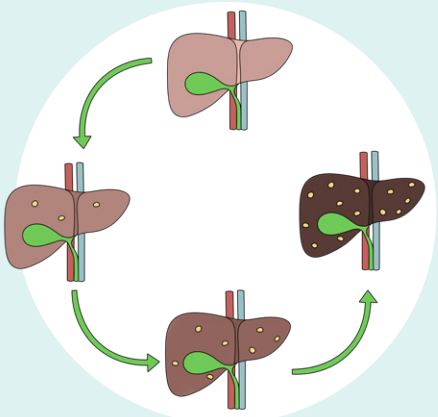
Cirrhosis

- Liver cells are **DESTROYED** and replaced with fibrotic (scar) tissue
- Normal function of the liver is compromised

Stages of Liver Damage

HEALTHY LIVER



FATTY LIVER
liver enlargement due to fat deposits

CIRRHOSIS LIVER
liver cell destruction

FIBROSIS LIVER
healthy liver tissue replaced with scar tissue

Functions of a healthy liver

1. **DETOXES** the body
2. Helps to **CLOT** the blood
3. Helps to **METABOLIZE** (break down) drugs
4. **SYNTHESIZES** (makes) albumin

If the function of the liver is disrupted, then none of these functions will work properly

Causes

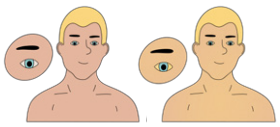
- **Alcoholic cirrhosis** (MOST COMMON)
  - Caused by excessive alcohol intake
- Nonalcoholic fatty liver disease (NAFLD)

- Viral hepatitis B or C
- Autoimmune disorders
- Hepatotoxic drugs
- Toxins & parasites
- Fat collection in the liver (obesity, diabetes, ↑ cholesterol)

Signs & Symptoms

- **Hepatic encephalopathy/coma**
  - When the liver is unable to filter toxins like ammonia, they build up in the blood and eventually reach the brain
- Asterixis "liver flap"

- Jaundice
  - Yellow discoloration of the eyes & skin
- Itchy skin
  - From buildup of toxins
- Ascites
- Edema
- Abdominal pain
- Heartburn



normal jaundice

Labs

- ↑ Bilirubin & ammonia
- ↓ Platelets
  - Risk for bleeding
- ↓ WBCs
  - Risk for infection

BLEEDING PRECAUTIONS
- Use electric razor
- Use soft-bristled toothbrush
- Hold pressure on scrapes/cuts to minimize bleeding

Medications

- ✓ **Lactulose** ★
  - ↓ Serum ammonia through the stool

MEMORY TRICK: Lactulose think LactuLOOSE because it LOOSENS the bowels

- ✓ Acid reducers (antacids & Histamine (H2) Receptor Antagonists)
- ✓ Diuretics
- ✓ Vitamins

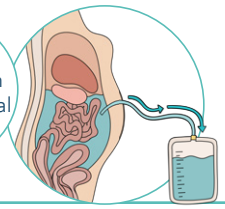
AVOID:
- ✗ Narcotics
- ✗ Acetaminophen

Treatment

TREAT UNDERLYING CAUSE OF CIRRHOSIS

IRREVERSIBLE

- Fibrosis (scarring) of liver cannot be reversed, but progression can be slowed
- Possibly a **paracentesis** for those with ascites
  - Removal of fluid from the peritoneal cavity



- In severe cases, a liver transplant may be needed

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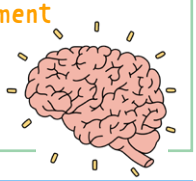
Neurological Assessments

LEVEL OF CONSCIOUSNESS (LOC)

LOC is always **#1** with neurological assessment

A change in LOC may be the only sign that there is a problem!



PUPILLARY CHANGES

MNEMONIC **PERRLA:**

- **P**upils
- **E**qual
- **R**ound
- **R**eactive to **L**ight
- **A**ccommodation

NORMAL PUPIL SIZE: 2-6 mm

GLASGOW COMA SCALE

Tool for assessing a patient's response to stimuli & assessing their level of consciousness

EYE-OPENING RESPONSE	Spontaneous	4
	To speech	3
	To pain	2
	No response	1
VERBAL RESPONSE	Oriented	5
	Confused	4
	Inappropriate words	3
	Unclear sounds	2
	No response	1
MOTOR RESPONSE	Obeys command	6
	Moves to localized pain	5
	Flexes to withdraw from pain	4
	Abnormal flexion	3
	Abnormal extension	2
	No response	1
TOTAL	3 - 15	

INTERPRETATION:

- WORST **3** Severe impairment of neurological function, coma, or brain death
- **< 8** Unconscious patient
- BEST **15** Fully alert & oriented

MENTAL STATUS

- Are they aware of their surroundings?
- Are they oriented to person, place, time & situation?
- Do they have their short-term & long-term memory?

ASK these types of questions to assess mental status:

- What is your name?
- Do you know where you are?
- Do you know what month it is?
- Who is the current U.S. president?
- Do you know what brought you here?

DEEP TENDON REFLEX (DTR) RESPONSES

- **0** = No response ABSENT
- **1+** = Present, but sluggish or diminished
- **2+** = Active or expected response NORMAL
- **3+** = More brisk response
- **4+** = Very brisk, hyperactive, with intermittent or transient clonus

BABINSKI REFLEX (plantar reflex)

Elicited by stroking the lateral side of the foot

INTACT CNS

Toes contract & draw together when lateral sole of the foot is stroked.

BRAIN DYSFUNCTION OR INJURY

Toes fan out when lateral sole of the foot is stroked.

Remember this is **only normal up to 2 years of age**, but **abnormal in adults!**

MEMORY TRICK Babinski - think normal in Babies & the Big toe fans out

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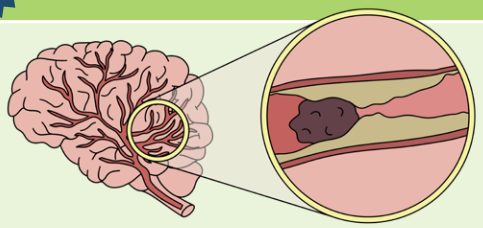
Cerebrovascular Accident (CVA) – Stroke

The sudden interruption of blood supply to the brain.
There are **2** types of strokes:

Ischemic STROKE

BLOCKAGE



Blood flow is cut off due to a **blockage,** which leads to **ischemia**

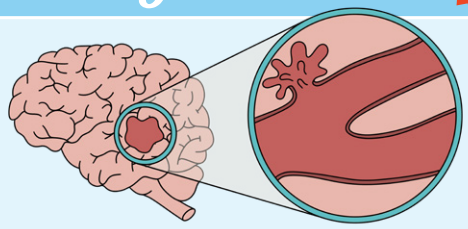
THROMBOSIS
A blood clot that forms on the artery wall

EMBOLISM
A blood clot, air, or fat travels within the body and gets stuck, blocking blood flow

TRANSIENT ISCHEMIC ATTACKS (TIAs)
"mini strokes"
No cerebral infarction or death occurs in a TIA, but it's a warning sign of an impending stroke

Hemorrhagic STROKE

BLEEDING



The **collection of blood** in the brain leads to **ischemia & increased ICP**

Can be caused by:
- Ruptured artery
- Aneurysm (weakening of the vessel)
- Uncontrolled hypertension

Classic symptom:
Sudden severe headache

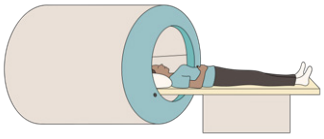
Risk Factors

MODIFIABLE
- **Hypertension**
- Atherosclerosis
- Anticoagulation therapy
- Uncontrolled diabetes mellitus
- Obesity
- Stress
- Oral contraceptives
- Smoking

NON-MODIFIABLE
- Family history of strokes
- Older age
- Male gender
- Black
- Hispanic

Diagnostic

CT scan to determine what type of stroke it is



Always identify the type of stroke BEFORE starting treatment

Treatment

ISCHEMIC STROKE

THROMBOLYTIC THERAPY ("clot buster")

SAVED BY THE SUFFIX

GENERIC	TRADE NAME
alteplase	Activase
reteplase	Retavase
streptokinase	–
tenecteplase	TNKase

SUFFIX: -ASE

Must be administered 3–4.5 hours after the onset of symptoms

HEMORRHAGIC STROKE

STOP THE BLEEDING!
- Prevent & treat increased intracranial pressure (ICP)
- Implement seizure precautions

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Cerebrovascular Accident (CVA) - Stroke

Signs & Symptoms

MNEMONIC

Act

F
A
S
T

Facial droop
(uneven smile)

Arm weakness
(arm numbness or weakness on one side)

Speech difficulty
(slurred speech)

Time to call 911

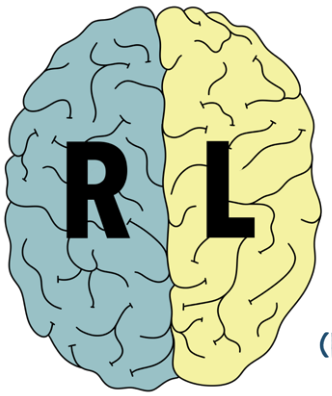
Complications

RIGHT BRAIN

MEMORY TRICK

Right = **R**eckless

- Impulsive behavior
- Rapid movements
- Impaired judgment



Paralysis (hemiparesis) on the **LEFT** side of the body

Paralysis (hemiparesis) on the **RIGHT** side of the body

LEFT BRAIN

MEMORY TRICK

Left = **L**anguages

Trouble speaking (**aphasia**), reading, or reasoning

TYPES OF APHASIA

RECEPTIVE: Unable to comprehend speech
(damage to the Wernicke's area)

EXPRESSIVE: Able to comprehend speech but unable to communicate back using speech
(damage to the Broca's area)

REMEMBER:
A stroke on one side of the brain will **NOT ALWAYS** produce negative effects on the opposite side. This depends on where the stroke occurs in the brain & is not a "rule of thumb" for all strokes.

Key Definitions

HEMIPARESIS

Weakness or paralysis on one side of the body, such as in the arms, legs, or face

HOMONYMOUS HEMIANOPSIA

Blindness in half of the visual field in both eyes
(either the left or right side)

Place the meal tray on the unaffected side so the patient can see their food

DIPLOPIA

Double vision

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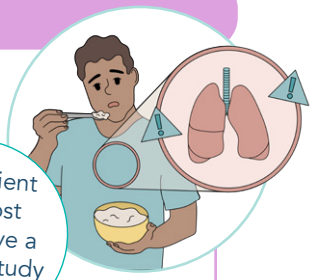
Cerebrovascular Accident (CVA) - Stroke

Nursing Considerations

PREVENTING ASPIRATION

- Do not feed the patient until the gag reflex has returned
- Keep suction at the bedside
- Crush medications if possible
- Provide sips of fluids with meals
- Place the food on the unaffected side of the mouth



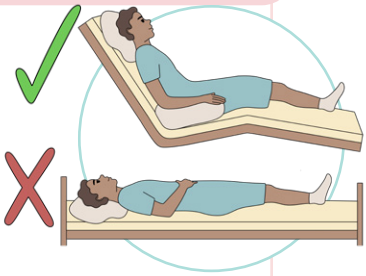
The patient will most likely have a swallow study completed

PREVENTING DVTs

- Place anti-embolism stockings on the patient
- Encourage passive range of motion (ROM) every 2 hours
- Encourage frequent position changes

PREVENTING INCREASED ICP

- Elevate the head of the bed > 30 degrees
- **EDUCATE** the patient to avoid coughing, sneezing, nose-blowing, and bending over/flexing at the hips
- Avoid endotracheal suction for >10 seconds
- Administer stool softeners to prevent straining during bowel movements



DIET MODIFICATIONS

- Will most likely start on a liquid diet and progress to a regular diet

LIQUID	SOLID
Thin	Pureed
Nectar-like	Mechanically altered
Honey-like	Mechanically softened
Spoon-thick	Regular

COMMUNICATION

- Use simple statements
- Ask simple questions
- Do not rush the patient (be patient!)

POSITIONING

- Place a pillow under the affected arm in a neutral position
- Relieve pressure on bony areas to avoid pressure injuries

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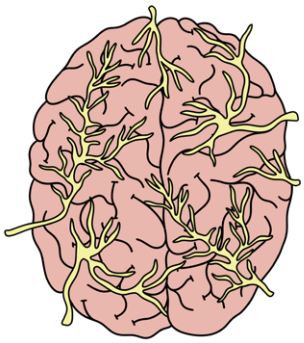
Seizures

WHAT IS A *seizure?*

Abnormal & sudden electrical activity of the brain



WHAT IS *epilepsy?*

Chronic disease of the brain (recurrent, unprovoked seizures)

Stages of a Seizure

PRODROMAL → **AURA** → **ICTUS** → **POST-ICTUS**

PRODROMAL

Symptoms that start before the actual seizure (can be days before the seizure happens)

AURA

Warning sign right before the seizure happens:

- Weird smell or taste
- Altered vision
- Dizziness

Not all patients experience an aura

ICTUS

SEIZURE!

Seizure activity of any type

Status epilepticus: a seizure that lasts > 5 minutes without any consciousness during the seizure

POST-ICTUS

Recovery after the seizure

- Headache
- Possible injury
- Confusion
- Extreme tiredness

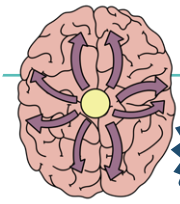
Causes

- ↑ Fever (febrile seizure in child)
- CNS infection
- Drug or alcohol withdrawal

- ABG imbalance
- Hypoxia
- Brain tumor

- Hypoglycemia
- Head injury
- Hypertension

Generalized Seizures



THE ENTIRE BRAIN IS AFFECTED

TONIC-CLONIC

- Used to be called "grand mal"
- May begin with an aura
- Stiffening (tonic) and/or rigidity (clonic) of the muscles

MYOCLONIC

- Sudden jerking or stiffening of the extremities (arms or legs)

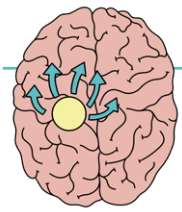
ABSENCE

- Usually looks like a blank stare that lasts seconds
- Often goes unnoticed

ATONIC

- Sudden loss of muscle tone
- May lead to sudden falls or dropping things

Partial (FOCAL) Seizures



ONE AREA OF THE BRAIN IS AFFECTED

SIMPLE PARTIAL

- Sensory symptoms with motor symptoms and continued awareness
- May be preceded by an aura

COMPLEX PARTIAL

- Altered behavior/awareness and loss of consciousness for a few seconds

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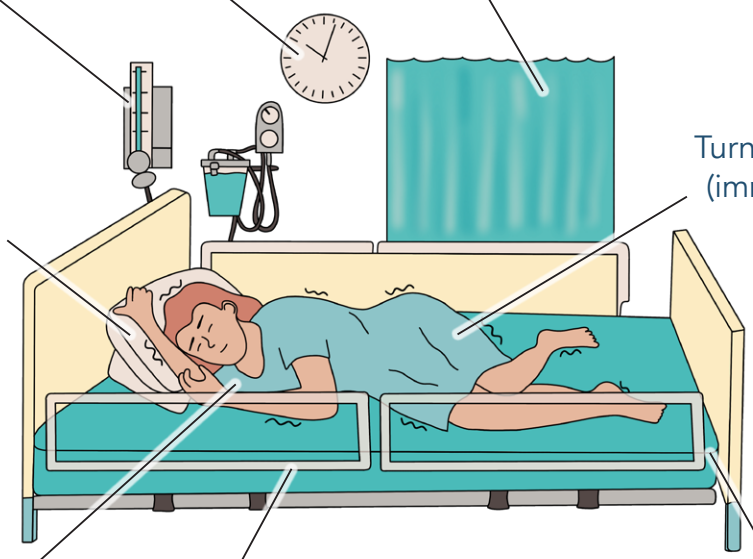
Seizure Precautions

DOs

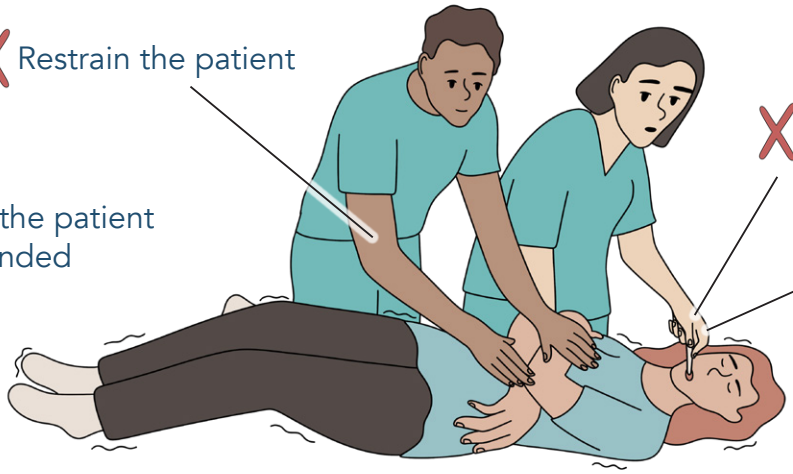
REMEMBER: *If the seizure lasts >5 minutes, it is status epilepticus. This needs IMMEDIATE attention*



- Maintain a patent airway: have oxygen & suction available
- Note the time and duration of seizure
- Provide privacy
- Place a pillow under their head for protection
- Turn to a side-lying position (immediately post-seizure)
- Loosen clothing, especially around their neck, to maintain a patent airway
- Put PADDED side rails UP
- Move furniture or dangerous edges & corners away from patient
- Place bed in LOWEST position

Aspiration Prevention
To prevent aspiration in the event that secretions are present in the oral cavity during the seizure

DON'Ts



- ✗ Move the patient (unless in imminent danger!)
- ✗ Restrain the patient
- ✗ Force the jaw open
- ✗ Leave the patient unattended
- ✗ Place anything in the mouth

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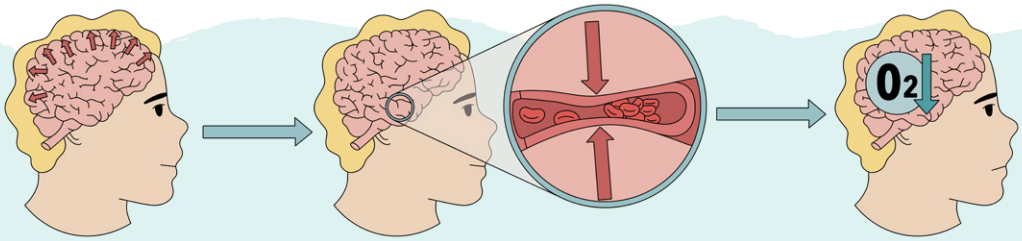
Increased Intracranial Pressure (ICP)

Pathology

Normally there is some pressure in the skull, which is called intracranial pressure (ICP). But when there is increased ICP, it's considered a medical emergency.

NORMAL INTRACRANIAL PRESSURE
10 – 15 mmHg



Increased pressure in the brain

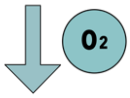
Pressure on the blood vessels

Decreased oxygen flow to the brain

Signs & Symptoms

EARLY SYMPTOMS

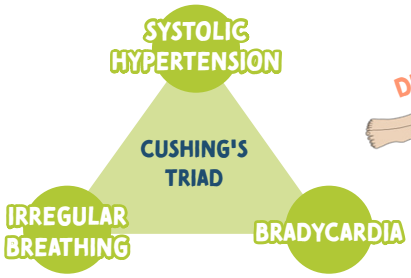
All from low levels of oxygen (cerebral hypoxia)



- Restlessness
- Irritability
- Agitation
- Altered level of consciousness (LOC)
- Headaches
- ★ **Sudden vomiting without nausea**

LATE SYMPTOMS

- Nuchal rigidity
- Fixed & dilated pupils
- Babinski reflex
- Abnormal posturing



MEMORY TRICK: DeCORticate = arms are at the CORE of the body

Risk Factors

- Infection (*Example:* meningitis)
- Aneurysm
- Seizures
- Tumors
- Stroke

Nursing Considerations

- Immobilize the head if a neck injury is suspected
- ★ **Avoid increasing intracranial pressure (ICP) further**
  - Maintain head of bed at 30° or *slightly* higher if necessary
  - Administer stool softeners to prevent straining during bowel movements
  - Avoid endotracheal suction for > 10 seconds

- **EDUCATE** the patient to avoid:
  - ✗ Coughing
  - ✗ Sneezing
  - ✗ Blowing their nose
  - ✗ Bending over/flexing at the hips

Medications

- ★ **Diuretic: IV mannitol**
  - To decrease ICP
- Anti-seizure drugs
  - To treat or prevent seizures (*Example:* phenytoin)
- Corticosteroids
  - To decrease inflammation

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Cranial Nerves

WHAT ARE *Cranial Nerves*?

These are nerves that originate from the brain stem.
They send information to & from various parts of the body.

START

SE SENSORY
M MOTOR
B BOTH

MNEMONICS

Ooh	Olfactory	Some	Sensory
Ooh	Optic	Say	Sensory
Ooh	Oculomotor	Marry	Motor
To	Trochlear	Money	Motor
Touch	Trigeminal	But	Both
And	Abducens	My	Motor
Feel	Facial	Brother	Both
Very	Vestibulocochlear/Acoustic	Says	Sensory
Good	Glossopharyngeal	Big	Both
Velvet	Vagus	Brains	Both
Such	Spinal Accessory	Matter	Motor
Heaven	Hypoglossal	More	Motor

XII: HYPOGLOSSAL — M

FUNCTION:
Tongue movement (swallowing & speech)

"Glosso" means tongue

TEST:
Inspect the patient's tongue & ask them to stick their tongue out

XI: SPINAL ACCESSORY — M

FUNCTION:
Strength of neck & shoulder muscles

TEST:
Ask the patient to rotate their head & shrug their shoulders

X: VAGUS — B

FUNCTION:
MOTOR: Swallowing, speaking & coughing
SENSORY: Facial sensation

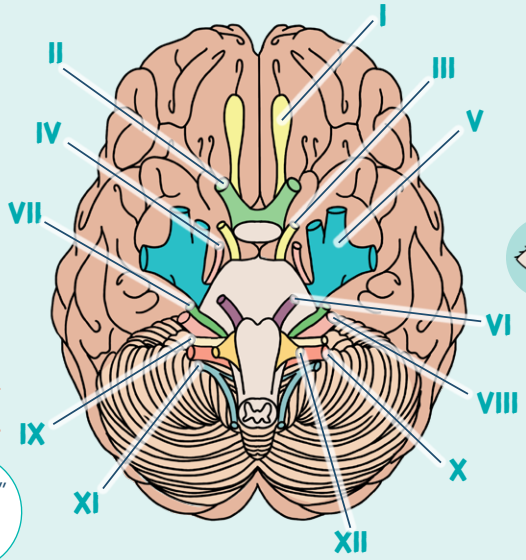
TEST:
Test the sensation coming from the skin around the patient's ear

IX: GLOSSOPHARYNGEAL — B

FUNCTION:
MOTOR: Tongue movement & swallowing
SENSORY: Taste (sour & bitter)

"Glosso" means tongue

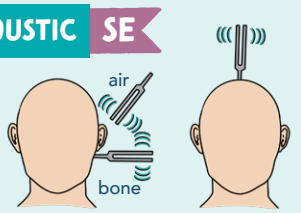
TEST:
Test tongue by giving patient sour, bitter & salty substances



VIII: VESTIBULOCOCHLEAR / ACOUSTIC — SE

FUNCTION:
Balance & hearing

TEST:
- Stand with eyes closed
- Otoscopic exam
- Rinne & Weber tests



VII: FACIAL — B

FUNCTION:
MOTOR: Facial expression
SENSORY: Taste (sweet & salty)

TEST:
- Ask the patient to make different facial expressions (frown, smile, raise eyebrows, close eyes, blow)
- Test tongue by giving the patient sour, sweet, bitter & salty substances

VI: ABDUCENS — M

FUNCTION:
Parallel eye movement
Abduction: moving laterally, away from midline

TEST:
- Look up, down & inward
- Ask the patient to follow your finger as you move it towards their face

V: TRIGEMINAL — B

FUNCTION:
MOTOR: Mastication (biting & chewing)
SENSORY: Facial sensation

TEST:
- Apply pressure on forehead, cheek & jaw with a cotton swab to check sensation
- Ask the patient to open their mouth & then bite down

I: OLFACTORY — SE

FUNCTION:
Sense of smell

TEST:
Smell substance with eyes closed (test each nostril separately)

II: OPTIC — SE

FUNCTION:
Vision

TEST:
- Snellen chart
- Ophthalmoscopic exam
- Confrontation to check peripheral vision

III: OCULOMOTOR — M

FUNCTION:
Ocular (eye) motor (movement)
Controls most eye movements, pupil constriction & upper-eyelid rise

TEST:
- Look up, down & inward
- Ask the patient to follow your finger as you move it towards their face

IV: TROCHLEAR — M

FUNCTION:
Downward & inward eye movement

TEST:
- Look up, down & inward
- Ask the patient to follow your finger as you move it towards their face

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Cranial Nerves

WHAT ARE *Cranial Nerves*?

These are nerves that originate from the brain stem.
They send information to & from various parts of the body.

START

LABEL THE FLAGS:
- SE SENSORY
- M MOTOR
- B BOTH

MNEMONICS

Ooh	O_____	Some	S_____
Ooh	O_____	Say	S_____
Ooh	O_____	Marry	M_____
To	T_____	Money	M_____
Touch	T_____	But	B_____
And	A_____	My	M_____
Feel	F_____	Brother	B_____
Very	V_____/A_____	Says	S_____
Good	G_____	Big	B_____
Velvet	V_____	Brains	B_____
Such	S_____	Matter	M_____
Heaven	H_____	More	M_____

XII:
FUNCTION:
TEST:

XI:
FUNCTION:
TEST:

X:
FUNCTION:
TEST:

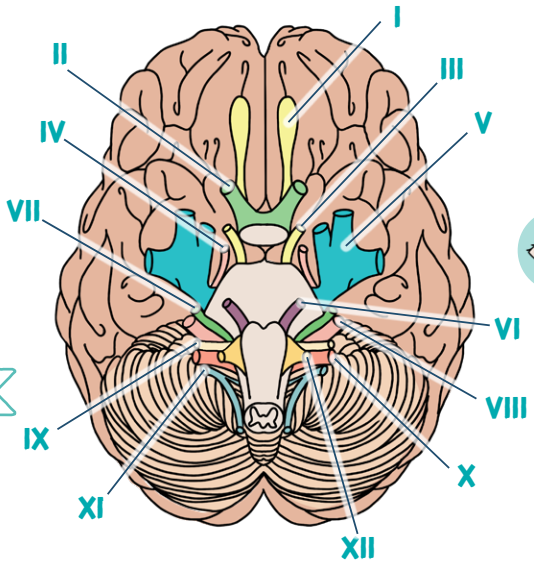
IX:
FUNCTION:
TEST:

VIII:
FUNCTION:
TEST:

VII:
FUNCTION:
TEST:

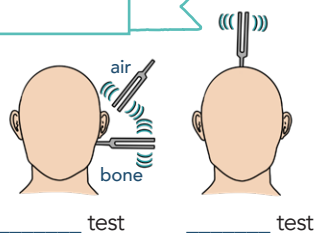
I:
FUNCTION:
TEST:

II:
FUNCTION:
TEST:



III:
FUNCTION:
TEST:

IV:
FUNCTION:
TEST:



VI:
FUNCTION:
TEST:

V:
FUNCTION:
TEST:

WANT MORE WORKSHEETS?
Check out The Complete Laminated Study Templates!

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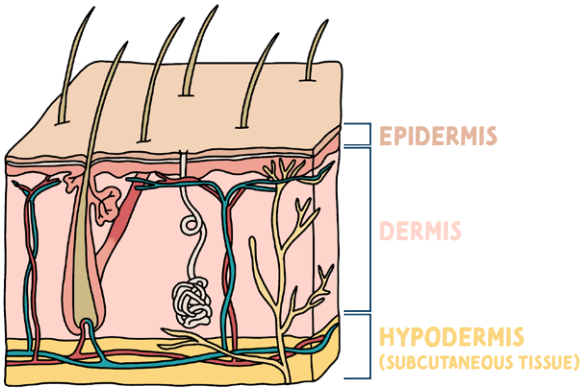
Burns

What is a burn? Damage to skin integrity through contact with extreme temperatures or other sources

Burn Injury Stages

1ST DEGREE — Superficial	▪ Epidermis ▪ Pink & painful (still has nerves) ▪ No scarring	**BLANCHING**: present **HEALS**: a few days
2ND DEGREE — Superficial Partial-Thickness	▪ Epidermis & dermis ▪ Blistered, shiny & moist ▪ Painful	**BLANCHING**: present **HEALS**: 2–6 weeks
3RD DEGREE — Full-Thickness	▪ Epidermis, dermis & hypodermis ▪ May look black, yellow, red & wet ▪ No pain/limited pain (nerve fibers are destroyed) ▪ Skin will not heal (need skin grafting) ▪ **Eschar**: dead tissue, leathery; must be removed!	

Layers of the Skin



Types of Burns

MOST COMMON — Thermal	Caused by superficial heat *Examples: liquid, steam, fire*
Chemical	Caused by a toxic substance (alkalotic or acidic) *Examples: bleach, gasoline, paint thinner*
Radiation	Caused by UV radiation (sunburns) & cancer treatment (radiation therapy)
Inhalation	Caused by inhaling smoke, which can cause flame injury or **carbon monoxide poisoning**
Friction	Caused when an object rubs off the skin *Examples: road rash, scrapes, carpet burn*
Cold	Caused when skin has been overexposed to cold *Examples: frostbite*
Electric	Caused when electrical current passes through the body, causing damage within

INHALATION INJURY

Damage to the respiratory system; happens mostly in a **closed area**

SIGNS OF INHALATION INJURY:

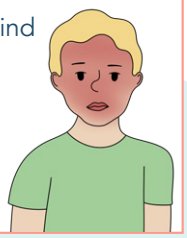
- Hair singed around the face, neck, or torso
- Trouble talking
- Soot in the nose or mouth
- Confusion or anxiety

CARBON MONOXIDE POISONING

Carbon monoxide travels faster than oxygen, allowing it to bind to Hgb first. Now oxygen cannot bind to Hgb = **HYPOXIA**

Classic symptom: cherry red skin

Treatment: 100% O_2



POTENTIAL COMPLICATIONS

Dysrhythmias, fracture of bones; release of **myoglobin & hemoglobin** into the blood, which can clog the kidneys

Complications

DISABILITY in the:

- Hands
- Feet
- Joints
- Eyes

TROUBLE HEALING

- Poor blood supply
- Diabetes

INFECTION

In any open area where bacteria can easily enter

- Perineum
- Ears
- Eyes

COMPARTMENT SYNDROME

- In the extremities

Tightened skin acts like a band around the skin, cutting off blood supply (ex: eschar)

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Phases of Burn Management

MNEMONIC "EAR" = Emergent, Acute, Rehabilitative

E Emergent Phase — FROM ONSET OF INJURY UP TO 48 HOURS

From the onset of injury to the restoration of capillary permeability

PATHOLOGY

↑ Capillary permeability (leaky vessels), causing:
- Plasma leaves the intravascular space
  - Albumin & sodium follow
- Fluids shift to the interstitial tissue

LEADS TO EDEMA

→ **Leads to fluid volume deficit (FVD) in the intravascular space**

VITAL SIGNS

- ↑ Pulse
- ↓ Blood pressure
- ↓ Cardiac output

Think HYPOVOLEMIC SHOCK!

- ↓ Urine output *(from ↓ perfusion to the kidneys)*

LABS
- ↑ Potassium (K^+)
- ↑ Hematocrit (HCT)
- ↑ BUN/creatinine
- ↓ White blood cells (WBCs)

NURSING CONSIDERATIONS

ABCs
- Airway monitoring/maintenance
- Establish IV access (preferably 2 large-bore)
- Fluids (Lactated Ringer's, crystalloids)
- Parkland formula
- Foley catheter to monitor urinary output (UOP)
  - **GOAL:** > 30 mL/hr
- ↓ Edema
  - Elevate extremities above heart level

A Acute Phase — 48 – 72 HOURS after burn & until wounds have healed

From the stabilization of capillary permeability to wound closure

PATHOLOGY

Capillary permeability is restored, which leads to diuresis (increased urine production). All the excess fluid that previously shifted from the interstitial tissue shifts back into the intravascular space.

GOALS

- **PREVENT INFECTION**
  - Systemic antibiotic therapy
- **ENSURE PROPER NUTRITION**
  - ↑ Calories
  - ↑ Protein & vit C to promote healing
- **ALLEVIATE PAIN**
- **WOUND CARE**
  - Premedicate before wound care
  - Debridement or grafting

NURSING CONSIDERATIONS

RENAL
- Diuresis is occurring
- Foley catheter to monitor UOP

RESPIRATORY
- Possible intubation if respiratory complications occur

GASTROINTESTINAL
- Since the client is in FVD, there is ↓ perfusion to the stomach
  - Paralytic ileus
  - Curling's ulcer
- Medication to ↓ chance of ulcers
  - H2 histamine blocking agent (↓ HCl)

MONITOR bowel sounds
- May need NG tube for suctioning

R Rehabilitative Phase — COULD BE WEEKS TO YEARS

Burn has healed and the patient is functioning mentally & physically

GOALS

- Psychosocial
- Activities of daily living (ADLs)
- Physical therapy (PT)
- Occupational therapy (OT)
- Cosmetic corrections

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Fluid Resuscitation for Burns

The Parkland Formula

Used to calculate the total volume of fluids (mL) that a patient needs **24 hours** after experiencing a burn

Apply only to 2nd & 3rd degree burns

4 mL X TBSA (%) X BODY WEIGHT (KG) = TOTAL ML OF FLUID NEEDED

↓

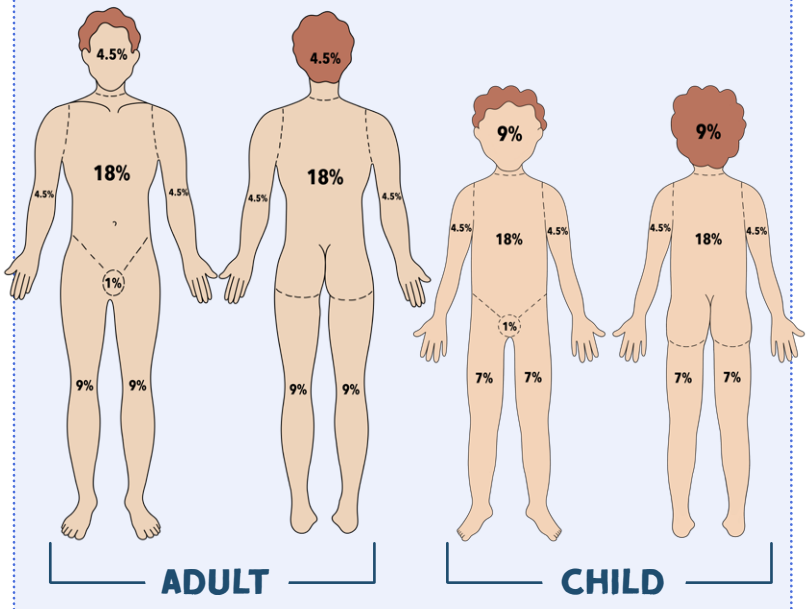
Give the first half of the solution in the **FIRST 8 HOURS**

↓

Give the second half of the solution over the **NEXT 16 HOURS**

Rule of Nines

Quick estimate of the % of **total body surface area (TBSA)** that has been affected by a partial- or full-thickness burn



Practice Question

PART 1: CALCULATING TBSA (%)

A 25-year-old male patient who weighs **79 kg** has sustained burns to the back of the right arm, posterior trunk, front of the left leg, and anterior head and neck. Using the **Rule of Nines**, calculate the total body surface area percentage that is burned.

Back of right arm: 4.5%
Posterior trunk: 18%
Front of left leg: 9%
Anterior head & neck: 4.5%

ANSWER: 36%

NOTE: The formula uses TBSA (%). Make sure to calculate using the percentage (36%) rather than the decimal equivalent (0.36)

PART 2: USING THE PARKLAND FORMULA

Use the Parkland formula to calculate the total amount of lactated Ringer's solution that will be given over the next 24 hours.

ANSWER: 11,376 mL

4 mL X 36% X 79 kg = 11,376 mL

↓

11,376 ÷ 2 = 5,688 mL
FIRST 8 HOURS

↓

11,376 ÷ 2 = 5,688 mL
NEXT 16 HOURS

Keep in mind: the question could ask you for mL given over varying time periods, such as the first 8 hours or the first 24 hours, so read the question carefully.

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Shock (Hypovolemic & Cardiogenic)

What is Shock?

A life-threatening condition resulting from **INADEQUATE TISSUE PERFUSION**, which leads to possible cell dysfunction, cell death, and even organ failure

HYPOVOLEMIC SHOCK

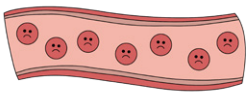
ETIOLOGY

MOST COMMON TYPE OF SHOCK

↓ Intravascular volume

HYPOVOLEMIC

low volume in the blood



CAUSES

NONHEMORRHAGIC (not from bleeding)

- Fluid shift (edema or ascites)
- Severe dehydration (vomiting, diarrhea, burns)

HEMORRHAGIC (from bleeding)

- Trauma
- GI bleed
- Postpartum

SIGNS & SYMPTOMS

Pulse	CO	HR	BP
Weak, thready pulse	↓ Not a lot of blood being pumped by the heart	↑ **Tachycardia** Compensating to increase blood flow	↓ **Hypotension**

Skin	CVP	SVR	O_2 Sat
Cyanosis *(Bluish tint of the lips, tongue & fingertips)* Cool, pale skin ↓ Capillary refill (> 3 seconds)	↓	↑ **Vasoconstriction**	↓ Blood being perfused to the body = low O_2

Other Signs & Symptoms

LABS CAN BE:

- ↑ HCT hemoconcentration
- ↓ HCT actually hemorrhaging the RBCs
- Oliguria (urine output of < 30 mL/hr)
- Confusion, agitation

due to decreased blood flow to the brain

TREATMENT

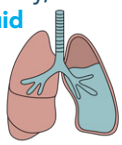
- **2 large-bore IVs**
- **Fluids & blood replacement**
  - Crystalloids *(normal saline or lactated Ringer's)*
  - Colloids *(albumin)*
  - Blood products *(plasma, PRBCs & PLTs)*

CARDIOGENIC SHOCK

ETIOLOGY

The heart can't pump enough blood to meet the perfusion needs of the body

NOTE:
There is enough blood, but the heart just can't pump it to the body, which causes **fluid accumulation** in the lungs!



CAUSES

- Damage from an acute MI
- Severe hypoxemia
- Acidosis
- Hypoglycemia
- Cardiomyopathy
- Cardiac tamponade
- Dysrhythmias

SIGNS & SYMPTOMS

Pulse	CO	HR	BP
Weak peripheral pulses	↓ Not a lot of blood being pumped by the heart	↑ **Tachycardia** Compensating to increase blood flow	↓ **Hypotension**

Skin	CVP	SVR	O_2 Sat
Cool, clammy skin ↓ Capillary refill (> 3 seconds)	↑	↑ **Vasoconstriction**	↓ blood being perfused to the body = low O_2

Other Signs & Symptoms

- Jugular vein distention (JVD)
- Chest pain
- Oliguria (urine output of < 30 mL/hr)
- Confusion, agitation

due to decreased blood flow to the brain

From fluid accumulation in the lungs:
- Dyspnea
- Pulmonary edema

TREATMENT

- **For an MI:**
  - Angioplasty
  - Thrombolytics

- **Oxygen**
- **Vasopressors** *(example: epinephrine, dobutamine, dopamine)*

Vasopressors cause vasoconstriction, which ↑ BP and ↑ perfusion to the organs

- **Diuretics**
  - ↓ Workload of the heart
  - ↓ Extra blood volume
- **Intra-aortic balloon pump** *(helps to improve coronary artery blood flow & increase cardiac output)*

BP = blood pressure **HR** = heart rate **CO** = cardiac output **SVR** = systemic vascular resistance **CVP** = central venous pressure

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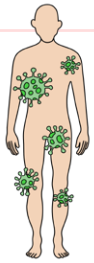
Distributive Shock (Septic, Neurogenic)

Distributive Shock: Leaky blood vessels → Excessive **vasodilation** (widening of vessels) → Intravascular volume pools in the peripheral blood vessels → Since the blood is in the peripherals, it is NOT perfusing the vital organs, which causes relative hypovolemia

SEPTIC SHOCK (SEPSIS)



ETIOLOGY

Caused by widespread infection or sepsis

MOST COMMON TYPE OF DISTRIBUTIVE SHOCK

CAUSES

- Pneumonia
- Urosepsis
- Bacterial, fungal, or viral infection
- Intra-abdominal infections
- Wound infection
- Invasive procedures
- Indwelling medical devices (catheters)

TREATMENT

CORRECT THE UNDERLYING CAUSE

- Fluid replacement
- **Broad-spectrum antibiotics**
- Vasopressors (norepinephrine & dopamine)
- Neuromuscular blockade agents & sedation
  - ↓ Metabolic demands & provides comfort
- Medications to prevent stress ulcers
  - H2 blocking agents
  - Proton pump inhibitors (PPIs)

Broad spectrum antibiotics are used when the organism that is causing the infection is not yet known/determined. Once the organism is known, the patient can be put on more specific antibiotics.

SIGNS & SYMPTOMS

Pulse	CO	HR	BP	Other Signs & Symptoms

Bounding pulses	↑	↑ Tachycardia	↓ Hypotension	▪ Hyperthermia & fever
				▪ Increased respiratory rate
				▪ GI upset: Nausea, vomiting, diarrhea, decreased gastric motility
				▪ ↑ Inflammatory markers
				  ▪ ↑ WBCs
				  ▪ ↑ C-reactive protein (CRP)

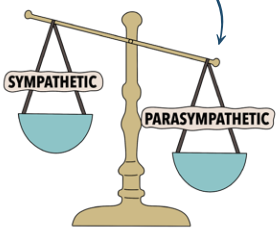
Skin	**CVP**	**SVR**	**O_2 Sat**	
Initially warm & flushed, but as BP drops, the skin becomes cool, pale & mottled	↓	↓ Vasodilation	↓	

NEUROGENIC SHOCK (VASOGENIC)

ETIOLOGY

Vasodilation due to a loss of balance between

In neurogenic shock, the patient mainly experiences parasympathetic stimulation, which causes VASODILATION for an extended period



MEMORY TRICK: P think Peaceful

PARASYMPATHETIC STIMULATION (Rest & digest) → Causes dilation (relaxing) of the smooth muscles

SYMPATHETIC STIMULATION (Fight or flight) → Causes constriction (tightening) of the smooth muscles

CAUSES

- Spinal cord injury (above T6, cervical)
- Spinal anesthesia
- Nervous system damage
- Insulin reaction

MEMORY TRICK: *NEUR*ogenic = Issue with *NER*vous system

SIGNS & SYMPTOMS

EVERYTHING IS DECREASED

Remember that parasympathetic means **RELAXED EVERYTHING**

CO	HR	BP
↓	↓	↓ Hypotension

The sympathetic NS is not working to compensate to increase the HR

RELATIVE HYPOVOLEMIA: There is enough blood volume. However, the vascular space is **dilated,** so blood volume is displaced, causing hypovolemia.

Skin	CVP	SVR	O_2 Sat
Dry, warm extremities (venous blood pooling) Hypothermia: warm/dry extremities, cold body	↓	↓ Vasodilation	↓

TREATMENT DEPENDS ON THE CAUSE OF THE SHOCK

- Protect the spine (for spinal cord injuries) → **SPINAL CORD INJURY:** Keep spine immobilized (cervical collar, backboards, log-rolling)
- Assess & manage airway (may need intubation or mechanical ventilation)
- Elevate the head of the bed
- Adm. IV fluids ⚠ Watch for fluid volume overload
- Watch for signs of a clot (increased risk for clots due to pooling of blood) → **S&S OF BLOOD CLOTS:**
  - Pain in the extremities
  - Redness
  - Tenderness
  - Warmth
- Use compression devices
- Adm. antithrombotic agents (heparin)
- Vasopressors (*example*: epinephrine, dobutamine, dopamine)

BP = blood pressure **HR** = heart rate **CO** = cardiac output **SVR** = systemic vascular resistance **CVP** = central venous pressure

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Distributive Shock (Anaphylactic)

Distributive Shock: Leaky blood vessels → Excessive **vasodilation** (widening of vessels) → Intravascular volume pools in the peripheral blood vessels → Since the blood is in the peripherals, it is NOT perfusing the vital organs, which causes relative hypovolemia

ANAPHYLACTIC SHOCK

ETIOLOGY

SEVERE ALLERGIC REACTION

Foreign substance (antigen)
↓
Systemic antigen-antibody reaction (IgE)
↓
Mast cells release potent vasoactive substance (histamine/bradykinin)
↓
Activates inflammatory cytokines
↓
Causes **VASODILATION** & capillary permeability

CAUSES/TRIGGERS

Often unknown (idiopathic)

- Foods (example: peanuts)
- Medications
- Insects (example: bee sting)
- Latex
- Exercise-induced anaphylaxis (EIA)

Signs & symptoms usually occur within 2–30 minutes of exposure to antigen

TREATMENT

REMOVE THE ALLERGEN!

★ **First-line drug: epinephrine**
- Causes vasoconstriction & bronchodilation

- High-flow oxygen
- Other possible medications:
  - Antihistamines
    - Diphenhydramine (Benadryl)
  - Albuterol (Proventil)
  - Corticosteroids
- Fluids
- Continue to monitor the patient

BIPHASIC ANAPHYLAXIS:
A recurrence of anaphylaxis after appropriate treatment

SIGNS & SYMPTOMS

Pulse	CO	HR	BP
Rapid, weak pulse	↓	↑ Tachycardia	↓ Hypotension

CAPILLARY PERMEABILITY:
Fluid is leaving the intravascular space

Skin	CVP	SVR	O_2 Sat
Generalized flushing	↓	↓ Vasodilation	↓

Other Signs & Symptoms

- **CARDIAC**
  - Cardiac dysrhythmias or cardiac arrest
- **GI**
  - Nausea/vomiting
  - Acute abdominal pain
- **FEELING OF IMPENDING DOOM**
- **RESPIRATORY**
  - Bronchoconstriction
    - Difficulty breathing
    - Wheezing
    - Coughing
    - Inability to speak
- **SKIN**
  - Itching, generalized flushing, redness, hives, or a rash

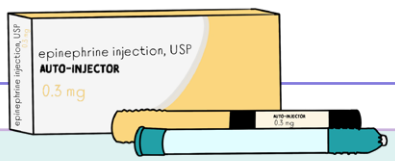
HOW TO USE AN EPINEPHRINE AUTO-INJECTOR (EAI)

EDUCATION POINTS:

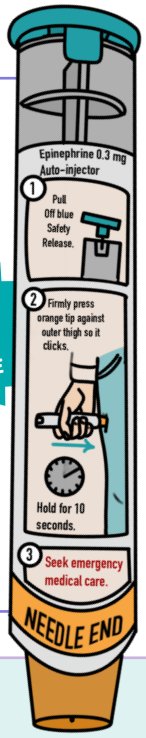
- Store in a dark room
- Administer EAI immediately after the first sign of an allergic reaction
- Inject through clothing layers (pants, jeans, etc)

EXPECTED SYMPTOMS AFTER ADMINISTRATION:

- Tachycardia
- Palpitations
- Dizziness



INJECT IN THE OUTER THIGH AT A 90° ANGLE



BP = blood pressure **HR** = heart rate **CO** = cardiac output **SVR** = systemic vascular resistance **CVP** = central venous pressure

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ABGs

A B G
ARTERIAL BLOOD GAS

ABGs MEASURE HOW ACIDIC or ALKALOTIC THE BLOOD IS IN THE ARTERIAL CIRCULATION

*They're also a measure of gases such as O_2 & CO_2

4 Must-Know COMPONENTS

pH	Measurement of how acidic or alkalotic the blood is	Regulated by the lungs & kidneys	7.35 - 7.45
$PaCO_2$	Measurement of carbon dioxide in the blood CO_2 think aCid	Regulated by the lungs	35 - 45 mm Hg
HCO_3	Measurement of bicarbonate in the blood Bicarbonate think Base	Regulated by the kidneys	22 - 26 mmol/L
PaO_2	Measurement of oxygen in the blood	Regulated by the lungs	80 - 100 mm Hg

This value is not needed to interpret alkalosis or acidosis. It just tells you if the patient is hypoxic or not.

ABG Interpretation

1 KNOW YOUR LAB VALUES!

	ACIDOSIS	NORMAL	ALKALOSIS
pH	< 7.35 ↓	7.35 - 7.45	> 7.45 ↑
CO_2	> 45 ↑	35 - 45	< 35 ↓
HCO_3	< 22 ↓	22 - 26	> 26 ↑

2 RESPIRATORY OR METABOLIC PROBLEM?

There are 2 ways to analyze the information

ROME METHOD

Respiratory	pH ↑	CO_2 ↓	Alkalosis
Opposite	pH ↓	CO_2 ↑	Acidosis
Metabolic	pH ↑	HCO_3 ↑	Alkalosis
Equal	pH ↓	HCO_3 ↓	Acidosis

TIC-TAC-TOE METHOD

ACID	NORMAL	BASE

3 UNCOMPENSATED, PARTIALLY COMPENSATED, OR FULLY COMPENSATED?

If the **pH** is out of range & **CO_2** or **HCO_3** is in range = **UNCOMPENSATED**

If **CO_2**, **HCO_3** & **pH** are **ALL** out of range = **PARTIALLY COMPENSATED**

If **pH** is in range (7.35-7.45) = **FULLY COMPENSATED**

7.35 — 7.40 — 7.45
Acidosis ← Absolute Normal → Alkalosis

pH IN RANGE? Even if the pH is "normal," it can still fall on the acidosis side or alkalosis side

How do the organs compensate?

KIDNEYS

MEMORY TRICK
B think BASE
BICARB HYDROGEN

Excreting excess **ACID** & **BICARB (HCO_3)**
OR
Retaining **HYDROGEN** & **BICARB (HCO_3)**

hours - days to compensate

LUNGS

MEMORY TRICK
CO_2 think ACID
CO_2

HYPERventilation = ↓ CO_2 = **ALKALOSIS**

HYPOventilation = ↑ CO_2 = **ACIDOSIS**

minutes to compensate

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ABG Practice Question

QUESTION:

A patient with a bowel obstruction has been treated with gastric suctioning for 4 days. The nurse notices an increase in nasogastric drainage. Which acid-base imbalance does that nurse correctly identify?

The patient labs are the following →

pH: **7.50**
$PaCO_2$: **50** mm Hg
PaO_2: **90** mm Hg
HCO_3: **32** mEq/L

This value is not needed to interpret alkalosis or acidosis. It just tells you if the patient is hypoxic or not.

TIC-TAC-TOE METHOD

1 What does the problem give you?

pH	7.50	○ ACIDIC	✓ ALKALOTIC	○ NORMAL
CO_2	50	✓ ACIDIC	○ ALKALOTIC	○ NORMAL
HCO_3	32	○ ACIDIC	✓ ALKALOTIC	○ NORMAL

2

ACID	NORMAL	BASE
CO_2		pH
		HCO_3

- ○ RESPIRATORY ACIDOSIS
- ○ RESPIRATORY ALKALOSIS
- ○ METABOLIC ACIDOSIS
- ✓ METABOLIC ALKALOSIS

3 UNCOMPENSATED, PARTIALLY COMPENSATED, or FULLY COMPENSATED?

If CO_2, HCO_3 & pH are ALL out of range

Is the pH in range? ○ YES ✓ NO
Is the CO_2 in range? ○ YES ✓ NO
Is the HCO_3 in range? ○ YES ✓ NO

- ○ UNCOMPENSATED
- ✓ PARTIALLY COMPENSATED
- ○ FULLY COMPENSATED

FINAL ANSWER:
METABOLIC ALKALOSIS, PARTIALLY COMPENSATED

ROME METHOD

1 What does the problem give you?

pH	7.50	○ ACIDIC	✓ ALKALOTIC	○ NORMAL
CO_2	50	✓ ACIDIC	○ ALKALOTIC	○ NORMAL
HCO_3	32	○ ACIDIC	✓ ALKALOTIC	○ NORMAL

2 Which of the four scenarios from the ROME method matches the information given in your problem?

Respiratory	pH ↑	CO_2 ↓	Alkalosis
Opposite	pH ↓	CO_2 ↑	Acidosis
Metabolic	pH ↑	HCO_3 ↑	Alkalosis
Equal	pH ↓	HCO_3 ↓	Acidosis

- ○ RESPIRATORY ACIDOSIS
- ○ RESPIRATORY ALKALOSIS
- ○ METABOLIC ACIDOSIS
- ✓ METABOLIC ALKALOSIS

3 UNCOMPENSATED, PARTIALLY COMPENSATED, or FULLY COMPENSATED?

If CO_2, HCO_3 & pH are ALL out of range

Is the pH in range? ○ YES ✓ NO
Is the CO_2 in range? ○ YES ✓ NO
Is the HCO_3 in range? ○ YES ✓ NO

- ○ UNCOMPENSATED
- ✓ PARTIALLY COMPENSATED
- ○ FULLY COMPENSATED

FINAL ANSWER:
METABOLIC ALKALOSIS, PARTIALLY COMPENSATED

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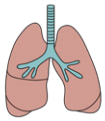
Respiratory Acidosis vs. Respiratory Alkalosis

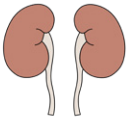
RESPIRATORY ACIDOSIS

PATHOPHYSIOLOGY



LUNG PROBLEM

The lungs are **RETAINING** too much **CO_2**



KIDNEYS COMPENSATE

The kidneys excrete excess **HYDROGEN** & retain **BICARB** (HCO_3)

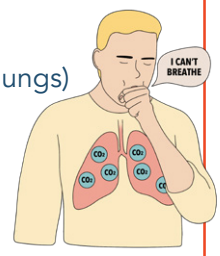
pH	CO_2
< 7.35	> 45

CAUSES

RETAINING CO_2: "Depress" breathing

MNEMONIC

- **D**rugs (opioids & sedatives)
- **E**dema (fluid in the lungs)
- **P**neumonia (excess mucus in the lungs)
- **R**espiratory center of the brain is damaged
- **E**mbolism (pulmonary embolism)
- **S**pasms of the bronchi (asthma)
- **S**ac elasticity damage (COPD & emphysema)
  - (alveolar sacs)



All these things cause impaired gas exchange

SIGNS & SYMPTOMS

- ↑ Respiration rate
- Restlessness
- Confusion
- Headache
- Sleepiness/coma

INTERVENTIONS

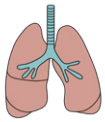
- Administer O_2
- Put patient in Semi-Fowler's position
- Have patient turn, cough, & deep-breathe (TCDB)
- Pneumonia: ↑ fluids to thin secretions & administer antibiotics
- If CO_2 > 50, they may need an endotracheal tube

MONITOR K^+ levels

NORMAL K^+
3.5 - 5.0 mmol/L

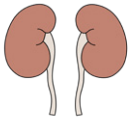
RESPIRATORY ALKALOSIS

PATHOPHYSIOLOGY



LUNG PROBLEM

The lungs are **LOSING** too much **CO_2**



KIDNEYS COMPENSATE

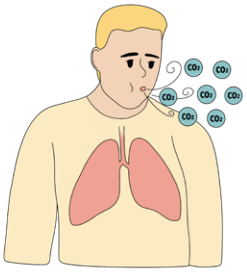
The kidneys excrete excess **BICARB** (HCO_3) & retain **HYDROGEN**

pH	CO_2
> 7.45	< 35

CAUSES

LOSING CO_2

- Hyperventilation
- Tachypnea
- ↑ Temperature
- Aspirin toxicity



SIGNS & SYMPTOMS

- ↑ Heart rate
- Confusion & tiredness
- Tetany
- EKG changes
- (+) Chvostek's sign

Twitching of the facial muscles when tapping the facial nerve in response to **HYPOCALCEMIA**

INTERVENTIONS

- **Rebreathe into a paper bag**
- Provide emotional support
- Fix the breathing problem
  - Encourage good breathing patterns
  - Give anti-anxiety medications or sedatives to ↓ breathing rate

This helps to put some of the lost CO_2 back into the lungs

MONITOR K^+ & Ca^- levels

NORMAL Ca
9 - 11 mg/dL

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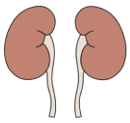
Metabolic Acidosis vs. Metabolic Alkalosis

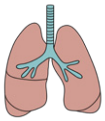
METABOLIC ACIDOSIS

PATHOPHYSIOLOGY



KIDNEY PROBLEM

Too much **HYDROGEN**
Too little **BICARB** (HCO_3)



LUNGS COMPENSATE

The lungs will blow off **CO_2**

pH	HCO_3
< 7.35	< 22

CAUSES

- Diabetic ketoacidosis → Not enough insulin = ↑ fat metabolism = excess **ketones (acid)**
- Acute/chronic kidney injury
- Malnutrition
- Severe diarrhea → Breaking down of fats = excess **ketones (acid)**

MEMORY TRICK
Bicarb comes out of your Base

SIGNS & SYMPTOMS

- ↑ Respiratory rate → **Kussmaul breathing:** Deep, rapid breathing > 20 breaths per minute
- Hyperkalemia
  - Muscle twitching
  - Weakness
  - Arrhythmias
- Confusion

INTERVENTIONS

NORMAL K^+
3.5 – 5.0 mmol/L

MONITOR
- Potassium (K^+) levels
- Intake & output
- Vital signs & EKG

- Administer IV solution of sodium bicarb to ↑ bases & ↓ acids
- Initiate seizure precautions

DIABETIC KETOACIDOSIS (DKA)
- Give insulin (this stops the breakdown of fats, which stops the production of ketones)

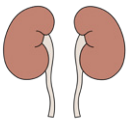
MONITOR for hypovolemia due to polyuria

KIDNEY DISEASE
- Dialysis to remove toxins

DIET MODIFICATIONS
- ↑ Calories
- ↓ Protein

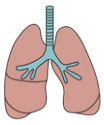
METABOLIC ALKALOSIS

PATHOPHYSIOLOGY



KIDNEY PROBLEM

Too much **BICARB** (HCO_3)
Too little **HYDROGEN**



LUNGS COMPENSATE

The lungs will retain **CO_2**

pH	HCO_3
> 7.45	> 26

CAUSES

- Too many antacids → Too much sodium bicarbonate (acid)
- Diuretics
- Excess vomiting → Excess loss of hydrochloric acid (HCL) from the stomach
- Hyperaldosteronism

SIGNS & SYMPTOMS

- ↓ Respiratory rate → **HYPOventilation** < 12 breaths per minute
- ↓ Potassium (K^+)
  - Dysrhythmias
  - Muscle cramps/weakness
  - Vomiting
  - Tetany
  - Tremors
  - EKG changes

Metabolic **ACIDOSIS** = ↑ serum potassium
Metabolic **ALKALOSIS** = ↓ serum potassium

INTERVENTIONS

NORMAL Ca
9 – 11 mg/dL

MONITOR
- Potassium (K^+) & calcium (Ca) levels
- Vital signs & EKG

- Administer IV fluids to help the kidneys get rid of bicarbonate
- Replace potassium (K^+)
- Give antiemetics for vomiting
  - Ondansetron (Zofran)
  - Metoclopramide (Phenergan)

- Watch for signs of respiratory distress

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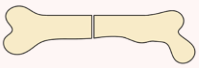
Fractures & Compartment Syndrome

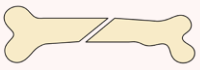
A fracture is a complete or incomplete disturbance in the progression of bone structure

Types of Fractures

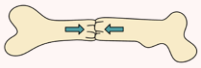
COMMINUTED
The bone is crushed, causing lots of little fragments



TRANSVERSE
The bone is fractured straight across



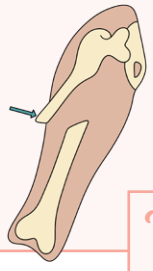
OBLIQUE
The fracture runs at an angle across the bone



IMPACTED
Segments of bone are wedged into each other at the fracture line

SPIRAL
The break partially encircles the bone

GREENSTICK
One side of the bone is bent & the other is broken



OPEN/COMPOUND
The bone breaks through the skin
→ Increased risk for infection

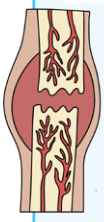
Post-Fracture

Neurovascular assessments

6 Ps

- **P** Pain
- **P** Pallor
- **P** Pulselessness
- **P** Paresthesia → burning or tingling sensation
- **P** Paralysis
- **P** Pressure

Stages of Bone Healing

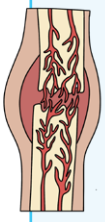


STAGE I

Hematoma formation

- Occurs first 1–2 days from injury
- Bleeding into the injured site occurs

MNEMONIC
He **F**ell **B**ecause he was **R**unning



STAGE II

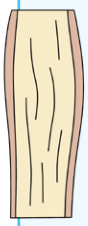
Fibrocartilaginous callus formation

- Occurs 3–14 days from injury
- Granulation tissue forms
- Reconstruction of bone begins
- Still not strong enough to bear weight

STAGE III

Bony callus formation (ossification)

- Occurs 3–4 weeks from injury
- Mature bone begins replacing callus



STAGE IV

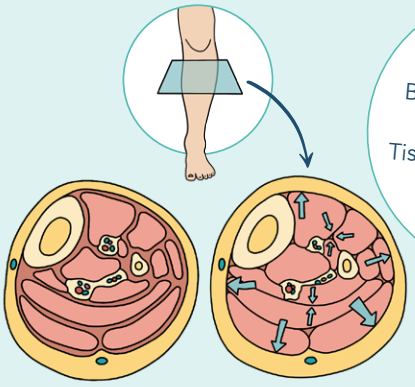
Remodeling

- May take months to years
- Compact bone replaces spongy bone
- X-rays are used to monitor the progress of bone healing

Compartment Syndrome

PATHOLOGY

Compartment syndrome cause increased pressure within a limited space, which compromises blood vessels and nerves. This can result in cell death, so prompt diagnosis is critical!



Pressure increases
↓
Blood flow is cut off
↓
Tissue damage due to **HYPOXIA** (lack of oxygen)

Normal

Compartment Syndrome
Muscle swelling causing compression of nerves & vessels

RISK FACTORS

- Restrictive dressings
- Casts or splints
- Increased pressure within a closed compartment (bleeding, edema, inflammation)

SIGNS & SYMPTOMS

- Deep, throbbing, unrelenting pain
- Pain unrelieved by medications
- Pain that is disproportional to the injury
- Pain that intensifies with passive ROM
- 6 Ps: (pain, pallor, pulselessness, paresthesia, paralysis & pressure)

TREATMENT/NURSING CONSIDERATIONS

IMMEDIATE

- Place the extremity at heart level (not above heart level)
- Open the cast or splint
- Loosen & remove restrictive clothing
- Avoid using cold compresses (this can cause vasoconstriction, which can worsen compartment syndrome)

⚠ **MEDICAL EMERGENCY**

FASCIOTOMY

Fascia is cut to relieve tension & pressure

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Gout

Pathology

Gout is a form of arthritis characterized by increased uric acid levels.

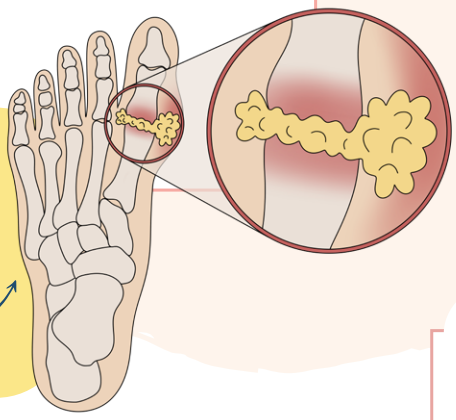
HYPERURICEMIA
high uric acid in the blood

This causes deposits of uric acid crystals in the joints.

Tophi

Accumulation of sodium urate crystals in joints such as the big toe and hands, or other areas such as the ears

MEMORY TRICK: Tophi think Toe



WHAT IS URIC ACID?

Uric acid is created from purine breakdown during digestion. It's produced by the liver and is mostly excreted by the kidneys.

EXPECTED RANGE:
F: 2.7 – 7.3 mg/dL
M: 4 – 8.5 mg/dL

Signs & Symptoms

Can be **ACUTE** or **CHRONIC**

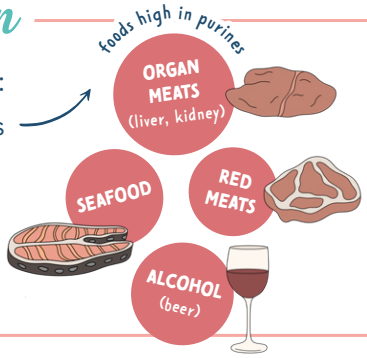
- Acute gouty arthritis
- Pain (severe)
- Swelling
- Warmth at the site
- Bone deformity
- Joint damage
- Tophi
- Renal calculi

Causes

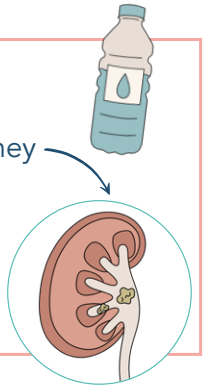
- Diet high in purines
- Certain medications
  - Diuretics (cause dehydration)
  - Aspirin
  - Cyclosporine
- Disorder of purine metabolism
- Kidney problems
  - Inadequate excretion of uric acid by the kidneys

Patient Education

- **EDUCATE** on avoiding:
  - ✗ Foods high in purines
  - ✗ Certain medications (aspirin)
  - ✗ Alcohol
  - ✗ Dehydration



- Stay hydrated: 2–3 liters per day
- Uric acid deposits can cause kidney stones; fluids help prevent this!
- Participate in a weight loss program if overweight



Medications

GENERIC	TRADE NAME
allopurinol	Aloprim, Zyloprim, Lopurin

MEMORY TRICK: AlloPurinol → Prevents gout

GENERIC	TRADE NAME
colchicine	Mitigare, Colcrys

MEMORY TRICK: Colchicine → for aCute gout attacks

For more information about gout medications, see the musculoskeletal section in the Pharmacology Bundle

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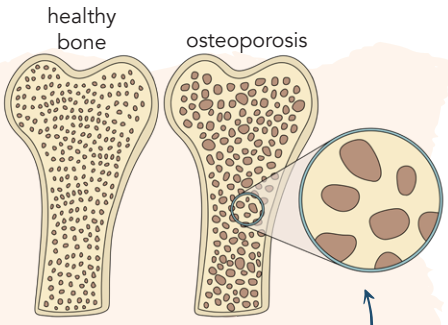
Osteoporosis

Pathology

OSTEOPOROSIS
- relating to bone
- porous

Osteoporosis essentially means: **HAVING POROUS BONES**

The rate of **BONE RESORPTION** (osteoCLASTS) is greater than the rate of **BONE FORMATION** (osteoBLASTS)
= **DECREASED TOTAL BONE MASS**

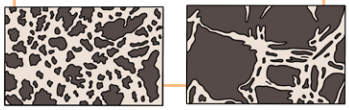


Normal bone marrow has small holes in it, but osteoporosis causes much larger holes

Diagnostic

Bone density test:
Dual-energy X-ray absorptiometry **(DEXA)**

This process takes X-ray images measuring calcium and other minerals in the bones



Risk Factors

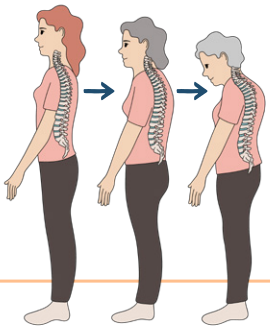
- **Low intake of calcium & vitamin D**
- **Long-term use of corticosteroids, levothyroxine, proton pump inhibitors (PPIs)**
- **Women after menopause**
  - Lifestyle factors:
    - Smoking, excessive alcohol intake, sedentary lifestyle, immobility
  - Caucasian or Asian women
  - Malabsorption disorders/being underweight (Celiac disease, bariatric surgery, eating disorders)
  - Advanced age > 65 yrs

↓ estrogen during menopause = ↑ bone resorption

Signs & Symptoms

- May be asymptomatic until a fracture occurs
- **Pathological fractures** (hips, spine, wrist)
- Low back, neck, or hip pain
- Rounded back "hunchback", causing height loss

Patients often think they fell and broke something, BUT bones may break first, causing them to fall



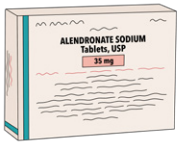
Patient Education

DIET MODIFICATIONS

- ↑ Calcium & vitamin D

MEDICATIONS

- Bisphosphonates
  - End in "**-dronate**"



MEMORY TRICK: Bisphosphonates Build up Bone

PREVENTION of osteoporosis

- Do weight-bearing exercises (weights, hiking, squats)
- Limit alcohol & coffee intake
- Smoking cessation

PREVENTION of falls

AT HOME
- Avoid area rugs (risk for falling)
- Watch out for pets
- Keep glasses nearby
- Keep rooms well lit

AT THE HOSPITAL
- Use call light
- Wear non-slip socks
- Communicate fall risk to the healthcare team
- Maintain clutter-free environment

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Osteoarthritis (OA) & Rheumatoid Arthritis (RA)

OSTEOARTHRITIS (OA)

RHEUMATOID ARTHRITIS (RA)

PATHOLOGY

Noninflammatory degenerative joint disorder caused by the breakdown of articular cartilage between the joints from "wear and tear"

NOT autoimmune or inflammatory

Chronic inflammatory type of arthritis classified as an autoimmune disease

Patient will have a (+) rheumatoid factor

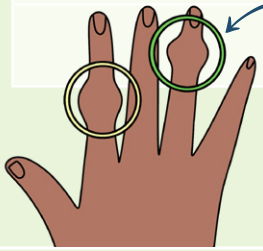
STAGES OF RHEUMATOID ARTHRITIS

1. **SYNOVITIS**
   - Synovium becomes inflamed
   - Synovial membrane thickens
2. **PANNUS FORMATION**
   - Layer of vascular fibrous tissue forms
3. **FIBROUS ANKYLOSIS**
   - Joint is invaded by fibrous connective tissue
4. **BONY ANKYLOSIS**
   - Bones become fused together

SIGNS & SYMPTOMS

SLOW (years)	ONSET	RAPID (weeks to months)
Asymmetrical (affects ONE joint)	PAIN	Symmetrical pain, redness, & swelling (affects both parts on opposite sides of the body)
Pain/stiffness after weight-bearing activities (subsides within 30 minutes)	STIFFNESS	Pain/stiffness in the morning (lasts > 1 hour)
Weight-bearing joints (knees, hips, spine)	JOINTS AFFECTED	Small joints of hands & feet (also affects other body systems: heart, lungs, skin)
Bony enlargements	ALERERATIONS OF THE HANDS	Deformity of the fingers

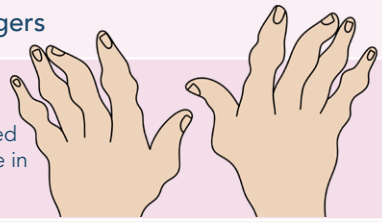
MEMORY TRICK
- Symmetrical joints
- Swelling (inflammatory condition)
- Stiffness in the morning



DISTAL
Distal interphalangeal (DIP), called **HEBERDEN'S NODES**

PROXIMAL
Proximal interphalangeal (PIP), called **BOUCHARD'S NODES**

The joints become deformed and are especially noticeable in the fingers and toes



RISK FACTORS

- **Obesity**
- Older age
- Female gender
- Occupational risks (frequent lifting, kneeling, or stooping)
- Genetics
- Wear and tear

- Environmental factors (pollution)
- Bacterial or viral illness
- Cigarette smoking
- Family history

May cause an inflammatory response & destruction of synovial fluid

TREATMENT

- Weight loss
- Orthotic devices (splints, knee braces)
- Walking aids (canes)
- Aerobic exercise & strength training (shown to prevent progression)
- Injection of a gel-like substance into the affected joint
- **Surgery**
  - Arthroplasty: replacement or reconstruction of the joint

- **Surgery**
  - Synovectomy: removal of synovium
  - Joint replacement
  - Arthrodesis: joint fusion

HEAT OR COLD?

HEAT for stiffness

COLD for pain/ inflammation

• Analgesics (NSAIDs) • Glucosamine • Chondroitin	**MEDICATIONS**
• Corticosteroids (To help with inflammation) • Analgesics (NSAIDs) • Disease Modifying Anti-Rheumatic Drugs (DMARDs)	

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Notes

How we **RESPOND** to **ADVERSITY** helps us to be **SUCCESSFUL.**

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Pharmacology

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Tips for Pharmacology Class

Saved by the suffixes & prefixes

Memorizing your suffixes and prefixes allows you to easily recognize which medication class the drug falls into just by looking at the ending or beginning. Note that it doesn't work for every medication, and there are some outliers.

Turn the page for an entire section listing all the most common suffixes and prefixes!

	DEFINITION	EXAMPLE
SUFFIXES	Seen at the **end** of the words	alen**dronate**
PREFIXES	Seen at the **beginning** of words	**ceph**alexin

Lab values

Labs? In pharmacology? YES. Lab values connect to everything, not only diagnoses and disease processes. Particular medications can make certain lab values increase or decrease, and many drugs have therapeutic ranges. Many medications are used to treat diseases that will present with abnormal lab values. Know how these connect, because you'll see them every day as a nurse, and they'll DEFINITELY be on your exams.

Ask for help

It may be scary, but asking your professor for guidance is okay. This is one of the most difficult classes for a lot of students, so don't get shy now—ask for help! After all, they are your resources for teaching and learning.

- Ask your professor if the exam will test on generic names, brand names, or both. Most nursing schools try to mimic the NCLEX, which tests only on generic medication names.
- Ask if they have any specific tips regarding studying, or how they personally studied pharmacology while first in nursing school.

Memorize the unique & alarming side effects

Many medications have unique side effects. Even if they rarely occur, the NCLEX & nursing exams like to test on these.

For example:

- The antibiotic class, fluoroquinolones, can cause tendon rupture, while the most common side effects are GI upset and photosensitivity.
- Phenytoin can cause gingival hyperplasia, nausea, vomiting, and hypotension. Remember, the most unique side effect associated with this medication is gingival hyperplasia, so it's one you can expect to see.

Study medication classes rather than individual medications

Common suffixes and prefixes will help you learn medication classes in a more efficient way, as well as learning how that specific medication class works to change the body's processes. Instead of zeroing in on one medication's use, look at the mechanism of action for the entire class. This will broaden your scope of knowledge in pharmacology!

Familiarize yourself with patient education for common medications

A good majority of the medications you will be learning about are taken in patients' homes. It's important you know how to educate them on how to properly take these medications.

Ask yourself these questions to cover all necessary education:

- When should the patient take it?
- Should they make any diet modifications?
- Should they make any modifications to their lifestyle?
- Does it affect their hormonal birth control?
- It is safe for pregnant or breastfeeding women?

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Drug Names, Suffixes & Prefixes

Drug Names

CHEMICAL NAME

Scientific name; describes the molecular structure of a drug

Not commonly used when referring to the medication

GENERIC NAME

"Official" name of a drug;
NON-PROPRIETARY
(not owned by a company)

Contains the suffixes discussed in this section

EXAMPLE: pantoprazole

The generic name is lowercase

TRADE/BRAND NAME

Name selected by the pharmaceutical company that made the drug;
PROPRIETARY
(owned by a company)

Followed by a trademark symbol: ™ or ®

EXAMPLE: Protonix®

The trade/brand name is capitalized

Ask your pharmacology professor if they will give you the generic name, trade name, or both on the exam, as this can help guide your studies. They may not give you an answer, but it can't hurt to try. As for the NCLEX, it tests only on generic medications!

Suffixes & Prefixes

It's nearly impossible to remember every single medication from your pharmacology class and as a new nurse on the floor.

That's why it's helpful to learn common suffixes and prefixes for certain medication classes.

This allows you to easily recognize a medication just by looking at the ending or beginning.

The NCLEX only tests on generic meds, **THIS IS GOOD NEWS!**

SAVED BY THE SUFFIX

	DEFINITION	EXAMPLE
SUFFIXES	Seen at the **end** of words	alen**dronate**
PREFIXES	Seen at the **beginning** of words	**ceph**alexin

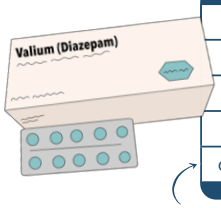
EXAMPLES:

Bisphosphonates end in the suffix "**-dronate**"

GENERIC	TRADE NAME
alen**dronate**	Binosto, Fosamax
eti**dronate**	Didronel
iban**dronate**	Boniva

SUFFIXES & PREFIXES only apply to the generic names

Benzodiazepines end in the suffixes "**-zolam**" & "**-zepam**"



GENERIC	TRADE NAME
alpra**zolam**	Xanax
lora**zepam**	Ativan
dia**zepam**	Valium
clona**zepam**	Klonopin
chlordiazepoxide	Librium

This is considered an **outlier**. This medication may not have the same suffixes, but it's still considered a benzodiazepine. Just commit this and other outliers to memory!

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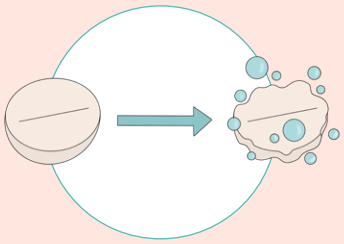
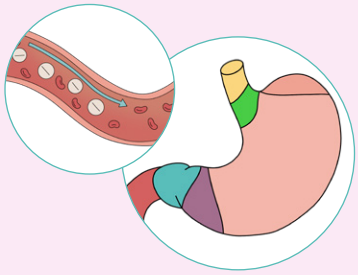
Pharmaceutic, Pharmacokinetic & Pharmacodynamic Phases

Pharmaceutic Phase



Change from a solid form (pill) by dissolving into a liquid form

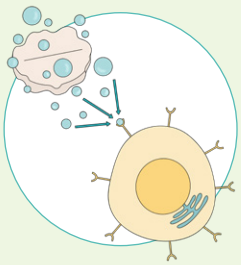
Pharmacokinetic Phase



- Absorption
- Distribution
- Metabolism
- Excretion

MEMORY TRICK **ADME** some medications

Pharmacodynamic Phase

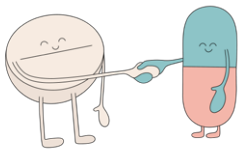


Movement of the drug into a cell & potential of the drug to bind to cellular receptors

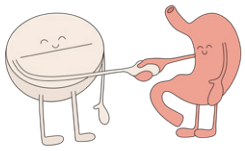
Think about the **DYNAMICS** existing between drugs & the body

EXAMPLES:

1. How different medications interact with each other (drug-drug interactions)



2. How a medication impacts the body



Terms to Know

ANTAGONISTS:
Work *AGAINST* the effect of another drug or body system

AGONISTS:
Help work *WITH* the effect of another drug or body system

LOADING DOSE:
High initial dose used to reach therapeutic levels more quickly

CONTRAINDICATION:
Any reason to *AVOID* or *HOLD* a medication, usually to avoid harm

ADVERSE REACTION:
Harmful or opposite reaction to a medication

THERAPEUTIC EFFECT:
Desired effect or outcome

THERAPEUTIC INDEX/LEVEL:
Desired level of medication in the body (usually measured by serum levels in the blood)

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Pharmacokinetics

MEMORY TRICK

ADME some medications

Pharmacokinetics: The study of how drugs are moved throughout the body

A ABSORPTION

Medication going from the location of administration to the bloodstream

ORAL

Slowest absorption time

SUBQ & IM

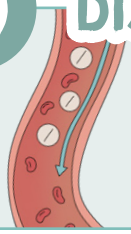
Depends on blood perfusion at injection site

↑ blood perfusion = ↑ rate of absorption

IV

Most rapid absorption time

D DISTRIBUTION



Transportation of the medication by bodily fluids to where it needs to go

INFLUENCING FACTORS

- Permeability of the cell membrane
- Plasma protein binding
- Circulation

↓ Circulation (ex: hemorrhage) = ↓ carrying capacity to take the medication to where it needs to be

M METABOLISM

The break down of the medication

INFLUENCING FACTORS

- Age
  Infants & elderly have a limited medication-metabolizing capacity
- Medication type
- Nutritional status
- **First-pass effect**
  As medications go through the liver or gut, they are inactivated by the enzymes in the liver or gut. This means that by the time they reach the systemic circulation, they are LESS POWERFUL & will not produce a full therapeutic effect.

FIRST-PASS EFFECT

The drug is ingested **orally**

↓

The drug is metabolized (usually by the liver or gut; can occur elsewhere like the lungs)

These act as filters

↓

The effect of the drug is **reduced** as only part of the drug reaches the systemic circulation

↓

The drug may need to be administered via **parenteral route (subQ, IM, or IV)** since these routes bypass the liver and gut

E EXCRETION

The removal of the unmetabolized medication from the body

INFLUENCING FACTORS

- Kidney dysfunction
  Leads to ↑ in the duration and intensity of a medication response

If the kidneys aren't working to excrete waste, the medication will stay in the body, which leads to toxic levels

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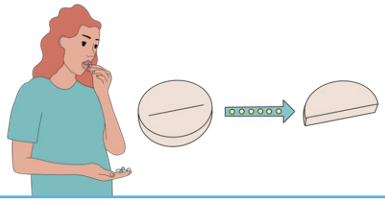
Half-Life & Therapeutic Index (TI)

Half-Life

The time it takes for the drug concentration in your body to be reduced by half (50%)



SHORT Half-Life

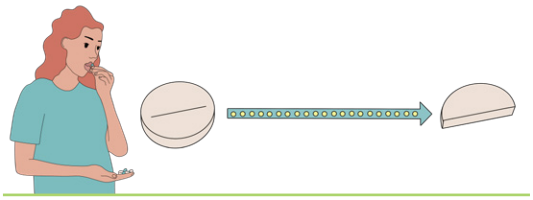
Is **quickly** reduced by half, decreasing the risk for toxicity!

- Quick-acting
- Ideal for fast relief
- Higher risk for **dependency**
- Lower risk for **toxicity**

Especially for pain or psychiatric medications

EXAMPLES:

- Rapid-acting insulin (Humalog)
- Oxycodone (Roxicodone)
- Midazolam (Versed)



LONG Half-Life

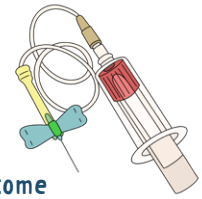
Takes a **longer** time to be reduced by half

- May take longer to "kick in"
- Ideal for long-term relief
- Lower risk for **dependency**
- Higher risk for **toxicity**

EXAMPLES:

- Long-acting insulin (Lantus)
- Fluoxetine
- Clonazepam (Klonopin)

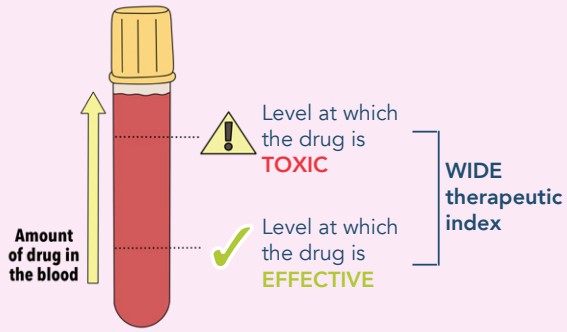
Therapeutic Index (TI)



Also called **"THERAPEUTIC LEVEL"**

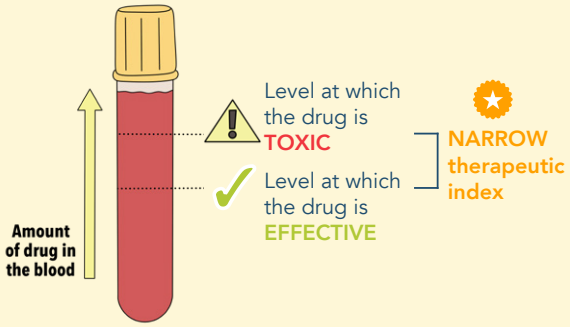
The level of medication in the body needed to produce the desired effect or outcome (usually measured by serum concentration in the bloodstream)

HIGH THERAPEUTIC INDEX



These drugs require less monitoring than those with a low therapeutic index because there is a wide range between effectiveness & toxicity

LOW/NARROW THERAPEUTIC INDEX



These drugs need to be closely monitored because they have a greater risk for toxicity

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Medication Safety

Joint Commission "DO NOT USE" List

✗ DO NOT USE	? POTENTIAL PROBLEM	✓ INSTEAD, WRITE:
U	Mistaken for "0" (zero) or "cc"	unit
IU	Mistaken for "IV" (intravenous) or "10" (ten)	international unit
Q.D., QD, q.d., qd, Q.O.D.,QOD, q.o.d, qod	Mistaken for each other	daily ***or*** every other day
Trailing zero (X.0 mg) Lack of leading zero (.X mg)	Decimal point is missed	X mg ***or*** 0.X mg
MS, MSO4, MgSO4	Can mean morphine sulfate or magnesium sulfate	morphine sulfate ***or*** magnesium sulfate
@	Mistaken for "2" (two)	at
cc	Mistaken for "U" (units)	mL ***or*** milliliters

Rights Of Medication Administration

The rights of medication administration are in place to catch and prevent errors that can cause harm to the patient

Right CLIENT/ PATIENT	Verify you are administering the medication to the right patient with unique identifiers: *First & last name, date of birth, medical record number*
Right TIME	Are you giving the medication at the ordered time? ▪ **Too early:** causes toxicity or unwanted side effects ▪ **Too late:** causes a missed dose & may require readjustment of dosing times
Right DOSE	▪ Verify the ordered dose (numbers and unit of measurement) ▪ Use double verification for dosage calculation
Right MEDICATION	Some medications have very similar names, so be sure you are administering the correct one!
Right ROUTE & FORM	How are you administering this medication? How will the medication ENTER the body? (orally, intravenously, rectally)
Right DOCUMENTATION	▪ Verify you have documented your checks ***and*** administration properly ▪ May include scanning or cosigning
Right REASON	▪ Are you giving this medication for the appropriately ordered REASON? ▪ Watch especially closely with PRN administration (such as varying medication for different pain levels)
Right EDUCATION	▪ Have you provided the patient with appropriate education about the medication they are receiving? ▪ Do they understand the key facts about the medication? ▪ Is the patient well-informed?
Right RESPONSE	▪ Continue to monitor & assess the patient for the desired response to the medication, as well as for possible undesired responses ▪ Monitor BP, weight, I&Os, lab values
Right TO REFUSE	If a patient refuses a medication, do not force it! Determine the reason, document it well, and notify the HCP.

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Herbal Therapy & Supplementation

Herbal products can alter medication function, therapeutic effect, and absorption, leading to unwanted outcomes or negative side effects.

HERBAL SUPPLEMENT	COMMON USES	DRUG INTERACTION	SIDE EFFECTS
St. John's Wort	▪ Depression ▪ Anxiety	▪ Statins ▪ Serotonin syndrome ▪ Anticoagulants ▪ Digoxin ▪ Antidepressants ▪ Warfarin	▪ Hypertension & serotonin syndrome when taken with antidepressants ▪ ↓ Effect of warfarin
▪ Ginkgo Biloba, ▪ Garlic, ▪ Ginger, ▪ Ginseng	Helps to improve memory & has antioxidant effect; Helps relieve nausea & has detoxifying effects; Helps boost the immune system & improves mental performance	▪ Anticoagulants ▪ Antiplatelets MEMORY TRICK: GOING to WAR → WARfarin makes you bleed	↑ Risk for bleeding
Saw Palmetto	Benign prostatic hyperplasia (BPH)	▪ Anticoagulants ▪ Antiplatelets	↑ Risk for bleeding
Milk Thistle	Hepatic dysfunction (gallbladder & liver issues)	▪ Anticoagulants ▪ Anxiolytics ▪ Lipid-lowering agents	▪ Interference with liver enzyme activity ▪ Impact on breakdown of medications

Nursing Considerations:

STOP the herbal supplement: 2–3 weeks before surgery

They can alter the body's response to anesthesia

ALWAYS perform a complete medication reconciliation upon admission

Include prescription drugs, over-the-counter medications, and all vitamins & supplements with doses

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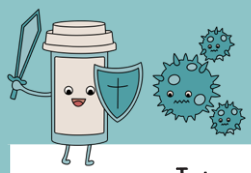
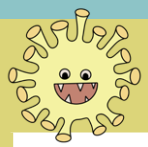


ANTIBIOTICS / ANTIBACTERIALS

	PREFIXES / SUFFIXES	EXAMPLES
Tetracyclines	-CYCLINE	doxycycline, tetracycline
Sulfonamides	SULF-	sulfasalazine, sulfamethaxazole
Cephalosporins	CEF-, CEPH-	cefazolin, cephalexin
Penicillins	-CILLIN	ampicillin, amoxicillin
Aminoglycosides & macrolides	-MICIN, -MYCIN	gentamicin, erythromycin
Fluoroquinolones	-FLOXACIN	ciprofloxacin, levofloxacin



ANTIVIRALS

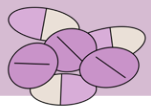
	PREFIXES / SUFFIXES	EXAMPLES
Antivirals (undefined group)	-VIR-, -VIR	oseltamivir, zanamivir
Antivirals (anti-herpes viral agents)	-CLOVIR	acyclovir, famciclovir
Antiretrovirals (protease inhibitors)	-NAVIR	atazanavir, nelfinavir
Antiretrovirals (nucleoside reverse transcriptase inhibitors)	-VUDINE	zidovudine, stavudine

ANTIFUNGALS

	PREFIXES / SUFFIXES	EXAMPLES
Antifungals	-AZOLE	fluconazole, voriconazole

ANESTHETICS / ANTIANXIETY

	PREFIXES / SUFFIXES	EXAMPLES
Local anesthetics	-CAINE	lidocaine, bupivacaine
Barbiturates (CNS depressants)	-BARBITAL	phenobarbital, secobarbital
Benzodiazepines (for anxiety/sedation)	-ZOLAM, -ZEPAM	alprazolam, lorazepam



ANTIDEPRESSANTS

	PREFIXES / SUFFIXES	EXAMPLES
Selective serotonin reuptake inhibitors (SSRIs)	-OXETINE, -TALOPRAM -ZODONE	fluoxetine, escitalopram, vilazodone
Serotonin-norepinephrine reuptake inhibitors (SNRIs/DNRIs)	-FAXINE, -ZODONE - NACIPRAN	venlafaxine, nefazodone, milnacipran
Tricyclic antidepressants (TCAs)	-TRIPTYLINE, -PRAMINE	amitriptyline, clomipramine

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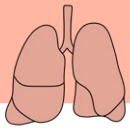
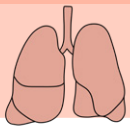
ANALGESICS / OPIOIDS

	PREFIXES / SUFFIXES	EXAMPLES
Opioids	-DONE, -ONE	oxycodone, hydromorphone
NSAIDs (anti-inflammatories)	-PROFEN	ibuprofen, fenoprofen
Salicylates	-	aspirin (ASA)
Nonsalicylates	-	acetaminophen



UPPER RESPIRATORY

	PREFIXES / SUFFIXES	EXAMPLES
H1 antagonists (second-generation antihistamines)	-TADINE, -TIRIZINE	loratadine, desloratadine, cetirizine, levocetirizine
Nasal decongestants	-EPH-, -ZOLINE	phenylephrine, pseudoephedrine, oxymetazoline



LOWER RESPIRATORY

	PREFIXES / SUFFIXES	EXAMPLES
Beta2-agonists (bronchodilators)	-TEROL	albuterol, levalbuterol
Xanthine derivatives (bronchodilators)	-PHYLLINE	aminophylline, dyphylline
Cholinergic blockers (anticholinergics)	-TROPIUM	tiotropium, ipratropium
Immunomodulators & leukotriene modifiers	-ZUMAB, -LUKAST	reslizumab, montelukast

GASTROINTESTINAL

	PREFIXES / SUFFIXES	EXAMPLES
Histamine H2 antagonists (H2 blockers)	-TIDINE	cimetidine, famotidine
Proton pump inhibitors (PPIs)	-PRAZOLE	omeprazole, pantoprazole

ANTIDIABETICS

	PREFIXES / SUFFIXES	EXAMPLES
Thiazolidinediones	-GLITAZONE	pioglitazone
Inhibitors of the DPP-4 enzyme	-GLIPTIN	sitagliptin, linagliptin

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CARDIAC: ANTIHYPERTENSIVES

	PREFIXES / SUFFIXES	EXAMPLES
ACE inhibitors	-PRIL	enalapril, captopril
Beta blockers	-OLOL	metoprolol, nadolol
Angiotensin II receptor antagonists	-SARTAN	losartan, olmesartan
Calcium channel blockers	-PINE, -AMIL	amlodipine, verapamil
Vasopressin receptor antagonists	-VAPTAN	conivaptan, tolvaptan
Alpha-1 blockers	-OSIN	prazosin, doxazosin
Loop diuretics	-IDE, -SEMIDE	furosemide, bumetanide
Thiazide diuretics	-THIAZIDE	hydrochlorothiazide, chlorothiazide
Potassium-sparing diuretics	-ACTONE	spironolactone

CARDIAC: ANTIHYPERLIPIDEMICS

	PREFIXES / SUFFIXES	EXAMPLES
HMG-CoA reductase inhibitors	-STATIN	simvastatin, rosuvastatin

CARDIAC: OTHER

	PREFIXES / SUFFIXES	EXAMPLES
Anticoagulants (factor Xa inhibitors)	-XABAN	apixaban
Low-molecular-weight heparins (LMWHs)	-PARIN	enoxaparin, dalteparin
Thrombolytics (clot busters)	-TEPLASE	alteplase
Antiarrhythmics	-ARONE	amiodarone

MISCELLANEOUS

	PREFIXES / SUFFIXES	EXAMPLES
Corticosteroids	-ASONE, -OLONE, -NIDE	betamethasone, prednisolone, budesonide
Triptans (anti-migraines)	-TRIPTAN	almotriptan, sumatriptan
Ergotamines (anti-migraines)	ERGOT-, -ERGOT-	dihydroergotamine, ergotamine
Antiseptics	CHLOR- MEMORY TRICK Chlor = Clean	chlorhexidine, chloroxylenol
Bisphosphonates	-DRONATE	risedronate, alendronate
Neuromuscular blockers	-NIUM	vecuronium, rocuronium
Retinoids	RETIN-, -RETIN-	tretinoin, retin-A
Phosphodiesterase 5 inhibitors	-AFIL	sildenafil, tadalafil
Carbonic anhydrase inhibitors	-LAMIDE, -AMIDE	acetazolamide, diclofenamide
Antiemetics	-SETRON	dolasetron, ondansetron

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COMMON THERAPEUTIC LEVELS

Digoxin	**0.5 - 2.0** ng/mL
Lithium	**0.6 - 1.2** mEq/L
Theophylline	**10 - 20** mcg/mL
Dilantin (Phenytoin)	**10 - 20** mcg/mL
Magnesium sulfate	**4 - 7** mg/dL
Acetaminophen (Tylenol)	**10 - 20** mcg/mL
Gentamicin	**5 - 10** mg/L
Salicylate (aspirin)	**100 - 300** mcg/mL
Vancomycin	Peak: **20 - 40** mcg/mL Trough: **5 - 15** mcg/mL
Valproic acid	**50-100** mcg/mL

Digoxin & lithium have two of the narrowest therapeutic index ranges

ANTIDOTES

Antidotes work to reverse the toxicity of a certain medication

✘ **ANTI**-dote = **ANTI**-drug

Opioids/narcotics	**NALOXONE** (Narcan)	NO more Opioids Narcan → Opioids
Warfarin (Coumadin)	**VITAMIN K**	During WAR, Vitamin K Kills WARfarin
Heparin	**PROTAMINE SULFATE**	You will need HELP from a PRO to stop bleeding out
Digoxin	**DIGIFAB**	DIGifab → DIGoxin
Anticholinergics	**PHYSOSTIGMINE**	
Benzodiazepines	**FLUMAZENIL** (Romazicon)	I FLU fast in my Mercedes-BENZ
Cholinergic crisis	**ATROPINE** (Atropen)	We don't have time to CHAT - we have a toxic situation CHolinergic → ATropine
Acetaminophen (Tylenol)	**ACETYLCYSTEINE** (Acetadote)	ACETylcysteine → ACETaminophen
Magnesium sulfate	**CALCIUM GLUCONATE**	MAGgie CALLs for help! MAGnesium → CALcium
Iron	**DEFEROXAMINE** (Desferal)	DeFEROxamine → FERrous means "containing iron"
Lead	**SUCCIMER** OR **CALCIUM DISODIUM EDETATE**	These are chelation agents
Alcohol withdrawal	**CHLORDIAZEPOXIDE** (Librium)	
Beta blockers	**GLUCAGON** (GlucaGen)	Beta blockers be GONe with GlucaGON
Calcium channel blockers	**GLUCAGON, INSULIN,** OR **CALCIUM**	
Aspirin	**SODIUM BICARBONATE**	You take ASPIRIN when you have a headache. You may also want a SALTY snack when you have a headache.
Insulin reaction	**GLUCAGON**	If you want your insulin GONe, you give GlucaGON
Pyridoxine	**DEFEROXAMINE** (Desferal)	
Tricyclic antidepressants	**SODIUM BICARBONATE**	
Cyanide	**HYDROXOCOBALAMIN** (also known as vitamin B12)	

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Antihypertensive Medications: Overview

MEMORY TRICK — As easy as... AABCDD

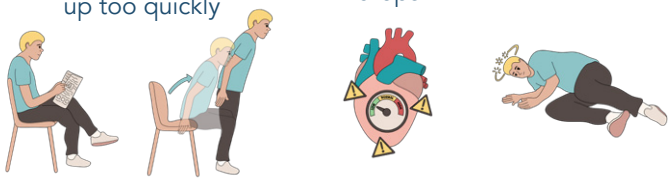
	MEDICATION CLASS	SUFFIX (SAVED BY THE SUFFIX)	EXAMPLES
A	ACE INHIBITORS	-pril	lisino**pril** enala**pril**
A	ARBs	-sartan	lo**sartan** cande**sartan** val**sartan**
B	BETA BLOCKERS	-olol	metopr**olol** aten**olol**
C	CALCIUM CHANNEL BLOCKERS	-dipine	amlo**dipine** nife**dipine**
D	DIGITALIS Cardiac glycosides	-oxin	dig**oxin** (Lan**oxin**, Digox)
D	DIURETICS	Loop: **-ide** Thiazide: **-thiazide** Potassium-sparing: **-actone**	furosem**ide** hydrochloro**thiazide** spironol**actone**

Common Side Effects of Antihypertensives

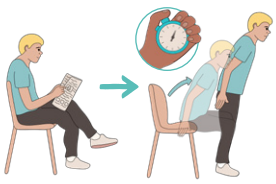
ORTHOSTATIC HYPOTENSION

- **ortho** → relating to posture & position
- **static** → low blood pressure

Changing positions from lying down or sitting to standing up too quickly → Blood pressure drops → ↑ Risk for falls



Change positions slowly



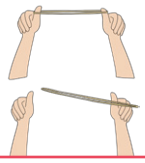
Sit on the edge of the bed & dangle your feet for 30–60 seconds before standing

REBOUND HYPERTENSION

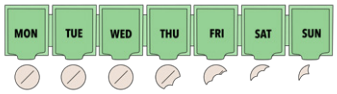
- **re** → to bounce or return back
- **bound** → high blood pressure

Suddenly or abruptly stopping a medication that controls blood pressure → Causes blood pressure to BOUNCE BACK to a high value

Think of a rubber band:
If a rubber band is released SUDDENLY, it will SNAP back sharply in the opposite direction
Stopping antihypertensives suddenly will cause a controlled blood pressure to INCREASE sharply and quickly



NEVER stop blood pressure medications abruptly—they must be tapered



Speak to your healthcare provider about tapering the medication down & weaning off slowly

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Angiotensin-Converting Enzyme Inhibitors
(ACE Inhibitors)

MEDICATION CLASS & NAME

Antihypertensives

GENERIC	TRADE NAME
captopril	–
enalapril	Vasotec
fosinopril	–
lisinopril	Prinivil
ramipril	Altace
SUFFIX: -PRIL	

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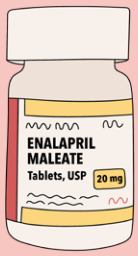


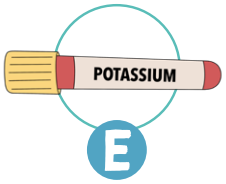
MUST-KNOW SIDE EFFECTS

- Orthostatic hypotension
- Dizziness
- Angioedema
- Cough (dry)
- Elevated K^+

MEMORY TRICK: ACE

A Angioedema
DANGEROUS: rapid swelling of the face & neck

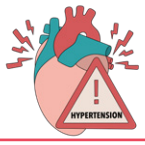
C Cough (dry)
NORMAL



E Elevated K^+
DANGEROUS: can cause cardiac dysrhythmias

THERAPEUTIC USES

- Hypertension
- Heart failure
- Myocardial infarction



NURSING CONSIDERATIONS

MONITOR renal function because...
- ACE inhibitors are excreted by the kidneys & can worsen kidney failure or cause renal impairment

NORMAL RENAL VALUES:
BUN: 7–10 mg/dL
Creatinine: 0.6–1.3 mg/dL

SAFE DURING PREGNANCY?
✗ NO!
ACE inhibitors are **TERATOGENIC**

PATIENT EDUCATION

DIET MODIFICATIONS
- Avoid potassium-containing foods & potassium pills

Salt substitutes contain potassium

REPORT to HCP if swelling of the face or mouth occurs

EDUCATE
- Do not suddenly stop the medication; it needs to be tapered — *This can cause rebound hypertension*
- Change positions slowly & sit on the side of the bed for a few minutes before standing — *Antihypertensives cause orthostatic hypotension*

MECHANISM OF ACTION

ACE

Inhibits ACE (angiotensin-converting enzyme), which converts angiotensin I to angiotensin II and allows the blood vessels to remain dilated, thus keeping BP managed

Angiotensin II constricts the vessels

ALDOSTERONE

ACE inhibitors also prevent the secretion of aldosterone, which prevents the body from holding on to water and sodium, thus keeping BP managed

Aldosterone causes our bodies to retain water & sodium, which makes our blood pressure HIGH

ACE Inhibitors VS. Angiotensin II Receptor Blockers (ARBs)

ACE INHIBITORS & ARBs are very similar medications. They have similar side effects, uses, nursing considerations & patient education. However, there are some differences to note.

ACE INHIBITORS

GENERIC	TRADE
captopril	–
enalapril	Vasotec
fosinopril	–
lisinopril	Prinivil
ramipril	Altace
SUFFIX: -PRIL	

SAVED BY THE SUFFIX

MAIN DIFFERENCES:

	ACE INHIBITORS	ARBs
ACTION	ACE, ALDOSTERONE	ANGIOTENSIN II
BLOOD PRESSURE	↓ BP	↓ BP
HEART RATE	Do not directly affect the HR	↓ HR
SIDE EFFECTS	ARBs are known to produce fewer side effect than ACE inhibitors	

ARBs

GENERIC	TRADE
losartan	Cozaar
olmesartan	Benicar
valsartan	Diovan
azilsartan	Edarbi
candesartan	Atacand
SUFFIX: -SARTAN	

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Alpha-2 Adrenergic Agonists

MEDICATION CLASS & NAME

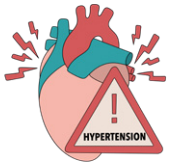
Antihypertensives

GENERIC	TRADE NAME
clonidine	Catapres
methyldopa	Aldomet
guanfacine	Tenex

Potent antihypertensive

THERAPEUTIC USES

- Hypertension & hypertensive crisis
- Alcohol & controlled drug withdrawal
- Anxiety
- ADHD



MECHANISM OF ACTION

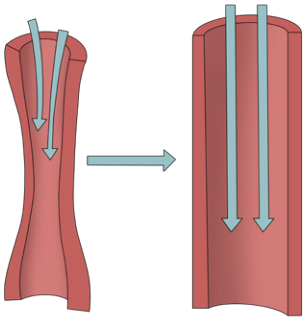
Alpha-2 adrenergic receptors are stimulated in the PRESYNAPTIC phase

↓

Blood vessels are opened (dilated)

↓

Vessels that were constricted are now OPEN, lowering blood pressure



PATIENT EDUCATION

DIET MODIFICATIONS

- Reduce salt intake
- Suck on hard candy & sip on liquids to reduce dry mouth

EDUCATE

- Do not suddenly stop the medication; it must be tapered
  - *This can cause rebound hypertension*
- Change positions slowly & sit on the side of the bed for a few minutes before standing
  - *Antihypertensives cause orthostatic hypotension*

MUST-KNOW SIDE EFFECTS

MEMORY TRICK

Ds of cloniDine

- **D**ecreased heart rate (bradycardia)
- **D**ecreased blood pressure (orthostatic hypotension)

- **D**ry mouth
- **D**rowsiness
- **D**izziness

Avoid the use of alcohol or central nervous system depressants, which could cause further drowsiness/dizziness

- **D**epression of central nervous system (CNS)

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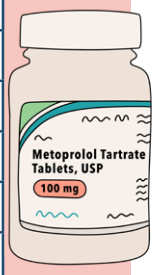
Beta-Adrenergic Blockers (Beta Blockers)

MEDICATION CLASS & NAME

Antihypertensives

GENERIC	TRADE NAME
acebut**olol**	Sectral
metopr**olol**	Lopressor, Toprol XL
propran**olol**	Inderal
nad**olol**	Corgard
aten**olol**	Tenormin
carvedi**lol**	Coreg



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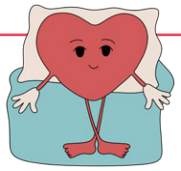
SUFFIX: -OLOL

THERAPEUTIC USES



- Hypertension
- Tachycardia & SVT
- Stable angina
- Chronic & compensated heart failure
- Dysrhythmias (*example:* A-fib)
- Anxiety

Beta blockers can be **selective**, meaning they block only beta-1 receptors, or **non-selective**, meaning they block both beta-1 and beta-2 receptors

BETA-1 RECEPTORS are found in cardiac muscle

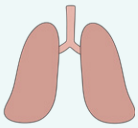
MEMORY TRICK

BETA-1 (one heart)

BETA-2 RECEPTORS are found in bronchial & smooth muscle

MEMORY TRICK

BETA-2 (two lungs)



MECHANISM OF ACTION

- **BLOCKS** beta receptors, thereby blocking adrenaline and epinephrine
- **SLOWS** the heart and provides a more efficient beat

MUST-KNOW SIDE EFFECTS

MEMORY TRICK

Bs of Beta Blockers

- **B** **B**radycardia & heart blocks
- **B** **B**reathing problems (bronchospasms or bronchoconstriction—bad for asthma patients)
- **B** **B**ad for heart failure patients (in acute setting)
- **B** **B**lood sugar masking (masks S&S of hypoglycemia/low blood sugar)
- **B** **B**lood pressure lowered (hypotension)
- **B** **B**locks beta-1 & beta-2 receptors

PATIENT EDUCATION

EDUCATE

- Do not suddenly stop the medication; it must be tapered — *This can cause rebound hypertension*
- Change positions slowly & sit on the side of the bed for a few minutes before standing — *Antihypertensives cause orthostatic hypotension*

NURSING CONSIDERATIONS

Certain medications are known to cause bronchospasms in clients with asthma. We want to "**BAN**" these medications from asthma patients.

MEMORY TRICK

- **B** Beta Blockers
- **A** Aspirin
- **N** NSAIDs

Remember: Non-selective works on beta-1 & beta-2 = lung constriction

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Calcium Channel Blockers

MEDICATION CLASS & NAME

Antihypertensives

MEMORY TRICK

Very
Nice
Drugs

GENERIC	TRADE NAME
amlo**dipine**	Norvasc
nicar**dipine**	Cardene
Verapamil	Calan SR, Verelan
Nife**dipine**	Procardia
Diltiazem	Cardizem

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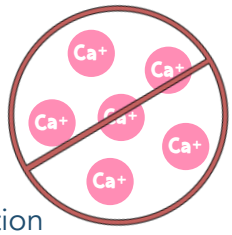
SUFFIX: -DIPINE

MECHANISM OF ACTION

- Inhibits calcium ions from entering the cells of the heart & arteries
- Inhibiting calcium causes the blood vessels to relax and heart stimulation to decrease which therefore decreases the workload of the heart



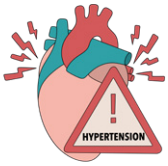
Calcium ↑ the contraction of the heart & vascular smooth muscle

MEMORY TRICK

Calcium = Contract

THERAPEUTIC USES

- Hypertension
- Stable angina
- Dysrhythmias



PATIENT EDUCATION

DIET MODIFICATIONS

Do not drink grapefruit juice

- This can cause severe hypotension

EDUCATE

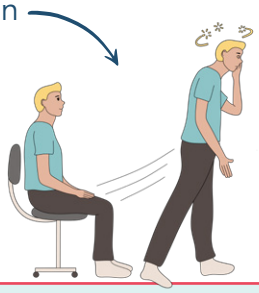
- Do not suddenly stop the medication; it must be tapered — *This can cause rebound hypertension*
- Change positions slowly & sit on the side of the bed for a few minutes before standing — *Antihypertensives cause orthostatic hypotension*
- Elevate the legs & use compression to reduce edema
- Reduce **constipation**:

MEMORY TRICK

Fruits, Fiber & Fluids
Fill up the toilet

MUST-KNOW SIDE EFFECTS

- Orthostatic hypotension & dizziness
- Constipation
- Flushing
- Headache
- Peripheral edema



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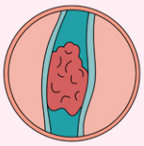
Heparin vs. Warfarin

	HEPARIN	WARFARIN
MEDICATION CLASS	Anticoagulants	
MEDICATION NAME	 **HEPARIN** — NAME: heparin sodium; SUFFIX: –; PROS/CONS: ▪ Requires more monitoring ▪ Greater chance of bleeding **LOW MOLECULAR WEIGHT HEPARIN (LMWH)** — NAME: dalte**parin** (Fragmin), enoxa**parin** (Lovenox); SUFFIX: **-parin**; PROS/CONS: ▪ Stable response ▪ Lower chance of bleeding **ANTIDOTE:** protamine sulfate	 **GENERIC:** warfarin; **TRADE NAME:** Coumadin **ANTIDOTE:** Vitamin K
MECHANISM OF ACTION	THROMBIN — Prevents thrombin activation, which inhibits the conversion of fibrinogen to fibrin	VITAMIN K — ▪ Interferes with the production of vitamin K ▪ ↓ Clotting factors: II (prothrombin), VII, IX and X **Vitamin K,** a fat-soluble vitamin, is stored in the liver & body tissue. It aids the blood in clotting & coagulation. By giving warfarin, we SUPPRESS this function & prevent clotting factors from being made
ROUTE	IV or SubQ (**NOT** orally) — *Heparin is inactivated by gastric acids in the stomach*	Orally (PO)
TERATOGENIC	Safe for pregnancy *(does not cross the placenta or enter breast milk)*	Teratogenic & unsafe in pregnancy *(crosses the placenta, but does not enter breast milk)*
ONSET	Rapid (takes effect in minutes) MEMORY TRICK: **H**eparin **H**appens fast	Slow (24–72 hours to take effect) MEMORY TRICK: **W**arfarin – you have to **W**ait
DURATION	Short-term therapy	Long-term therapy
MONITORING	aPTT	PT/INR
THERAPEUTIC USES	 **Prevents new clots** → Those with a risk for developing clots (prophylactic) *Example:* atrial fibrillation **&** **Prevents current clots from getting bigger** → Those who currently have clots *Example:* deep vein thrombosis (DVT)	
SIDE EFFECTS	 **Bleeding (can be mild or severe)** ▪ Bruising ▪ Petechiae ▪ Bloody stool ▪ Vomiting (looks like coffee grounds)	
PATIENT EDUCATION	Administration: ▪ SubQ in the belly ▪ 2 inches from the umbilicus ▪ 90° angle ▪ Do not massage the injection site afterward 	**DIET MODIFICATIONS** **EDUCATE patient to be consistent with their vitamin K intake** ▪ Do not increase, decrease, or be inconsistent with the intake of foods rich in vitamin K **VITAMIN K RICH FOODS:** Liver & green leafy vegetables (broccoli, spinach, brussels sprouts, cabbage)

Both medications will be given for several days until warfarin (INR) reaches a therapeutic level. Then the patient will "BRIDGE" from heparin to warfarin.

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Anticoagulants (Warfarin & Heparin)

EDUCATE on *Bleeding Precautions:*

AVOID NSAIDs, aspirin, antibiotics, alcohol (including alcohol-based mouthwash)

AVOID flossing

AVOID contact sports

AVOID straining of the bowels

MEMORY TRICK: Fruits, Fiber & Fluids Fill up the toilet

Remove throw rugs & ensure rooms are well lit (↓ risk for falls)

Use an electric razor only

Gently brush teeth with a soft-bristled toothbrush

Wear a MedicAlert device or medical bracelet

MONITORING

	HEPARIN	WARFARIN
MEASURED WITH	aPTT: activated Partial Thromboplastin Time	INR: International Normalized Ratio (Both PT & INR are used for monitoring. INR is more commonly used.)
MEMORY TRICK	HEPARIN: 7+3 =10	WaRfaRin → INR; WaRfaRin has more Rs than HepaRin – think INR
PATIENTS NOT ON ANY ANTICOAGULANTS	**aPTT:** 30–40 seconds	**INR:** < 1
PATIENTS ON ANTICOAGULANTS	▪ **aPTT:** 1.5–2 times the normal value ▪ **aPTT:** 47–70 seconds	▪ **INR:** 1.5–2 times the normal value ▪ **INR:** 2–3 ▪ **INR:** 2.5–3.5 (for patients with heart valve replacement)
INTERPRETATION	↑ Numbers are too **HIGH** = Patient will **DIE** (HIGH risk for bleeding)	↓ Numbers are **LOW** = Clots will **GROW**

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HMG-CoA Reductase Inhibitors (Statins)

MEDICATION CLASS & NAME

Antihyperlipidemics

GENERIC	TRADE NAME
atorva**statin**	Lipitor
fluva**statin**	Lescol
lova**statin**	Mevacor
pitava**statin**	Livalo
simva**statin**	Zocor
rosuva**statin**	Crestor
prava**statin**	Pravachol

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SUFFIX: -STATIN

Ny**statin** is an *antifungal* medication, *not a statin*

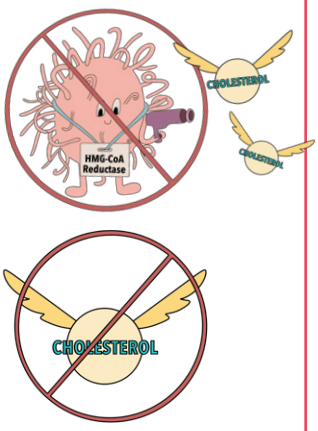
MEMORY TRICK: NyStatin is Not a Statin

MECHANISM OF ACTION

Inhibits the enzyme HMG-CoA Reductase, which produces LDL (bad cholesterol)

↓

LOWERS the amount of CHOLESTEROL in the blood



THERAPEUTIC USES

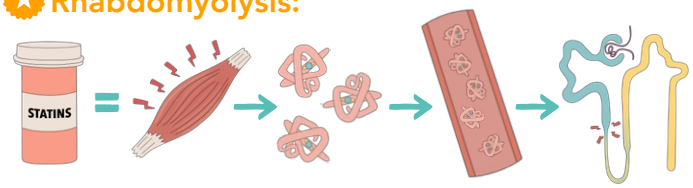
Antihyperlipidemic → against high or elevated fat in the blood

- To help lower cholesterol in patients with hyperlipidemia

Statins are NOT a CURE

MUST-KNOW SIDE EFFECTS

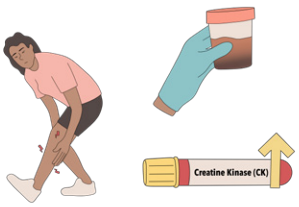
Rhabdomyolysis:



Muscles break down → Myoglobin is released → Myoglobin leaks into the bloodstream → Myoglobin CLOGS the kidneys = kidney damage

SIGNS & SYMPTOMS OF RHABDOMYOLYSIS:

- Muscle pain, tenderness, or weakness
- Malaise
- ↑ Creatine kinase (CK) levels
- Dark urine color (tea- or cocoa-like urine)



NURSING CONSIDERATIONS

MONITOR

- Lipid levels
- Kidney function
- Creatine kinase levels

SAFE DURING PREGNANCY? ✗ NO!

PATIENT EDUCATION

REPORT any feelings of muscle aches or weakness

EDUCATE

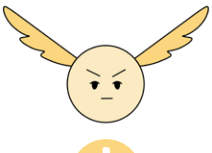
- Take the medication in the evening or at bedtime for maximum effectiveness (cholesterol is synthesized at night)

AVOID

- ✗ Grapefruit juice
- ✗ St. John's wort
- ✗ Alcohol
- ✗ Direct sun exposure (wear sunscreen when outside as statins can cause photosensitivity)
- ✗ Pregnancy (use barrier contraceptives)

MEMORY TRICK: Statins think avoid Sunburn (photosensitivity) & take before Sleep

THERAPEUTIC RESPONSE



↓ LDL
< 100 mg/dL

↓ Triglycerides
< 150 mg/dL

↓ Total cholesterol
< 200 mg/dL

MEMORY TRICK: LDL think we want Low levels because it's a "bad" fat

↑ HDL
> 40 mg/dL

MEMORY TRICK: HDL think we want High levels because it's a Happy cholesterol

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Bile Acid Resins

MEDICATION CLASS & NAME

Antihyperlipidemics

GENERIC	TRADE NAME
cholestyramine	Prevalie
colestipol	**Cole**stid
colesevelam	Wel**chol**

PREFIX: COLE-, CHOLE-

Cholestyramine comes as a powder. Mix it with water, water-based soups, applesauce, and shakes. Do not mix it with carbonated beverages.

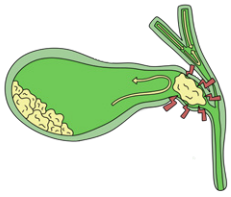
MUST-KNOW SIDE EFFECTS

- Constipation
- Increased risk for bleeding
- Vitamin A & D deficiencies

Due to vitamin K malabsorption

THERAPEUTIC USES

- Hyperlipidemia
- Gallstone dissolution
- Pruritus (itching) associated with biliary obstruction



PATIENT EDUCATION

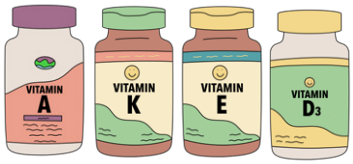
This medication interferes with the digestion of **fats**

↓

Patient should supplement with *fat-soluble vitamins*

MEMORY TRICK

All Kids Eat Donuts



To help with constipation:

↓

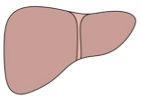
- Exercise regularly
- Use stool softeners

MEMORY TRICK

Fruits, Fiber & Fluids Fill up the toilet

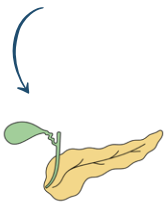
MECHANISM OF ACTION

Bile is made & secreted by the **LIVER**



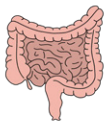
↓

Then, it's stored in the **GALLBLADDER**



↓

Once emulsified, the fats & lipids are absorbed in the **INTESTINES**



↓

BILE ACID RESINS bind to the bile acid to form an insoluble substance

↓

This is excreted in the **FECES**

↓

Bile acids decrease

↓

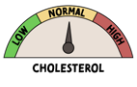
The **LIVER** uses cholesterol to make more bile

MEMORY TRICK

Bile acid resins Bind to Bile = Bye Bye Bile

↓

↓ **CHOLESTEROL**



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Nitrates

MEDICATION CLASS & NAMES

Vasodilators
- vessels
- enlarge or make bigger

GENERIC	TRADE NAME
nitroglycerin	**Nitro**stat, **Nitro**-bid
isosorbide mononi**trate**	Imdur

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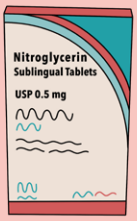
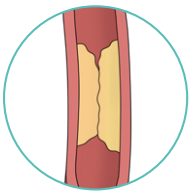
SUFFIX: -NITRA
PREFIX: NITRO-



MECHANISM OF ACTION

Blood flow is occluded due to atherosclerosis (fatty plaque buildup in the vessels)



↓

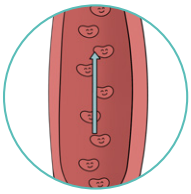
Decreased blood flow = Chest pain (angina)

↓

Nitrates work to dilate or open up the vessel

↓

Increased blood flow



MUST-KNOW SIDE EFFECTS

The Hs

- **H**eadache
- **H**ot, flushed face — Blood is rushing through the body and to the head
- **H**ypotension — Opening the blood vessels = ↓ in blood pressure

THERAPEUTIC USES

- Treatment or prevention
  - Angina (acute chest pain)
  - Acute coronary syndrome

ROUTES OF ADMINISTRATION

QUICK ONSET	SLOW ONSET
Intravenous (IV)	Patch
Sublingual tablets	Topical ointment
Buccal tablets	Sustained-release tablets
Oral spray	

PATIENT EDUCATION

TOPICAL & TRANSDERMAL PATCH

- Remove prior patches before applying new doses
- Rotate sites
- Place patch over a clean, hairless area
- Patches can be worn in the shower
- Wipe off any excess medication prior to applying more
- Do not rub nitro ointment into the skin
- Avoid getting nitro ointment on the skin (wear gloves!)

CONTRAINDICATIONS:

✗ Phosphodiesterase inhibitors
- Erectile dysfunction drugs
- Suffix: "-**afil**"
  - silden**afil** (Viagra)

hypotension

✗ Increased intracranial pressure (ICP)

SUBLINGUAL, BUCCAL, OR SPRAY

- Do not swallow or chew tablets

SUBLINGUAL: Place under the tongue

BUCCAL: Place between the cheek & gum

STORAGE

- Keep medication in original container (dark glass bottle)
- Store medication in a dry, cool place
- Keep medication with patient at all times

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Antacids

MEDICATION NAME

GENERIC	TRADE NAME
sodium bicarbonate	Alka-Seltzer
calcium carbonate	Tums, Rolaids
aluminum hydroxide	Amphojel
magnesium hydroxide	Milk of Magnesia

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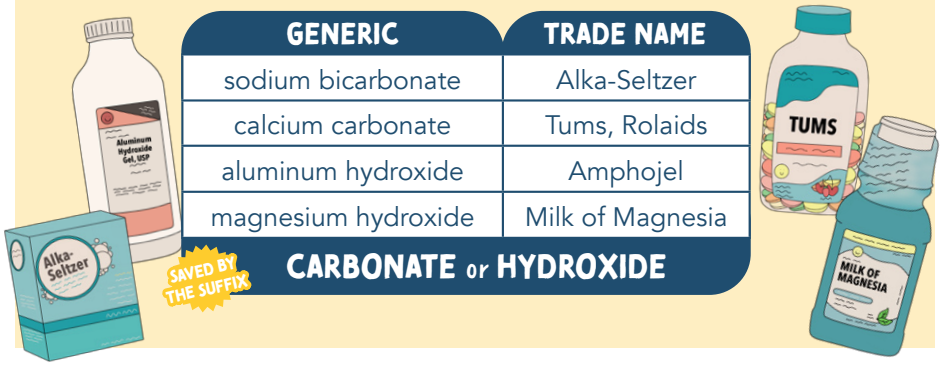
CARBONATE or **HYDROXIDE**



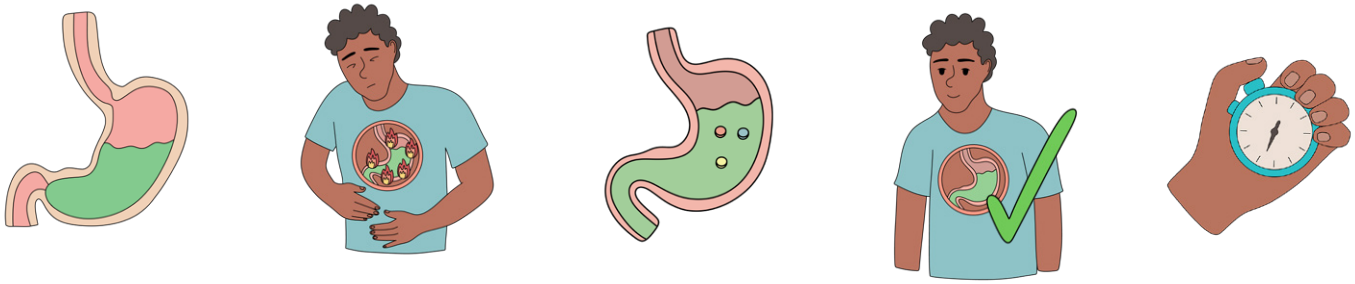
MEDICATION CLASS:

Antacids

against → acid

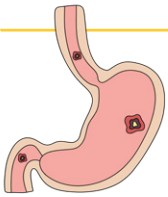
MECHANISM OF ACTION

Excess stomach acid (hydrochloric acid) → Stomach pain, heartburn & indigestion → Antacids work as a base to neutralize the hydrochloric acid → Symptoms are relieved once the acid is neutralized → This happens quickly, but the effect doesn't last long



THERAPEUTIC USES

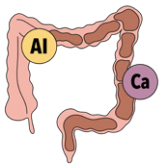
- Peptic ulcers
- Heartburn
- Upset stomach
- Gastroesophageal reflux disease (GERD)



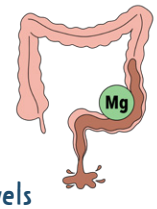
MUST-KNOW SIDE EFFECTS

Depends on antacid used:

CALCIUM & ALUMINUM
- ConstipAtion



MAGNESIUM
- GI upset
- Diarrhea



MEMORY TRICK

Magnesium Moves the bowels

PATIENT EDUCATION

TAKE

✓ 1 hour before or after other medications

✓ At bedtime

DO NOT

✗ Take with meals
- Should be administered on an empty stomach

✗ Give to children
- Age 2 or younger
- Who have had a recent virus

Antacids that contain aspirin + Viral illness = Reye's syndrome
Encephalitis

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Proton Pump Inhibitors (PPIs)

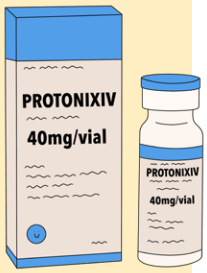
MEDICATION CLASS & NAME

Proton Pump Inhibitors

GENERIC	TRADE NAME
dexlanso**prazole**	Dexilant
esome**prazole**	Nexium
lanso**prazole**	Prevacid
ome**prazole**	Prilosec
panto**prazole**	Protonix

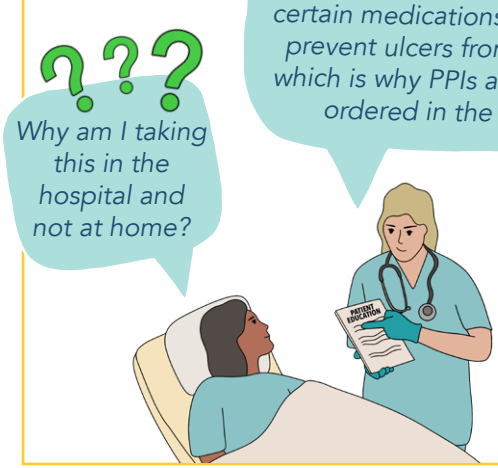
SAVED BY THE SUFFIX

SUFFIX: -PRAZOLE



PATIENT EDUCATION

- Take with calcium and Vitamin D supplements to prevent **osteoporosis**
- Take 30 minutes prior to first meal of the day

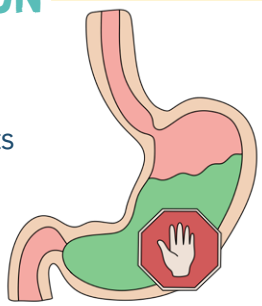


MECHANISM OF ACTION

Inhibits (stops) the proton pump in the stomach from producing excessive amounts of **acid**

MEMORY TRICK

PPIs Pause acid Production

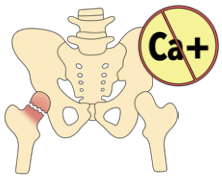


The Ps of PPIs

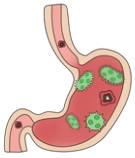
MEMORY TRICK

- P "-Prazole"
- P Pauses acid Production
- P Prevents ulcers
- P H. Pylori
- P Porous & sPongy bones
- P Potential infection
- P Prior to meals
- P Prevents damage of Pain meds

Long-term use of PPIs can cause fractures & osteoporosis, because they ↓ calcium absorption in the body



↓ in acid production
=
↑ in susceptibility to infection (C. diff)

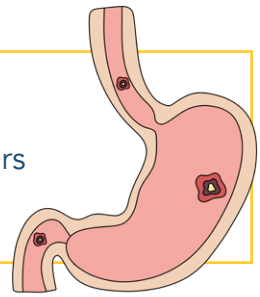


Aspirin & NSAIDs irritate the stomach

THERAPEUTIC USES

Prevents & treats ulcers

- Gastric & duodenal ulcers
- H. pylori



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Histamine (H2) Receptor Antagonists

MEDICATION CLASS & NAME

Histamine (H2) Receptor Antagonists

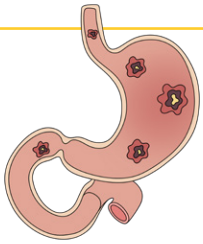
GENERIC	TRADE NAME
cime**tidine**	Tagamet
famo**tidine**	Pepcid
niza**tidine**	Axid

SAVED BY THE SUFFIX

SUFFIX: -TIDINE

THERAPEUTIC USES

- Treats & prevents duodenal & gastric ulcers
- Treats gastroesophageal reflux disease (GERD)



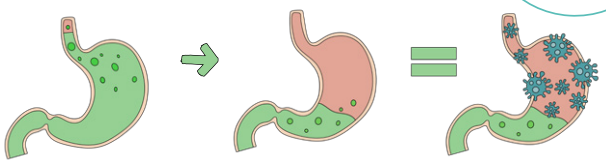
MECHANISM OF ACTION

- Decreases gastric secretions by blocking H2 receptors in the stomach

MUST-KNOW SIDE EFFECTS

- High risk for stomach infection
  - GI upset
  - Diarrhea
- Anxiety or mood change

Not enough stomach acid to fight infection



INVERSE RELATIONSHIP!

As stomach acid ↓, risk for GI infection ↑

PATIENT EDUCATION

DIET MODIFICATIONS

- Limit irritants (smoking, spicy foods)
- Limit caffeine (coffee is acidic!)
- Reduce citrus/acidic fruit
- Take this medication 30–60 minutes before eating a trigger food

MEMORY TRICK take -tiDINE before you DINE

NURSING CONSIDERATIONS

- **AVOID** NSAIDs (higher risk for GI bleed & ulcers)

Quick Overview

	ANTACIDS	H2 BLOCKERS Histamine (H2) Receptor Antagonists	PPIs Proton Pump Inhibitors
ACTION	**NEUTRALIZES** the stomach acid	**BLOCKS** secretion receptors in the stomach	**STOPS** the production of stomach acid
TIMING	Works quickly, but does not last long (Within seconds to minutes)	Works quickly and lasts up to 12 hours (Within 30–90 minutes)	Works slowest, but can last up to several days (Within a few hours)
RELIEF	**Immediate** relief	**Immediate** relief & **long-term** relief	**Long-term** relief

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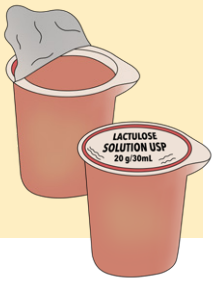
Lactulose (Cholac)

MEDICATION CLASS & NAME

Osmotic Laxatives, Ammonia Detoxicants

GENERIC	TRADE NAME
lactulose	Cholac



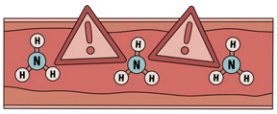
MUST-KNOW SIDE EFFECTS

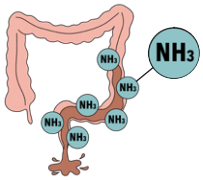
- **RAPID onset of diarrhea**
  - **Should have 2–3 soft/loose stools per day**
- Low urine output
- Muscle twitching & cramping
- Change in mental status
- Abdominal pain

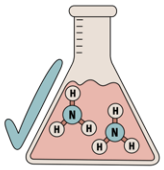
MEMORY TRICK
Lactulose think
LactuLOOSE because it
LOOSENS the bowels

MECHANISM OF ACTION

Ammonia (NH_3) is pulled from the bloodstream into the colon → Rapid onset of diarrhea → Normal serum ammonia (NH_3) levels







THERAPEUTIC USES

- Helps relieve constipation
- **Decreases ammonia (NH_3) levels in those with liver cirrhosis**

Liver damage or cirrhosis
Can't excrete toxins such as ammonia

↓

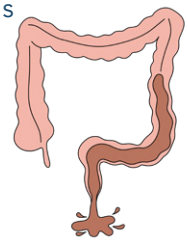
Ammonia (NH_3) stays in the blood
Ammonia is a toxin

↓

Brain dysfunction
- Hepatic encephalopathy
- Confusion
- Sleepiness
- Coma

NURSING CONSIDERATIONS

MONITOR
- Ammonia levels
  - Lactulose should decrease ammonia levels
- Number of stools
  - **Lactulose should produce 2–3 soft bowel movements each day**
- Mental status
  - Lactulose should improve mental status
- For dehydration



PATIENT EDUCATION

DIET MODIFICATIONS
- ↑ Natural dietary fiber
- ↑ Fluids

Remember these patients are losing a lot of fluids through loose stools, so we want to replace those fluids!

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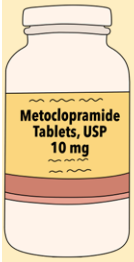
Metoclopramide (Reglan)

MEDICATION NAME



GENERIC	TRADE NAME
Metoclopramide	Reglan

MEDICATION CLASS:

Dopamine Receptor Antagonist

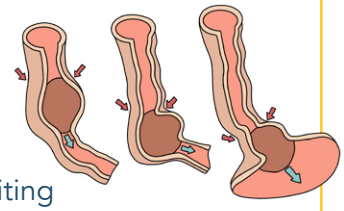
- Prokinetic → encourages movement
- Antiemetic → works against vomiting

MECHANISM OF ACTION

MEMORY TRICK

Metoclopramide think Motility!

- **↑ GI motility & promotes stomach emptying**
  - Stimulates the smooth muscle of GI tract with anticholinergic effects, allowing for better motility & movement
  - Impacts dopamine receptors, helping to reduce nausea/vomiting



THERAPEUTIC USES

Nausea & vomiting related to
- Chemotherapy & radiation
- Anesthesia
- Pain medication (opioids)

- Gastroparesis
  - Slowed gastric emptying
- Heartburn
  - Gastroesophageal reflux disease (GERD)

MUST-KNOW SIDE EFFECTS

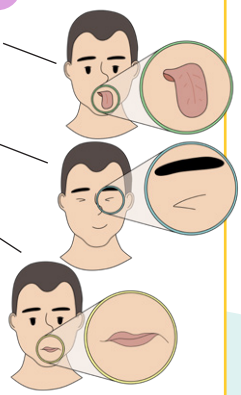
- Fatigue & sedation
- Headache
- Dry mouth
- Constipation

These are called extrapyramidal symptoms (EPS)

Are due to LONG-TERM use & can be irreversible

TARDIVE DYSKINESIA (TD)

- Erratic movements of tongue
- Excessive blinking
- Lip-smacking
- Jerking of extremities
- Puffing of cheeks
- Involuntary spasm of limbs
- Muscle rigidity
- Tremors

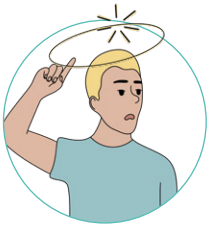


NEUROLEPTIC MALIGNANT SYNDROME (NMS)

- Muscle rigidity
- Tachycardia
- High fever
- Sweating

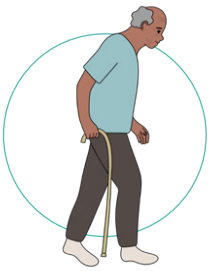
PATIENT EDUCATION

- **Drink fluids**
  - Aids with motility
  - Prevents dehydration from vomiting



- **Avoid driving or operating heavy machinery**
  - Can cause drowsiness & dizziness

- **Rinse mouth frequently & suck on sugar-free candy**
  - Helps with dry mouth associated with metoclopramide



- **Do not take if affected by Parkinson's disease**
  - May worsen symptoms

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Nonsteroidal Anti-Inflammatory Drugs (NSAIDs)

MEDICATION CLASS & NAME:

Analgesics, Anti-Inflammatories

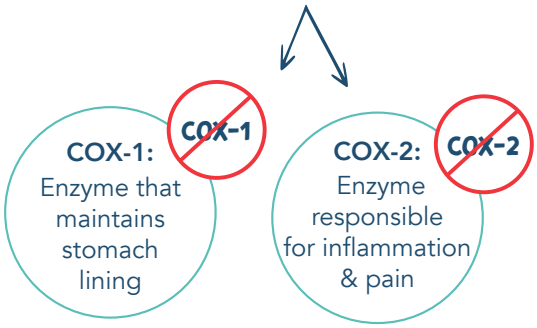
GENERIC	TRADE NAME
ibu**profen**	Advil
feno**profen**	Nalfon
flurbi**profen**	–
diclofenac	Voltaren
celecoxib	Celebrex
keto**rolac**	Toradol
naproxen	Aleve
indomethacin	Indocin

SAVED BY THE SUFFIX — SUFFIXES: -PROFEN, -OLAC

Inhibits COX-2 without inhibiting COX-1

MECHANISM OF ACTION

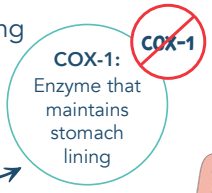
- Inhibits prostaglandin synthesis by blocking the **cyclooxygenase (COX) enzyme**



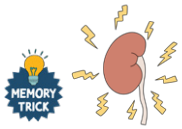
- Inhibiting or blocking these enzymes BLOCKS pain and inflammation as well!

MUST-KNOW SIDE EFFECTS

- Gastroesophageal upset
  - Nausea/diarrhea/vomiting
  - Abdominal discomfort
  - Acid reflux
- ★ Possible GI bleed



- Impaired renal function (nephrotoxicity)
  - Blocking prostaglandins causes vasoconstriction, which impairs renal blood flow
- Hypertension
- Clot formation (non-aspirin-containing NSAIDs)
  - ↑ Risk for strokes & myocardial infarction (MI)



NSAIDs think Nephrotoxic

PATIENT EDUCATION

EDUCATE

- Take **with** food to decrease stomach upset
- Take proton pump inhibitors (PPIs) such as ome**prazole** or panto**prazole** to decrease acid reflux

REPORT

- Signs of a GI bleed:
  - Black, tarry stools
  - Coffee ground emesis (vomit)
  - Unresolved abdominal cramping

THERAPEUTIC USES

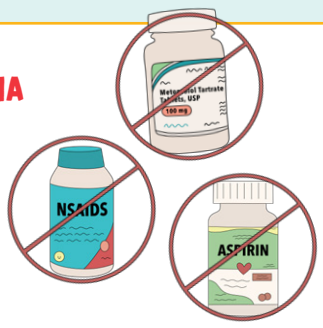
- Mild to moderate pain
- Menstrual cramps
- Fever (antipyretic)
- Musculoskeletal disorders like OA & RA

CONTRAINDICATIONS

Do not give these medications to those with **ASTHMA**

MEMORY TRICK — "**BAN**" these medications from asthma patients

- **B** Beta blockers
- **A** Aspirin
- **N** NSAIDs



Risk for bronchospasms

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Aspirin (Bayer Aspirin)

MEDICATION CLASS & NAME:

Acetylsalicylic Acid (ASA)

GENERIC	TRADE NAME
aspirin	Bayer Aspirin

ANTIDOTE: SODIUM BICARBONATE

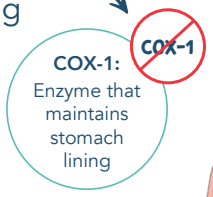
THERAPEUTIC USES

- Mild to moderate pain
- Fever (antipyretic)
- Inflammatory conditions (RA, OA & rheumatic fever)
- Cardiac health & clot prophylaxis

Aspirin is used to ↓ the risk of MI & CVA because of its antiplatelet effect

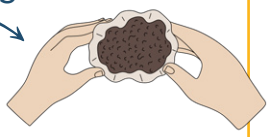
MUST-KNOW SIDE EFFECTS

- Gastroesophageal upset
- Nausea/diarrhea/vomiting
- Mental status changes could indicate toxicity



INTERNAL BLEEDING

- Abdominal pain & cramping
  - Coffee ground emesis
    Dark red = old blood
    Bright red = active bleed

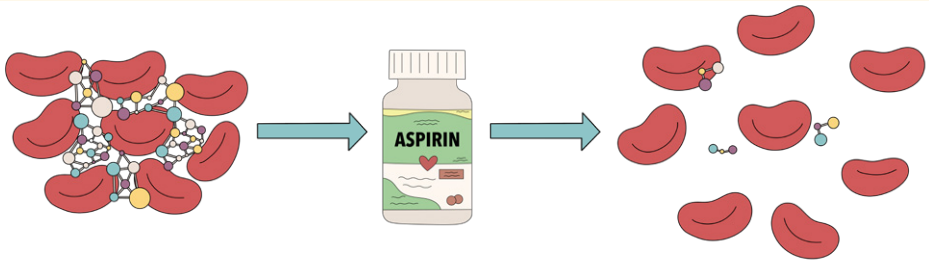


- Hematuria
  - Blood in the urine

- Hemoptysis
  - Coughing up blood
- Black, tarry stools

MECHANISM OF ACTION

- Blocks the production of prostaglandins to decrease pain & inflammation
- Blocks platelets from clumping together



PATIENT EDUCATION

STOP:

Taking aspirin one week prior to any major surgery

- Do not give aspirin to children with a recent or current viral infection

Aspirin + Viral illness =

Reye's syndrome

CONTRAINDICATIONS

Do not give aspirin to patients...

- ✗ With bleeding tendencies or bleeding disorders
- ✗ With an ACTIVE bleed
  - GI bleeds
  - Bleeding ulcers
- ✗ Currently taking anticoagulants like warfarin or heparin

- ✗ With a vitamin K deficiency
  - Vitamin K helps the body clot

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Acetaminophen (Tylenol)

MEDICATION CLASS & NAME:

Nonsalicylates, Analgesics, Antipyretics

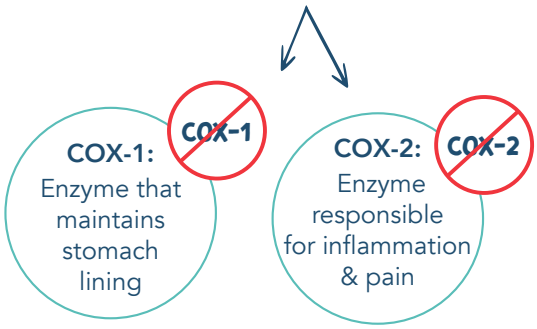
GENERIC	TRADE NAME
acetaminophen	Tylenol

ANTIDOTE: ACETYLCYSTEINE (ACETADOTE)

MEMORY TRICK: ACETylcysteine think ACETaminophen

MECHANISM OF ACTION

- Inhibits prostaglandin synthesis by blocking the **cyclooxygenase (COX) enzyme**



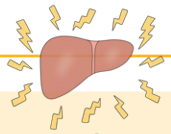
- Inhibiting or blocking these enzymes BLOCKS pain and inflammation as well!

THERAPEUTIC USES

- Mild to moderate pain
- Fever or flu-like symptoms in children
- Aspirin substitute for those:
  - Allergic to aspirin
  - With bleeding tendencies

MUST-KNOW SIDE EFFECTS

- GI upset
  - Nausea/diarrhea/vomiting
  - Anorexia
  - Abdominal discomfort
- Impaired liver function (hepatotoxicity)



High dose or long-term use of acetaminophen → Body is unable to metabolize → **Accumulation is toxic to the liver**

WHO IS AT RISK FOR HEPATOTOXICITY? Chronic alcohol use & malnourished patients are at ↑ risk for liver damage

PATIENT EDUCATION & NURSING CONSIDERATIONS

Assess alcohol use before administration & educate to limit or discontinue alcohol use

→ Risk for hepatotoxicity

Acetylcysteine is the antidote for acetaminophen

It's normal for acetylcysteine to smell like rotten eggs

Limit dosage to 3,000 mg or less per day

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Opioids

MEDICATION CLASS & NAME:

Opioid Analgesics

GENERIC
hydromorph**one**
codeine
oxyco**done**
fentanyl
morphine sulfate

SAVED BY THE SUFFIX

SUFFIXES: -DONE, -ONE

ANTIDOTE: NALOXONE (NARCAN)

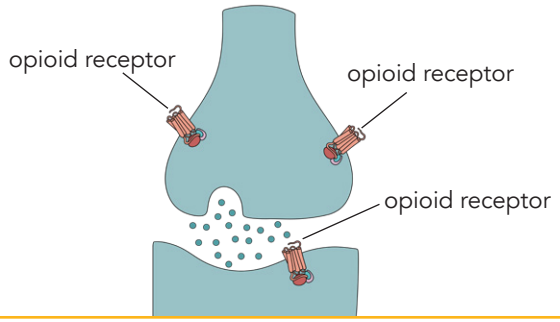
THE GOLD STANDARD

Most commonly used opioid for chronic pain

Can be given in many forms: (PO, nasally, subcut, IM, IV & suppository)

MECHANISM OF ACTION

- Binds to opioid receptors in the CNS, which causes an analgesic, sedative & euphoric effect



THERAPEUTIC USES

- ↓ Anxiety & sedates post-operatively
- ↓ Anxiety in those with dyspnea
- Relieves pain (myocardial infarction)
- Relieves diarrhea & intestinal cramping
- Provides end-of-life comfort & respiratory care

MUST-KNOW SIDE EFFECTS

MEMORY TRICK

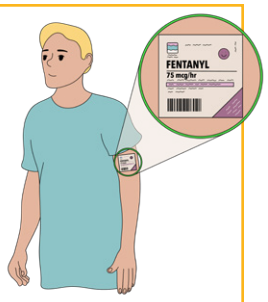
Think "SLOW" when remembering all the signs & symptoms of Opioids

SLOW SLOW GI function
- Constipation → Patient will NOT build tolerance

SLOW SLOW vital signs
- ↓ Heart rate (bradycardia)
- ↓ Blood pressure (hypotension)
- ↓ Respiratory rate → Respiratory depression

SLOW SLOW central nervous system
- Weakness
- Dizziness
- Sedation

TRANSDERMAL FENTANYL PATCH



- Used for chronic pain (*Example:* cancer)
- Provides relief of pain for up to 72 hours

DOs

✓ Rotate the application site

✓ Discard appropriately
- Fold with the adhesive side together
- Place in a designated bin or trash can
- Keep away from children

✓ Place on clean, dry, hair-free area

✓ The patient can swim or shower with the patch in place

DON'Ts

✗ Don't apply a new patch while an old patch is still applied

✗ Don't alter or cut the patch

✗ Don't be exposed to prolonged heat
- Fever
- Hot tubs
- Heating pads
- Saunas

↑ release of fentanyl = extremely dangerous!

PATIENT EDUCATION & NURSING CONSIDERATIONS

Have naloxone (Narcan) available when administering

When to give naloxone?
- For respiratory depression
  Respiratory rate less than 12
- If the patient is unarousable

To help with constipation
- Administer stool softeners or laxatives
- Encourage daily exercise
- Encourage patient to defecate when they feel the urge (do not wait)

MEMORY TRICK
Fruits, Fiber & Fluids Fill up the toilet

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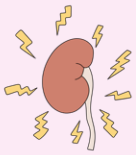
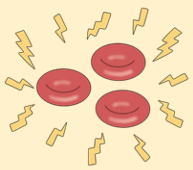
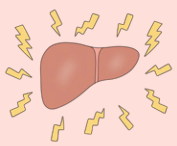
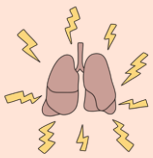
Quick Comparison of
NSAIDs, Aspirin, Acetaminophen & Opioids

	NSAIDs Example: ibuprofen (Advil)	**ASPIRIN**	**ACETAMINOPHEN** Example: Tylenol	**OPIOIDS**
ANALGESIC	✓	✓	✓	✓
ANTI-INFLAMMATORY	✓	✓	✗	✗
ANTIPYRETIC	✓	✓	✓	✗
ANTIPLATELET	✗ *While NSAIDs may have a blood thinning/antiplatelet effect, it is not classified as an antiplatelet medication itself.*	✓	✗	✗
SAFE FOR CHILDREN?	✓	✗ *Risk for Reye's syndrome* **Exceptions to this rule may apply to children with a cardiac background or congenital heart defects*	✓	✗ *Do not give unless prescribed by a doctor*
MAJOR COMPLICATION	Nephrotoxicity (hurts the kidneys)  MEMORY TRICK: NSAIDs think Nephrotoxic	Bleeding 	Hepatotoxicity (hurts the liver) Only in high doses  MEMORY TRICK: Acetaminophen (tylenol) think A +	Respiratory depression 

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Selective Serotonin Reuptake Inhibitors (SSRIs)

MEDICATION CLASS & NAME:

Antidepressants

Sertraline is an outlier to the suffix

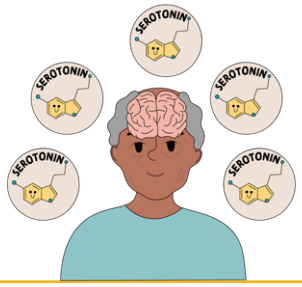
MEMORY TRICK
Sertraline think SSRI

GENERIC	TRADE NAME
sertra**line**	Zoloft
c**italopram**	Celexa
esc**italopram**	Lexapro
fluoxe**tine**	Prozac
vila**zodone**	Viibryd
fluvoxam**ine**	Luvox

SAVED BY THE SUFFIX
SUFFIXES: -TALOPRAM, -INE, -ZODONE

MECHANISM OF ACTION

- Inhibits (blocks) the reabsorption of serotonin into neurons, allowing more **serotonin** to remain in the brain and improving communication within the brain



MEMORY TRICK
SSRIs increase serotonin in the brain – Smiley Serotonin

MUST-KNOW SIDE EFFECTS

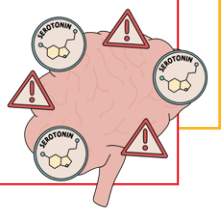
4 Ss of SSRIs

- **S**exual dysfunction
- **S**tomach issues
- **S**wollen (weight gain)
- **S**erotonin syndrome

ADVERSE REACTION:

Serotonin syndrome
(too much serotonin in the brain)

- Mental changes
- Muscle rigidity/tightness
- ↑ HR, BP & temperature
- Tremors



PATIENT EDUCATION

EDUCATE

- May take 4–6 weeks to take effect

Educate on the importance of compliance!

- Take in the morning
- Do not abruptly stop the medication; it must be tapered
- Do not combine with:
  - ✗ MAOIs
  - ✗ St. John's wort

MEMORY TRICK
SSRIs think take at Sunrise

MEMORY TRICK
SSRIs + St. John's wort = Serotonin Syndrome

THERAPEUTIC USES

- **First-line drug for depression/anxiety**
- Obsessive-compulsive disorder (OCD)
- Eating disorders

NURSING CONSIDERATIONS

MONITOR FOR

- Behavior change
- Suicidal ideation
- Worsening depression
- Serotonin syndrome

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SNRIs & DNRIs

MEDICATION CLASS & NAME:

Antidepressants

GENERIC	TRADE NAME
bupropion	Wellbutrin
duloxetine	Cymbalta
venlafaxine	Effexor XR
milnacipran	Savella
nefazodone	–

MUST-KNOW SIDE EFFECTS

Worsening depression/ suicidal ideation

- Changes in weight (loss or gain depending on the medication)
- Drowsiness & lethargy
- Dizziness
- GI upset

MECHANISM OF ACTION

SNRIs

Act on **S**erotonin & **N**orepinephrine to limit reuptake and allow for increased levels in the brain

DNRIs

Act on **D**opamine & **N**orepinephrine to limit reuptake and allow for increased levels in the brain

THERAPEUTIC USES

- Depression & depressive episodes
- Fibromyalgia & related pain
- Anxiety disorders
- Neuropathic pain
- Bupropion may be used for smoking cessation

PATIENT EDUCATION

EDUCATE

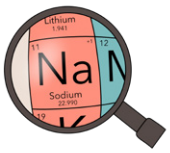
- May take 4–6 weeks to take effect
- Do not take with MAOIs or TCAs
- Bupropion may cause weight loss

Educate on the importance of compliance!

NURSING CONSIDERATIONS

MONITOR FOR

- Electrolyte levels
  - Risk for sodium imbalance
- Suicidal ideation or worsening depression



- Do not give these medications to those with an eating disorder as it may cause a decrease in appetite

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Tricyclic Antidepressants (TCAs)

MEDICATION CLASS & NAME:

Antidepressants

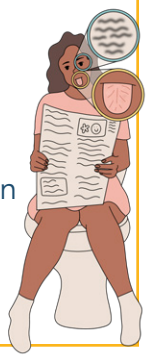
GENERIC	TRADE NAME
ami**triptyline**	Elavil
amoxapine	–
clomi**pramine**	Anafranil
pro**triptyline**	Vivactil
nor**triptyline**	Pamelor

SAVED BY THE SUFFIX

SUFFIXES: -TRIPTYLINE, -PRAMINE

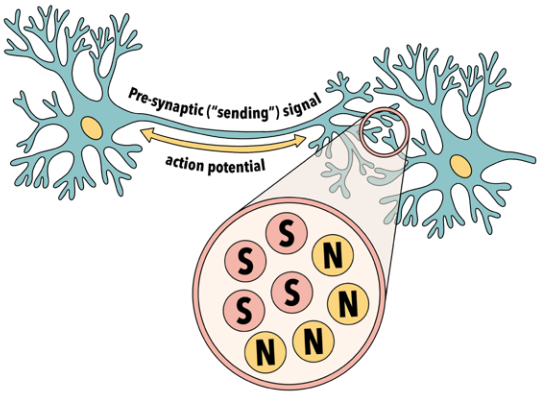
MUST-KNOW SIDE EFFECTS

- Dizziness
- Orthostatic hypotension
- Anticholinergic effects:
  - **CAN'T SEE** Blurred vision
  - **CAN'T SPIT** Dry mouth
  - **CAN'T PEE** Urinary retention
  - **CAN'T POOP** Constipation



MECHANISM OF ACTION

- Forces higher levels of **serotonin** and **norepinephrine** to collect in the SYNAPSE by inhibiting (blocking) the reuptake into presynaptic areas of the brain



THERAPEUTIC USES

- Depression & depressive episodes
- Bipolar disorder
- Obsessive-compulsive disorder (OCD)
- Neuropathy & neuropathic pain
- Enuresis
- Migraines

PATIENT EDUCATION

EDUCATE

- May take 2-3 weeks to take effect
- Wait 14 days after being off MAOIs to start taking TCAs

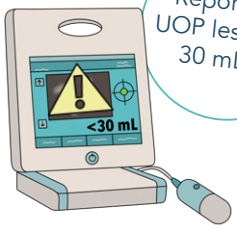
Called a **WASHOUT period**

- Change position slowly to avoid falls
- To help with dry mouth:
  - ↑ fluid & suck on sugar free hard candies

NURSING CONSIDERATIONS

MONITOR FOR

- Mental status/confusion
- Elderly patients
- Decreased kidney function (can lead to toxicity)
- Bladder distention
- Urinary output (UOP)



Report any UOP less than 30 mL/hr

DO NOT ADMINISTER TO:

- ✗ Those with glaucoma
- ✗ Those having difficulty passing urine

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Monoamine Oxidase Inhibitors (MAOIs)

MEDICATION CLASS & NAME:

Antidepressants

MEMORY TRICK Let me give you a **TIP** about these...

	GENERIC	TRADE NAME
T	tranylcypromine	Parnate
I	isocarboxazid	Marplan
P	phenelzine	Nardil

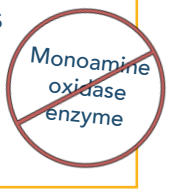
MUST-KNOW SIDE EFFECTS

- Sleep difficulties (insomnia)
- Elevated blood pressure
- Drowsiness
- Dizziness
- **Worsening depression**
- **Suicidal ideation**

MEMORY TRICK
MAOI think
Morning Administration
Otherwise Insomnia

MECHANISM OF ACTION

- Keeps the positive neurotransmitters present in the brain by blocking monoamine oxidase, which is known to REMOVE these from the brain



THERAPEUTIC USES

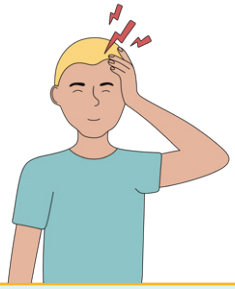
- Depression & depressive episodes
- Bipolar disorder
- Obsessive-compulsive disorder (OCD)
- Neuropathy & neuropathic pain
- Enuresis (bedwetting)

NURSING CONSIDERATIONS

- SSRIs and TCAs should never be taken with MAOIs
- MAOIs stay in the body even after the dose is stopped
  - Wait 14 days before starting an SSRI

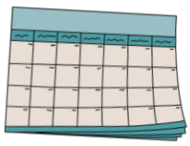
MONITOR FOR

- Signs of hypertensive crisis:
  - Severe headache
  - Dizziness
  - Blurred vision



PATIENT EDUCATION

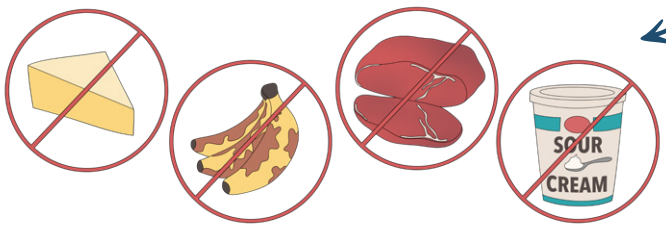
- May take 4 weeks to take effect

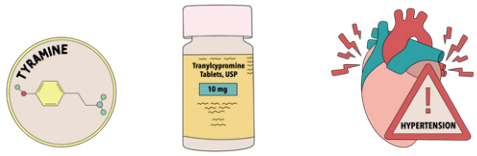


DIET MODIFICATIONS

Restrict tyramine

- X Aged cheeses
- X Sour cream
- X Fermented meats & liver
- X Yogurt
- X Over-ripened fruit





Tyramine + MAOIs = Risk for **HYPERTENSIVE CRISIS!**

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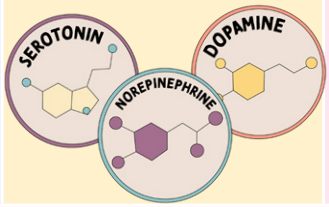
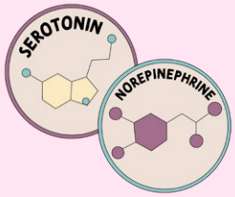
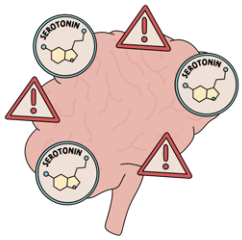
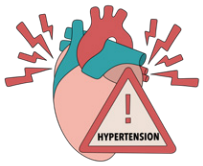
Quick Comparison of Antidepressants

	SSRIs Selective Serotonin Reuptake Inhibitors	SNRIs/DNRIs Serotonin/Norepinephrine & Dopamine/Norepinephrine Reuptake Inhibitors	TCAs Tricyclic Antidepressants	MAOIs Monoamine Oxidase Inhibitors
ACTION	↑ Serotonin 	↑ Serotonin, norepinephrine & dopamine 	↑ Serotonin & norepinephrine 	
USES	Treat major depression & anxiety disorders			
RISK FOR	 **SEROTONIN SYNDROME** (too much serotonin in the brain)			 **HYPERTENSIVE CRISIS**
EDUCATE	▪ Mental changes ▪ Muscle rigidity/tightness ▪ Tremors ▪ ↑ Heart rate (HR) ▪ ↑ Blood pressure (BP) ▪ ↑ Temperature (T)			★ Restrict tyramine ✗ Aged cheeses ✗ Sour cream ✗ Fermented meats & liver ✗ Yogurt ✗ Over-ripened fruit
SUICIDE WARNING	⚠ **All antidepressants have a SUICIDE WARNING** ⚠ A patient who had suicidal plans may now have the energy, due to the medication, to carry out those plans			

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First-Generation Antipsychotics (FGAs)

MEDICATION CLASS & NAME:

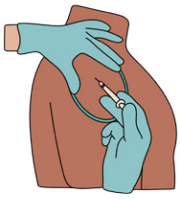
Antipsychotics

Also called ***typical*** or ***conventional antipsychotics***

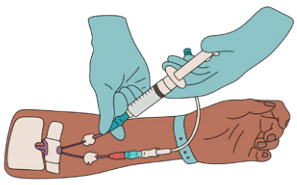
GENERIC	TRADE NAME
chlorpromazine	Thorazine
haloperidol	Haldol
loxapine	Adasuve

Haloperidol

✓ Administered intramuscularly

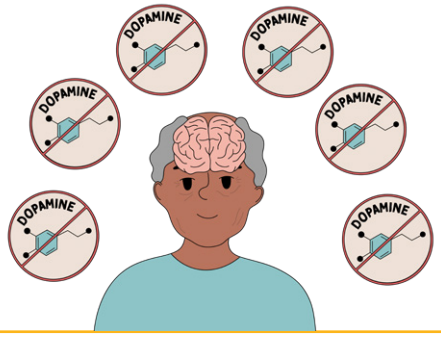


✗ Never administered IV



MECHANISM OF ACTION

- Inhibits (blocks) **dopamine** from being released in the brain, relaxing and reducing excitatory behaviors

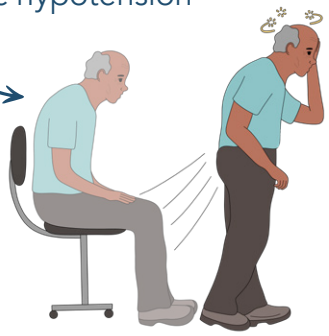


MUST-KNOW SIDE EFFECTS

HIGHER RISK for:

- **Extrapyramidal symptoms (EPS)**
- **Tardive dyskinesia (TD)**
- **Neuroleptic malignant syndrome (NMS)**

- Orthostatic hypotension



PATIENT EDUCATION

EDUCATE

- May take 6–10 weeks to take effect
- Change position SLOWLY to prevent falling
- DO NOT stop taking without clear directions from HCP

THERAPEUTIC USES

- Reduces **positive (excitatory)** symptoms of schizophrenia

NURSING CONSIDERATIONS

MONITOR FOR

- Symptoms of EPS, TD & NMS

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Second-Generation Antipsychotics (SGAs)

MEDICATION CLASS & NAME:

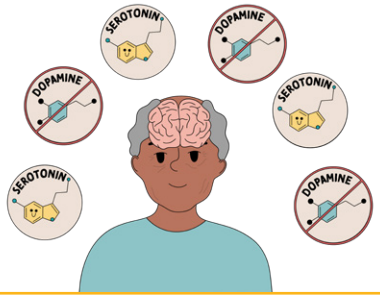
Antipsychotics

Also called ***atypical antipsychotics***

GENERIC	TRADE NAME
risperidone	Risperdal
clozapine	Clozaril
quetiapine	Seroquel
ziprasidone	Geodon
aripiprazole	Abilify

MECHANISM OF ACTION

- Inhibits (blocks) **dopamine** from being released in the brain, relaxing and reducing excitatory behaviors
- ↑ **Serotonin** in the brain



PATIENT EDUCATION

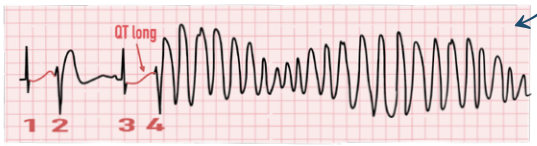
EDUCATE

- May take 6–10 weeks to take effect
- To ↓ the risk of gaining weight:
  - Exercise regularly
  - Follow a low-calorie diet

Educate on the importance of compliance!

MUST-KNOW SIDE EFFECTS

- Lower risk of TD, EPS & NMS than FGAs
- ↑ Weight
- ↑ Cholesterol
- ↑ Triglycerides
- ↑ Blood sugar
- Anticholinergic effects
- Photosensitivity
- Sedation/lethargy
- **Clozapine** may cause agranulocytosis (↑ risk for infection)
- **Ziprasidone** may cause prolonged QT intervals (can lead to Torsades de Pointes)



THERAPEUTIC USES

- Helps diminish **positive & negative** symptoms of schizophrenia

NURSING CONSIDERATIONS

MONITOR FOR

- Weight increase
- Labs:
  - Blood sugar
  - LDL
  - Triglycerides

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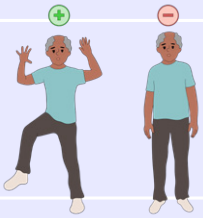
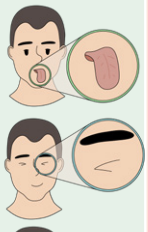
Quick Comparison of FGAs & SGAs

	FIRST-GENERATION ANTIPSYCHOTICS (FGAs) Also called *typical* or *conventional antipsychotics*	SECOND-GENERATION ANTIPSYCHOTICS (SGAs) Also called *atypical antipsychotics*
EXAMPLES	▪ haloperidol (Haldol) ▪ chlorpromazine	▪ risperidone (Risperdal) ▪ clozapine (Clozaril) ▪ ziprasidone (Geodon)
MECHANISM OF ACTION	D2 antagonists	D2 & 5-HT2A antagonists
USES	Helps diminish **positive** symptoms of schizophrenia	Helps diminish **positive & negative** symptoms of schizophrenia 
SIDE EFFECTS	**Higher risk** of developing TD, EPS & NMS	**Lower risk** of developing TD, EPS & NMS

TD, EPS & NMS

These are serious complications of antipsychotic medications

	TARDIVE DYSKINESIA (TD)	EXTRAPYRAMIDAL SYMPTOMS (EPS)	NEUROLEPTIC MALIGNANT SYNDROME (NMS)
SYMPTOMS	Involuntary movements of the face, tongue, or limbs that may be irreversible  **Examples:** ▪ Lip-smacking ▪ Tongue protrusion ▪ Excessive blinking ▪ Puffing of cheeks	Parkinson's-like symptoms + Akathisia (restlessness) + Dystonia (muscle twitching)	EPS + High fever + Autonomic disturbance
TREATMENT	Withdrawal and safe tapering of the antipsychotic medications **Medications:** ▪ Ingrezza ▪ Austedo	**Medications:** ▪ Anticholinergics ▪ Dopamine agonists (amantadine) ▪ Benzodiazepines ▪ To calm the restlessness	▪ Treatment of the fever ▪ **Medications:** ▪ Muscle relaxants

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Benzodiazepines

MEDICATION CLASS & NAME:

Anxiolytic (Anti-Anxiety)

GENERIC	TRADE NAME
alpra**zolam**	Xanax
lora**zepam**	Ativan
dia**zepam**	Valium
clona**zepam**	Klonopin

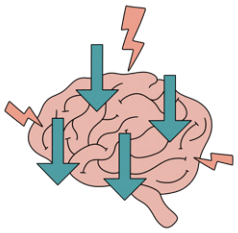
SAVED BY THE SUFFIX

SUFFIXES: -ZOLAM, -ZEPAM

ANTIDOTE: FLUMAZENIL

MEMORY TRICK: I FLU fast in my Mercedes-BENZ

MUST-KNOW SIDE EFFECTS



MEMORY TRICK: BenZO think LOW & SLOW

- ↓ Respiratory rate (bradypnea) → Respiratory depression
- ↓ Blood pressure (hypotension)
- ↓ GI motility (constipation)
- ↓ Energy levels (sedation)
- ↓ Secretions in the mouth (dry mouth)

MECHANISM OF ACTION

- Binds to GABA receptors in the brain, ultimately creating a calming effect by DEPRESSING the CNS

GABA (inhibitory neurotransmitter) slows/calms the activity of the nerves in the brain

THERAPEUTIC USES

- Anxiety
- Panic disorders
- Seizure activity
- Decreasing muscle tension
  - For those with muscle spasms & musculoskeletal pain
- Lorazepam (Ativan) is used for alcohol (ETOH) withdrawal

Benzodiazepines are also referred to as "muscle relaxants"

MEMORY TRICK: Ativan think Alcohol withdrawal

PATIENT EDUCATION

EDUCATE

- Change positions slowly
  - Can cause dizziness or falls
- Take at bedtime

MEMORY TRICK: BEnzos think BEdtime

- To help with constipation:

MEMORY TRICK: Fruits, Fiber & Fluids Fill up the toilet

DO NOT

- ✗ Drive or operate heavy machinery
  - May take several weeks for the body to adapt
- ✗ Drink alcohol
- ✗ Take illicit drugs
- ✗ Take any unprescribed CNS depressants

Can lead to sedation or respiratory depression

- ✗ Abruptly stop the medication
  - Must be tapered (decrease dose gradually)
- ✗ Take as a long-term treatment
  - Leads to **tolerance** and/or **dependency**

TOLERANCE
Larger doses of the drug are required to achieve the desired outcome

DEPENDENCY
The body goes through "withdrawals" & experiences negative effects when the medication is STOPPED

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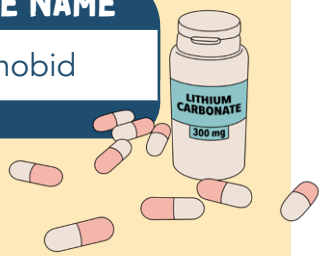
Lithium

MEDICATION CLASS & NAME:

Mood Stabilizer

GENERIC	TRADE NAME
lithium carbonate	Lithobid



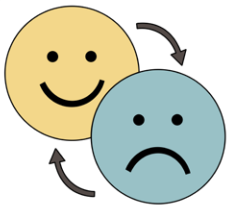
MEMORY TRICK

Lithium think
Long-term management

THERAPEUTIC USES

Manages & treats bipolar disorder

Regulates "mood swings" (depression & mania)



THERAPEUTIC RANGE

✓ **THERAPEUTIC RANGE**
0.6 - 1.2 mEq/L

⚠ **LITHIUM TOXICITY**
> 1.5 mEq/L

Lithium is known for its narrow therapeutic range

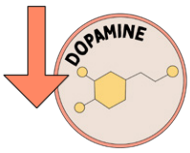
MECHANISM OF ACTION

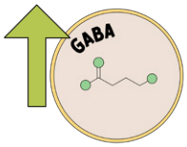
Alters the release of chemicals in the **brain** to CONTROL mood & expression

↓ Dopamine

↑ GABA

↑ GABA = calming effect on the body





PATIENT EDUCATION

DOs

✓ Hydrate
- 2,000–3,000 mL of fluids each day

✓ Go for routine monitoring of serum lithium levels
- Every 2–3 weeks until therapeutic range is achieved
- Every 2–3 months after that

DON'Ts

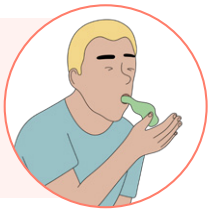
✗ Restrict sodium
✗ Restrict water
✗ Stop medication abruptly

TOXICITY

EXPECTED SYMPTOMS *not a sign of toxicity
- Dry mouth

EARLY SYMPTOMS

GASTROINTESTINAL
- Nausea
- Vomiting
- Diarrhea



LATER SYMPTOMS

NEUROLOGICAL
- Confusion
- Neuromuscular excitability
- Uncoordinated movements (ataxia)
- Agitation

Reasons TOXICITY OCCURS:

4 Ds of toxicity:

D Dehydration
Lithium is a salt & needs to be dissolved in water

D Diet low in sodium

D Decreased renal function

D Drug interactions
(NSAIDs or thiazide diuretics)

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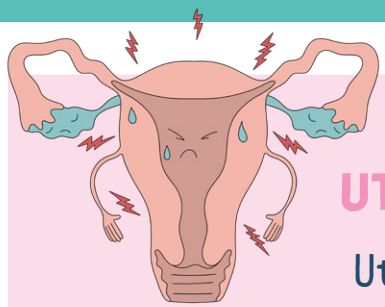
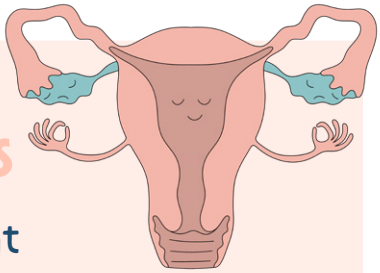
Drugs Given During Labor



UTEROTONICS VS. TOCOLYTICS



UTEROTONICS — Uterine stimulant

TOCOLYTICS — Uterine relaxant

Used to induce labor

EFFECTS:
- ↑ Contractions
- ↑ Muscle tone in the uterus

Used to slow or delay labor

MEMORY TRICK: Tocolytics think let's Talk about it first before I deliver this baby

		USES	NURSING CONSIDERATIONS	MEMORY TRICK
UTEROTONICS	**MISOPROSTOL** (Cytotec)	• Used to ripen the cervix and induces labor	**CONTRAINDICATIONS:** • History of uterine surgery (cesarean birth) • Abnormal fetal heart rate • Uterine tachysystole: > 5 contractions in 10 min	Misoprostol can cause a Miscarriage Misoprostol is also used to prevent stomach ulcers. Pregnant women should not take this medication unless they are ready to go into labor.
UTEROTONICS	**OXYTOCIN** (Pitocin)	**DO NOT ADM. MISOPROSTOL & OXYTOCIN TOGETHER!** • Induces labor & stimulates uterine contractions • Promotes delivery of the placenta • Treats postpartum hemorrhage (PPH)	• D/C if contractions last > 60 sec • D/C if the frequency of the contractions is more than 2–3 minutes apart • D/C if abnormal FHR patterns develop • Normal FHR: 110–160 bpm • Piggyback oxytocin into the main IV fluid • Can cause painful contractions & uterine rupture	OxyTOcin think TO contract
TOCOLYTICS	**TERBUTALINE** (Brethine)	HALTS uterine contractions (delays labor) • Used to prevent preterm labor	• May be given for 48 hours to suppress preterm labor • Position mother on side, not back **MONITOR** contraction frequency & duration	Terbutaline think Turbulence Turbulence delays airplane arrival times. Same goes for labor—Terbutaline delays labor!
TOCOLYTICS	**NIFEDIPINE** (Procardia XL)	HALTS uterine contractions (delays labor) • Used to prevent preterm labor	**MONITOR** • For orthostatic hypotension • Contraction frequency & duration	Nifedipine think No more contractions
TOCOLYTICS	**MAGNESIUM SULFATE**	• Prevents & controls seizures in mothers with preeclampsia/eclampsia • Used to prevent preterm labor	• Always administer by IV infusion via an infusion monitoring device **MONITOR** • For signs & symptoms of magnesium toxicity Respiratory depression (< 12 bpm), ↓ deep tendon reflex, ↓ urine output (< 30 mL/hr) Stop the infusion & give calcium gluconate!	Magnesium Sulfate is for Moms at risk for Seizures (preeclampsia)

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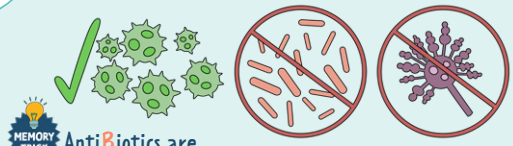
Antibiotics Overview

Saved by the Suffixes!

There are so many types of antibiotics that it can be very hard to remember all the medication names! Remember, suffixes and prefixes will really help you!

MEDICATION CLASS	SUFFIX/PREFIX	EXAMPLE
Tetracyclines	-CYCLINE	Doxycycline
Sulfonamides	SULF-	Trimethoprim/ sulfamethoxazole
Cephalosporins	CEF- CEPH-	Cefadroxil Cephalexin
Aminoglycosides & macrolides	-MYCIN -MICIN	Azithromycin gentamicin
Fluoroquinolones	-FLOXACIN	Ciprofloxacin

Antibiotics are only used for treatment or prevention (prophylaxis) of bacterial infections, not viral or fungal infections



MEMORY TRICK: AntiBiotics are used for Bacterial infections ONLY!

Viral infections

Fungal infections

Take a daily probiotic

Keeps the gut balanced with healthy bacteria and helps reduce negative symptoms

Remember: antibiotics reduce healthy gut flora and kill beneficial microbes.

Finish the entire prescription of antibiotics
(even if the patient is feeling better)

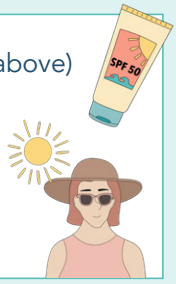
This is to prevent a superinfection. Antibiotics disrupt the body's "normal flora," which can cause a super infection (secondary infection).

Some antibiotics cause photosensitivity

MEMORY TRICK: "Fear The Sun"
Fluoroquinolones, Tetracyclines, Sulfa drugs

PATIENT EDUCATION:

- Wear sunscreen (SPF 30 or above) even if cloudy
- Avoid direct sun exposure
- Protect skin & eyes with clothing, sunglasses & hats



Some antibiotics make oral contraceptives ineffective
(use additional contraception)

MEMORY TRICK: Penicillin bumps the Pill
TetraCyclines & Cephalosporins require Child care

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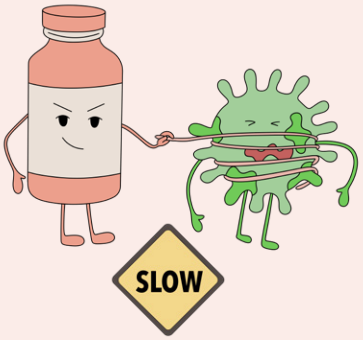
Antibiotics Overview

Antibiotics are either...

Bacteriostatic



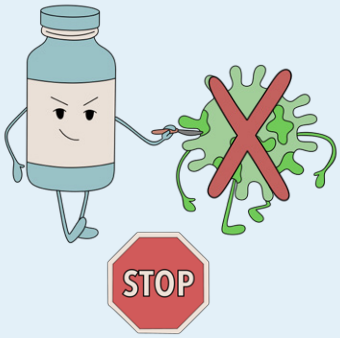
Antibiotics that are bacteriostatic **SLOW** or prevent the growth of bacteria

MEMORY TRICK: Bacterio**S**tatic think **S**low down

EXAMPLES
- **T**etracyclines
- **M**acrolide antibiotics
- **S**ulfonamides (sulfa antibiotics)
- **E**rythromycin

MEMORY TRICK: **T**hese **M**edications **S**low **E**verything

Bactericidal



Antibiotics that are bactericidal **KILL** bacteria

MEMORY TRICK: Bactericidal think "-cide" or "-cidal" means "killer"

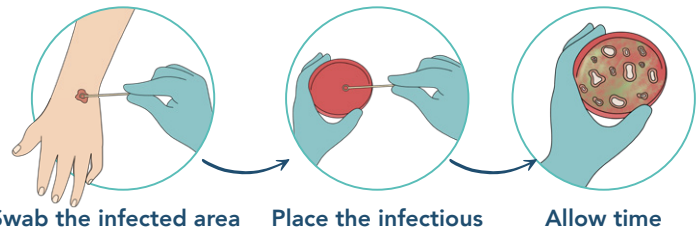
EXAMPLES
- **A**minoglycosides
- **P**enicillin
- **C**ephalosporins
- **F**luoroquinolones
- **V**ancomycin

MEMORY TRICK: **A**ntibiotic **P**ills **C**rush **F**ierce **V**illains

Culture

MEMORY TRICK: **C**ulture think **C**ollecting data

Identify the type of germs/infection

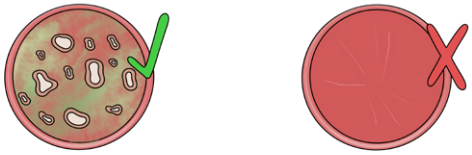


Swab the infected area for a sample
(Skin or body fluid such as blood, urine, saliva, vaginal secretions, tissue)

Place the infectious material on the culture plate
(The plate has a special growing medium that makes the bacteria grow quickly)

Allow time for bacteria to grow

INTERPRETATION



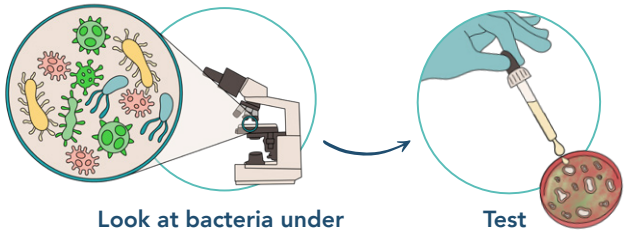
Positive culture
(germ growth present)

Negative culture
(no germ growth)

Sensitivity

MEMORY TRICK: **S**ensitivity think **S**earching for the correct medication

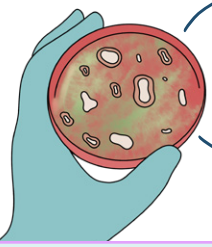
Identify what type of medication needs to be given to treat/eradicate the infection



Look at bacteria under a microscope

Test different antibiotics to see what will work best for that specific infection

INTERPRETATION



Areas that have no growth
=
bacteria are sensitive to that antibiotic, so we can use it to fight the infection

Areas that have growth
=
bacteria are NOT sensitive to that antibiotic, so it will not be effective

Always obtain a culture & sensitivity test BEFORE administering an antibiotic

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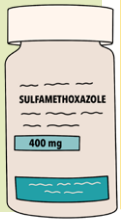
Sulfonamides & Fluoroquinolones

Sulfonamides (SULFA DRUGS)

MEDICATION CLASS & NAME:

Antibiotics

GENERIC	TRADE NAME
sulfadiazine	–
trimethoprim/ **sulfa**methoxazole	Bactrim
sulfasalazine	Azulfidine

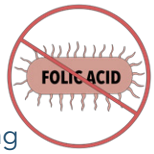


SAVED BY THE SUFFIX

PREFIX: SULFA-

MECHANISM OF ACTION

BACTERIOSTATIC



Sulfas inhibit folic acid production by the bacteria thereby slowing the growth of the bacteria

Bacteria needs folic acid to continue growing

THERAPEUTIC USES

- Urinary tract infections (UTIs)
- Acute otitis media
- Ulcerative colitis
- Topical: used for burn wounds

MUST-KNOW SIDE EFFECTS

MEMORY TRICK **THE Ss OF SULFAS:**

Sore **S**tomach
- Nausea, vomiting, diarrhea, loss of appetite & abdominal tenderness

Sunburn (photosensitivity)

Stones/crystalluria
- Sulfas = dehydration = crystal formation = kidney stones

Stevens-Johnson syndrome (SJS)
- Red, blistering rash
- Flu-like symptoms

PATIENT EDUCATION

STOP the medication if a rash, skin lesion, or fever occurs → Symptoms of SJS

- To prevent sunburn:
  - Wear sunscreen (SPF 30 or above) even if cloudy
  - Avoid direct sun exposure
  - Protect skin & eyes with clothing, sunglasses & hats

DIET MODIFICATIONS

↑ Fluids

Take folic acid daily

CROSS SENSITIVITY
- Assess for allergies to:
  - **Sulfa** drugs
  - **Sulf**onylurea medications
    - Glyburide
  - **Sulfa** derivatives:
    - Diuretics (thiazide & loop diuretics such as furosemide)

Fluoroquinolones

MEDICATION CLASS & NAME:

Antibiotics

GENERIC	TRADE NAME
levo**floxacin**	–
cipro**floxacin**	Cipro
gemi**floxacin**	Factive
o**floxacin**	–

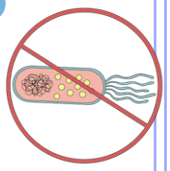
SAVED BY THE SUFFIX

SUFFIX: -FLOXACIN

MEMORY TRICK Fluoroquinolones end in Floxacin

MECHANISM OF ACTION

BACTERICIDAL



Obstructs bacterial DNA reproduction

Prevents the cells from multiplying

THERAPEUTIC USES

- Lower respiratory infections
- Bone & joint infections
- Urinary tract infections (UTIs)
- Sexually transmitted infections (STIs)
- Skin & eye infections

MUST-KNOW SIDE EFFECTS

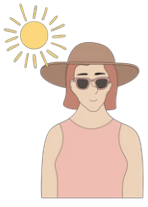
Tendonitis
- Tendon **rupture** may occur

MEMORY TRICK Your Achilles Tendon (Tendonitis) is near the Floor & can rupture due to FLUOROquinolones

- Crystalluria
  - Dehydration = crystal formation = kidney stones
- Photosensitivity

PATIENT EDUCATION

- To prevent sunburn:
  - Wear sunscreen (SPF 30 or above) even if cloudy
  - Avoid direct sun exposure
  - Protect skin & eyes with clothing, sunglasses & hats



DIET MODIFICATIONS

↑ Fluids

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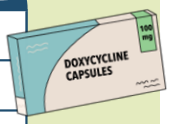
Tetracyclines & Aminoglycosides

Tetracyclines

MEDICATION CLASS & NAME:

Antibiotics

GENERIC	TRADE NAME
tetra**cycline**	–
doxy**cycline**	Acticlate
mino**cycline**	Minocin
demeclo**cycline**	Declomycin



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SUFFIX: -CYCLINE

MECHANISM OF ACTION

BACTERIOSTATIC

Inhibits (blocks) growth & replication of new bacteria by disturbing their ability to bind to cells

THERAPEUTIC USES

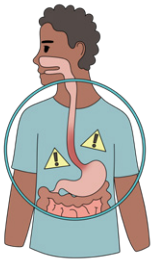
- For the skin:
  - Soft tissue infection
  - Severe acne
- Rocky Mountain spotted fever
- Helicobacter pylori (H. pylori)

MUST-KNOW SIDE EFFECTS

- Photosensitivity
- Pill-induced esophagitis
  - Heartburn
  - Scarring of the esophagus

MEMORY TRICK

TEtracyclines are Terrible for the Esophagus



PATIENT EDUCATION

PREVENTION

To prevent esophagitis:

- ✓ Sit upright for 30 minutes
- ✓ Take with a glass of water
- ✗ Do not lie down or go to sleep immediately after taking

- Avoid direct sun exposure
- Use additional contraception
  - Tetracyclines make oral contraceptives ineffective

MEMORY TRICK

TetraCyclines require Childcare

⚠ CONTRAINDICATIONS ⚠

✗ **Kids under 8 years old**

- Causes tooth discoloration

MEMORY TRICK Tetra think Teeth

✗ **During pregnancy**

- Impairs bone mineralization & can cause permanent tooth discoloration in the developing fetus

MEMORY TRICK Tetra think Toxic to developing fetus

Aminoglycosides

MEDICATION CLASS & NAME:

Antibiotics

GENERIC
genta**micin**
kana**mycin**
neo**mycin**
strepto**mycin**

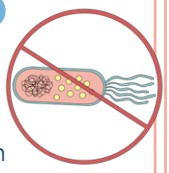
SAVED BY THE SUFFIX

SUFFIXES: -MYCIN, -MICIN

MECHANISM OF ACTION

BACTERICIDAL

Interferes with protein synthesis, stopping bacterial multiplication which leads to bacterial cell death

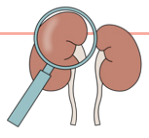


THERAPEUTIC USES

- Aids in bowel preparation
  - ↓ normal flora in the gut for those having abdominal surgery
- Manages hepatic coma
  - ↓ ammonia in the intestines

PATIENT EDUCATION

MONITOR kidney function



- ↑ BUN
- ↑ Creatinine

Impaired kidney function

Urinary Output (UOP)

⚠ NCLEX ALERT

< 30 mL per hour must be reported right away

Give carefully to those with renal failure & to the elderly (as age ↑, GFR ↓)

MUST-KNOW SIDE EFFECTS

MEMORY TRICK

AMINOglycosides are A MEAN antibiotic because they have very harmful side effects

NEPHROTOXICITY

hurts the **kidneys**

- Blood & protein in the urine
- ↑ BUN & creatinine

OTOTOXICITY

hurts the **ears**

- Tinnitus
- Vertigo
- Permanent hearing loss

NEUROTOXICITY

hurts the **brain**

- Numbness
- Convulsions
- Muscular paralysis

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Penicillin & Cephalosporins

Penicillin

MEDICATION CLASS & NAME:

Antibiotics

GENERIC	TRADE NAME
peni**cillin** G	–
peni**cillin** V	–
amoxi**cillin**	–
ampi**cillin**/sulbactam	Unasyn
pipera**cillin**/tazobactam	Zosyn
oxa**cillin**	Bactocill

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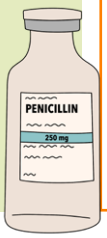
SUFFIX: -CILLIN



MECHANISM OF ACTION

BACTERICIDAL

Is a β-lactam antibiotic, which means it destroys the bacteria cell wall, eventually killing it

THERAPEUTIC USES

- Urinary tract infections (UTIs)
- Septicemia
- Meningitis
- Intra-abdominal infections
- Sexually transmitted infections (STIs)
- Respiratory infections (pneumonia)

PATIENT EDUCATION

- Use additional contraception
  - Penicillin makes oral contraceptives ineffective

MEMORY TRICK Penicillin bumps the Pill

CROSS SENSITIVITY

Ask about allergy to penicillin or cephalosporins before administering the first dose! A patient who is allergic to penicillin may be allergic to cephalosporins.

MUST-KNOW SIDE EFFECTS

- GI upset
  - Nausea, vomiting, diarrhea
- Allergic reaction
  - Urticaria (hives)
  - Pruritus (itching)
  - Skin rash
  - Wheezing & spasms in the airway
  - Hypotension

Many patients don't know if they have a penicillin allergy, so monitor for S&S of allergic reaction after administration

Cephalosporins

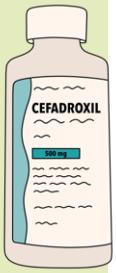
MEDICATION CLASS & NAME:

Antibiotics

GENERIC	TRADE NAME
cefadroxil	-
cefazolin	–
cephalexin	Keflex
cefaclor	Ceclor
cefoxitin	–
cefotetan	**Cef**otan
cefdinir	–
ceftriaxone	–
cefotaxime	–
ceftazidime	Fortaz

SAVED BY THE SUFFIX

PREFIX: CEF-, CEPH-



MECHANISM OF ACTION

BACTERICIDAL

Is a β-lactam antibiotic, which means it destroys the bacteria cell wall, eventually killing it

THERAPEUTIC USES

- Otitis media
- Respiratory infections
- Bone infections
- Urinary tract infections (UTIs)
- Prophylactic measure to prevent infection during surgery

MUST-KNOW SIDE EFFECTS

- Stevens-Johnson syndrome (SJS)
  - Red, blistering rash
  - Flu-like symptoms
- GI upset
  - Nausea, vomiting, diarrhea
- Nephrotoxicity
  - Hurts the kidneys

PATIENT EDUCATION

Do NOT drink alcohol while on this medication

Disulfiram-like reaction may occur: vomiting, sweating, low blood pressure, flushing of the skin

- Use additional contraception
  - Penicillin makes oral contraceptives ineffective

STOP the medication if a rash, skin lesion, or fever occurs → Symptoms of SJS

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Vancomycin

MEDICATION CLASS & NAME:

Glycopeptide Antibiotics

GENERIC	TRADE NAME
vancomycin	Vancocin

MECHANISM OF ACTION

BACTERICIDAL

Inhibits the synthesis of the cell wall & blocks growth

MEMORY TRICK

Vancomycin is a first-line medication because it Vanquishes bacteria the best

THERAPEUTIC USES

Strong antibiotic for strong infections

- Methicillin-resistant Staphylococcus aureus (MRSA)
- Clostridium difficile (C. diff)
- Infections of skin or bone

MUST-KNOW SIDE EFFECTS

Vancomycin flushing syndrome

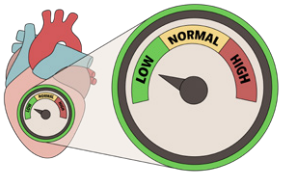
Previously known as:
- *Red man syndrome*
- *Red neck syndrome*

Vancomycin is infused too quickly

Red rash on face, chest, and/or extremities

Flushing & itching

↓ Blood pressure (hypotension)



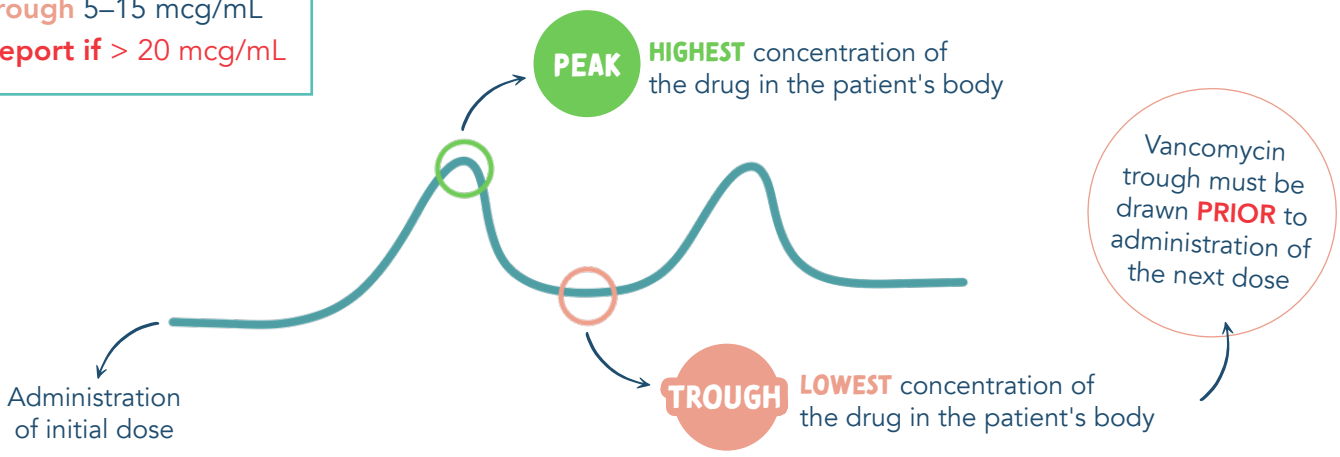
NURSING CONSIDERATIONS

MONITOR
Peak & trough

INFUSE
- **Over 60 minutes**
- Over 100 minutes if infusing > 1 gram

STOP the infusion if any symptoms of vancomycin flushing syndrome occur

Peak 20–40 mcg/mL
Trough 5–15 mcg/mL
Report if > 20 mcg/mL



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Macrolides

MEDICATION CLASS & NAME:

Antibiotics

GENERIC	TRADE NAME
azithro**mycin**	Zithromax
clarithro**mycin**	–
erythro**mycin**	Ery-tab

SAVED BY THE SUFFIX

SUFFIXES: -MYCIN, -MICIN

MECHANISM OF ACTION

BACTERIOSTATIC

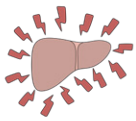
Blocks multiplication of bacteria by cutting off the transportation tract needed for protein synthesis

THERAPEUTIC USES

- Clostridium difficile (C. diff)
- Skin infections
- Upper respiratory infections caused by Haemophilus influenzae
- Ear and eye infections

MUST-KNOW SIDE EFFECTS

- Prolonged QT interval/ cardiac arrhythmia
- Hepatotoxicity



NURSING CONSIDERATIONS

MONITOR liver function

Signs of liver damage:

Jaundice | Itchy skin | Pale stools | Liver function tests (LFTs) will show ↑ AST & ↑ ALT

Drugs with the same suffixes: -MYCIN -MICIN

All these medications end in "**-mycin**" or "**-micin**," so don't get them mixed up!

MACROLIDES

Antibiotics

GENERIC	TRADE NAME
azithro**mycin**	Zithromax
clarithro**mycin**	–
erythro**mycin**	Ery-tab

AMINOGLYCOSIDES

Antibiotics

GENERIC
genta**micin**
kana**mycin**
neo**mycin**
strepto**mycin**

GLYCOPEPTIDES

Antibiotics

GENERIC	TRADE NAME
vanco**mycin**	Vancocin

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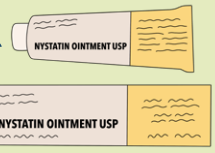
Nystatin

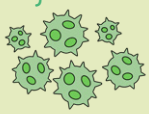
Nystatin

MEDICATION NAME & CLASS:

GENERIC	TRADE NAME
nystatin (topical)	Nystop
nystatin (oral)	–

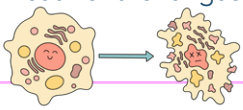


Antifungals
- against
- fungal infections



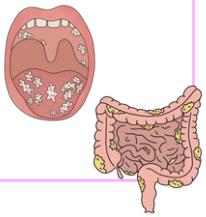
MECHANISM OF ACTION

Creates holes in the cell, which causes infectious content to leak out
=
Death of the fungus



THERAPEUTIC USES

Eliminates Candida (thrush) in the...
- Mouth
- Intestines
- Vagina
- Skin



MUST-KNOW SIDE EFFECTS

BY MOUTH (PO)
- GI upset
  - Nausea
  - Vomiting
  - Diarrhea

LIQUID SUSPENSION
- Irritation to the inside of the mouth

PATIENT EDUCATION

Take the medication every day & avoid missing doses
- Even if symptoms subside

Remove dentures & soak them in liquid suspension
- Dentures harbor bacteria

For liquid suspension:
- Shake bottle well
- Swish medication around in mouth for several minutes & swallow the medication

Helps treat any esophageal Candida

Drugs with the same suffix: -STATIN

All these medications end in "**-statin**," so don't get them mixed up!

ANTIHYPERLIPIDEMIC MEDICATIONS

Hmg-CoA Reductase Inhibitors

GENERIC	TRADE NAME
atorva**statin**	Lipitor
fluva**statin**	Lescol
lova**statin**	Mevacor
pitava**statin**	Livalo
simva**statin**	Zocor
rosuva**statin**	Crestor
prava**statin**	Pravachol

SAVED BY THE SUFFIX

SUFFIX: -STATIN

ANTIFUNGAL MEDICATIONS

Nystatin

GENERIC	TRADE NAME
ny**statin** (topical)	Nystop
ny**statin** (oral)	–

MEMORY TRICK NyStatin is Not a Statin

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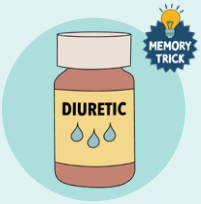
Diuretics: Overview

	LOOP	THIAZIDE	POTASSIUM-SPARING
EXAMPLES	Furosemide (Lasix)	Hydrochlorothiazide (Microzide)	Spironolactone (Aldactone)
IMPACT ON POTASSIUM	MEMORY TRICK Loop think Losing Potassium — Potassium-wasting diuretics		Potassium-sparing diuretic — MEMORY TRICK S think Sparing
MECHANISM OF ACTION	Inhibits reabsorption of sodium & chloride in **3** parts of the kidneys	Inhibits reabsorption of sodium & chloride in **2** parts of the kidneys	Blocks the effects of aldosterone
MONITOR	For **hypo**kalemia (< 3.5 mEq/L)	NORMAL K^+ VALUES: **3.5–5.0** mEq/L	For **hyper**kalemia (> 5.0 mEq/L)



3 Ds OF DIURETICS:

Decrease blood pressure (BP)
- Diuretics cause the body to lose fluids, which causes BP to go DOWN

Diuresis
- Diuretics drain fluids from the body by making the patient urinate more

MEMORY TRICK DIURESIS
Diuretics = Diuresis = Dry Inside

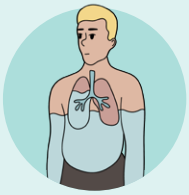
Dehydrate
- Since the body is being flushed of fluid, it becomes dry inside, leading to dehydration

TO PREVENT ORTHOSTATIC HYPOTENSION

- Change positions slowly
- Sit on the side of the bed for a few minutes before standing

MONITOR THE PATIENT'S WEIGHT

- Same time
- Same scale
- Same clothes
- Before breakfast
- After voiding



HEART FAILURE (HF)

- Diuretics are the most commonly used drug for HF
- Remember that HF patients have too much fluid backed up by the heart. This causes edema, weight gain, fluid in the lungs, and increased abdominal girth. Diuretics help to pull off some of this excess fluid.

TO HELP WITH CONSTIPATION

- ↑ Fibers & fruits

★ Don't increase fluids
- Increasing fluid intake in a patient with heart failure is contraindicated due to the risk for fluid volume overload & increased workload on the heart

GIVE DIURETICS IN THE MORNING, NOT AT NIGHT

MEMORY TRICK Diuretics think take during Daytime

This helps the patient avoid peeing all night (nocturia)

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Types of Diuretics

Loop Diuretics

Potassium-wasting diuretics

GENERIC	TRADE NAME
furose**mide**	Lasix
bumeta**nide**	Burinex
tors**emide**	Demadex

SAVED BY THE SUFFIX

SUFFIX: -MIDE, -NIDE

Potent diuretic

MECHANISM OF ACTION

Loop diuretics INHIBIT reabsorption of sodium & chloride in **3 parts** of the kidneys (distal tubules, proximal tubules & the loop of Henle)

Most potent type of diuretic because they inhibit reabsorption in **3** parts

THERAPEUTIC USES

- Hypertension
- Heart failure (HF)
- Renal disease
- Peripheral edema & pulmonary edema

MUST-KNOW SIDE EFFECTS

- **Hypokalemia**
- Hypotension
- Dehydration
- Hyperglycemia
- Photosensitivity
- Hyponatremia

NURSING CONSIDERATIONS

MONITOR potassium levels

- Watch out for **hypokalemia**

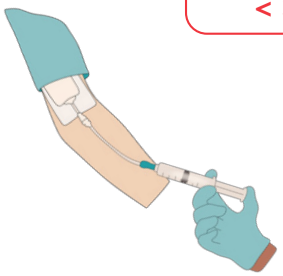
Adm. furosemide SLOWLY (rapid adm. can cause ototoxicity)

MEMORY TRICK

Fast **F**urosemide = **F**uzzy hearing (hearing loss or ringing in the ears)

NORMAL K^+ VALUES: 3.5–5.0 mEq/L

HYPOKALEMIA: < 3.5 mEq/L



Thiazide Diuretics

Potassium-wasting diuretics

GENERIC	TRADE NAME
hydrochloro**thiazide**	Microzide
chloro**thiazide**	Diuril
methyclo**thiazide**	Aquatensen

SAVED BY THE SUFFIX

SUFFIX: -THIAZIDE

MECHANISM OF ACTION

Thiazide diuretics INHIBIT reabsorption of sodium & chloride in **2 parts** of the kidneys (ascending portion of the loop of Henle & the early distal tubule)

THERAPEUTIC USES

- Hypertension
- Heart failure (HF)
- Renal disease
- Cirrhosis
- Peripheral edema & pulmonary edema

MUST-KNOW SIDE EFFECTS

- **Hypokalemia**
- Hypotension
- ↓ Libido
- Dehydration
- Hyperglycemia
- Photosensitivity
- Hyperuricemia
- ↑ Risk for Stevens-Johnson syndrome (SJS)

May cause or worsen gout attacks; do not give to patients with gout

MONITOR for rash

NURSING CONSIDERATIONS

Ask about sulfa allergies

- Patients with a sulfa allergy should avoid thiazide diuretics

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Types of Diuretics

K+ Sparing Diuretics

Potassium-sparing diuretics

GENERIC	TRADE NAME
spironolactone	Aldactone
amiloride	–

MECHANISM OF ACTION

Blocks the effects of **aldosterone** causing sodium & water to be eliminated from the body. Potassium is not eliminated; the body holds on to potassium.

↑ **Aldosterone** = sodium & water **retention**
↓ **Aldosterone** = sodium & water **excretion**

THERAPEUTIC USES

- Hypertension
- Heart failure (HF)
- Edema
- Hyperaldosteronism
- Cross-sex hormonal therapy
- Hypokalemia

MUST-KNOW SIDE EFFECTS

- **Hyperkalemia**
- Constipation
- Dehydration
- Drowsiness
- Erectile dysfunction
- Gynecomastia (enlargement of the breasts in men)

MEMORY TRICK
Spironolactone think Sparing

This usually goes away after therapy has stopped

NURSING CONSIDERATIONS

MONITOR potassium levels
- Watch out for **hyperkalemia**

NORMAL K⁺ VALUES:
3.5–5.0 mEq/L

HYPERKALEMIA:
> 5.0 mEq/L

PATIENT EDUCATION

- Avoid potassium-containing foods & potassium pills

Salt substitutes contain potassium

Osmotic Diuretics

GENERIC	TRADE NAME
mannitol	Osmitrol

MECHANISM OF ACTION

Increases osmotic pressure in the kidneys and, as a result, limits water reabsorption into the bloodstream; water is then eliminated via urine, and solute reabsorption is inhibited

THERAPEUTIC USES

- **Reduces intraocular pressure (IOP) & intracranial pressure (ICP)**
- Treats cerebral edema
- Used as an irrigation solution in prostate surgical procedures
- Promotes diuresis in acute renal failure

MUST-KNOW SIDE EFFECTS

- Blurred vision
- Dizziness
- Nausea/vomiting
- Fluid volume overload
  - Swelling of the body, feet, face
  - ↑ Heart rate (HR)

NURSING CONSIDERATIONS

- Must be given IV
- **Inspect the solution before administering**
  - The solution can **crystallize** when exposed to low temperatures
  - Never put it in the fridge; must be kept at room temperature

MEMORY TRICK
Cold = Crystals

MONITOR FOR
- Neurological status & LOC
- Fluid volume overload
- Worsening heart failure
- Acute weight gain
- Crackles (indicate fluid in the lungs)
- Edema

WHEN NOT TO USE MANNITOL:
- X Severe renal disease
- X Dehydration
- X Intracranial bleeding
- X Severe pulmonary edema
- X Cardiac failure

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Corticosteroids

MEMORY TRICK

The Ss of Steroids

S Suffixes: -sone, -asone, -ide

GENERIC	TRADE NAME
predni**sone**	Rayos
hydrocorti**sone**	Cortef
dexameth**asone**	Decadron
flutica**sone**	Flovent HFA
beclometh**asone**	QVAR
flunisol**ide**	–
ciclesoni**de**	Zetonna

SAVED BY THE SUFFIX

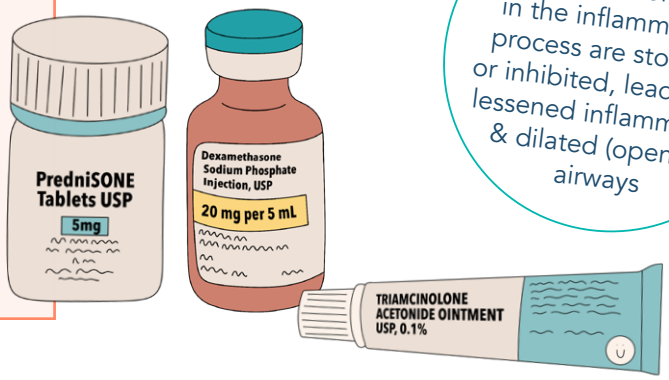
SUFFIX: -SONE, -ASONE, -IDE

MECHANISM OF ACTION

- Decrease inflammation
- S Suppress the immune system

Also called "anti-inflammatory"

Genes involved in the inflammatory process are stopped or inhibited, leading to lessened inflammation & dilated (opened) airways



S Several uses & routes

INHALED CORTICOSTEROIDS (ICS)

- Asthma
- COPD (chronic inflammation)
- Nasal polyps
- Rhinitis

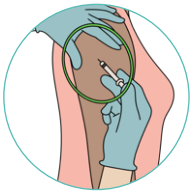
ORAL (PO)

- Systemic inflammation
- Chronic pain
- Autoimmune disease
- Respiratory infections

TOPICAL

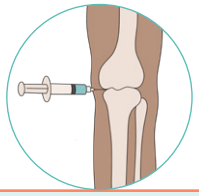
- Eczema
- Contact dermatitis
- Rash (allergic, systemic, generalized)
- Insect bites
- Pruritus (itching)

INJECTION (IV, IM & INTRA-ARTICULAR)



- IV & IM
  - Anaphylaxis & allergic reaction
  - Throat swelling
  - Respiratory infections & pneumonia

IV & IM are often used for rapid administration during medical emergencies



- Intra-articular
  - Chronic joint pain
  - Chronic arthritis
  - After an injury

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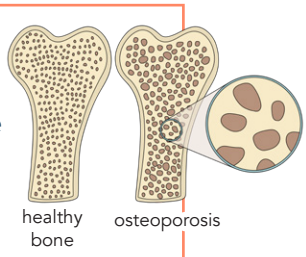
Corticosteroids

Soft bones

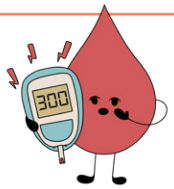
Corticosteroids can cause **osteoporosis**, which is when the body's bones become porous & spongy



EDUCATE
- Increase calcium in the diet

Sugar

Corticosteroids can cause blood sugar levels to spike (**hyperglycemia**)

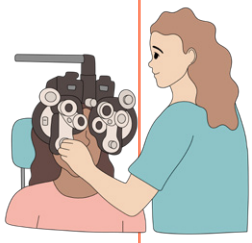


MONITOR
- Glucose levels; sliding scale insulin may be necessary for diabetic patients

Sight

Corticosteroids can cause cataracts

MEMORY TRICK: **C**orticosteroids think **C**ataracts



EDUCATE
- Get an annual eye exam while on these medications

Sick

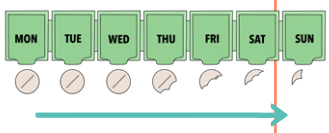
Corticosteroids suppress the immune system, making the patient immunocompromised, which means it's easier for them to become sick or get an infection

Since they are anti-inflammatory, they may hide the fact that the patient has an infection

Stop

DO NOT stop corticosteroids abruptly, as this can cause an **adrenal crisis**

EDUCATE
- Taper this medication when it is time to discontinue use



Stress

When patients are on long-term corticosteroids, the body's normal function & adrenal function are suppressed

EDUCATE
- Manually INCREASE steroid dose in times of stress

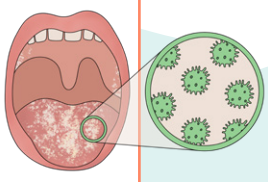
Salt

Corticosteroids can cause water & salt retention (hypertension)

EDUCATE
- This can cause swelling in the cheeks and face with long-term use

Slimy tongue

Inhaled corticosteroids can cause **oral candidiasis (THRUSH)**, which is a fungal infection



This appears as white spots on the tongue & mucus that can resemble thickened milk

EDUCATE
- Use a spacer when using an inhaler, then rinse the mouth & spit afterwards

Sad & Swings

Corticosteroids can cause:
- Mood swings
- Feelings of sadness
- Irritability
- Restlessness & anxiety
- Spikes in energy

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Bronchodilators (SABAs & LABAs)

SABAs
Short-Acting Beta2-Agonists

MEDICATION NAMES

GENERIC	TRADE NAME
albuterol	Proventil HFA
epinephrine	Adrenalin
levalbuterol	Xopenex
terbutaline	–

SAVED BY THE SUFFIX

SUFFIX/PREFIX: -TEROL, TER-

LABAs
Long-Acting Beta2-Agonists

GENERIC	TRADE NAME
salmeterol	Serevent Diskus
arformoterol	Brovana
budesonide/formoterol/ glycopyrrolate	Breztri Aerosphere
olodaterol	Striverdi Respimat

SAVED BY THE SUFFIX

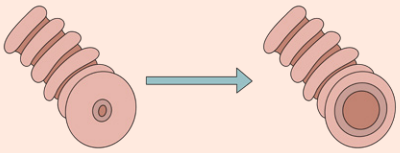
SUFFIX: -TEROL

MEDICATION CLASS

Bronchodilators

↓ Airway

↓ Stretch or enlarge an opening



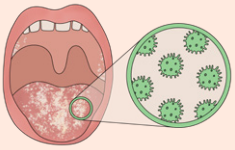
SIDE EFFECTS

Fight or Flight

- ↑ Heart rate > 100 bpm
  - Tachycardia
- ↑ Palpitations
- ↑ Tremors
  - Feeling "jittery"
- ↑ Cardiac arrhythmia
- ↑ Energy = insomnia/anxiety

Activates the sympathetic nervous system!

Thrush
(oral candidiasis)
Fungal infection of the mouth



PREVENTION OF THRUSH:

- Rinse mouth with water after administration
- Use a spacer with an inhaler, which will also help children get the full dose

MEMORY TRICK

Albuterol is for Acute Asthma Attacks

Rescue inhaler

Salmeterol is for Slow and Steady, working a LONG time

THERAPEUTIC USES

Best for **QUICK RELIEF** of **ACUTE** symptoms

- Bronchospasms (treatment)
- Asthma exacerbation

Better for **LONG-TERM** management

- COPD
- Chronic bronchitis
- Bronchospasms (prevention)

PATIENT EDUCATION

- Shake before administration
- Do not exceed 3 doses of 2–4 puffs every 20 minutes

EDUCATE on taking a bronchodilator & a corticosteroid:

MEMORY TRICK **B comes before C in the alphabet** A B C

1 Use Bronchodilator first to help open the airways
2 WAIT 5 minutes
3 Then administer Corticosteroid after airways are **open enough to allow it through!**

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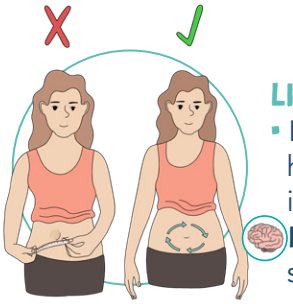
Insulin

MUST BE GIVEN SUBCUTANEOUSLY OR IV

Insulin is destroyed by the GI tract, so it cannot be given by mouth (PO). It must be given subQ or IV



LIPOATROPHY

- Loss of subcutaneous fat, which happens when an injection is made in the same site too many times

EDUCATE your patient to rotate the site 1 inch from the previous site

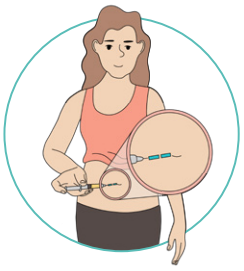
HYPOGLYCEMIA (↓ BLOOD SUGAR)

This is the most common complication of INSULIN, which works to decrease blood sugar but can lead to dangerously low blood sugar levels if given without meals or at an improper dosing

HIGH ALERT MEDICATION

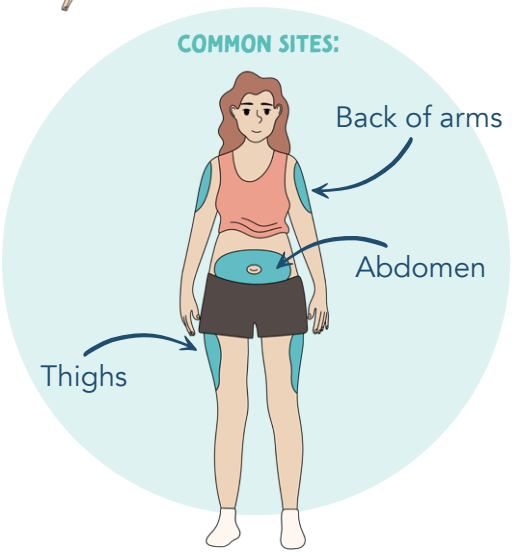
Insulin requires double verification, meaning two nurses should check the insulin order, dose, and time

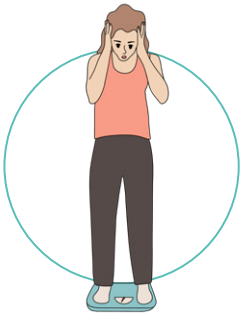
Verification MUST match and be documented prior to administration



ADMINISTRATION

- Remove all air bubbles
- Administer at least 2 inches away from the belly button





WEIGHT GAIN

Insulin is a growth hormone, which may cause weight gain

Sliding Scale

RAPID OR REGULAR INSULIN IS GIVEN ON A SLIDING SCALE

The amount of insulin given is based on the patient's blood glucose measurement

You don't need to memorize this, NCLEX questions and hospital facilities will provide you with a reference range depending on the medication used. You will just need to understand how to use it.

EXAMPLE OF WHAT A SLIDING SCALE INSULIN ORDER LOOKS LIKE:

Blood glucose level (mg/dL)	Mild dose rapid-acting insulin	Moderate dose rapid-acting insulin	Aggressive dose rapid-acting insulin
< 70	Initiate hypoglycemia protocol		
71–149	0 units	0 units	0 units
150–199	0 units	3 units	4 units
200–249	2 units	5 units	6 units
250–299	4 units	7 units	10 units
300–349	8 units	10 units	12 units

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Insulin Types

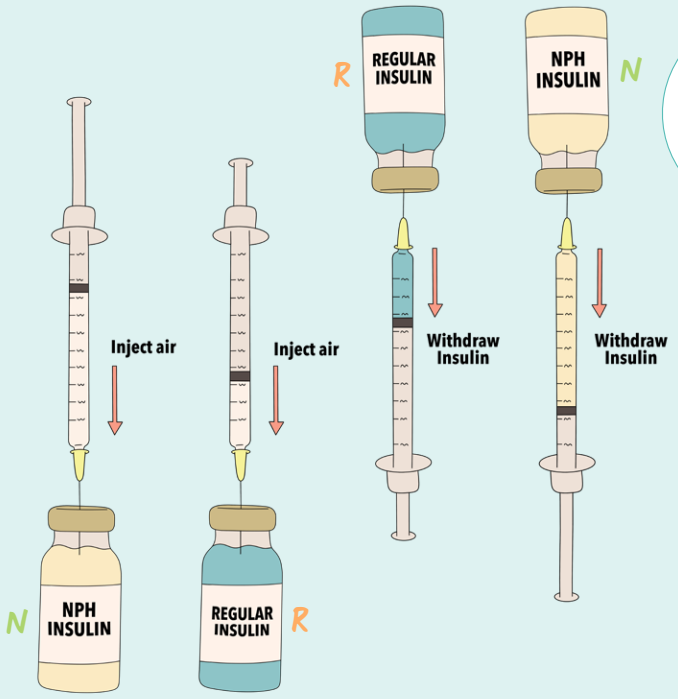
	GENERIC NAMES	TRADE NAMES		
RAPID	lispro	Humalog	ONSET: 5–30 min PEAK: 30–90 min DURATION: 3–5 hrs	Highest risk for hypoglycemia
	aspart	Novolog		
	glulisine	Apidra		
SHORT	regular	Humulin R Novolin R	ONSET: 30–60 min PEAK: 2–4 hrs DURATION: 5–7 hrs	The only form of insulin which can be given IV MEMORY TRICK: Regular goes Right into the vein
INTERMEDIATE	NPH	Humulin N Novolin N	ONSET: 1–2 hrs PEAK: 4–12 hrs DURATION: 18–24 hrs	Never give IV
LONG	glargine	Lantus	ONSET: 3–4 hrs PEAK: none DURATION: 24 hrs+	Lowest risk for hypoglycemia DO NOT MIX WITH ANY OTHER INSULIN MEMORY TRICK: Long think Lonely
	detemir	Levemir	ONSET: 3–4 hrs PEAK: 3–14 hrs DURATION: 6–24 hrs	
	degludec	Tresiba	ONSET: 2 hrs PEAK: 12 hrs DURATION: up to 42 hrs	

Mixing Regular insulin with NPH insulin

NPH = cloudy
REGULAR = clear

MEMORY TRICK

How to remember this order?
"You are Not Retired;
you are an RN"



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Antithyroid Drugs

Methimazole

MEDICATION CLASS & NAME:

First-Line Antithyroid Drug

GENERIC

methimazole

MEMORY TRICK

ME thimazole ME lts away the thyroid hormones

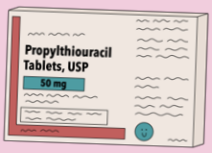
Propylthiouracil (PTU)

MEDICATION CLASS & NAME:

First-Line Antithyroid Drug

GENERIC

propylthiouracil (**PTU**)



MEMORY TRICK

Prevents **T**hyroid from being **U**p

MECHANISM OF ACTION

- Inhibits the production of thyroid hormones
  - Does not affect existing thyroid hormones circulating in the blood or stored in the thyroid gland

THERAPEUTIC USES

- Treats hyperthyroidism
- Treats thyrotoxicosis
- Treats Graves' disease (autoimmune disease that causes hyperthyroidism)
- Used before thyroidectomy surgery (shrinks the thyroid before the surgery)

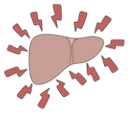
MUST-KNOW SIDE EFFECTS

- Fever
- Skin rash
- Headache
- Nausea & vomiting
- Paresthesia

SYSTEMIC ADVERSE REACTIONS

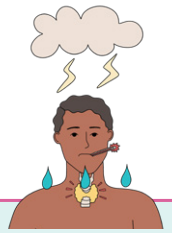
Risk for:

- Agranulocytosis → Increased risk for infection
- Drug-induced hepatitis → Monitor liver values



PATIENT EDUCATION

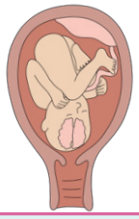
- It may take 1–2 weeks to see the full effect → Educate on the importance of compliance!
- **REPORT** signs of **HYPOTHYROIDISM** (bradycardia, weight gain, lethargy, cold intolerance, depression)
- **REPORT** signs & symptoms of an infection (fever, sore throat)
- Do not abruptly stop the medication (could cause **THYROID STORM**)



PREGNANCY CONSIDERATIONS

- Use with extreme caution during pregnancy because it can cause **hypothyroidism** in the fetus
  - If it's necessary, **PROPYLTHIOURACIL** is the preferred drug (does not cross the placenta)

Thyroid hormone is needed for fetal brain development in utero



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Levothyroxine

MEDICATION CLASS & NAME:

Synthetic Hormones

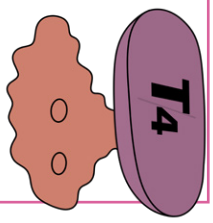
GENERIC	TRADE NAME
levothyroxine	Synthroid

MEMORY TRICK: eLEVates THYroid levels

MEMORY TRICK: SYNthetic THYroid

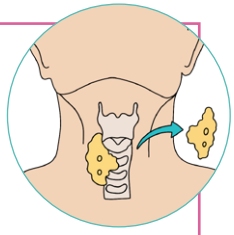
MECHANISM OF ACTION

Levothyroxine replaces or mimics the hormone thyroxine, also called T4



THERAPEUTIC USES

- **Hypo**thyroidism
- Thyroidectomy



Thyroidectomy → removal of → thyroid gland

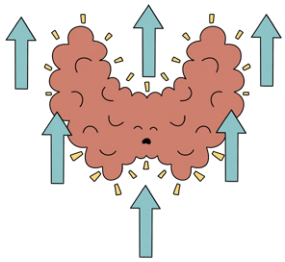
✗ Should not be used as a weight loss regimen

MUST-KNOW SIDE EFFECTS

Everything **INCREASES** ↑

Same symptoms as hyperthyroidism:

- ↑ Energy (nervousness/tremors)
- ↑ Blood pressure
- ↑ Pulse
- ↑ GI function (diarrhea)
- ↑ Metabolism (weight loss)
- ↑ Temperature (hot sensation)



MEMORY TRICK: LEVothyroxine think eLEVated (increased)

THERAPEUTIC RESPONSE

- **No longer shows signs of hypothyroidism**
  - Normal heart rate (60–100 bpm)
  - Improved energy levels (not fatigued)
  - Normal skin (not cool or pale)

PATIENT EDUCATION

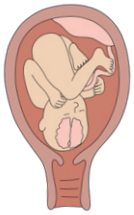
REPORT signs of **HYPERTHYROIDISM** ⚠
- Tachycardia, heart palpitations, weight loss, insomnia, anxiety

- Do not stop the medication even if symptoms resolve
- May take 8 weeks to see the full effect → Educate on the importance of compliance!
- **Take the medication:**
  - Once a day in the morning at the same time every day
  - 30 minutes before a meal, on an empty stomach
  - For the rest of your life

MEMORY TRICK: Levothyroxine think Lifelong therapy

SAFE DURING PREGNANCY? ☑ YES!

Thyroid hormone is needed for fetal brain development in utero



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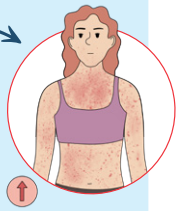
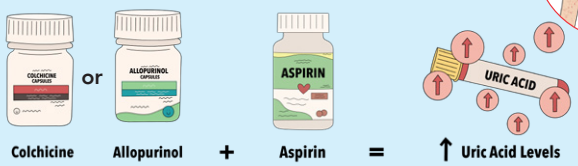
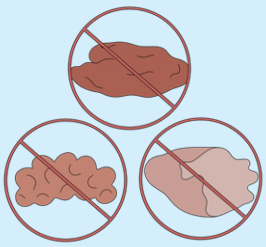
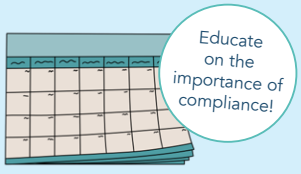
Allopurinol vs. Colchicine

	ALLOPURINOL	COLCHICINE
MEDICATION CLASS	Uric acid inhibitors	Antigout agents
MEDICATION NAMES	**GENERIC:** allopurinol; **TRADE NAME:** Aloprim, Zyloprim, Lopurin	**GENERIC:** colchicine; **TRADE NAME:** Mitigare, Colcrys
THERAPEUTIC USES	**Prevents gout attacks** ▪ Does **not** help with acute attacks MEMORY TRICK → AlloPurinol → Prevents gout	**Relieves acute gout attacks** ▪ Decreases inflammation and pain in acute attacks of gout ▪ Also **prevents** gout attacks MEMORY TRICK → Colchicine → for aCute gout attacks

MUST-KNOW SIDE EFFECTS	▪ GI upset
	  ▪ Nausea, vomiting, abdominal pain, diarrhea
	▪ Skin rash
	
		Same GI side effects!
		▪ GI upset
		  ▪ Nausea, vomiting, abdominal pain, diarrhea
		Adverse reaction:
		▪ Risk for bone marrow suppression

PATIENT EDUCATION	**STOP** the medication if a rash occurs ▪ This may indicate a **hypersensitivity reaction** (Stevens-Johnson syndrome) ▪ Do not take aspirin while on these medications due to ↑ uric acid levels ▪ Instead, take NSAIDs or acetaminophen for acute attacks  	**G** Gulp a lot of fluid during the day (2–3 L/day) Take the medication with a glass of water **O** No Organ meats  **U** ▪ Urine output up to 2 L/day ▪ Uric acid deposits can cause kidney stones **T** Time: These medications take several months to take effect 

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Bisphosphonates vs. Calcitonin-Salmon

MEDICATION CLASS	BISPHOSPHONATES	CALCIUM METABOLISM MODIFIERS

MEDICATION NAMES

GENERIC	TRADE NAME
alen**dronate**	Binosto, Fosamax
pami**dronate**	Aredia
iban**dronate**	Boniva

SAVED BY THE SUFFIX

SUFFIX: -DRONATE

GENERIC	TRADE NAME
calcitonin-salmon	Miacalcin

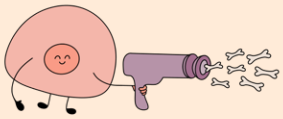
MECHANISM OF ACTION

Slows the activity of osteoclasts

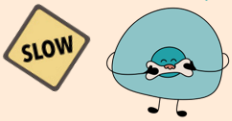
MEMORY TRICK

OsteoBlasts think Build new bone

Osteoblasts
Bone formation



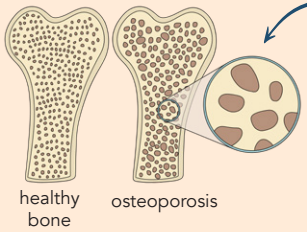
Osteoclasts
Bone resorption



MEMORY TRICK

OsteoClasts think Clear old bone

THERAPEUTIC USES



Treats & prevents osteoporosis
- Commonly postmenopausal osteoporosis
- Treats Paget's disease
- Treats hypercalcemia
  - Too much calcium in the bloodstream

We want the majority of calcium in the bones, not in the bloodstream

Osteoporosis is when the rate of bone resorption (**osteoCLASTS**) is greater than the rate of bone formation (**osteoBLASTS**)
=
↓ Total bone mass

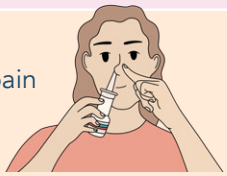
MEMORY TRICK

Bisphosphonate think Builds Bone

CalciTONin helps TONe down calcium levels in the blood

MUST-KNOW SIDE EFFECTS

- GI upset
  - Nausea, diarrhea, dyspepsia, acid reflux, abdominal pain
- Intranasal route
  - Nasal irritation & nasal dryness



PATIENT EDUCATION

PREVENTION

PREVENTION OF OSTEOPOROSIS
- Encourage weight-bearing exercises to preserve bone mass

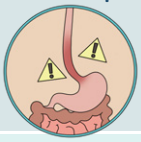
DIET MODIFICATIONS
- ↑ Calcium
- ↑ Vitamin D

PREVENTION

PREVENTION OF PILL-INDUCED ESOPHAGITIS
- Take with a full glass of water on an empty stomach
- Stay upright for 30 minutes

MEMORY TRICK

Bisphosphonates Burn the esophagus



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Notes

Today I will
NOT STRESS over
what I can't
CONTROL.

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Templates & Planners

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HOURLY Planner

MONTH: _____

☐ SUNDAY	☐ MONDAY	☐ TUESDAY	☐ WEDNESDAY	☐ THURSDAY	☐ FRIDAY	☐ SATURDAY
PRIORITIES	PRIORITIES	PRIORITIES	PRIORITIES	PRIORITIES	PRIORITIES	PRIORITIES
6 AM	6 AM	6 AM	6 AM	6 AM	6 AM	6 AM
7 AM	7 AM	7 AM	7 AM	7 AM	7 AM	7 AM
8 AM	8 AM	8 AM	8 AM	8 AM	8 AM	8 AM
9 AM	9 AM	9 AM	9 AM	9 AM	9 AM	9 AM
10 AM	10 AM	10 AM	10 AM	10 AM	10 AM	10 AM
11 AM	11 AM	11 AM	11 AM	11 AM	11 AM	11 AM
12 PM	12 PM	12 PM	12 PM	12 PM	12 PM	12 PM
1 PM	1 PM	1 PM	1 PM	1 PM	1 PM	1 PM
2 PM	2 PM	2 PM	2 PM	2 PM	2 PM	2 PM
3 PM	3 PM	3 PM	3 PM	3 PM	3 PM	3 PM
4 PM	4 PM	4 PM	4 PM	4 PM	4 PM	4 PM
5 PM	5 PM	5 PM	5 PM	5 PM	5 PM	5 PM
6 PM	6 PM	6 PM	6 PM	6 PM	6 PM	6 PM
7 PM	7 PM	7 PM	7 PM	7 PM	7 PM	7 PM
8 PM	8 PM	8 PM	8 PM	8 PM	8 PM	8 PM
9 PM	9 PM	9 PM	9 PM	9 PM	9 PM	9 PM
PRODUCTIVITY METER	PRODUCTIVITY METER	PRODUCTIVITY METER	PRODUCTIVITY METER	PRODUCTIVITY METER	PRODUCTIVITY METER	PRODUCTIVITY METER
BAD GOOD GREAT!	BAD GOOD GREAT!	BAD GOOD GREAT!	BAD GOOD GREAT!	BAD GOOD GREAT!	BAD GOOD GREAT!	BAD GOOD GREAT!

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WEEKLY *Planner*

MONDAY:

TUESDAY:

WEDNESDAY:

THURSDAY:

FRIDAY:

SATURDAY:

SUNDAY:

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NOTES

TESTS / EXAMS

PROJECTS / ASSIGNMENTS

SELF-CARE ♥

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MONTHLY Planner

MONTH: _____ YEAR: _____

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NCLEX Study Schedule

Month: _____

NCLEX Date: _____

SUNDAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:	Subject: Body System: # of Practice Questions: Self Care:
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Course Tracker

COURSE:

SUBMITTED	ASSIGNMENT/PROJECT	DUE DATE	SCORE
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Test / Quiz Tracker

COURSE:

TEST DATE	CHAPTERS / TOPICS COVERED	GRADE	PASSED?
			YES ○ NO ○
			YES ○ NO ○
			YES ○ NO ○
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			YES ○ NO ○
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			YES ○ NO ○
			YES ○ NO ○

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NURSING DIAGNOSIS

SUPPORTING DATA

GOALS

NURSING DIAGNOSIS

SUPPORTING DATA

GOALS

PATIENT INFO

MEDICAL HISTORY

NURSING DIAGNOSIS

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GOALS

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DISEASE:

PATHOLOGY

SIGNS & SYMPTOMS

RISK FACTORS

COMPLICATIONS

DIAGNOSIS

TREATMENT

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Pharmacology Template

DRUG CLASS: _____

GENERIC NAME	TRADE NAME

SUFFIXES OR PREFIXES:

ANTIDOTE:

MECHANISM OF ACTION

THERAPEUTIC USES

SIDE EFFECTS

CONTRAINDICATIONS

NURSING CONSIDERATIONS

PATIENT EDUCATION

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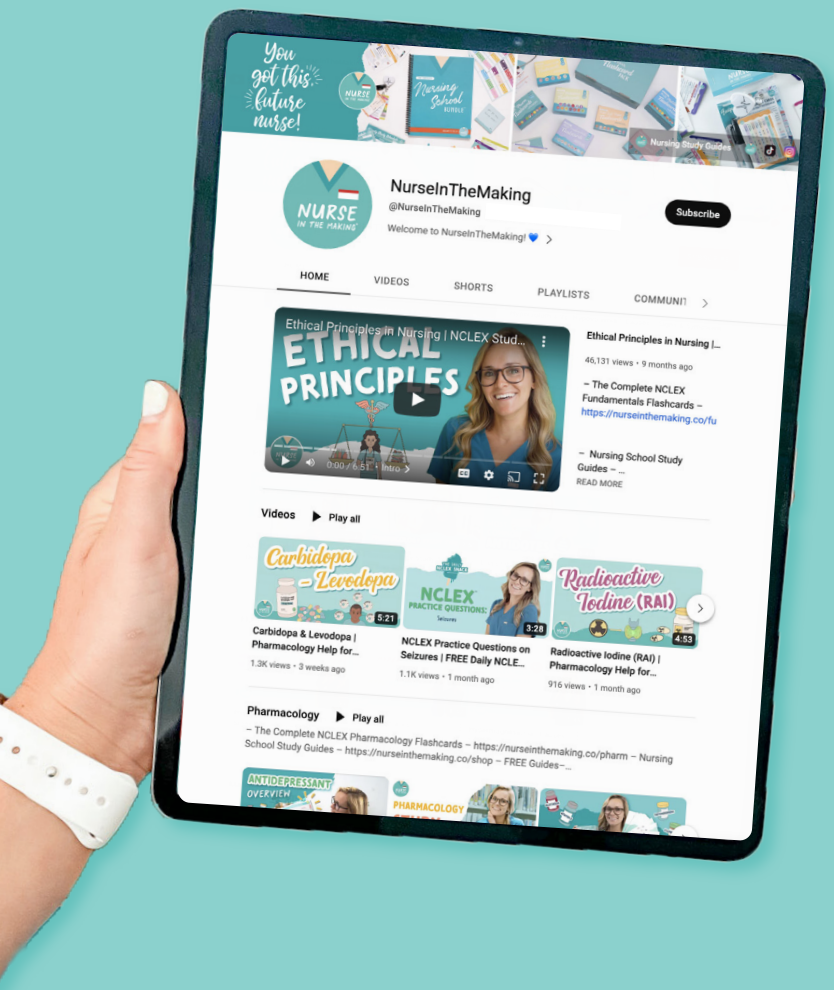

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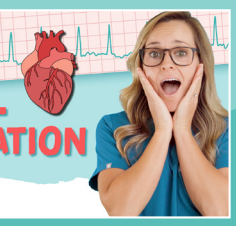

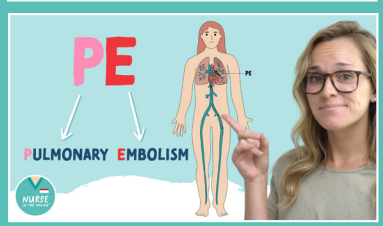

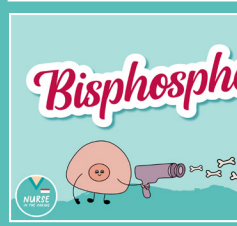

DEAR FUTURE NURSE,

You may be stressed, you may feel tired, and you may want to give up. Nursing school is hard, there's no doubt about it. Everyone cries, everyone has meltdowns, and there will be moments you don't feel qualified for the task at hand. But take heart, the challenge only makes you stronger. Put in the work, show up on time, and find an amazing study group. You got this! ☺

– Kristine Tuttle, BSN, RN

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